

# Automating Induction-Reduction for Concurrent Object Proofs

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A principal challenge in reasoning about concurrent objects is to develop an intuitive correctness argument that nonetheless applies to unboundedly many threads. Two key strategies for tackling this unbounded domain are induction (e.g. over a trace or over threads) and Lipton’s reduction (re-ordering a trace based on commutativity) and recent works are increasingly incorporating both strategies.

In this paper we integrate induction over traces with unbounded left/right reductions at each step, and show how to reduce this combination to standard building blocks: Hoare logic and contextual commutativity, along with a few automaton operations. The upshot is that concurrent object proofs can be organized around a few user-provided scenarios written as “induction-reduction” specifications or “IRSpecs” (which extend to unbounded thread interleaving) and, through new proof rules, they can be automatically verified. We implemented a prototype tool that automatically verifies all of the semantic and syntactic conditions and show that IRSpecs reduce reasoning about difficult concurrent objects (e.g. the Michael-Scott Queue, the Hendler *et al.* Elimination Stack, the Harris *et al.* RDCSS) to a few intuitive canonical executions.

## 1 Introduction

Lock-free concurrent implementations of standard abstract data types, often called “concurrent objects”, leverage efficient hardware primitives for performance. Formalizing their correctness proofs, e.g., establishing the standard linearizability property [22], continues to be challenging and requires substantial effort.

A large and successful body of work tackles such proofs using program logics like Owicki-Gries [38], Rely/Guarantee [23], or extensions of concurrent separation logic [3, 37]. These logics are powerful and highly developed: they enable thread-local reasoning, wherein the effects of threads are captured in state invariants, and provide expressive mechanisms, e.g., additional ghost variables, to track the relative order between steps of different threads when necessary.

In this paper, we pursue a complementary direction, inspired by a different style of reasoning. The informal proofs or intuitive arguments accompanying these implementations (typically in the distributed computing literature) establish global properties—e.g., invariants or trace properties—by explicitly analyzing the orders in which steps of concurrent threads can interleave. Such analysis stays tractable because it silently relies on a reduction argument: only certain “canonical” orderings are considered, while the rest are implicitly treated as equivalent because of commutativity arguments. Consider, for instance, a scenario of the Michael and Scott [34] queue (paraphrasing Herlihy and Shavit [21]): many threads read concurrently, one enqueue takes a read path that finds the tail pointer outdated and then succeeds at a compare-and-swap (CAS), causing the others’ CASs to fail. This description fixes no order among the concurrent reads or among the failed CASs, since these steps commute. It even permits reading the failed CASs as occurring immediately after the successful one: a failed CAS that actually occurs later in the execution can be commuted left, past intervening steps, to directly follow the CAS that caused it to fail.

Proofs based on program logics can also rely on commutativity-based reduction arguments to simplify assertions, but these arguments are mostly limited to a local form of Lipton’s reduction [33], which makes it possible to soundly assume that certain sequences of steps of a single thread execute

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```

int increment() {
    while (true) {
        int c = ctr;
        if (cas(ctr,c,c+1)) return c;
    }
}

int decrement() {
    while (true) {
        int c = ctr;
        if (cas(ctr,c,c-1)) return c;
    }
}

```

Fig. 1. A lock-free concurrent counter; both methods return the previous value of `ctr`. Each runs a retry-loop that reads `ctr` and tries to update it with a compare-and-swap. A `cas(ctr,c,c')` atomically tests whether `ctr` still equals its second argument `c` and, if so, assigns the third argument `c'` and returns `true`; otherwise it leaves `ctr` unchanged and returns `false`, and the method retries.

without interruption (as an atomic section) [12, 17, 19]. In contrast, the informal proofs mentioned above use a more global form of reduction that concerns interleavings between steps of multiple threads (see the example above).

Recent work [13] formalizes such global reductions, introducing a small context-free “quotient language” for describing the set of canonical interleaved executions, e.g., by arranging the actions of any number of threads into a small number of canonical patterns. As a result, the whole set of executions of a concurrent object can be described as a relatively small collection of “scenarios” that more closely resemble the intuitive, operational correctness arguments given by algorithm designers. That work, however, provides no clear path to automation, leaving users to write *ad hoc* proofs.

To make this concrete, consider our running example: the lock-free concurrent counter of Fig. 1, whose `increment` and `decrement` methods each read a shared variable `ctr` and attempt to update it with a CAS, retrying on failure. Modulo commutativity, every interleaved execution of the counter collapses to a canonical sequence of fragments we call *layers*: in each layer, many threads read `ctr` concurrently, then one increment or decrement succeeds its CAS, and immediately afterward all the threads that read `ctr` earlier fail theirs. Reasoning about correctness then reduces to analyzing this *quotient* set of executions rather than every interleaving; for the counter, for instance, linearizability follows because each such layer has the same net effect as a single atomic increment or decrement.

The main contribution of this paper is an automatable *induction-reduction* scheme for establishing the soundness of such quotient sets, i.e., that every execution is equivalent, modulo reorderings of commutative actions, to one in the quotient set. Induction-reduction combines two ingredients: a *reduction*, which rearranges commutative actions to bring an execution closer to the canonical form defining the quotient, and an *induction* over the occurrences of a chosen action, which applies such a rearrangement repeatedly—once for each occurrence. Induction is essential because an execution may contain unboundedly many threads and steps, so these reorderings must be applied an unbounded number of times.

Our scheme realizes this idea through a collection of *Induction-Reduction Specifications* (IRSpecs). Each IRSpec captures a single inductive proof that soundly reorders commutative actions around each occurrence of a chosen induction action, and thus reads as a lemma. Composing several of them—subject to side conditions presented later—builds a stronger, still-sound reduction, i.e., one that yields a smaller set of executions. For the counter, four IRSpecs suffice: two *atomic* IRSpecs, inducting over the successful CASs, group each successful read-CAS-return into an atomic block, and two *cooperative* IRSpecs arrange the reads and failed CASs of the remaining threads around the successful update that caused their failure. Their composition reduces every interleaved execution of the counter to a sequence of layers, i.e., to the quotient above. We develop this example in full in Sec. 2.

**Challenges and Contributions.** Turning induction-reduction into an automated method raises two main challenges: expressing induction-reduction arguments in a form that supports sound composition, and validating each argument automatically, without reasoning directly about unboundedly many threads. Our contributions address these challenges and demonstrate the resulting method:

1. *Induction-Reduction.* (Sec. 5) We define schemas for induction-reduction in the form of IRSpecs. IRSpecs come in two classes: *atomic* (atom) IRSpecs, which group a single thread’s steps into an atomic section, and *cooperative* (coop) IRSpecs, which group the steps of other threads around such a section. We show that a collection of IRSpecs can be *composed* together in a calculus that applies a series of IRSpec reduction steps, taking into account that some steps are to be taken in a specific order (an atom step before the coop steps that build on it). Finally, we define *orthogonality* of IRSpecs to prevent them from “competing” with each other, and *coverage* to ensure enough IRSpecs have been provided to cover all paths of the object’s implementation.

2. *Proof rules.* (Sec. 6) Validating an IRSpec involves reasoning about the behavior of unboundedly many threads. We introduce proof rules for atom and coop IRSpecs that reduce induction-reduction to automation-friendly primitives: Hoare logic, contextual commutativity (commutativity conditioned on a thread’s local context), predicate transformers, and syntactic conditions reducible to finite-state automata operations.

3. *Implementation and user-interactivity.* (Sec. 7) We provide an end-to-end automated prototype QREF, written in OCaml that takes an input concurrent object implementation (with support for arrays and heap-allocated structures) and user-provided IRSpecs. QREF automatically validates (i) all semantic obligations by translating them into programs and constructing queries to the Veracity commutativity verifier [6] and (ii) all syntactic obligations by translating them to automaton queries dispatched to libmata [7]. These checks allow for some interactivity: as a user starts to write candidate IRSpecs QREF will report whether they are incorrect, e.g., pairs of them are not orthogonal or left/right-moverness does not hold. If the user has not provided enough IRSpecs, QREF will provide a path through the implementation that has not yet been covered (which it obtains from a failed coverage check). If left/right moverness does not hold, the user can make paths more fine-grained.

4. *Evaluation.* (Sec. 8) We finally provide an evaluation showing that IRSpecs reduce reasoning to a canonical representation for a variety of notorious concurrent objects (including the Michael-Scott queue [34], Treiber’s Stack [42], the Hendler *et al.* Elimination Stack [20], and the Harris *et al.* Restricted Double-Compare, Single-Swap (RDCSS) [18]). We discuss user-interactivity on the MSQ and report that QREF can validate nearly all of the 3,205 semantic and syntactic conditions from across all those objects’ IRSpecs within a few minutes.

*Supplement.* Included with the supplement is the source code of our prototype tool QREF, which we plan to submit for artifact evaluation and publicly release.

## 2 Overview

We now develop the counter of Fig. 1 in full, showing how four IRSpecs capture its unboundedly many interleavings.

### 2.1 From unbounded interleavings to a layered quotient

The executions of the concurrent counter are interleavings of an arbitrary number of increment and decrement invocations from an arbitrary number of threads. Recall from the introduction that, modulo commutativity, they reduce to a quotient set whose elements are sequences of layers. To formalize this quotient, we represent the counter methods as Kleene Algebra with Tests (KAT)

expressions, with a symbol for each atomic statement and the CAS operation split into two cases depending on success:

$$\left( \begin{array}{l} c := \text{ctr}_R \left( \text{cas}(\text{ctr}, c, c+1)/\text{true}_I \text{ return } c_X + \text{cas}(\text{ctr}, c, c+1)/\text{false}_F \right) \\ c := \text{ctr}_S \left( \text{cas}(\text{ctr}, c, c-1)/\text{true}_D \text{ return } c_Y + \text{cas}(\text{ctr}, c, c-1)/\text{false}_G \right) \end{array} \right)^*$$

Above we have given shorthand names for each symbol such as  $R$  for  $c := \text{ctr}$ . We intentionally distinguish between identical statements occurring in different program locations (e.g.  $R$  versus  $S$ ).

The quotient set can be written as the following KAT expression:

$$((R + S)^*(RIX + SDY)(F + G)^*)^* . \quad (1)$$

Reading this expression from the inside out, each layer  $(R + S)^*(RIX + SDY)(F + G)^*$  begins with a block  $(R + S)^*$  of concurrent reads by increment and decrement threads, followed by exactly one successful operation—a complete increment  $RIX$  or decrement  $SDY$ —and then a block  $(F + G)^*$  of failed CASs from the threads that read  $\text{ctr}$  earlier. The outer Kleene star concatenates such layers into a complete execution.

The principal benefit of quotients is that reasoning about correctness can be reduced to analyzing only executions in the quotient. In the counter example above, proving linearizability reduces to showing that executions in this quotient can be simulated by an *atomic* specification where each method increments or decrements the counter in a single indivisible step. This becomes straightforward since every layer contains exactly one successful CAS—the increment’s  $I$  or the decrement’s  $D$ —which is the only step that modifies  $\text{ctr}$ . The surrounding reads ( $R, S$ ) and failed CASs ( $F, G$ ) leave the counter’s value unchanged, so each layer as a whole has the same effect as a single atomic increment or decrement. The successful CAS thus serves as the linearization point, and the simulation follows by mapping each layer to the corresponding atomic step. Similar correctness arguments for a wide range of objects can be found in [13].

We now continue by showing how to establish the soundness of this quotient—that every interleaved execution is equivalent, modulo commutativity, to one of its layer sequences—through induction-reduction.

Each IRSpec has the form  $(a, \text{atom/coop}(R \triangleright N \triangleleft L))$  and denotes *an induction together with a reduction applied at each inductive step*:

- (i)  $a$  is an *induction action* (or a set of actions, for cooperative IRSpecs) that may be taken by unboundedly many threads; the induction proceeds over its occurrences;
- (ii) the tag *atom* or *coop* fixes the kind of reduction: *atom* forms atomic sections within a single thread, whereas *coop* forms “layers” of optimistic concurrency in which several threads attempt an update but only one succeeds;
- (iii) at each step  $k$  of the induction—the  $k$ th occurrence of  $a$ —the reduction is given by three KAT expressions: an anchor  $N$  of actions that stay in place, actions  $R$  that may move right up to the anchor, and actions  $L$  that may move left up to it.

A *bijection* side-condition ensures that each occurrence of the induction action pins down exactly which step sequences (matching  $N$ ,  $R$ , or  $L$ ) are handled at that step. Separate proof rules state when these reorderings are sound and how IRSpecs compose.

An IRSpec can thus be read as an *operator on sets of traces* that returns a smaller set (i.e., a reduction). For the counter, we use four:

$$\begin{aligned}\alpha_{incS} &\equiv (I, \text{atom}(R \triangleright I \triangleleft X)), \\ \alpha_{decS} &\equiv (D, \text{atom}(S \triangleright D \triangleleft Y)), \\ \alpha_{incF} &\equiv (\{I, D\}, \text{coop}(R \triangleright (RIX + SDY) \triangleleft F)), \text{ and} \\ \alpha_{decF} &\equiv (\{I, D\}, \text{coop}(S \triangleright (RIX + SDY) \triangleleft G)).\end{aligned}$$

The first IRSpec,  $\alpha_{incS}$ , inducts over each occurrence  $I_k$  of an increment's successful CAS operation `cas(ctr, c, c+1)/true`. At step  $k$  it moves the matching read  $R$  right up to the anchor  $I_k$  and the matching return  $X$  left up to it, so that every successful increment's read-CAS-return is grouped together. The second,  $\alpha_{decS}$ , does the same for decrements' successful CASs. These are called atom reductions because they form atomic sections.

What remains are the iterations that *fail* their CAS because another thread succeeded—now grouped as  $RIX$  or  $SDY$ . The coop IRSpec  $\alpha_{incF}$  handles failed increments: inducting over the successful  $I$  (or  $D$ ) in these groups, at step  $k$  it moves each failed thread's read  $R$  right and its failure  $F$  left, up to the anchor  $(RIX)_k$  or  $(SDY)_k$ ; the bijection condition guarantees this is well-defined. Symmetrically,  $\alpha_{decF}$  moves failed decrements' reads  $S$  right and failures  $G$  left, up to each  $(RIX)_k$  or  $(SDY)_k$ . These are called coop reductions because they group one thread's actions around another thread's successful update—the update that causes the failure. Composed together, these four IRSpecs reduce every unbounded, arbitrarily interleaved execution to the form in Eq. (1).

## 2.2 Automating soundness of IRSpecs

The crux of this paper is a set of proof rules for establishing that IRSpecs such as the four above are indeed sound reductions, and for composing them. The rules reduce the combination of induction & reduction to primitives that are amenable to automated verification: Hoare triples, contextual commutativity, predicate transformers, and syntactic conditions on control paths. Roughly, the two main rules have the shape:

$$\begin{array}{c} \text{ATOM} \quad \frac{\alpha \leftrightarrow \text{RNL} \quad \{Q\} \text{N} \{Q'\} \quad R \triangleright Q \quad Q' \triangleleft L \quad \dots}{O[N] \stackrel{\alpha}{\sqsubseteq} O[\text{RNL}]} \\[2ex] \text{COOP} \quad \frac{\alpha^U \leftrightarrow \text{N} \quad Q = \text{sp}(R) \quad Q' = \text{wp}(L) \quad \{Q\} \text{N} \{Q'\} \quad R \triangleright Q \quad Q' \triangleleft L \quad \dots}{O[N] \stackrel{\alpha}{\sqsubseteq} O[\langle\langle R[N] L \rangle\rangle]}\end{array}$$

In the conclusions,  $O[k]$  describes executions in which each control path of  $k$  is run by a single thread as an uninterrupted block, while  $O[\langle\langle R[N] L \rangle\rangle]$  describes the cooperative layers of Eq. (1): several threads run paths of  $R$ , one thread runs the anchor  $N$ , and the former threads continue with paths of  $L$ . A conclusion  $O[r] \stackrel{\alpha}{\sqsubseteq} O[r']$  states that restricting the object's executions to the block structure  $r'$  is a sound reduction of the executions with block structure  $r$ . These rule sketches are for intuition only: they elide several premises as well as the quantification over threads; we provide the formal rules (five in total) and proofs in Sec. 6.

The ATOM rule reduces the process of forming atomic blocks out of successful “happy paths” to Hoare logic and contextual commutativity (along with some syntactic automata conditions discussed below). When this rule is instantiated for  $\alpha_{incS}$ , it proceeds by induction over the occurrences of the successful CAS action  $\alpha = I$ : at each occurrence, the matching read and return are moved next to the anchor, grouping  $RIX$  as an atomic section, thus reducing interleavings, yet maintaining trace-refinement. The Hoare triple  $\{Q\} \text{N} \{Q'\}$  summarizes the effect of the anchor on the local view of the thread executing it, and the commutativity conditions move the read operations (specified

by R) to the right up to context  $Q$ , and the return statement (specified by L) to the left down to context  $Q'$ . For the counter,  $Q$  is  $c = ctr$ , stating that the thread's local copy of the counter is still current when the anchor executes, and  $Q'$  states that  $ctr$  has just been incremented while  $c$  still holds the pre-update value. Note that these are “big-step” movers (see also [16]) in the sense that an action can move to the right/left across multiple actions of other threads *even if* it does not pair-wise commute with those actions, as long as it ultimately commutes with their combined effect, captured by the context  $Q/Q'$ .

The second COOP rule works in tandem with ATOM, and lets us further group the other threads (such as those that fail their CAS in  $\alpha_{incF}$ ) around the previously-grouped successful atomic sections, inducting over the set  $\mathbf{a}^U$  of their induction actions (for the counter  $\mathbf{a}^U = \{I, D\}$ ). In this rule the contexts are not free parameters:  $C$  is the strongest postcondition of the failing thread's prefix R,  $C'$  is the weakest precondition of its fallback L, and the Hoare triple  $\{C\}N\{C'\}$  states that it is precisely the anchor—another thread's successful update—that switches the failing thread from  $C$  to  $C'$ . For the counter,  $C$  is  $c = ctr$  and  $C'$  is  $c \neq ctr$ , i.e., the failing thread's earlier read has become stale. The elided premises additionally require the context  $C$  to be stable under all interference other than the anchors', and the moved paths to commute among themselves. Because COOP groups threads around the atomic sections formed by ATOM, IRSpecs are applied following a precedence order: every cooperative IRSpec is preceded by the atomic IRSpecs that provide its anchors, e.g.,  $\alpha_{incS} \prec \alpha_{incF}$  and  $\alpha_{decS} \prec \alpha_{incF}$  for the counter.

To use these rules, there are three remaining conditions and we show that they are all amenable to syntactic automaton reasoning. First, each IRSpec must be well-formed in the sense that there is a *bijection* between the occurrences of its induction action  $\mathbf{a}$  and the occurrences of the paths it pins down (the pattern RNL of an atomic IRSpec, the anchor N of a cooperative one). The induction action does not need to be exactly that path but rather, intuitively, whenever the induction action occurs, there must inevitably be a corresponding path in the implementation (and vice versa), even if it starts earlier or ends later. Second, IRSpecs should not “compete” with each other so they must be *orthogonal*: a prefix of one IRSpec's pattern should not be a suffix of another's (nor of its own). Nor should one be a KAT-factor of another. Finally, a collection of IRSpecs should have enough *coverage* so that all of the object's behaviors are captured. Below we discuss automation for all of these conditions.

If a collection of IRSpecs satisfies all of these conditions, then its IRSpecs can be *composed* together, following the precedence order. This composition of the IRSpecs *constructs* the quotient in Eq. (1), which, per Theorem 6.4, represents the set of all unbounded interleaved executions of all increment/decrement invocations. This sound, automated derivation of quotient expressions via IRSpecs was not possible in prior work [13]. Moreover, IRSpecs result in a quotient in the form of a starred-union of paths, but can also be further refined and written as a “quotient automaton” [13] in which control states are introduced for ADT abstract states, and IRSpec paths form transitions only when actions are feasible. Finally, IRSpecs resulting quotients drastically reduce interleavings to a canonical form, enabling straight-forward linearizability proofs.

### 2.3 Writing IRSpecs with QREF

We implemented an end-to-end automated prototype QREF<sup>1</sup> that takes a concurrent object implementation written in a C-like language with arrays and heap allocation and an IRSpec for the object, and verifies all semantic conditions (left/right mover, Hoare triples, stability) and all syntactic conditions (coverage, orthogonality, bijection). Semantic conditions are translated into programs and dispatched to the Veracity commutativity verifier [6]. For the syntactic conditions,

<sup>1</sup>Short for “Quotient Refiner.”

QREF constructs a number of automata including RNL path descriptions, suffix/prefix-closure, automata of the forms  $\Sigma^* \cdot A \cdot \Sigma^*$  and  $\Sigma_{\neg A}^* \cdot A \cdot \Sigma_{\neg A}^*$  for bijectivity, and a starred-union of IRSpec paths for coverage. QREF constructs the necessary automaton operations such intersection, emptiness, containment for each of the conditions and invokes libmata [7] to perform the operations and produce counterexamples to coverage.

A user can work with QREF to interactively create the IRSpecs necessary for a concurrent object. A user can begin by trying to define some IRSpecs with long control-flow paths. QREF then checks whether bijection holds, and whether this IRSpec is orthogonal w.r.t. those previously provided. Also, if not enough IRSpecs have been given, QREF will detect this via a failed coverage check reporting a counterexample that illustrates a path through a method not yet accounted for. Once coverage is reached, if the semantic conditions such as left/right moverness do not hold, the user can revise the IRSpecs to build up atomicity in more fine-grained pieces.

We used QREF to create IRSpecs for several canonical and difficult concurrent objects including the Michael-Scott queue [34], Treiber's Stack [42], the Hendler *et al.* Elimination Stack [20], and the Harris *et al.* Restricted Double-Compare, Single-Swap (RDCSS) [18]. As an example, the 9 IRSpecs for the Michael-Scott queue are shown in Fig. 5 (and the rest are in Apx. B of the supplement). For example, the IRSpec  $(f, \text{atom}(\text{abcde} \triangleright f \triangleleft g))$  covers the behavior of a successful dequeue operation, with abcdefg being the “happy path.” IRSpecs like  $(\{f\}, \text{coop}((\text{abcde}) \triangleright \text{abcdefg} \triangleleft \text{hs}))$  then cover the paths that are consequently impacted. In total, 15 IRSpecs are needed to cover enqueue, dequeue, dequeue on an empty queue, and the “helping” feature of MSQ where other threads discover a stale tail pointer and advance it. Overall, our experimental results show that across all of the objects, nearly all of the 3,205 syntactic and semantic conditions could be checked within a few minutes.

### 3 Preliminaries

*Kleene Algebra with Tests (KAT)*. KAT [26] offers an algebraic framework for modeling and reasoning about programs, which we adopt in this work to model concurrent object operations.

*Definition 3.1.* A KAT is a pair  $(\mathcal{K}, \mathcal{B})$ , where  $\mathcal{K} = (K, +, \cdot, *, 0, 1)$  is a Kleene Algebra and  $\mathcal{B} = (B, +, \cdot, \neg, 0, 1)$  is a boolean algebra of tests, together with a homomorphic embedding of  $\mathcal{B}$  into  $\mathcal{K}$ . The set of KAT terms is generated from a finite alphabet of primitive actions  $A$  and primitive tests  $B$ , its syntax being given by the following grammar  $KAT \ni k ::= a \in A \mid b \in B \mid 0 \mid 1 \mid k + k \mid k \cdot k \mid k^*$ .

KAT expressions model the behavior of concurrent objects by representing statements as actions and conditionals as tests. Multiplication corresponds to sequential composition, addition to nondeterministic choice, and the Kleene star to iteration. We further conservatively extend KAT with the notation  $[b \cdot a]$  to syntactically encode atomic operations consisting of a primitive test  $b$  followed immediately by a primitive action  $a$ , with no other actions in between. While it introduces a restriction of interleavings (defined below), by considering an identity relation for the two symbols  $[$  and  $]$ , the semantics is not impacted.

We use  $\preceq_{seq}$  to denote the sequential sub-expression relation on KAT terms, that is,  $k_1 \preceq_{seq} k_2$  whenever  $k_2$  accepts a factorization  $k'k_1k''$  for some  $k', k''$ .

*Concurrent objects.* A concurrent object  $O$  is defined by a finite set of operations that can be invoked concurrently by multiple threads. Each operation  $\text{op}(\vec{x}) \setminus \vec{r}.k_{op} \in O$  is given by its signature, consisting of a unique name  $op$ , a list of formal parameters  $\vec{x}$ , and a distinguished list of return variables  $\vec{r}$  used to record the result, and an implementation  $k_{op}$  encoded as a KAT expression. Let  $\mathcal{T}$  denote the set of thread identifiers. We write  $Var$  and  $Val$  for the countably infinite set of variables and the set of values, respectively, and we define a state over a finite set of variables  $V \subseteq Var$  as a function  $\sigma_V : V \rightarrow Val$ . The set of all such states is denoted by  $S_V$ . Let  $XG$

be the set of global variables and let  $XL_t$  be the set of local variables associated with thread  $t \in \mathcal{T}$ . We write  $XL^\cup = \bigcup_{t \in \mathcal{T}} XL_t$  for the set of all local variables. We assume w.l.o.g. that  $XG \cap XL^\cup = \emptyset$  and  $XL_{t_i} \cap XL_{t_j} = \emptyset$  for all distinct threads  $t_i \neq t_j$ . For convenience, we write  $\sigma_{X,Y}$  for a state projected onto variables  $X \cup Y$ ,  $\sigma_G$  for  $\sigma_{XG}$ . All projections we consider include global variables, and we write  $\sigma_t$  for  $\sigma_{XG, XL_t}$ .

We model the behavior of concurrent objects in terms of a labeled transition system  $\mathcal{M}_O = (C, C_{init}, \rightarrow, \mathbb{E})$ , under a most general client that makes an arbitrary number of invocations from an arbitrary number of threads. A configuration  $(\sigma_{XG, XL^\cup}, T) \in C$  consists of a state over global and local variables,  $\sigma_{XG, XL^\cup}$ , and a thread pool  $T : \mathcal{T} \rightarrow KAT \cup \{\perp\}$  capturing the remaining computation per thread, where  $\perp$  represents the end of code. When clear from the context, we omit the subscript of the state component and simply write  $\sigma$  for  $\sigma_{XG, XL^\cup}$ . We write  $st(C)$  and  $threads(C)$  for the state  $\sigma$  and the thread pool  $T$  components of  $C$ , respectively. The transition relation  $\rightarrow \subseteq C \times \mathbb{E} \times C$  is nondeterministic and it involves a set of events  $\mathbb{E} \subseteq \mathcal{T} \times \mathcal{L}$ . An event  $e$  is a thread-label pair  $(t, x)$ , written as  $x_t$ , with labels  $x$  being drawn from the set  $\mathcal{L} = A \cup B \cup \{inv(op, \vec{v}), res(op, \vec{r})\} \cup \{[b \cdot a] \mid b \in B, a \in A\}$  of actions, tests, invocation of operations, their responses, and atomic read-modify-write operations (e.g., compare-and-swap). Throughout this paper, we refer to any sequence  $\pi \in \mathcal{L}^*$  of labels as a *path*. We write  $C \xrightarrow{e} C'$  whenever  $(C, e, C') \in \rightarrow$ , and denote by  $\rightarrow^*$  the transitive closure of the transition relation. For brevity, we use  $C \xrightarrow{e_1 \dots e_n} C'$  to mean that there exist configurations  $C_0, C_1, \dots, C_n$  such that  $C = C_0$ ,  $C' = C_n$ , and  $C_{i-1} \xrightarrow{e_i} C_i$  for every  $1 \leq i < n$ . An execution of  $\mathcal{M}_O$  is a sequence of alternating configurations and events,  $\rho = C_0 e_1 C_1 e_2 \dots$ , starting from an initial configuration  $C_0 \in C_{init}$ , such that  $C_i \xrightarrow{e_{i+1}} C_{i+1}$  for all  $i$ . A configuration  $C$  is reachable, denoted by  $Reach(C)$ , if there exists an execution ending in  $C$ . An execution is complete if it ends in a configuration  $C$  such that  $threads(C)(t) = \perp$  for all  $t \in \mathcal{T}$ . By projecting out the configurations of an execution, we obtain a *trace*, usually denoted by  $\tau$ . Throughout this paper, we restrict attention to finite executions, and hence to finite traces. A *trace fragment* refers to any contiguous sub-sequence of a trace, and we write  $\mathcal{F}$  for the set of all trace fragments. We write  $Traces$  for the universe of all traces and  $Traces(O) \subseteq Traces$  for the trace language of an object  $O$ .

*Thread-indexed languages of paths.* Given a language  $L \subseteq \mathcal{L}^*$  of paths and a thread identifier  $t$ , we write  $L_t$  for the image of  $L$  under the homomorphism  $\mathcal{L}^* \rightarrow (\mathcal{T} \times \mathcal{L})^*$  that tags every label in  $\mathcal{L}$  with  $t$ . For a finite set of threads  $J \subseteq \mathcal{T}$ , assuming a fixed total order on  $\mathcal{T}$ , we denote by  $\prod_{t \in J} L_t$  (resp.  $\overleftarrow{\prod}_{t \in J} L_t$ ) the ordered (resp. reverse-ordered) product of the thread-indexed languages  $L_t$ , with the product taken to be language concatenation. Furthermore, for any language  $L$  of either paths or thread-tagged paths, we write  $L^k$  for its  $k$ -fold product,  $\text{pref}(L)$  for its prefix closure, and  $\text{suf}(L)$  for its suffix closure.

*Contexts.* A context is a state predicate  $B_V : S_V \rightarrow \mathbb{B}$  whose free variables are a finite set  $V \subseteq \text{Var}$ . We write  $B_{X,Y}$  to denote a context over the union of variables  $X \cup Y$ . The projection of a context  $B_V$  onto  $W \subseteq V$  is defined by existential abstraction  $\text{proj}_W(B_V) = \exists U. B_V$ , where  $V = W \uplus U$ . We assume that context variables always include the global variables  $XG$ , and for brevity we omit them. Accordingly, we write  $B_t$  instead of  $B_{XG, XL_t}$ , and  $\text{proj}_t(B_V)$  for the projection of  $B_V$  onto  $XG \cup XL_t$ . We write  $\sigma_t \models B_t$  when the state  $\sigma_t$  satisfies the context  $B_t$ .

## 4 Reductions

A well-established paradigm in concurrency verification, reduction is based on identifying a representative subset of execution interleavings that soundly preserve the behaviors of concurrent



systems. Our method builds upon this principle, proposing a new class of reductions inspired by semi-commutative Mazurkiewicz traces [10].

In the standard setting, we assume an irreflexive, not necessarily symmetric, semi-independence (also called semi-commutativity) relation  $I \subseteq \mathcal{L} \times \mathcal{L}$  on labels, which we expect to be semantically sound, meaning that  $(x, y) \in I$  only if:

$$\forall t, t' \in \mathcal{T}. t \neq t' \implies \forall C_1, C_2, C_3 \in \mathcal{C}. C_1 \xrightarrow{x_t} C_2 \xrightarrow{y_{t'}} C_3 \implies \exists C_4. C_1 \xrightarrow{y_{t'}} C_4 \xrightarrow{x_t} C_3.$$

Drawing on the insight that trace equivalence admits coarser formulations [16], we generalize semi-commutativity from pairs of labels to pairs consisting of a label and a *language* of trace fragments.

*Definition 4.1.* Let  $x \in \mathcal{L}$ ,  $t \in \mathcal{T}$ , and let  $Z \subseteq \mathcal{F}$  be a language of trace fragments. The label  $x$  executed by  $t$  *semi-commutes to the right* (resp. *left*) of  $Z$  if for every  $\tau \in Z$  that contains no events of thread  $t$  and all  $C_1, C_2, C_3 \in \mathcal{C}$ :

$$C_1 \xrightarrow{x_t} C_2 \xrightarrow{\tau} C_3 \implies \exists C_4. C_1 \xrightarrow{\tau} C_4 \xrightarrow{x_t} C_3 \quad (\text{resp. } C_1 \xrightarrow{\tau} C_2 \xrightarrow{x_t} C_3 \implies \exists C_4. C_1 \xrightarrow{x_t} C_4 \xrightarrow{\tau} C_3).$$

We use  $x \bullet_t Z$  (resp.  $Z \bullet_t x$ ) as shorthand notations for right (resp. left) semi-commutations of  $x$  and  $Z$  for  $t$ . The set  $\mathcal{U} = \{(x, Z) \mid \forall t \in \mathcal{T}. x \bullet_t Z\} \cup \{(Z, x) \mid \forall t \in \mathcal{T}. Z \bullet_t x\}$  defines all such pairs for a concurrent object. It is immediate to see that the independence relation  $I$  is a specialization of  $\mathcal{U}$ , which is obtained by restricting  $Z$  to single-event words.

The set  $\mathcal{U}$  induces a preorder on traces obtained by directionally swapping thread-indexed labels with trace fragments. Concretely,  $\preceq$  is the smallest preorder on traces such that for every pair  $(x, Z) \in \mathcal{U}$  (resp.  $(Z, x) \in \mathcal{U}$ ), thread identifier  $t$ , and  $\tau \in Z$  containing no events of  $t$ , we have  $\tau' x_t \tau \tau'' \preceq \tau' \tau x_t \tau''$  (resp.  $\tau' \tau x_t \tau'' \preceq \tau' x_t \tau \tau''$ ). Given a language of traces  $Tr$ , its *downward closure* with respect to  $\preceq$  is defined as:

$$\downarrow Tr = \{\tau' \mid \exists \tau \in Tr. \tau' \preceq \tau\}.$$

*Definition 4.2 (Reduction).* Let  $O$  and  $O'$  be two concurrent objects with their respective trace languages  $Traces(O)$  and  $Traces(O')$ . We say that  $O'$  is a *reduction* of  $O$ , denoted by  $O' \sqsubseteq O$ , if  $Traces(O) \subseteq \downarrow Traces(O')$ .

Intuitively, a coarse definition of semi-commutativity allows for reasoning about concurrent executions that are semantically equivalent but that fall beyond the scope of traditional  $I$ -based reductions. The approach is rooted in the key insight that an arbitrary thread  $t$  can observe an identical local context at multiple points across an execution. This occurs when the environment (i.e., the partial execution of other threads) either performs purely thread-local actions, or updates the global state and later restores it, a phenomenon known in the concurrency literature as the ABA problem. Consider our running counter example, where a thread executes the `increment()` operation. It reads a global counter value (say 42) into its local counter `int c = ctr`. The thread is then suspended for an extended period, during which other threads increment and decrement the global counter multiple times. When the suspended thread awakens, it finds the original value (42) still in the global counter and successfully updates it via `cas(ctr, c, c+1)`. A trace preorder restricted to adjacent label swaps fails here, as the initial counter read cannot commute with any intermediate CAS operation that updates the global state.

While  $\mathcal{U}$  is theoretically appealing, establishing membership in it directly is intractable. To recover tractability, we introduce predicate-based abstractions that symbolically represent sets of trace fragments.

*Definition 4.3 (Sound Context Abstractions).* Let  $Z \subseteq \mathcal{F}$  be a set of trace fragments, and let  $P$  and  $Q$  be two contexts, i.e., assertions over the global state and the local state of some thread.

- $P$  is a *sound pre-abstraction* of  $Z$ , summarizing the reachable configurations from which the fragments in  $Z$  can execute, if and only if:

$$\forall C, C' \in C. \forall \tau \in Z. \text{Reach}(C) \wedge C \xrightarrow{\tau} C' \implies st(C) \models P.$$

- $Q$  is a *sound post-abstraction* of  $Z$ , summarizing the reachable configurations in which the fragments in  $Z$  can terminate, if and only if:

$$\forall C, C' \in C. \forall \tau \in Z. \text{Reach}(C) \wedge C \xrightarrow{\tau} C' \implies st(C') \models Q.$$

Since the soundness conditions are closed under unions of sets of trace fragments, largest such sets exist: we write  $\llbracket P \triangleright \rrbracket$  for the largest set of trace fragments of which  $P$  is a sound pre-abstraction, and  $\llbracket \triangleright Q \rrbracket$  for the largest set of which  $Q$  is a sound post-abstraction. The context  $\text{True}$  acts as the least informative abstraction: it is a sound pre-abstraction and post-abstraction of any set of trace fragments, vacuously satisfying the corresponding soundness condition, and thus  $\llbracket \text{True} \triangleright \rrbracket = \llbracket \triangleright \text{True} \rrbracket = \mathcal{F}$ .

**Definition 4.4.** [Contextual Movers] Let  $x$  be a label, let  $t$  be a thread identifier, and let  $P$  and  $Q$  be two contexts.

- $x_t$  is a *right-mover w.r.t.  $Q$* , written  $x_t \triangleright Q$ , if and only if  $x \bullet_t \llbracket \triangleright Q \rrbracket$ .
- $x_t$  is a *left-mover w.r.t.  $P$* , written  $P \triangleleft x_t$ , if and only if  $\llbracket P \triangleright \rrbracket_t \bullet x$ .

This definition requires a thread's action to semi-commute with environment transitions only under a contextual guard: a right-mover must commute across environment fragments that end in a configuration satisfying  $Q$ , and a left-mover across fragments that start from a configuration satisfying  $P$ .

This contextual formulation enables algorithmic verification of semi-commutativity. For instance, consider a thread executing an atomic compare-and-swap (CAS). Every trace prefix that ends immediately before a successful CAS terminates in a configuration satisfying the CAS's test, so this test is a sound post-abstraction of all such prefixes. Moreover, a sound pre-abstraction of all trace fragments following the CAS update can be obtained via a strongest postcondition predicate transformer. Checking that a label preceding the CAS in program order is a right-mover w.r.t. the former context (or that a label following the CAS in program order is a left-mover w.r.t. the latter context) can then be discharged by existing tools, where the effect of environment fragments is encoded as usual by havocking (i.e., existentially quantifying) the global variables and assuming the context.

## 5 Induction-Reduction Specifications: IRSpecs

In this section, we introduce *induction–reduction specifications* (IRSpecs), the central construct of our methodology. IRSpecs describe reductions that correspond to recurring thread interaction, providing a compositional framework for selecting maximal traces under the trace preorder. Each IRSpec connects inductively quantified actions recurring across traces of a concurrent object with a pattern expression that contracts right/left-moving control paths to a non-moving anchor. Throughout this section, we write  $\llbracket k \rrbracket \subseteq \mathcal{L}^*$  for the language of paths of a KAT expression  $k$ , obtained by reading  $k$  as a regular expression over  $\mathcal{L}$ , with primitive actions, primitive tests, and atomic operations as letters. We order KAT expressions by inclusion of their path languages, writing  $k_1 \leq k_2$  whenever  $\llbracket k_1 \rrbracket \subseteq \llbracket k_2 \rrbracket$ . We distinguish two main classes of IRSpecs that we discuss next.

The first is the *atomic* (atom) class. It specifies a block of instructions that may be executed by an arbitrary thread without interference. An atomic IRSpec  $\alpha = (\mathbf{a}, \text{atom}(\mathbf{R} \triangleright \mathbf{N} \triangleleft \mathbf{L}))$  consists of an inductively quantified label, typically a primitive action or primitive atomic operation,  $\mathbf{a} \in \hat{A} = A \cup \{[b \cdot a] \mid b \in B, a \in A\}$ , and a reduction pattern described by three KAT expressions  $\mathbf{R}$ ,

N, L, coarsely representing right-moving paths, anchor paths, and left-moving paths, respectively. The  $\alpha$ -induced reduction selects traces where indivisible sequences of labels in the language of RNL are executed in isolation with each such sequence corresponding to an occurrence of the recurring action  $\alpha$ .

The second, called the *cooperative* (coop) class, specifies an interaction pattern amongst threads that participate in the optimistic synchronization protocol inherent in lock-free concurrent objects. In this pattern, many threads attempt an update of the shared state with one of them succeeding, leading the others to restart the operation. The focus of a cooperative IRSpec is on the paths executed by the later failing threads. The specification  $\alpha = (\alpha^\cup, \text{coop}(R \triangleright N \triangleleft L))$  carries the reason(s) for failure in the anchor N. A thread may fail an update for multiple reasons. For example, a thread attempting to increment a concurrent counter might fail because another thread has successfully updated it, either via an increment operation or decrement operation. To model this, the inductive component of the cooperative specification now takes the form of a set of actions  $\alpha^\cup \subseteq \hat{A}$ , where each action is associated with a specific failure reason. The KAT expressions, R, N, and L, whose trace semantics captures the interplay between threads, describe again a reduction pattern. The anchor N has the shape of a nondeterministic choice among subexpressions representing atomic segments, each corresponding to global-state update, and the movers, R and L, represent paths in the failing threads. Any such thread executes a R-path, and, after observing the effect of a N-path, continues with a fallback L-path, possibly leading to a restart of the retry loop.

*IRSpecs order.* Since atomic IRSpecs describe paths within an object's methods that are scheduled to run without interruption, they can contribute to other IRSpecs by enforcing atomicity patterns of paths occurring in either their moving or non-moving parts. Grounded in this observation, we define a *precedence order* of IRSpecs determined by the inherited structural relation between the constituents of their reduction patterns.

*Definition 5.1.* Let  $\alpha_1 = (\alpha_1, \text{atom}(R_1 \triangleright N_1 \triangleleft L_1))$  be an atomic IRSpec, and let  $\alpha_2$  be either an atomic or cooperative IRSpec. We say that  $\alpha_1$  *precedes*  $\alpha_2$  and we write  $\alpha_1 \prec \alpha_2$ , if either of the two following cases hold, depending on where the atom reduction pattern occurs:

- (i) *Anchor:*  $\alpha_2 = (\alpha_2^\cup, \text{coop}(R_2 \triangleright N_2 \triangleleft L_2))$  and  $\alpha_1 \in \alpha_2^\cup$  and  $R_1 N_1 L_1 \leq N_2$
- (ii) *Mover:*  $\alpha_2 = (\alpha_2, \text{atom}/\text{coop}(R_2 \triangleright N_2 \triangleleft L_2))$  and either  $R_1 N_1 L_1 \leq R_2$  or  $R_1 N_1 L_1 \leq L_2$

Given a finite collection  $\mathcal{A}$  of IRSpecs, we assume that the precedence order  $\prec$  on  $\mathcal{A}$  is acyclic. An *admissible linear extension*  $\prec_1$  of  $\prec$  is a total order on  $\mathcal{A}$  such that the identity map on  $\mathcal{A}$  is order-preserving from  $(\mathcal{A}, \prec)$  to  $(\mathcal{A}, \prec_1)$ ; acyclicity guarantees such an extension always exists.

*Example 5.2.* Recall the counter example from Section 2, where we informally related the *atomic* and *cooperative* IRSpecs. For the cooperative increment IRSpec  $\alpha_{incF}$ , we require  $\alpha_{incS} \prec \alpha_{incF}$  and  $\alpha_{decS} \prec \alpha_{incF}$ . These constraints state that the atomic blocks of  $\alpha_{incS}$  and  $\alpha_{decS}$  are admissible candidates for anchors in the induction-reduction steps of  $\alpha_{incF}$ . In this way, the order relates a cooperative IRSpec to the atomic IRSpecs that may justify its reductions.

*Language of reduction patterns.* We have seen that IRSpecs provide pattern expressions that determine reductions. We call them *reduction patterns* and we give next a language for describing their more composite forms. Later, we define their trace semantics. Let *Red* denote the set of expressions generated by the grammar:

$$\text{Red} \ni r ::= [k] \mid \langle k[k]k \rangle \mid r + r \mid r \cdot r \mid r^*$$

where  $k$  ranges over KAT expressions. The two basic constructs correspond to the two classes of IRSpecs. The pattern  $[k]$  corresponds to the *atomic fragments*, and  $\langle k_1[k_2]k_3 \rangle$  corresponds to

the *cooperative interaction patterns*. Composite expressions are formed using the standard regular expression operators of choice (+), sequencing ( $\cdot$ ), and Kleene star (\*). It is straightforward to represent the reduction pattern of IRSpecs. An atomic specification  $\alpha = (\mathbf{a}, \text{atom}(\mathbf{R} \triangleright \mathbf{N} \triangleleft \mathbf{L}))$  has the corresponding pattern  $\llbracket \mathbf{RNL} \rrbracket$ , while the reduction pattern of a cooperative specification  $\alpha = (\mathbf{a}^\cup, \text{coop}(\mathbf{R} \triangleright \mathbf{N} \triangleleft \mathbf{L}))$  is  $\llbracket \mathbf{R[N]L} \rrbracket$ . For IRSpec  $\alpha$ , we write  $\text{red}(\alpha)$  to refer to its reduction pattern.

*Semantics of reduction patterns.* We next provide a semantics  $\llbracket r \rrbracket$  for reduction patterns  $r$  in terms of sets of thread-indexed paths that conform to a particular discipline prescribed by the shape of  $r$ . The semantics builds on the path semantics  $\llbracket k \rrbracket$  of KAT expressions and it is defined as:

$$\begin{aligned} \llbracket [k] \rrbracket &\triangleq \bigcup_{t \in \mathcal{T}} \llbracket [k] \rrbracket_t & \llbracket \langle k_1[k_2]k_3 \rangle \rrbracket &\triangleq \bigcup_{t \in \mathcal{T}} \bigcup_{J \subseteq \text{fin } \mathcal{T} \setminus \{t\}} (\overrightarrow{\prod}_{t_j \in J} \llbracket [k_1] \rrbracket_{t_j}) \llbracket [k_2] \rrbracket_t (\overleftarrow{\prod}_{t_j \in J} \llbracket [k_3] \rrbracket_{t_j}) \\ \llbracket r_1 + r_2 \rrbracket &\triangleq \llbracket r_1 \rrbracket \cup \llbracket r_2 \rrbracket & \llbracket r_1 \cdot r_2 \rrbracket &\triangleq \llbracket r_1 \rrbracket \llbracket r_2 \rrbracket & \llbracket r^* \rrbracket &\triangleq \bigcup_{n \in \mathbb{N}} \llbracket r \rrbracket^n \end{aligned}$$

We interpret an atomic expression  $[k]$  as the set of all thread-indexed paths, where the paths are described by  $k$  and each label within the path is indexed by the same thread identifier. The semantics of cooperative expressions  $\llbracket \langle k_1[k_2]k_3 \rangle \rrbracket$  captures interactions where a distinguished thread executes a path  $k_2$  while an arbitrary finite set of other threads execute surrounding paths  $k_1$  and  $k_3$ . Lastly, composite expressions follow the standard language interpretation of regular expression operators. We write  $\text{Any} = \mathbb{E}^*$  for the set of all event sequences and, for any reduction pattern  $r \in \text{Red}$ ,  $\text{Any}_{\neg r} = \text{Any} \setminus \llbracket r \rrbracket$  for the complement of the set of sequences described by  $r$ .

**LEMMA 5.3 (DISTRIBUTION OVER ANCHOR CHOICE).** *Let  $\mathbf{N}_U$  be a finite set of KAT expressions, and let  $R$  and  $L$  be KAT expressions. Then*

$$\llbracket R[\sum_{N \in \mathbf{N}_U} N]L \rrbracket = \sum_{N \in \mathbf{N}_U} \llbracket R[N]L \rrbracket$$

**PROOF.** By unfolding the semantics of cooperative reduction pattern and using that concatenation distributes over union.  $\square$

**Definition 5.4 (Traces  $r$ -rank).** Let  $r$  be a reduction pattern with  $\varepsilon \notin \llbracket r \rrbracket$ , and let  $L = \text{Any}_{\neg r} \cdot \llbracket r \rrbracket$  be the language of sequences formed of an  $r$ -block preceded by an arbitrary non- $r$  prefix. The  $r$ -rank of a trace  $\tau$ , denoted  $\text{rank}_r(\tau)$ , counts the maximal number of  $r$ -occurrences within a prefix:

$$\text{rank}_r(\tau) = \max\{k \mid \tau \in L^k \text{Any}\}$$

The order  $\preceq_r$  induced by  $r$ -rank is called a *preference order*. Concretely,  $\tau_1 \preceq_r \tau_2$  if and only if  $\text{rank}_r(\tau_1) \leq \text{rank}_r(\tau_2)$ .

We write  $\tau_1 \prec_r \tau_2$  for the corresponding strict order. Trace orderings are widely used [10, 14] in model-checking for trace normalization, where they are traditionally defined using some form of (contextual) lexicographic order. In our setting, trace normalization is instead formulated in terms of  $r$ -rank maximality. We later establish the conditions under which the  $r$ -rank maximal traces coincide precisely with the  $\alpha$ -induced traces introduced below.

*Semantics of IRSpecs.* We give an interpretation of IRSpecs as reduction operators  $\llbracket [\alpha] \rrbracket : 2^{\text{Traces}(O)} \rightarrow 2^{\text{Traces}(O)}$ . Let  $\alpha$  be an induction-reduction specification, and let  $r = \text{red}(\alpha)$  be its reduction pattern. Further, let  $T \subseteq \text{Traces}(O)$  be a subset of traces of  $O$ . The operator  $\llbracket [\alpha] \rrbracket$  discards a trace whenever reordering its events, as allowed by the trace preorder  $\preceq$ , can strictly improve its  $r$ -rank. Formally:

$$\llbracket [\alpha] \rrbracket(T) = \{\tau \in T \mid \neg \exists \tau' \in T. \tau \preceq \tau' \wedge \text{rank}_r(\tau) < \text{rank}_r(\tau')\}$$

**REMARK 1.** *Let  $O$  be a concurrent object, and let  $O'$  be the object whose trace set is  $\llbracket [\alpha] \rrbracket(\text{Traces}(O))$  for some IRSpec  $\alpha$ . Then  $O' \sqsubseteq O$ .*

Indeed, every discarded trace starts a chain of strict  $r$ -rank improvements along  $\preceq$ . Since the swaps generating  $\preceq$  preserve the events of a trace, the  $r$ -rank along the chain is bounded by the trace length, so the chain is finite and ends in a retained trace  $\tau'$  with  $\tau \preceq \tau'$  by transitivity. Hence  $Traces(O) \subseteq \downarrow \llbracket \alpha \rrbracket (Traces(O))$ .

We write  $O' \stackrel{\alpha}{\sqsubseteq} O$  for the reduction of  $O$  via  $\alpha$ .

*IRSpec-induced restrictions.* The operator  $\llbracket \alpha \rrbracket$  selects traces by comparing them against their reorderings. It is convenient to also have an *intrinsic* characterization of the traces it aims to retain, stated on each trace in isolation. To this end, we consider *well-formed* IRSpecs, whose occurrences of induction actions along each thread projection are in bijective correspondence with occurrences of the reduction pattern (for atomic IRSpecs) or of the anchor (for cooperative ones); this bijection is formalized in Sec. 6. Let  $|\tau|_{\mathfrak{a}^\cup}$  denote the number of events of  $\tau$  labeled by induction actions of  $\alpha$ , where we take  $\mathfrak{a}^\cup = \{\mathfrak{a}\}$  for an atomic IRSpec. Well-formedness makes  $|\tau|_{\mathfrak{a}^\cup}$  a ceiling for the  $r$ -rank, i.e.,  $\text{rank}_r(\tau) \leq |\tau|_{\mathfrak{a}^\cup}$ : each block counted by the rank contains exactly one occurrence of the bijection's target—the whole pattern for atomic IRSpecs, the anchor for cooperative ones—and distinct blocks thus correspond to distinct occurrences of induction actions. For a concurrent object  $O$  and a well-formed IRSpec  $\alpha$ , we write  $O[\alpha]$  for the object obtained by restricting  $Traces(O)$  to the traces whose  $r$ -rank attains this ceiling:

$$Traces(O[\alpha]) = \{\tau \in Traces(O) \mid \text{rank}_r(\tau) = |\tau|_{\mathfrak{a}^\cup}\}.$$

Intuitively, in such a trace, the reduction pattern associated with every occurrence of an induction action is realized as a contiguous block. Note that the definition constrains only the blocks: it makes no commitment about the position of the induction events themselves, which may lie outside the blocks they are paired with. Traces of  $O[\alpha]$  are referred to as  $\alpha$ -induced traces.

While both  $Traces(O[\alpha])$  and  $\llbracket \alpha \rrbracket (Traces(O))$  are determined by the same IRSpec  $\alpha$ , their outcomes differ in general. The former is an intrinsic restriction, retaining the traces whose  $r$ -rank attains the ceiling given by the induction actions. The latter is relative, retaining the traces whose  $r$ -rank cannot be strictly improved by reordering events along the trace preorder. The equality between the two outcomes requires additional conditions and does not hold in general. For example, when the expression  $R$  (and/or  $L$ ) of the IRSpec represents control paths that cannot be soundly moved immediately before (and/or after) the anchor, occurrences of induction actions may never be organized into contiguous blocks: the corresponding traces are retained by  $\llbracket \alpha \rrbracket$ , being  $r$ -rank maximal, without attaining the ceiling. We introduce in Sec. 6 sufficient conditions on IRSpecs that guarantee  $\llbracket \alpha \rrbracket (Traces(O)) = Traces(O[\alpha])$ .

A degenerate case arises when the right- and left-mover components of an IRSpec are both  $\epsilon$  and its anchor is a single primitive action or atomic operation, such as a CAS. The  $r$ -rank of a trace then equals its number of induction events and, since the swaps generating  $\preceq$  preserve events, no reordering can improve it. Hence  $\llbracket \alpha \rrbracket$  acts as the identity on trace sets and  $O[\alpha]$  coincides with  $O$ . Such IRSpecs do not yield progress and serve only as trivial specifications.

*Composition of IRSpecs.* Since IRSpecs act on sets of traces their sequential composition has a natural interpretation as function composition. Specifically, we define the semantics of  $\alpha_1; \alpha_2$  as  $\llbracket \alpha_1; \alpha_2 \rrbracket = \llbracket \alpha_2 \rrbracket \circ \llbracket \alpha_1 \rrbracket$ . The composition is partial:  $\alpha_1; \alpha_2$  is defined only when  $\alpha_2 \not\prec \alpha_1$ , i.e., when the order of application is consistent with the precedence order.

## 6 Proof Rules for IRSpec Reductions

This section introduces automation-amenable rules for soundly proving quotients of concurrent objects expressed in terms of reduction patterns. Specifically, given a IRSpec  $\alpha$  with its reduction pattern  $r = \text{red}(\alpha)$ , the proof rules provide sufficient obligations to satisfy so that the equality

$[[\alpha]](Traces(O)) = Traces(O[r])$  holds, i.e., the IRSpec-induced reduction coincides with the IRSpec-induced restriction. The rules reduce the induction-reduction argument to automation-friendly primitives: Hoare logic, contextual commutativity, and syntactic automaton operations on control-path structure.

### 6.1 Contextual Semi-Commutativity over KAT

We begin by lifting the contextual-mover relation, introduced in Sec. 3, to KAT Expressions. To this end, we define contextual right KAT-mover terms of the form  $(k)_t \triangleright Q$  that relate a thread-indexed KAT expression  $k$ , representing paths executed by thread  $t$ , and a contextual guard  $Q$ . Fig. 2 presents the set of sound rules used to inductively define this relation. The contextual left-KAT mover terms  $P \triangleleft (k)_t$  are defined symmetrically.

Firstly, we discuss the predicate transformers used in these rules. To that purpose, we extend the contextual mover relation from single thread-indexed labels to trace fragments  $\tau$ , that is,  $\tau \triangleright Q$  (resp.  $P \triangleleft \tau$ ). Since  $Q$  is a state-predicate we can apply Dijkstra predicate transformers over trace fragments, propagating the context  $Q$  along  $\tau$ . There are two types depending on the direction of propagation: weakest precondition for backward moving and strongest postcondition for forward moving. Specifically, the weakest precondition is applicable for right-movers, while the strongest-postcondition for left-movers.

We assume semantic functions for actions and tests denoted by  $[[a]] : S_{XG,XL} \rightarrow S_{XG,XL}$  and  $[[b]] : S_{XG,XL} \rightarrow \mathbb{B}$ , respectively, where the action/test are defined over global variables  $XG$  and local variables  $XL$ , and  $\mathbb{B}$  is the semantical Boolean domain. We naturally extend these functions to thread-indexed actions and tests using thread-local variables  $XL_t$ , leaving the global variables  $XG$  unchanged. Given a finite sequence of events  $\tau = e_1 e_2 \dots e_n$  that can be executed by distinct threads, its semantic function is their sequential composition  $[[\tau]] = [[e_1]] \circ \dots \circ [[e_n]]$ . Lastly, as expected, for  $\tau$ , we take the weakest precondition transformer to be defined by  $wp(\tau, Q)(\sigma) = \forall \sigma' \in S_V. \sigma' = [[\tau]](\sigma) \Rightarrow Q(\sigma')$ , and the strongest postcondition transformer by  $sp(\tau, B)(\sigma') = \exists \sigma \in S_V. B(\sigma) \wedge [[\tau]](\sigma) = \sigma'$ .

Then, in order to define when a language of trace fragments  $Z$  is a contextual right/left mover, we consider a formulation of predicate transformers in terms of demonic nondeterministic choice of traces in  $Z$ . Concretely,  $wp(Z, Q) = \bigwedge_{\tau \in Z} wp(\tau, Q)$  and  $sp(Z, P) = \bigvee_{\tau \in Z} sp(\tau, P)$ . We abuse notation, and use  $wp((k)_t, Q)$  for  $wp([k]_t, Q)$ .

Finally, we introduce rules that extend basic contextual commutativity facts from primitive actions to full KAT expressions. We assume a given set  $\mathcal{M}$  of basic commutativity facts of the form  $(\triangleright, a, Q)$  or  $(\triangleleft, a, Q)$ , indicating that primitive action  $a$  is a right-mover or left-mover, respectively, under context  $Q$ .

The term  $k_t \triangleright Q_t$  means that, given a KAT expression  $k$ , any  $k$ -path executed by thread  $t$  is *right-moving with respect to context  $Q_t$* : any path represented by  $k$  can be permuted one event at a time to the right of other threads' events up to the context  $Q_t$ . (Recall that  $Q_t$  represents a context projection onto the variables  $XL_t \cup XG$ .)

The following rules derive right-mover judgments for composite KAT expressions from  $\mathcal{M}$ :

Tests do not modify state, so the Rule TEST establishes that a test  $b_t$  is right-moving whenever the context  $Q_t$  entails the test (i.e.,  $Q_t \subseteq b_t$ ). In the case of a sequence  $(k_1 k_2)_t$ , the Rule SEQ captures a backward reasoning. The rule relates each expression to its own context, while the weakest-precondition constraint ensures that the states required for right-commuting  $(k_1)_t$  preserve the enabledness of  $(k_2)_t$ . The nondeterministic choice Rule CHOICE requires that each branch of  $(k_1 + k_2)_t$  is right-moving up to the same context. Lastly, the case of iteration  $(k^*)_t$  is handled in the Rule STAR. It is right-moving with respect to a context  $Q_t$  when each  $k_t$  is right-moving to  $Q_t$  and all initial states from which  $k$  can be executed by thread  $t$  are within the same footprint as  $Q_t$ .

$$\begin{array}{c}
\text{PRIMITIVE} \quad \frac{(\triangleright, a, Q) \in \mathcal{M}}{a_t \triangleright Q_t} \quad \text{TEST} \quad \frac{Q_t \subseteq b_t}{b_t \triangleright Q_t} \quad \text{SEQ} \quad \frac{(k_2)_t \triangleright Q_t \quad Q'_t \subseteq \text{wp}((k_2)_t, Q_t) \quad (k_1)_t \triangleright Q'_t}{(k_1 k_2)_t \triangleright Q_t} \\
\\
\text{CHOICE} \quad \frac{(k_1)_t \triangleright Q_t \quad (k_2)_t \triangleright Q_t}{(k_1 + k_2)_t \triangleright Q_t} \quad \text{STAR} \quad \frac{k_t \triangleright Q_t \quad Q_t \subseteq \text{wp}((k)_t, Q_t)}{(k^*)_t \triangleright Q_t}
\end{array}$$

Fig. 2. Contextual Semi-Commutativity over KAT expressions, derived from basic facts in  $\mathcal{M}$ 

**THEOREM 6.1 (SOUNDNESS).** *The rules  $(k)_t \triangleright Q_t$  are sound.*

Soundness follows by a straightforward structural induction.

## 6.2 Proof Rules

We now present the proof rules that establish soundness of IRSpec-induced reductions and show how to compose them. The rules validate IRSpecs by checking two global conditions—coverage and orthogonality—that ensure an object’s full specification is sound, followed by local conditions that verify each IRSpec induces a valid reduction.

In order to formally define control-path-related checks, we introduce the notion of reduction scope. Given an IRSpec  $\alpha$ , its *reduction scope*, denoted by  $[\alpha]$ , is the KAT expression describing all control paths on which  $\alpha$  acts. For atomic IRSpecs, the reduction scope includes the full atomic block: prefix, anchor, and suffix. For cooperative IRSpecs, the scope includes only the prefix and suffix, excluding the anchor (which is covered by an atomic IRSpec). Concretely:

$$\begin{aligned}
[(\alpha, \text{atom}(R \triangleright N \triangleleft L))] &= \text{RNL} \\
[(\alpha^\cup, \text{coop}(R \triangleright N \triangleleft L))] &= \text{RL}
\end{aligned}$$

**Coverage.** We assume a set  $\mathcal{A}$  of IRSpecs for a concurrent object  $O$ . We say that  $\mathcal{A}$  is **covering** for  $O$  if

- (i) for each operation implementation  $k_{\text{op}}$ , there exists  $\mathcal{A}_{\text{op}} = \{\alpha_1, \dots, \alpha_l\} \subseteq \mathcal{A}$  that *covers* it, i.e.,  $k_{\text{op}} \leq ([\alpha_1] + \dots + [\alpha_l])^*$
- (ii) for every cooperative IRSpec  $\alpha = (\alpha^\cup, \text{coop}(R \triangleright N \triangleleft L))$  in  $\mathcal{A}_{\text{op}}$ , there exists a set of atomic IRSpecs  $C_a \subseteq \mathcal{A}_{\text{op}}$  such that  $\forall \alpha_a \in C_a. \alpha_a \prec \alpha$  and

$$\alpha^\cup = \{\alpha \mid \exists (\alpha, \text{atom}(\dots)) \in C_a\} \wedge N = \sum \{[\alpha_a] \mid \alpha_a \in C_a\}$$

- (iii) for every cooperative IRSpec  $\alpha = (\_, \text{coop}(R \triangleright N \triangleleft L))$ , there exists an atomic IRSpec  $\alpha_a \in \mathcal{A}$ , such that  $\alpha_a \prec \alpha$  and  $R = \text{red}(\alpha_a)$

The first condition requires that any repetitive partial control-flow path of the implementation be described by the reduction scope of some IRSpec in  $\mathcal{A}_{\text{op}}$ . The second condition requires that every cooperative specification is anchored in a collection of preceding atomic specifications, so that its failure scenarios are aligned with concrete successful executions. The third condition requires that, for each  $R$ -mover part of a Coop IRSpec, there must exist an Atom whose reduction pattern describes it. We will explain later, when we give the proof rules, why the  $R$ -mover requires atomicity, while the  $L$ -mover can be reasoned about by the rules in Fig 2.

**Orthogonality.** We require that IRSpecs don’t overlap, a condition we call **orthogonality**.

Given two distinct induction-reduction specifications  $\alpha_i$  and  $\alpha_j$ , we say their reduction scopes are *orthogonal*, written  $[\alpha_1] \text{ orthog } [\alpha_2]$ , when the following conditions hold:

$$\begin{array}{c}
\text{ATOM} \frac{\mathbf{a} \xleftrightarrow{t_i} \text{RNL} \quad \text{non-self-ovl}(\text{RNL}) \quad \{Q_{t_i}\} N_{t_i} \{Q'_{t_i}\} \quad (\text{R})_{t_i} \triangleright Q_{t_i} \quad Q'_{t_i} \triangleleft (\text{L})_{t_i}}{O[\mathbf{N}] \overset{\alpha}{\sqsupseteq} O[\text{RNL}]} \quad [\forall t_i \in \mathcal{T}] \\
\\
\text{ASEQ} \frac{O[N_1] \overset{\alpha_1}{\sqsupseteq} O[R_1 N_1 L_1] \quad O[N_2] \overset{\alpha_2}{\sqsupseteq} O[R_2 N_2 L_2] \quad [R_1 N_1 L_1] \text{orthog} [R_2 N_2 L_2]}{O[N_1 + N_2] \overset{\alpha_1; \alpha_2}{\sqsupseteq} O[R_1 N_1 L_1 + R_2 N_2 L_2]}
\end{array}$$

Fig. 3. Proof rules for reductions induced by Atom IRSpecs.

- (i) *non-overlapping*:  $\text{suf}([\alpha_i]) \cap \text{pref}([\alpha_j]) = \emptyset$  and  $\text{pref}([\alpha_i]) \cap \text{suf}([\alpha_j]) = \emptyset$
- (ii) *non-factoring*:  $[\alpha_i] \not\sqsubseteq_{\text{seq}} [\alpha_j]$  and  $[\alpha_j] \not\sqsubseteq_{\text{seq}} [\alpha_i]$

Intuitively, orthogonality ensures that distinct induction-reductions do not compete for the same control path. The first condition requires that any two control-paths subject to two IRSpecs respectively, do not partially overlap, while the second condition excludes cases in which one control-path is subpath of another. A reader familiar with string rewriting systems may notice the analogy with the sufficient condition for confluence.

Orthogonality also requires that a single IRSpec does not *overlap* with itself. Given a specification  $\alpha$ , we write  $\text{non-self-ovl}([\alpha])$  to mean that  $\text{suf}([\alpha]) \cap \text{pref}([\alpha]) = \emptyset$ .

*IRSpec proof rules.* In the remainder of this section, we introduce rules for IRSpec-induced reductions that coincide with IRSpec-induced restrictions, and rules for their composition. Given a concurrent object  $O$ , an IRSpec  $\alpha$ , and a reduction pattern  $r$  such that  $r = \text{red}(\alpha)$ , we write  $O[r] \overset{\alpha}{\sqsubseteq} O$  to indicate that  $O[r]$  is a reduction of  $O$  via  $\alpha$ . In general, the restriction  $\text{Traces}(O[\alpha])$  and the reduction  $[[\alpha]](\text{Traces}(O))$  need not coincide. The rules below capture the sufficient conditions under which they coincide.

Since our presentation starts from a concrete concurrent implementation and derives progressively reductions for it, we reason upward from less-reduced to more-reduced objects. Accordingly, in the next rules we use an inverse order  $\sqsupseteq$ , where the less-reduced object (higher degree of interleaving) appears on the left-hand side and the more-reduced object appears on the right-hand side.

**6.2.1 Atomic reductions.** The rule ATOM in Figure 3 establishes an *atomicity reduction* induced by  $\alpha = (\mathbf{a}, \text{atom}(\text{R} \triangleright \text{N} \triangleleft \text{L}))$ . The variable  $t_i$  in the rule is universally quantified. Intuitively, it captures the fact that all executions in which the control paths represented by RNL are performed by a thread  $t_i$  and interleaved with statements from concurrent threads can be simulated by executions in which these paths can be executed in isolation.

Given a trace  $\tau$  and a thread  $t$ , we write  $\tau|_t$  for the restriction of  $\tau$  to  $t$ , obtained by filtering out all events executed by other threads while preserving the program order of events of  $t$ . The premise  $\mathbf{a} \xleftrightarrow{t_i} \text{RNL}$  is a shorthand notation for a trace-indexed family of bijective and order preserving functions  $\{\forall \tau \in \text{Traces}(O). \text{occ}_{\mathbf{a}}(\tau|_{t_i}) \rightarrow \text{occ}_{\text{RNL}}(\tau|_{t_i})\}_{t_i \in \mathcal{T}}$  which map occurrences of induction action to occurrences of control paths described by RNL in any thread restriction. This premise ensures the well-formedness of an atom IRSpec. We define these functions in terms of two auxiliary functions:  $\text{occ}_{\mathbf{a}}(\cdot) : \mathbb{E}^* \rightarrow \mathbb{N}$  maps occurrences of  $\mathbf{a}$  in a trace to their index positions, while  $\text{occ}_k(\cdot) : \mathbb{E}^* \rightarrow (\mathbb{N} \times \mathbb{N})$  maps occurrences of paths represented by  $k$  to their interval positions (the



span from the first to last label of the trace fragment). Note that, by applying both functions to thread-restrictions of traces, we can define them in terms of control paths. Note that we can always define such family of bijections, provided the next three conditions hold:

- (i)  $RNL \leq \text{Any}_{\neg a} \mathbf{a} \text{Any}_{\neg a}$
- (ii)  $\forall \alpha' \in \mathcal{A}. \alpha' \neq \alpha \wedge \alpha < \alpha' \Rightarrow [\alpha'] \leq \text{Any}_{\neg a} \mathbf{a} \text{Any}_{\neg a} + \text{Any}_{\neg a}$
- (iii)  $\forall \alpha' \in \mathcal{A}. \alpha' \neq \alpha \wedge \alpha \not< \alpha' \Rightarrow [\alpha'] \cap (\text{Any} \mathbf{a} \text{Any}) = 0$

Condition (i) requires that the induction action  $\mathbf{a}$  occurs exactly once in the reduction pattern  $RNL$ . Condition (ii) requires that  $\mathbf{a}$  occurs at most once in the reduction scope of any  $IRSpec$  following  $\alpha$  in the precedence order. Condition (iii) requires that  $\mathbf{a}$  does not occur in the reduction scope of any distinct  $IRSpec$  unrelated by the precedence order. Together, these conditions guarantee the well-definedness of the bijection functions  $\text{occ}_a(\tau|_{t_i}) \rightarrow \text{occ}_{RNL}(\tau|_{t_i})$ . We defer the full details of the proof to the Appendix A.

The premise *non-self-ovl*( $RNL$ ) excludes  $IRSpec$ s whose reduction pattern  $RNL$  self-overlaps. Specifically, it rules out nonempty proper sequences of labels that are both suffixes and prefixes of  $RNL$ . This guarantees that occurrences of the atomic block do not overlap inside a trace.

The triple  $\{Q_{t_i}\}N_{t_i}\{Q'_{t_i}\}$  is called a *grounding Hoare triple*. It is required that  $Q_{t_i}$  is a context that holds in every execution immediately before every occurrence of  $N_{t_i}$ , and that the triple is a valid Hoare triple. In our tool, we use the following heuristic to construct such a triple. We take  $Q_{t_i} = \text{wp}(N_{t_i}, \text{True})$ , the weakest condition under which  $N_{t_i}$  is enabled, and then  $Q'_{t_i} = \text{sp}(N_{t_i}, Q_{t_i})$ , the strongest condition holding after executing  $N_{t_i}$  from a state satisfying  $Q_{t_i}$ . It is straightforward that  $Q_{t_i}$  satisfies the requirement stated above and that this forms a valid Hoare triple.

The last two premises,  $(R)_{t_i} \triangleright Q_{t_i}$  and  $Q'_{t_i} \triangleleft (L)_{t_i}$  establish the contextual commutativity conditions for the induction-reduction candidate's right- and left-mover components.

Consider, in the counter example, the atomic  $IRSpec$   $\alpha_{incs}$  that corresponds to a successful increment of the counter. The contexts  $Q_{t_i}$  and  $Q'_{t_i}$  capture the contexts corresponding to the grounding Hoare Triple for thread  $t_i$ . Concretely,  $Q_{t_i}$  is  $c_{t_i} = ctr$ , representing states where thread  $t_i$  observes a local copy of the global counter, and  $Q'_{t_i}$  is  $\exists oldCtr. ctr = oldCtr + 1 \wedge c_{t_i} = oldCtr$ , indicating that the global counter has been incremented while the local value  $c_{t_i}$  observed by thread  $t_i$  still reflects the pre-update state. See Apx B for the contexts used in the rules for counter.

The rule  $ASEQ$  in Figure 3 composes two atomic reductions into a single one. Each premise states that executions centered around an anchor  $N_i$  with  $i \in \{1, 2\}$  can be reduced to the corresponding atomic block  $R_iN_iL_i$ . The orthogonality side condition ensures that the two atomic patterns do not overlap, so that the corresponding reductions can be applied independently. Under this assumption, the conclusion combines them by nondeterministic choice. Specifically, an execution whose anchor is drawn from  $N_1 + N_2$  is reduced to the corresponding reduction pattern drawn from  $R_1N_1L_1 + R_2N_2L_2$ . Thus,  $ASEQ$  expresses composition of orthogonal atomic reductions.

**LEMMA 6.2 (SOUNDNESS OF ATOMIC REDUCTIONS).** *Let  $\mathcal{A}$  be a collection of  $IRSpec$ s for a concurrent object  $O$ . The rules  $ATOM$  and  $ASEQ$  are sound for all atomic  $IRSpec$ s in  $\mathcal{A}$ .*

The proofs are relegated to Appendix A.

**6.2.2 Cooperative reductions.** The Rule  $COOP$  introduces the conditions for reductions induced by cooperative  $IRSpec$ s  $\alpha = (\mathbf{a}^\cup, \text{coop}(R \triangleright N \triangleleft L))$ . The conclusion states that executions where the anchor  $N$  occurs atomically can further be normalized into a cooperative block, where an arbitrary number of threads have their  $R$ -paths and  $L$ -paths surrounding an  $N$ -path executed by a distinct thread. The rule captures the interplay between a thread  $t_j$ , which executes an  $R$ -path and is then suspended, and a thread  $t_i$ , which executes an anchor  $N$ -path in the meantime; afterwards,  $t_j$

$$\begin{array}{c}
\begin{array}{c}
\mathbf{a}^\cup \xleftrightarrow{t_i} N \quad t_j \neq t_i \quad Q_{t_j} = \text{sp}((R)_{t_j}, \text{True}) \quad Q'_{t_j} = \text{sp}((N)_{t_i}, Q_{t_j}) \\
\text{IStable}(Q_{t_j}, \overline{bk(N)}) \quad bk(N) \xleftrightarrow{t_i} N \\
\text{non-self-ovl}(RL) \quad (R)_{t_j} \triangleright Q_{t_j} \quad Q'_{t_j} \triangleleft (L)_{t_j} \quad R \rightrightarrows R \quad L \rightrightarrows L
\end{array} \\
\text{Coop} \frac{}{O[N] \overset{\alpha}{\sqsupseteq} O[\langle R[N]L \rangle]} \quad [\forall t_i, t_j \in \mathcal{T}] \\
\\
\begin{array}{cc}
\begin{array}{c}
\text{CSTRUCT} \\
[ \sum_{N \in N^\cup} ] \overset{\alpha}{\sqsupseteq} \langle R[ \sum_{N \in N^\cup} N ] L \rangle
\end{array}
&
\begin{array}{c}
\text{CSEQ} \\
[N] \overset{\alpha_1}{\sqsupseteq} \langle R_1[N]L_1 \rangle \quad [N] \overset{\alpha_2}{\sqsupseteq} \langle R_2[N]L_2 \rangle \\
R_1L_1 \text{ orthog } R_2L_2 \quad R_1 \rightrightarrows R_2 \quad L_1 \rightrightarrows L_2
\end{array}
\end{array} \\
\hline
[N] \overset{\alpha}{\sqsupseteq} \langle R[N]L \rangle \quad [N] \overset{\alpha_1, \alpha_2}{\sqsupseteq} \langle (R_1 + R_2)[N](L_1 + L_2) \rangle
\end{array}$$

Fig. 4. Proof rules for reduction induced by Coop IRSpecs

resumes and continues with an  $L$ -path. The variables  $t_i$  and  $t_j$  in the rule are universally quantified and required to be distinct.

The premise  $\mathbf{a}^\cup \xleftrightarrow{t_i} N$  requires a bijective correspondence between occurrences of inductive actions in the set  $\mathbf{a}^\cup$  and occurrences of  $N$  in the restriction of the trace to thread  $t_i$ . The syntactic conditions guaranteeing the existence of such a bijection are analogous to those discussed for the **ATOM** rule. The anchor of a Coop IRSpec decomposes into atomic blocks, each subject to an Atom IRSpec whose reduction pattern is in one-to-one correspondence with its inductive action. Since  $\mathbf{a}^\cup$  collects all these inductive actions, the bijection for the Coop IRSpec is obtained piecewise from these correspondences.

The premise  $Q_{t_j} = \text{sp}((R)_{t_j}, \text{True})$  computes the contextual guard  $Q_{t_j}$  of right-mover paths via strongest postcondition. Recall that for right-mover  $R$  to semi-commute to the right up to the anchor, we require that the contextual guard abstract the trace fragments ending before the anchor. The argument that  $Q_{t_j}$  is a sound abstraction is twofold and it relies on the coverage property. Firstly, by coverage, a cooperative IRSpecs expects a preceding atomic IRSpec to induce a reduction where the  $R$ -paths are atomic. Therefore, for any paths in  $R$  executed by an arbitrary thread  $t_j$ ,  $Q_{t_j}$  is a sound characterization of states immediately after the execution of such paths. Secondly, the premise  $\text{IStable}(Q_{t_j}, \overline{bk(N)})$ , that we discuss next, guarantees that, for any thread-indexed path in  $R$  that occurs between two occurrences of  $N$ -paths,  $Q_{t_j}$  holds also immediately before the later  $N$ -path. Then, the conditions  $\text{IStable}(Q_{t_j}, \overline{bk(N)})$  and  $bk(N) \xleftrightarrow{t_i} N$ , where  $\text{IStable}$  is defined below, provide the stability of  $Q_{t_j}$  under interference from the set of actions and tests that excludes the *bijective support* of the anchor  $N$ , denoted by  $\overline{bk(N)}$ .

The helper  $bk(N)$  extracts the actions in  $N$  whose occurrences in a trace projection to a thread are in bijective correspondence with occurrences of  $N$ -paths in that projection. This correspondence is captured by the condition  $bk(N) \xleftrightarrow{t_i} N$ , and  $bk(N)$  can be computed automatically by enumerating the labels of  $N$  and checking which of them are in bijective correspondence with the  $N$ -paths. While not explicitly stated,  $\mathbf{a}^\cup$  is typically a subset of  $bk(N)$ , as the inductive actions of a cooperative IRSpec usually coincide with a distinguished subset of the anchor's bijective support.

Given a context  $Q_{t_j}$  and a set of actions  $Act$ , we define the interference stability condition as:

$$IStable(Q_{t_j}, Act) \triangleq \forall a \in Act, t_k \in \mathcal{T}. t_k \neq t_j \Rightarrow \{Q_{t_j} \wedge pre(a_{t_k})\} a_{t_k} \{Q_{t_j}\},$$

where  $pre(a_{t_k})$  denotes the state assertion characterizing the precondition under which action  $a$  is enabled for thread  $t_k$ . This condition ensures that once the context  $Q_{t_j}$  holds at some point between two consecutive occurrences of  $N$ -paths, it cannot be invalidated by unrelated interference. Consequently, any transition from  $Q_{t_j}$  to a distinct context  $Q'_{t_j}$  can only be caused by an action in  $bk(N)$ , and hence arises exclusively from occurrences of paths in  $N$ . Since  $Q_{t_j}$  is interference stable, after any occurrence of a path in  $R_{t_j}$  the context still holds right before the next occurrence of an  $N$ -path, which justifies commuting the  $R$ -path into the cooperative block.

The premise  $Q'_{t_j} = sp_{t_j}((N)_{t_i}, Q_{t_j})$  defines the sound context projection for thread  $t_j$  that observes the effect of the execution of anchor  $N_{t_i}$  by thread  $t_i$ .

For instance, in the case of the counter “Coop”  $IRSpec_{\alpha_{incF}}$ , the contexts  $Q_{t_j}$  and  $Q'_{t_j}$  capture the state local to thread  $t_j$ . Specifically,  $Q_{t_j}$  is  $c_{t_j} = ctr$ , representing states in which the thread  $t_j$  observes, via its local variable  $c_{t_j}$ , the current value of the global counter. The post-context  $Q'_{t_j}$  is  $c_{t_j} \neq ctr$ , indicating that the global counter has changed since that observation. The reason is given by the anchor executed by a different thread  $t_i$ . This anchor  $N$  represents a nondeterministic choice between the paths `int c = ctr; cas(ctr, c, c+1); return c` and `int c = ctr; cas(ctr, c, c-1); return c`, which, when executed by other threads, lead to the change in the global counter observed by  $t_j$ . See Apx. B for all contexts used in the rules for counter.

The premise *non-self-ovl*( $RL$ ) excludes self-overlapping reduction patterns  $RL$  corresponding to the cooperative specification. The contextual commutativity conditions  $(R)_{t_j} \triangleright Q_{t_j}$  and  $Q'_{t_j} \triangleleft (L)_{t_i}$  cover paths executed by threads  $t_j \neq t_i$  and establish their conditions under which they can be moved into position around the anchor. Finally, the conditions  $R \rightleftharpoons R$  and  $L \rightleftharpoons L$  require that actions occurring in paths of  $R$ , and respectively,  $L$ , commute pairwise, so that their interleavings can be freely reordered.

The rule CSTRUCT specializes a cooperative reduction whose anchor is given as a nondeterministic choice. It states that if a cooperative reduction holds for an anchor of the form  $\sum_{N \in N^U} N$ , then the same reduction may be understood component-wise for each  $N \in N^U$ . In this way, CSTRUCT reduces reasoning about a composite anchor to reasoning about its individual cases.

The rule CSEQ composes two cooperative reductions around the same anchor  $N$ . Each of the two reduction premises captures one class of surrounding behaviors, namely paths in  $R_i$  before the anchor, and paths in  $L_i$  afterwards. The orthogonality side condition ensures that their corresponding paths do not overlap, while the commutativity conditions  $R_1 \rightleftharpoons R_2$  (resp.  $L_1 \rightleftharpoons L_2$ ) require their respective actions to be swappable. The conclusion then merges them into a single cooperative block  $\langle\langle R_1 + R_2 \rangle\langle N \rangle\langle L_1 + L_2 \rangle\rangle$ , expressing closure of cooperative reductions under nondeterministic combination of compatible surrounding paths.

**LEMMA 6.3 (SOUNDNESS OF COOPERATIVE REDUCTIONS).** *Let  $\mathcal{A}$  be a collection of  $IRSpec$ s for a concurrent object  $O$ . The rules COOP, CSTRUCT, and CSEQ are sound for all cooperative  $IRSpec$ s in  $\mathcal{A}$ .*

The proofs are relegated to Appendix A.

### 6.3 Canonicalization of an object

Let  $\mathcal{A}$  be a collection of  $IRSpec$ s for a concurrent object  $O$ , and let

$$\prec_1 \triangleq \alpha_1; \alpha_2; \dots; \alpha_n$$

be an admissible linear extension of their precedence order. The *canonicalization* of  $O$  induced by  $\prec_1$  is the object  $O_{can}^{\prec_1}$  obtained by applying the corresponding reductions in sequence:

$$O \stackrel{\alpha_1}{\sqsupseteq} O_1 \stackrel{\alpha_2}{\sqsupseteq} \dots \stackrel{\alpha_n}{\sqsupseteq} O_{can}^{\prec_1}$$

The reduction  $O_{can}^{\prec_1}$  retains only the traces that survive canonicalization under the IRSpecs ordered by  $\prec_1$ .

**THEOREM 6.4 (CLOSED FORM OF THE CANONICAL TRACES).** *Let  $O$  be a concurrent object, and  $\mathcal{A}$  a collection of sound IRSpecs that is covering and orthogonal for  $O$ . Let  $r_1, \dots, r_m$  be the reduction patterns of the IRSpecs in  $\mathcal{A}$  that are maximal w.r.t. the precedence order  $\prec$ . Then, for any admissible linear extension  $\prec_1$  of  $\prec$ , we have  $O_{can}^{\prec_1} \sqsubseteq O$  and  $(r_1 + \dots + r_m)^*$  is an abstraction of the  $O_{can}^{\prec_1}$ , that is:*

$$\text{Traces}(O_{can}^{\prec_1}) \subseteq (r_1 + \dots + r_m)^*.$$

The proof details are available in Appendix A, but we summarize here. The covering condition gives a decomposition of every operation of path of  $O$  into IRSpec patterns, while orthogonality guarantees that this decomposition is unambiguous and non-interfering. The soundness of the reduction rules then let us select each such reduction pattern, one reduction at a time. Since reduction pattern expressions are closed under choice and concatenation, the whole trace belongs to the starred union of IRSpecs reduction patterns. IRSpecs allow for a sound and automated derivation of a quotient expression, which was not possible in prior work [13]. Furthermore, the quotient resulting from IRSpecs can further be refined into an automaton in which control states represent ADT abstract states, and transitions are labeled by IRSpecs and are enabled whenever those IRSpec atom paths are feasible. For lack of space, we refer the reader to Enea and Koskinen [13] for relevant quotient automaton definitions.

## 7 Implementation

The proof rules in Sec. 6 reduce IRSpec reasoning to obligations that are automation-friendly. We implemented an end-to-end prototype tool in OCaml called QREF<sup>2</sup> that implements all of the automation. QREF consumes an input object with method implementations written in a C-like .cl source language with arrays, references, and heap-allocated structures (see examples in Apx. B). As a first step, statements need to be given identifiers and the program needs to be translated to a KAT-friendly structure. Users can do this manually, or else QREF automatically translates the .cl source to a labeled .c11 version of the source. With the labels established, users then write ASCII IRSpecs in a corresponding .irspecs file (we will see an example in Fig. 5 and all examples' IRSpecs are given in Apx. B).

From the input program and IRSpec, QREF performs three main checks:

- (1) Calculates commutativity contexts ( $Q/Q$ 's and  $C/C$ 's in the atom and coop proof rules) using strongest postcondition and weakest precondition.
- (2) Verifies semantic conditions: semi-commutativity (left/right movers), Hoare triples, subsumption (corresponding to entailments in Fig 2 for proving commutativity of tests), and interference stability.
- (3) Checks automaton conditions: bijectivity, orthogonality (non-overlapping and non-factoring), and coverage.

QREF outputs a detailed report in ASCII, HTML, or LaTeX showing each obligation and its result.

The semantic conditions (left/right commutativity, Hoare triples, subsumption, and stability) are translated to the Veracity commutativity verification language [6]. Each query is translated into a

<sup>2</sup>We plan to publicly release QREF. The QREF source code and case study implementations are available in the supplemental materials.

|            |     |       | Semantic Queries |         |      |         |     |       |        | Automaton Queries |     |     |     | Total |           |        |
|------------|-----|-------|------------------|---------|------|---------|-----|-------|--------|-------------------|-----|-----|-----|-------|-----------|--------|
| Benchmark  | LOC | Specs | HTrp             | RMov    | LMov | Subs    | SGA | IStbl | Time   | Cov               | NO  | NSO | Bij | Time  | Queries   | Time   |
| Treiber    | 29  | 6     | 5                | 4       | 4    | 2       | 4   | 10    | 1.6s   | 2                 | 16  | 6   | 29  | 0.0s  | 83        | 1.6s   |
| MS-Queue   | 51  | 15    | 15               | (-3) 19 | 9    | 6       | 13  | 176   | 8.4s   | 3                 | 180 | 15  | 210 | 0.0s  | (-3) 647  | 8.4s   |
| Counter    | 24  | 5     | 4                | 4       | 2    | 2       | 3   | 10    | 1.1s   | 2                 | 12  | 5   | 21  | 0.0s  | 66        | 1.1s   |
| Elim-Stack | 100 | 19    | 19               | (-2) 34 | 20   | (-2) 4  | 12  | 451   | 256.3s | 2                 | 310 | 19  | 345 | 0.1s  | (-4) 1216 | 256.3s |
| RDCSS      | 64  | 19    | 19               | 19      | 6    | (-1) 7  | 19  | 439   | 16.3s  | 1                 | 316 | 19  | 348 | 0.0s  | (-1) 1193 | 16.3s  |
| Total      |     |       | 62               | (-5) 80 | 41   | (-3) 21 | 51  | 1086  | 283.8s | 10                | 834 | 64  | 953 | 0.1s  | (-8) 3205 | 283.8s |

Table 1. Results of applying QREF to verify IRSpecs for various canonical object implementations. Nearly all 3,205 queries were verified automatically, except for 8 as discussed below.

Veracity program, using assume, assert, havoc, and commute blocks. All thread-local variables of type  $\tau$  are translated to  $\text{Array}(\mathbb{N}, \tau)$  types so they can be thread-indexed. We extended Veracity's `commute_right` primitive with a new `commute_right_ctx` primitive meaning semi-commutativity up to a context (i.e. Def 4.4). This allowed right-mover queries  $R \triangleright_{t_j} Q$  to be discharged by constructing a program of the form

$$\text{commute\_right\_ctx } (global\ inv) \{ R \} \{ \text{havoc}(shared\ memory_{t_j}) \} \text{ context} : Q$$

The right-mover precondition *global inv* above involves simple facts about the ADT such as `Head ≠ null` && `Tail ≠ null` for MSQ. The shared memory and context operations are re-written so that they are indexed by the appropriate thread IDs (following the proof rules). QREF also generates a Veracity obligation to verify that all actions preserve *global inv*.

Veracity had support for arrays but not heap objects, so we extended Veracity's language and commutativity verification to support the heap structures and allocation used by our benchmarks such as linked-lists, two-dimensional arrays, and arrays of pointers to descriptors (elimination stack). For the automaton conditions, QREF translates the programs to automata and constructs the various automaton conditions (prefix closure, suffix closure, inclusion in  $\text{Any}_{\neg a} a \text{Any}_{\neg a}$ , etc) before dispatching queries to libmata [7].

*Interactivity.* The automated checking can guide users (or AI agents) through the process of writing IRSpecs by reporting errors that indicate what to fix next. We illustrate one approach using MSQ in Sec. 8.

A user can organize work into two phases. In a first phase, write IRSpecs to cover all paths, initially allowing long control-flow paths without worrying about atomic sections. QREF reports uncovered paths as counterexamples and flags bijectivity violations (requiring spec refinement) or orthogonality violations (requiring narrowing). Once coverage is complete, a second phase focuses on semantic conditions, e.g., verifying left/right movers. If a path fails the atomicity check, the user creates a new atom rule. If movers cannot move far enough, the user may revisit the first phase to make control-flow paths more fine-grained.

## 8 Case Studies

We evaluated (i) whether IRSpecs could be written and (ii) whether the semantic commutativity/Hoare conditions and automaton orthogonality, bijectivity and coverage conditions could be automatically validated. To that end, we used QREF to guide the process of writing and validating IRSpecs for several canonical examples from the literature:

- Treiber's Stack [42]. Unboundedly many threads performing concurrent push/pop operations.
- Hendler *et al.* Elimination Stack [20]. Uses Treiber as a basic object and provides another layer of synchronization where a thread attempting a pop can consume the value arriving

from a concurrent thread attempting a push. This involves multiple CAS operations on two arrays (one to announce thread descriptors and another to perform the “collision”).

- Michael-Scott Queue [34]. A non-blocking queue that supports unboundedly many threads concurrently enqueueing/dequeueing.
- Harris *et al.* Restricted Double-Compare, Single-Swap (RDCSS) [34]. A non-blocking object consisting of a Data register and a Control register, with threads attempting to “write” by updating the Data register, but only if both Data & Control have an expected value. Threads may “help” if one thread has begun but not completed its write operation.

Our experiments were performed on a MacBook M1 Pro 2021 with 32 GB of RAM and 10 cores. The results are summarized in Table 1. For each object, we report lines of code, and the number of IRSpecs required. We next report the number of semantic queries performed of each type (Hoare triples **HTrp**, right-movers **RMov**, left-movers **LMov**, subsumption **Subs**, single-global-access **SGA**, and stability **IStbl**, respectively), along with the total time to perform all of them. We then report the automaton queries (coverage **Cov**, non-overlapping **NO**, non-self-overlapping **NSO**, and bijection **Bij**) total time for them, followed by the total number of queries and time for the object. The semantic queries are dispatched to Veracity, which is configured to use CVC5 [1] with a 120s timeout and, if it returns unknown, we configure Veracity to use Z3 [9] with a 60s timeout (although timeouts were rare).

All IRSpecs for these objects are given in Apx. B along with QREF’s output. These IRSpecs were written with the help of QREF (see example below) and each object required 5-20 specs, confirming research question (i).

As shown in Table 1, QREF verified *nearly all 3,205 queries* automatically and did so within a few minutes, confirming research question (ii). In 8 cases (noted “(-n)” in the table) the queries went beyond what CVC5 or Z3 or Veracity could support so we manually validated them. In a few cases, SMT solvers reported unknown due to heavy use of quantification and arrays, such as Elimination Stack which used two-dimensional arrays. MSQ needed to quantify over prior values of Tail.

*Example: MSQ.* Recall the implementation of MSQ shown in the top of Fig. 5. We label statements and conditions with black circles (e.g., **Ⓐ**) to mark key points referenced in the IRSpecs. (Some local variable definitions are omitted for brevity.)

The MSQ is a linked list maintained by global pointers `q.head` and `q.tail`. Values are stored in nodes between them, with `enq` adding nodes at `q.tail` and `deq` removing nodes at `q.head`.

During a successful `enq` CAS at **Ⓕ**, `q.tail→next` changes from null to a new node. However, that node cannot be dequeued until `adv` advances `q.tail` to point to it. A `deq` on an empty list (when `q.head=q.tail`) returns immediately; otherwise it advances `q.head` and returns the value. The original MSQ implementation includes the `adv` CAS inline within `enq` and `deq` loops. For expository purposes, we factor it into a separate `adv` method in Fig. 5.

*Interactively writing IRSpecs for MSQ.* The 15 IRSpecs at the bottom of Fig. 5 cover all unbounded thread behaviors of the MSQ. Below we describe how QREF guides a user to write and validate these specs using the proof rules from Sec. 6.

The IR-spec `aEnqS` covers the successful “happy path” of an enqueueing thread: the prefix **Ⓐ-Ⓔ** (captured in the previous IR-spec `aEnqPrep`), a successful CAS at **Ⓕ**, and return at **Ⓖ**. This IRSpec directs induction over all successful CAS **Ⓕ** instances; for each, we use movers *L* and *R* to make **Ⓐ-Ⓖ** atomic.

When a user writes `aEnqS`, QREF follows the atom proof rule and verifies the left-/right-mover and Hoare triple semantic conditions along with syntactic bijection. QREF also verifies orthogonality (trivially true with just one spec) but fails coverage, reporting the path “abcdehs” as uncovered. This path occurs when a concurrent enqueueer races against the first enqueueer’s successful CAS and

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> int enq(int v){ loop {   A node_t *node=...;   B tail=Q.tail; C next=tail→next;   if (D Q.tail==tail) {     if (E next==null) {       if (F CAS(&amp;tail→next,next,node)) G ret 1;     } else H[cas/f] S skip;   } else J[next≠null] S skip;   } else K[Q.tail≠tail] S skip; } }  void adv() { loop {   I tail=Q.tail; L next=tail→next;   if (M next≠null) {     if (N CAS(&amp;Q→tail,tail,next)) O ret 0;   } else P[cas/f] Q skip;   } else R[next==null] T skip; } } </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <pre> int deq(){ loop {   A int pval; B head=Q.head;   C tail=Q.tail; D next=head→next;   if (E Q.head==head) {     if (F head==tail) {       if (G next==null) H ret 0;     } else I[next≠null] J skip;   } else K[head≠tail] {     L pval=next→val;     if (M CAS(&amp;Q→head,head,next))       N ret pval;     else O[cas/f] P skip;   } } }   } else Q[Q.head≠head] R skip; } </pre> |
| <ol style="list-style-type: none"> <li>1. aEnqPrep = (<math>a_i:e</math>, <math>\text{atom}\{[ab]c \triangleright [de] \triangleleft \epsilon\}</math>)</li> <li>2. aEnqS = (<math>a_i:f</math>, <math>\text{atom}\{a\text{EnqPrep} \triangleright f \triangleleft g\}</math>)</li> <li>3. cEnqF = (<math>a_i:f</math>, <math>\text{coop}\{[a\text{EnqPrep}] \triangleright [a\text{EnqS}] \triangleleft \text{hs}\}</math>)</li> <li>4. cEnqFstale = (<math>a_i:f</math>, <math>\text{coop}\{[ab] \triangleright [a\text{EnqS}] \triangleleft \text{cdjs}\}</math>)</li> <li>5. aDeqPrep = (<math>a_i:K</math>, <math>\text{atom}\{[AB]CD \triangleright [EKL] \triangleleft \epsilon\}</math>)</li> <li>6. aDeqS = (<math>a_i:M</math>, <math>\text{atom}\{a\text{DeqPrep} \triangleright M \triangleleft N\}</math>)</li> <li>7. cDeqF = (<math>a_i:M</math>, <math>\text{coop}\{a\text{DeqPrep} \triangleright a\text{DeqS} \triangleleft OP\}</math>)</li> <li>8. cDeqFH = (<math>a_i:M</math>, <math>\text{coop}\{[AB] \triangleright [a\text{DeqS}] \triangleleft CDQR\}</math>)</li> <li>9. aEmp = (<math>a_i:G</math>, <math>\text{atom}\{[AB]CD \triangleright [EFG] \triangleleft H\}</math>)</li> <li>10. aOne = (<math>a_i:I</math>, <math>\text{atom}\{[AB]CD \triangleright [EFI] \triangleleft J\}</math>)</li> <li>11. aAdvPrep = (<math>a_i:m</math>, <math>\text{atom}\{i \triangleright [lm] \triangleleft \epsilon\}</math>)</li> <li>12. aAdvS = (<math>a_i:n</math>, <math>\text{atom}\{a\text{AdvPrep} \triangleright n \triangleleft o\}</math>)</li> <li>13. cAdvF = (<math>a_i:n</math>, <math>\text{coop}\{a\text{AdvPrep} \triangleright [a\text{AdvS}] \triangleleft pq\}</math>)</li> <li>14. cAdvF2 = (<math>a_i:n</math>, <math>\text{coop}\{[ab] \triangleright [a\text{AdvS}] \triangleleft \text{cks}\}</math>)</li> <li>15. aAdv = (<math>a_i:r</math>, <math>\text{atom}\{i \triangleright [lr] \triangleleft t\}</math>)</li> </ol> |                                                                                                                                                                                                                                                                                                                                                                                          |

Fig. 5. The Michael-Scott Queue and its 15 IRSpecs. Recall that brackets  $[\pi]$  create a sub-requirement that  $\pi$  be atomic, which is discharged either by one of the other IRSpec rules, or by a check that  $\pi$  contains only one global memory access. For simplicity we consider that the read of the `val` field is not a global access since this can only be written by the creator of the node.

fails its own CAS operation. The user writes `cEnqF` to cover the case of CAS failure at **H** and `QREF` verifies right-mover, Hoare triple, and subsumption conditions. However, a second coverage failure reveals the path “abcdjs”, where a concurrent enqueueer fails even earlier because it observes a stale tail; this is covered by `cEnqFstale`.

The dequeue operation follows a similar pattern. `aDeqPrep` captures the shared prefix **A–D** (reads of `pval`, `head`, `tail`, `next`) along with checks that `Q.head` remains stable and the queue is non-empty (at **E**, **K**, **L**). `aDeqS` extends this to the successful CAS at **M** and return at **N**. Two cooperative specs cover concurrent dequeuer races: `cDeqF` covers a dequeuer losing the race at **O**, and `cDeqFH` covers a dequeuer that reads `head/tail` early (at **A**, **B**) but finds the head already advanced when it re-reads (at **Q**), requiring a retry.

Further coverage checks reveal uncovered paths (e.g., ABCDEFGH). The user adds `aEmp` to cover the empty queue case (returns at **H**) and `aOne` to cover a single-element queue where the dequeuer must retry because the tail has not advanced.

The remaining IRSpecs cover the helper thread that advances the tail with `adv`. `aAdvPrep` captures the prefix **i** (reading `tail`) and the check that `next ≠ null` (at **l**, **m**). `aAdvS` extends this to the successful CAS at **n** and return at **o**. `cAdvF` covers other advancers losing a race at **p**, while `cAdvF2` covers enqueueers that read early (at **a**, **b**) and observe the advancer complete before their re-read of `next` (at **c**) finds `Q.tail` stale. Finally, `aAdv` covers the idle case where the advancer finds nothing to do because the tail is not lagged, **q**.

**Automatically discharging semantic obligations.** `QREF` reports that these 15 IRSpecs cover the entirety of `MSQ`. Discharging each IRSpec’s premises (Sec. 6) reduces to five kinds of semantic checks: contextual commutativity of *left/right movers*, a *Hoare triple* for the anchor, an *interference-stability* check on the surrounding context, a *subsumption* check (for cooperative IRSpecs) linking the referenced atomic event to the interfering thread’s path, and preservation of the object’s *global invariant*. Contexts (denoted  $Q_{t_i}$ ,  $Q'_{t_i}$ ,  $C_{t_j}$ ,  $C'_{t_j}$  in the proof rules in Sec. 6) are calculated using weakest pre / strongest post and those calculated contexts can be seen in the list of obligations in Apx. B.

These checks are all verified automatically using Veracity [6] as discussed in Sec. 7. For `MSQ`, the only exception is that in three `RMov` conditions, it was necessary to see that a read of `Tail` could read any possible earlier value, which required reasoning beyond Veracity’s current capabilities, so we verified this manually. We now provide some examples of the semantic checks for `MSQ`, with the full details in Apx. B.

*Left/right movers.* For `aEnqS`, we verify that the prefix `abcde` is a right-mover up to **f** and suffix `g` is a left-mover. For the move of `abcde` to the right, the precondition context  $Q_{t_i}$  records:  $tail_{t_i} \rightarrow next = next_{t_i}$  (thread  $t_i$ ’s earlier read matches the current value of `tail`’s `next`) and that  $tail_{t_i} = Q.tail$ . For the move of `g` to the left, the postcondition  $Q'_{t_i}$  records that  $tail_{t_i} \rightarrow next = node_{t_i}$  (the CAS succeeded). For the failing case `cEnqF`, we move failing prefix `abcde` up to the context  $C_{t_j}$  which describes a concurrent thread  $t_j$  that observed  $tail_{t_j} \rightarrow next = next_{t_j}$  before attempting its CAS. Post-context  $C'_{t_j}$  describes the failure state  $Q.tail \rightarrow next \neq node_{t_j}$  and we move `hs` left.

*Hoare triples.* For `aEnqS`’s anchor **f** (the CAS itself), `QREF` checks the Hoare triple  $\{Q.tail \rightarrow next = next_{t_i}\} \text{ f } \{Q.tail \rightarrow next = node_{t_i}\}$ : assuming thread  $t_i$ ’s locally-read `next` still matches the shared `tail`’s successor, performing the CAS re-links the `tail` to  $t_i$ ’s freshly-allocated node.

*Interference stability.* The context  $C_{t_j}$  for `cEnqF` records that thread  $t_j$  has read `node`, `val`, `tail`, `next` and observed `next = NULL`. This context must remain stable when any other thread  $t_k$  executes a single statement. For example, a concurrent dequeuer might execute **B** (`head=Q.head`) or **C** (`tail=Q.tail`). Since these reads touch disjoint global variables, they do not invalidate  $C_{t_j}$ .

*Subsumption.* For `cEnqF`, we verify that once  $t_i$ ’s `aEnqS` completes, its postcondition  $Q.tail \rightarrow next = node_{t_i}$  forces  $t_j$  into the failure case. When  $t_j$  re-evaluates its CAS guard at **f**, it will find that  $tail \rightarrow next$  no longer equals its locally-read `next`, so the CAS fails and  $t_j$  takes the failure path **h**.

*Global invariant.* `MSQ` maintains a global invariant:

$$Q.head \neq \text{NULL} \wedge Q.tail \neq \text{NULL} \wedge (Q.head = Q.tail \vee Q.head \rightarrow next \neq \text{NULL})$$

This is declared once and checked implicitly as part of every context. `QREF` verifies the invariant is preserved by each atomic anchor. For `aEnqS`, the CAS at **f** links a non-`NULL` node, so if  $Q.head = Q.tail$  held before, then  $Q.head \rightarrow next \neq \text{NULL}$  now holds, maintaining the invariant.

**RDCSS and Elimination Stack.** `RDCSS` was challenging because it involved a special compare-and-swap that returned the previous value in addition to returning a boolean success flag. We added this “`rCAS`” primitive to the language. There are a total of 8 CAS/`rCAS` instances due to the various possibilities of how a thread (or helping thread) updates the register in two steps.

Elimination Stack was the most challenging, having two-dimensional arrays. The more difficult `RMov` queries required CVC5 with its “`full-saturate-quant`” option to support extensionality over `havoc` (i.e. the fact that  $\text{store}(\text{havoc}(\text{arr}), \text{idx}, \text{val}) \supseteq \text{havoc}(\text{arr})$ ).



We also asked Claude Code to assist with these two objects. Claude was helpful in keeping the .c11 statement labels consistent with the IRSpec descriptions. It could also examine the output of QREF (such as coverage counter examples or semantic failures) and made some suggestions for how to repair candidate IRSpecs.

## 9 Related work

The core contribution of this paper is to show that scenario-based reasoning can be automated through induction-reduction specifications and proof rules that reduce the problem to Hoare logic, contextual commutativity and automaton operations (for orthogonality, bijectivity and coverage). To our knowledge no prior works provide such automation. We here build upon the “scenario-based” reasoning of Enea and Koskinen [13]. Although their work set out key definitions and described manual scenario-based proofs, it did not provide a means of mechanizing such proofs, and the implementation merely produces candidate (unsound) so-called quotient expressions.

Program *logics* for concurrent programs have been studied extensively, as mentioned in Sec. 1. Improving on classical Owicki-Gries [38] and Rely-Guarantee [23] logics, numerous extensions of Concurrent Separation Logic [2, 4, 36, 39] have been proposed to reason compositionally about different instances of fine-grained concurrency, e.g. [8, 11, 24, 25, 30, 31, 35, 40, 41, 43–45].

Other earlier approaches to concurrent reasoning have also been based on the *reduction theory* of Lipton [32], who introduced the concept of left/right *movers* that re-order operations to create atomic blocks of code. Tools such as QED [12] and CIVL [19] expanded upon Lipton’s theory and added invariant reasoning and refinement layers [28, 29]. The reduction theory in CIVL has been formalized in [17]. The soundness proof uses an induction on non-mover actions in atomic blocks which are determined statically. Reasoning via simplifying program transformations has also been adopted in the context of mechanized proofs, e.g., [5]. Inductive sequentialization [27] builds upon this prior work, and introduces a new scheme for reasoning inductively over unbounded concurrent executions.

The works of Farzan et al. [14, 15] define an algorithmic verification framework for concurrent programs that includes computing sound reductions. The syntactic representation of these reductions is somewhat complex, which is fine in the context of a fully automated, non-interactive proof. In our paper, we look at classes of reductions which are humanly readable and use simple syntactic constructs.

## 10 Conclusion

We have shown a way of expressing a combination of the powerful reasoning techniques of induction and reduction, and those so-called IRSpecs can be validated through known reasoning primitives: Hoare logic, contextual commutativity, and syntactic automaton conditions. A handful of IRSpecs allow unbounded thread behavior on a concurrent object to be expressed as a few tangible atom/coop “scenarios.” Furthermore the proof rules, which are embodied in our implementation QREF, allow a user to interactively write IRSpecs, receiving feedback when they are incorrect or insufficient.

*Future work.* IRSpecs reduce an input concurrent object into a far simpler form, where critical algorithm points (e.g. successful CAS operations) are isolated and impacted concurrent threads are organized around those points, making linearizability far more immediate. This automatic derivation of a sound quotient was not possible in prior works such as Enea and Koskinen [13].

In the future, we will extend QREF so that it refines the current starred-union representation of the quotient with an automaton (like the one described in Enea and Koskinen [13]), to obtain a more precise version that takes into account the ADT’s abstract state and what transitions are

possible. Furthermore, we will integrate QREF with tools for proving Linearizability to extend those tools' reach. We will also explore whether IRSpecs can be synthesized.

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## A Proofs

LEMMA A.1. *Let  $\mathcal{A}$  be a collection of IRSpecs that are covering and orthogonal for a concurrent object  $O$ , and let  $\alpha = (\mathbf{a}, \text{atom}(\mathbf{R} \triangleright \mathbf{N} \triangleleft \mathbf{L})) \in \mathcal{A}$ . For each trace  $\tau$  and each thread  $t$ , let*

$$f_{\tau,t} : \text{occ}_{\mathbf{a}}(\tau|_t) \rightarrow \text{occ}_{\text{RNL}}(\tau|_t)$$

*be an order-preserving bijection. Such a family  $\{f_{\tau,t}\}_{\tau,t}$  exists provided the following conditions hold:*

- (i)  $\text{RNL} \leq \text{Any}_{\neg \mathbf{a}} \mathbf{a} \text{Any}_{\neg \mathbf{a}}$
- (ii)  $\forall \alpha' \in \mathcal{A}. \alpha' \neq \alpha \wedge \alpha < \alpha' \Rightarrow [\alpha'] \leq (\text{Any}_{\neg \mathbf{a}} \mathbf{a} \text{Any}_{\neg \mathbf{a}} + \text{Any}_{\neg \mathbf{a}}) = 0$
- (iii)  $\forall \alpha' \in \mathcal{A}. \alpha' \neq \alpha \wedge \alpha \not< \alpha' \Rightarrow [\alpha'] \cap (\text{Any } \mathbf{a} \text{Any}) = 0$

PROOF. The proof is by construction. Fix an arbitrary trace subset  $T$  in  $\text{Traces}(O)$ . Given the two conditions, we define a family of bijections  $\forall \tau \in T. \forall t \in \mathcal{T}. \text{occ}_{\mathbf{a}}(\tau|_t) \leftrightarrow \text{occ}_{\text{RNL}}(\tau|_t)$  between occurrences of  $\mathbf{a}$  in a trace restriction to thread  $t$  and occurrences of control paths represented by the regular expression RNL in the same trace restriction.

Given any word  $w$ , we define the occurrence functions  $\text{occ}$  for a symbol  $\mathbf{a}$  and a regular expression  $k$  as follows:

$$\begin{aligned} \text{occ}_{\mathbf{a}}(w) &= \{i \in \mathbb{N} \mid w[i] = \mathbf{a}\} \\ \text{occ}_k(w) &= \{(i, j) \in \mathbb{N}^2 \mid w[i : j] \in \llbracket k \rrbracket\} \end{aligned}$$

As expected, for a symbol  $a$ ,  $\text{occ}_a(w)$  returns the set of positions at which  $a$  occurs in the string  $w$ , while for a regular expression,  $k$ ,  $\text{occ}_k(w)$  returns the set of index pairs identifying factors of  $w$  in the language of  $k$ .

Let  $\tau$  be an arbitrary trace in  $T$ , let  $t$  be an arbitrary thread ID in  $\mathcal{T}$ .

By the definition of covering and orthogonality of  $\mathcal{A}$ , there exists a factorization of the restriction of  $\tau$  to the thread  $t$ ,  $\tau|_t = \tau_0 \tau_1 \cdots$  such that, for every  $\tau_k$ , there exist an induction-reduction  $\alpha_k$  with its reduction pattern  $\text{pat}_k = [\alpha_k]$  and  $\tau_k \in \llbracket \text{pat}_k \rrbracket_t$ . That is,  $\tau_k$  is in the language defined by the reduction pattern of  $\alpha_k$  tagged by thread  $t$ . Since  $\alpha$  is an atomic specification,  $[\alpha] = \text{RNL}$ . Therefore, the set  $\text{occ}_{\text{RNL}}(\tau|_t)$  induces a total interval order. The interval order is defined in the usual way:  $(i_1, j_1) \leq (i_2, j_2) \Leftrightarrow j_1 \leq i_2$  and it's strict whenever  $j_1 < i_2$ .

Define the function  $f : \text{occ}_{\text{RNL}}(\tau|_t) \rightarrow \text{occ}_{\mathbf{a}}(\tau|_t)$  by:

$$f((i, j)) = l, \text{ where } i \leq l < j \text{ and } \tau|_t(l) = \mathbf{a}$$

We show that  $f$  is a bijection.

From (i) it follows any word in RNL has exactly one occurrence of  $\mathbf{a}$ , that is,  $l$  exists and it's unique. That ensures that the function  $f$  is well defined. It's immediate to see that  $f$  is injective, since there is no overlap between RNL-factors in  $\tau_t$ .

To prove surjectivity of  $f$  we proceed by contradiction. Assume that there exists  $l \in \text{occ}_{\mathbf{a}}(\tau|_t)$  that lies outside the RNL-factors of  $\tau|_t$ . Let  $\tau_l$  be such factor. By the factorization of  $\tau|_t$ , it must be that  $\tau_l$  is in the language of the reduction pattern  $\text{pat}' = [\alpha']$  of some IRSpec  $\alpha'$  that is distinct from  $\alpha$ . We proceed by case whether  $\alpha < \alpha'$  or  $\alpha \not< \alpha'$ . Both cases are immediately discharged by contradiction. The first contradicts (ii). This requires a case analysis on the structural dependency between the reduction pattern of  $\alpha$  and the reduction components of  $\alpha'$  that justifies the order between  $\alpha$  and  $\alpha'$ . The second contradicts (iii).

Lastly,  $f$  is order-preserving by definition. □

LEMMA A.2. *Let  $\mathcal{A}$  be a collection of IRSpecs that are covering and orthogonal for a concurrent object  $O$ . For any  $\alpha = (\mathbf{a}^\cup, \text{coop}(\mathbf{R} \triangleright \mathbf{N} \triangleleft \mathbf{L})) \in \mathcal{A}$ , there exists a family of trace- and thread-indexed, order-preserving bijections*

$$\{\text{occ}_{\mathbf{a}^\cup}(\tau|_t) \leftrightarrow \text{occ}_{\mathbf{N}}(\tau|_t)\}_{\tau,t}$$

*provided that*

- (i) for each  $\mathbf{a} \in \mathbf{a}^\cup$ , there exists an atomic IRSpec  $\alpha_a \in \mathcal{A}$  such that  $\alpha_a \prec \alpha$
- (ii)  $N = \sum k$  for  $k \in \{R_a N_a L_a \mid \exists (\mathbf{a}_a, \text{atom}(R_a \triangleright N_a \triangleleft L_a)) \prec \alpha\}$
- (iii) for each  $\alpha_a = (\mathbf{a}_a, \text{atom}(R_a \triangleright N_a \triangleleft L_a))$  such that  $\alpha_a \in \mathcal{A}$  and  $\alpha_a \prec \alpha$ , there exists a family of trace- and thread-indexed order-preserving bijections  $\{f_{\tau,t} : \text{occ}_{\mathbf{a}_a}(\tau|_t) \leftrightarrow \text{occ}_{R_a N_a L_a}(\tau|_t)\}_{\tau,t}$ .

PROOF. Fix a trace  $\tau$  and a thread  $t$ .

The set  $\text{occ}_{\mathbf{a}^\cup}(\tau|_t)$  admits a partition  $\bigcup_{\mathbf{a}_a \in \mathbf{a}^\cup} \text{occ}_{\mathbf{a}_a}(\tau|_t)$ .

Let  $N_\cup = \{R_a N_a L_a \mid \exists (\mathbf{a}_a, \text{atom}(R_a \triangleright N_a \triangleleft L_a)) \prec \alpha\}$ . By the condition (ii), i.e.,  $N = \sum R_a N_a L_a$ , and orthogonality  $\mathcal{A}$ , it must be that  $\text{occ}_N(\tau|_t)$  admits a partition  $\bigcup \text{occ}_{R_a N_a L_a}(\tau|_t)$ .

By assumption (iii), for each predecessor atomic IRSpec  $\alpha_a = (\mathbf{a}_a, \text{atom}(R_a \triangleright N_a \triangleleft L_a))$  there is an order-preserving bijection  $\text{occ}_{\mathbf{a}_a}(\tau|_t) \leftrightarrow \text{occ}_{R_a N_a L_a}(\tau|_t)$ . We can define a global bijection from the family of corresponding partitions related by componentwise order-preserving bijections, and this concludes the proof.  $\square$

LEMMA A.3 (ATOM SOUNDNESS). *Let  $\mathcal{A}$  be a collection of IRSpecs for a concurrent object  $O$ . For every atomic specification  $\alpha \in \mathcal{A}$ , the rule ATOM is sound.*

PROOF. Let  $\alpha = (\mathbf{a}, \text{atom}(R \triangleright N \triangleleft L))$  be an atomic IRSpec. By definition, its corresponding canonical expression  $r = \text{canon}(\alpha)$  is equal to  $[RNL]$ . Fix  $T = \text{Traces}(O[N])$  a subset of traces of  $O$  where the paths in  $N$  are executed in isolation.

Define  $T_0^\alpha = T$  and for any  $k \geq 1$ , let  $T_k^\alpha$  denote the following restriction

$$T_k^\alpha = T \cap \left( \left( \bigcup_{0 \leq j \leq k-1} (\text{Any}_{\neg r}(|r|))^j \text{Any}_{\neg r} \right) \cup (\text{Any}_{\neg r}(|r|))^k \text{Any} \right)$$

consisting of traces that admit a factorization  $\tau_0 \tau_{r_1} \tau_1 \cdots \tau_{r_j} \tau_j$  for  $j \leq k$  with  $\tau_{r_i} \in (|r|)$ , where  $(|r|) = \bigcup_{t \in \mathcal{T}} [[RNL]]_t$ .

The proof has two steps. We begin with showing by induction that  $\forall k \geq 0. T \rightarrow T_k^\alpha$ . This will let us infer that  $T \rightarrow \inf_{k \geq 0} T_k^\alpha$ , that is there is a reduction from  $T$  to the greatest lower bound, i.e.,  $\inf$  of all  $T_k^\alpha$ . Then we demonstrate that  $\inf_{k \geq 0} T_k^\alpha$  coincides with  $\text{Traces}(O[r])$ . By construction it equivalently describes  $[[\alpha]](T)$ . Because  $T$  stands for  $\text{Traces}(O[N])$ , together it will give us that  $O[N] \xrightarrow{\alpha} O[RNL]$ .

*Step 1.* We show by induction on  $k$  that  $T \rightarrow T_k^\alpha$  for all  $k \geq 0$ .

*Base case* ( $k = 0$ ). Since  $T = T_0^\alpha$ , the base case hold trivially.

*Inductive step.* Assume that  $T \rightarrow T_k^\alpha$ . We must prove  $T \rightarrow T_{k+1}^\alpha$ .

Observe that by the definition of  $T_k^\alpha$  it is the case that  $T_k^\alpha \supseteq T_{k+1}^\alpha$ . Furthermore, if we can establish that for any  $\tau \in T_k^\alpha$  there exists  $\tau' \in T_{k+1}^\alpha$  such that  $\tau \leq \tau'$ , then we can infer that  $T_k^\alpha \rightarrow T_{k+1}^\alpha$ . By transitivity of  $\rightarrow$  and the induction hypothesis we can conclude that  $T \rightarrow T_{k+1}^\alpha$ .

Let  $\tau$  be an arbitrary trace in  $T_k^\alpha$ . We proceed by case analysis on the number of occurrences of  $\mathbf{a}$  in  $\tau$ . We use  $\#(\mathbf{a}, \tau)$  to denote the function that inductively calculates such occurrences.

*Case  $\#(\mathbf{a}, \tau) \leq k$ .* The proof is immediate by choosing  $\tau' = \tau$ . The condition of  $\tau' \in T_{k+1}^\alpha$  holds, since for any  $l \leq k$ , the trace  $\tau$  has the first  $l$  occurrences of paths  $\pi$  in RNL already arranged in canonical form, and for  $(k+1)^{\text{th}}$  the condition holds vacuously.

*Case  $\#(\mathbf{a}, \tau) \geq (k+1)$ .* We show how to derive a trace  $\tau' \in T_{k+1}^\alpha$  from  $\tau$  such that  $\tau \leq \tau'$ . Let  $(\mathbf{a})_t$  be the  $(k+1)^{\text{th}}$  occurrence of  $\mathbf{a}$  in  $\tau$ ; the subscript  $t$  indicates the thread ID that executes it. By the existence of an bijective map between  $\text{occ}_{\mathbf{a}}(\tau|_t)$  and  $\text{occ}_{RNL}(\tau|_t)$  we know that there is a trace fragment  $\pi_t \in [[RNL]]_t$  in the trace restriction to thread  $t$  denoted by  $\tau|_t$ , that corresponds to

(a)<sub>t</sub>. We write  $\pi_t = (\pi(i) : t)_{i=1}^{|\pi|}$  as a shorthand for pointwise tagging the control path  $\pi$  with the thread identifier  $t$ . Furthermore,  $\pi_t$  admits a factorization  $\pi_t = (\pi_R)_t(\pi_N)_t(\pi_L)_t$ , where  $\pi_R \in \llbracket R \rrbracket$ ,  $\pi_N \in \llbracket N \rrbracket$ , and  $\pi_L \in \llbracket L \rrbracket$ .

Since  $\tau \in T$  and  $T$  is the trace restriction  $Traces(O[N])$ , then it must be that  $\tau$  admits the factorization  $\tau_0 \tau_{N_1} \tau_1 \tau_{N_2} \dots$ , where for all  $i$ ,  $\tau_{N_i} \in \llbracket N \rrbracket_{t_i}$  for some  $t_i \in \mathcal{T}$ . Therefore we can infer that  $\tau$  admits also the following factorization:

$$\exists \tau_1 \tau_2 \in \mathbb{E}^*. \tau = \tau_1 (\pi_N)_t \tau_2$$

such that the events in  $(\pi_R)_t$  occur in  $\tau_1$  and are interleaved with actions from other threads, and similarly, the events in  $(\pi_L)_t$  are interleaved with other actions in  $\tau_2$ .

We instantiate the rule **ATOM** for the thread  $t$  and we apply the premise giving us the contexts  $B_t$  and  $B'_t$  from the grounding Hoare triple for the path  $(\pi_N)_t \in \llbracket N \rrbracket$  executed by  $t$ :

$$\{B_t\}(\pi_N)_t\{B'_t\}$$

We can now exploit the contexts for the commutativity movers over **KAT**.

By the definition of the commutativity right mover,  $R \triangleright_{t_j} B$ , and the fact that  $\pi_R \in \llbracket R \rrbracket$ , we infer that:

$$\exists \tau'_1 \in \mathbb{E}^*. \tau \leq \tau'_1 (\pi_R)_t (\pi_N)_t \tau_2 \quad (2)$$

Similarly, by the definition of  $B'_{t_j} \triangleleft L$  and the fact that  $\pi_L \in \llbracket L \rrbracket$ :

$$\exists \tau'_1, \tau'_2 \in \mathbb{E}^*. \tau \leq \tau'_1 (\pi_R)_t (\pi_N)_t (\pi_L)_t \tau'_2 \quad (3)$$

Take  $\tau' = \tau'_1 (\pi_R)_t (\pi_N)_t (\pi_L)_t \tau'_2$ .

The final step of our induction step is to show that  $\tau' \in T_{k+1}^\alpha$ .

First, by the fact that the bijections  $\forall t \in \mathcal{T}. \text{occ}_a(\tau|_t) \leftrightarrow \text{occ}_{\text{RNL}}(\tau|_t)$  are also order-preserving, we infer that the first  $k$  occurrences of paths  $\pi \in \llbracket \text{RNL} \rrbracket$  must occur in the prefix  $\tau_1$  of  $\tau$ . That it,  $\tau$  admitted the factorization

$$\tau = \tau_{10}(\pi_1)_{t_1} \tau_{11}(\pi_2)_{t_2} \tau_{12} \dots (\pi_k)_{t_k} \tau_{1k} (\pi_N)_t \tau_2, \text{ where } \forall j \leq k. \pi_j \in \llbracket \text{RNL} \rrbracket$$

We proceed by case analysis on whether there exists  $j \leq k$ , s.t.  $t_j = t$ .

*Case  $\nexists j \leq k. t_j = t$ .* Then all previous  $(\pi_{\text{RNL}})_{t_j}$  occur in the prefix  $\tau'_1$  of  $\tau'$ , since their actions were not selected during the previous rewritings of constructing the witness  $\tau'$ . This concludes this case

*Case  $\exists j \leq k. t_j = t$ .* We prove by contradiction that the trace fragment  $(\pi_j)_{t_j}$  in  $\tau_1$  has not been affected by the previous rewritings. Let  $a_t$  be the first action encountered when traversing  $(\pi_R)_t$  in reverse that occurs in  $(\pi_{\text{RNL}})_{t_j}$ . This implies that suffix of  $(\pi_j)$  and prefix of  $(\pi_R \pi_N \pi_L)$  overlap. However this contradicts the non-self-overlapping condition *non-self-ovl*(*RNL*) of the rule.

This proves that  $\tau'$  admits a factorization:

$$\tau' = \tau'_{10}(\pi_1)_{t_1} \tau'_{11}(\pi_2)_{t_2} \tau'_{12} \dots (\pi_k)_{t_k} \tau'_{1k} (\pi_R \pi_N \pi_L)_t \tau'_2$$

where for all  $j \leq k$  it is the case that  $\pi_j \in \llbracket \text{RNL} \rrbracket$  and  $(\pi_R \pi_N \pi_L) \in \llbracket \text{RNL} \rrbracket$ . This concludes the proof that  $\tau' \in T_{k+1}^\alpha$ .

*Step 2.* We prove that  $\inf_{k \geq 0} T_k^\alpha = \text{Traces}(O[\text{RNL}]) = \llbracket \alpha \rrbracket(T)$ .

Consider the sequence  $(T_k^\alpha)_{k \geq 0}$ . Since we have established in Step 1. that  $T_k^\alpha \rightarrow T_{k+1}^\alpha$  for all  $k$ , this sequence forms an descending chain  $T = T_0^\alpha \supseteq T_1^\alpha \supseteq T_2^\alpha \dots$ . The limit of this chain is the restriction  $T \cap (\text{Any}_{\neg r} \llbracket r \rrbracket)^* \text{Any}_{\neg r}$ , which coincides with  $T[r]$ . Additionally, since the limit has representatives of equivalence classes that are  $\preceq_r$ -maximal, it is also equal to  $\llbracket \alpha \rrbracket(T)$ . Consequently,  $\text{Traces}(O[r]) = \llbracket \alpha \rrbracket(T)$ .



This concludes our proof. □

LEMMA A.4 (ASEQ SOUNDNESS). *Let  $\mathcal{A}$  be a collection of IRSpecs for a concurrent object  $O$ . For every two atomic IRSpec,  $\alpha_1, \alpha_2 \in \mathcal{A}$ , the rule ASEQ is sound*

PROOF. We fix two atomic IRSpecs  $\alpha_1 = (\mathbf{a}_1, \text{atom}(R_1 \triangleright N_1 \triangleleft L_1))$  and  $\alpha_2 = (\mathbf{a}_2, \text{atom}(R_2 \triangleright N_2 \triangleleft L_2))$ . Let  $P_1 \triangleq R_1 N_1 L_1$  and  $P_2 \triangleq R_2 N_2 L_2$ . We show that

$$[N_1 + N_2] \stackrel{\alpha_1; \alpha_2}{\sqsupseteq} [P_1 + P_2].$$

Let  $\tau \in \text{Traces}(O[N_1 + N_2])$ . We reason by an arbitrary path in  $[N_1 + N_2]$  occurring in  $\tau$  that is executed in isolation by some thread. By the semantics of choice, the anchor selected by  $[N_1 + N_2]$  is either an occurrence of  $N_1$  or an occurrence of  $N_2$ .

*Case 1:* The selected anchor is an occurrence of  $N_1$ . By the premise  $[N_1] \stackrel{\alpha_1}{\sqsupseteq} [P_1]$ , the trace  $\tau$  reduces, via  $\alpha_1$ , to a trace  $\tau'$  in  $\text{Traces}(O[P_1])$ .

*Case 2:* The selected anchor is an occurrence of  $N_2$ . By the premise  $[N_2] \stackrel{\alpha_2}{\sqsubseteq} [P_2]$ , the trace  $\tau$  reduces, via  $\alpha_2$ , to a trace  $\tau'$  in  $Traces(O[P_2])$ .

In both cases, it remains to show that the reduction is preserved in the presence of the other pattern. This follows from the orthogonality assumption

$$P_1 \text{ orthog } P_2,$$

which ensures that occurrences of  $P_1$  and  $P_2$  neither overlap nor factor through one another. Hence, applying the reduction associated with one IRSpec does not create, destroy, or partially consume occurrences governed by the other.

Therefore, in either case, the resulting trace  $\tau'$  belongs to

$$Traces(O[P_1 + P_2]).$$

Since  $\tau$  and the path in  $[N_1 + N_2]$  were arbitrary, we conclude that

$$[N_1 + N_2] \stackrel{\alpha_1; \alpha_2}{\sqsubseteq} [P_1 + P_2].$$

□

**LEMMA A.5 (COOP SOUNDNESS).** *Let  $\mathcal{A}$  be a collection of IRSpecs for a concurrent object  $O$ . For every cooperative IRSpec  $\alpha \in \mathcal{A}$ , the rule COOP is sound.*

**PROOF.** Fix a cooperative IRSpec  $\alpha = (\mathfrak{a}^\cup, \text{coop}(R \triangleright N \triangleleft L))$ . As in the previous proof we fix a subset of traces  $T = Traces(O[N])$  where the paths in  $N$  are executed in isolation. We need to prove that the conditions of the rule soundly prove a reduction  $T \rightarrow Traces(O[r])$ , where  $r = \langle\langle R[N]L \rangle\rangle$  is the canonical expression corresponding to  $\alpha$ .

We again introduce the restriction  $T_k^\alpha$ , where  $T_0^\alpha = T$  and for  $k \geq 1$  it is defined as follows:

$$T_k^\alpha = T \cap \left( \bigcup_{0 \leq j \leq k-1} (\text{Any}_{\neg r}(\llbracket r \rrbracket)^j \text{Any}_{\neg r}) \cup (\text{Any}_{\neg r}(\llbracket r \rrbracket)^k \text{Any}) \right)$$

Recall that the trace semantics of an cooperative canonical expression  $r = \langle\langle R[N]L \rangle\rangle$  is given by:

$$\langle\langle R[N]L \rangle\rangle = \bigcup_{t \in \mathcal{T}} \bigcup_{J \subseteq \mathcal{T} \setminus \{t\}} (\vec{\Pi}_{t_j \in J} \llbracket [R] \rrbracket_{t_j}) \llbracket [N] \rrbracket_t (\overleftarrow{\Pi}_{t_j \in J} \llbracket [L] \rrbracket_{t_j})$$

The proof has two parts, following a similar strategy as seen previously in the proof of Rule ATOM. The first part shows the induction on  $k$  that  $\forall k \geq 0. T \rightarrow T_k^\alpha$ . The key difference from the previous proof of Rule ATOM is the interplay between the thread executing the actions in  $N$  and the other threads executing the paths represented by  $R$  and  $L$ . To this point, at each inductive step, we must show that it exists a subset of threads running the paths in  $R$  and  $L$  and which interfere with the thread executing a control path in  $N$ . It turns out that we can define it by scanning the trace interval between two subsequence occurrences of  $N$ -paths. The second part is as before to show that the greatest lower bound of restrictions  $T_k^\alpha$  coincides with  $Traces(O[\langle\langle R[N]L \rangle\rangle])$ , and equivalently, to  $\llbracket [\alpha] \rrbracket(T)$ , therefore concluding that  $O[N] \rightarrow O[\langle\langle R[N]L \rangle\rangle]$ .

*Step 1.* We show by induction on  $k$  that  $T \rightarrow T_k^\alpha$  for all  $k \geq 0$ .

*Base case* ( $k = 0$ ). Since  $T = T_0^\alpha$ , the base case hold trivially.

*Inductive step.* Assume that  $T \rightarrow T_k^\alpha$ . We must prove  $T \rightarrow T_{k+1}^\alpha$ . By definition of  $T_k^\alpha$  we get that  $T_k^\alpha \supseteq T_{k+1}^\alpha$ . Next we prove that  $\forall \tau \in T_k^\alpha. \exists \tau' \in T_{k+1}^\alpha. \tau \leq \tau'$ .

Let  $\tau$  be an arbitrary trace in  $T_k^\alpha$ .

We use the same notation for the number of occurrences of the induction action  $\mathbf{a}^\cup$  in a trace, only this time, since the induction component  $\mathbf{a}^\cup$  is a set of actions, the helper function  $\#(\mathbf{a}^\cup, \tau)$  scans for any  $\mathbf{a} \in \mathbf{a}^\cup$  in a trace restriction. We proceed by case analysis on  $\#(\mathbf{a}^\cup, \tau)$ .

*Case  $\#(\mathbf{a}^\cup, \tau) = k$ .* The proof is immediate by taking  $\tau' = \tau$ .

*Case  $\#(\mathbf{a}^\cup, \tau) \geq (k + 1)$ .* We show next how to derive  $\tau' \in T_{k+1}^\alpha$  such that  $\tau \leq \tau'$ . We begin by identifying the  $k$ -th and  $(k + 1)$ -th occurrences of an (possibly different) action in  $\mathbf{a}^\cup$  in  $\tau$ , and then using the bijective maps  $f_{\tau, t_k}$  and  $f_{\tau, t_i}$ , where  $t_k$  and  $t_i$  are the threads executing the  $k$ -th and the  $(k + 1)$ -th occurrences of an (possibly different) action in  $\mathbf{a}^\cup$ , respectively, to obtain the following factorization of  $\tau$ :

$$\tau = \tau_1(\pi_{N_k})_{t_k} \tau_2(\pi_{N_i})_{t_i} \tau_3$$

where  $(\pi_{N_k})_{t_k}$ ,  $(\pi_{N_i})_{t_i}$  are the paths represented by  $N$  corresponding to  $(\mathbf{a}')_{t_k}$ , and  $(\mathbf{a}'')_{t_i}$ , respectively, with  $\mathbf{a}', \mathbf{a}'' \in \mathbf{a}^\cup$ .

We introduce some additional notation. For any trace  $\tau = e_0 e_1 \dots e_n$ , we write  $t[i]$  for the event at index  $i$ ,  $\tau[i : j]$  for the trace fragment  $e_i e_{i+1} \dots e_{j-1}$ ,  $\tau[: i]$  for the prefix  $e_0 \dots e_{i-1}$ , and  $\tau[j : ]$  for the suffix  $e_j \dots e_n$ . We also use the following auxiliary functions. Given a regular expression  $k$  over an alphabet  $\Sigma$ , we define  $first(k) \subseteq \Sigma$  and  $last(k) \subseteq \Sigma$  to be the sets of symbols that can appear, respectively, as first and last symbol of any word in the language of  $k$ . For any trace  $\tau$ , let  $suf(\tau) \subseteq Ev^*$  (resp.  $pref(\tau) \subseteq Ev^*$ ) denote the set of all prefix (resp. suffix) subtraces of  $\tau$ .

Let  $k_1, k_2, i_1, i_2$  be indices such that  $\tau[k_1 : k_2] = (\pi_{N_k})_{t_k}$  and  $\tau[i_1 : i_2] = (\pi_{N_i})_{t_i}$ . It follows from the factorization of  $\tau$  and that empty word is not in  $N$  that  $k_1 < k_2 < i_1 < i_2$ .

Next, we define the following set of thread identifiers:

$$J = \left\{ t_j \left| \begin{array}{l} \exists m \in \{k_2, \dots, i_1 - 1\}. \\ \tau[m] \in (\{t_j\} \times last(R)) \wedge suf(\tau[: i_1]|_{t_j}) \cap [[R]]_{t_j} \neq \emptyset \\ \wedge \\ \forall m' \in \{k_2, \dots, i_1 - 1\}. \\ \tau[m'] \notin (\{t_j\} \times first(L)) \\ \wedge \\ pref(\tau[i_2 : ]|_{t_j}) \cap [[L]]_{t_j} \neq \emptyset \end{array} \right. \right\}$$

and show that it characterizes precisely the threads contributing to the  $(k + 1)$ -th cooperative block of events represented by  $coop\{R, N, L\}$ .

A thread  $t_j$  satisfies the defining property of  $J$  if it executes a path  $\pi$  which can be decomposed as  $\pi = \pi_R \pi_L$ , where  $(\pi_R)_{t_j}$  is executed before  $(\pi_{N_j})_{t_i}$  and  $(\pi_L)_{t_j}$  is executed after. Moreover  $\pi_R \in [[R]]$  and  $\pi_L \in [[L]]$ . We define the existence of such a path split by the presence of an occurrence, within  $\tau_R$ , of the last event of some path in  $R$ , and by the absence, within  $\tau_L$ , of any occurrence of a first action of a path in  $L$ . Note that this condition excludes paths  $\pi$  where the last action in  $\pi_R$  coincides with the first action in  $\pi_L$ , since such a split would be ambiguous.

We prove that for every  $t_j \in J$ , there exists a trace  $\tau'$  with the following factorization:

$$\tau' = \tau'_1(\pi_{N_k})_{t_k} \tau'_2(\pi_R)_{t_j} (\pi_N)_{t_i} (\pi_L)_{t_j} \tau'_3$$

such that  $\tau \leq \tau'$ , where  $(\pi_R)_{t_j} \in [[R]]_{t_j}$  and  $(\pi_L)_{t_j} \in [[L]]_{t_j}$ .

Let  $t_j$  be an arbitrary thread identifier in  $J$ .

We proceed by instantiating the Rule Coop for the threads  $t_i$  and  $t_j$ .

By the defining property of  $t_j$ , there exists a path  $\pi = \pi_R \pi_L$  executed by thread  $t_j$  such that all events in  $(\pi_R)_{t_j}$  are interleaved with actions of other threads and executed before  $(\pi_{N_i})_{t_i}$ , while all events of  $(\pi_L)_{t_j}$  are executed after  $(\pi_{N_i})_{t_i}$  and are also interleaved with other threads' actions. Moreover  $\pi_R \in \llbracket R \rrbracket$  and  $\pi_L \in \llbracket L \rrbracket$ .

According to the rule premises, the contexts  $C_{t_j} = \text{sp}_{t_j}(R, \text{true})$  and  $C'_{t_j} = \text{sp}((N)_{t_i}, Q_{t_j})$  summarize the effect of the surrounding paths executed by thread  $t_j$ . Namely,  $C_{t_j}$  characterizes the states reachable after executing a path in  $R$ , while  $C'_{t_j}$  characterizes the states from which a path in  $L$  can be executed.

It must be that  $C_{t_j}$  holds for the state immediately after executing the last action in  $(\pi_R)_{t_j}$ , and  $C'_{t_j}$  holds for the state before the first action in  $(\pi_L)_{t_j}$ .

Consider now the last action, say  $c$ , in  $\pi_R$ , and the first action, say  $d$ , in  $\pi_L$ , both executed by  $t_j$ . This will give us the factorization:

$$\tau = \tau_1(\pi_{N_k})_{t_k} \tau_{21}(c_{t_j}) \tau_{22}(\pi_N)_{t_i} \tau_{31}(d_{t_j}) \tau_{32}$$

Observe that any actions in  $\tau_{22}$  are executed by threads in the environment, different from both  $t_i$  and  $t_j$ . Furthermore, no action in  $bk(N)$  can occur in the  $\tau_{22}$ . Indeed, suppose for contradiction, that there exists such an action occurring in  $\tau_{22}$ . Since traces in  $\text{Traces}(O[N])$  execute  $N$ -paths in isolation, and  $bk(N) \xleftrightarrow{t_i} N$ , it must be that any action in  $bk(N)$  can arise only from such paths. Therefore, such action would have to belong to another isolated occurrence of an  $N$ -path. By the bijection  $\alpha^\cup \leftrightarrow N$ , there would be an additional occurrence of an action from  $\alpha^\cup$  strictly between the  $k$ -th and the  $(k+1)$ -th ones, contradicting the choice of  $(\pi_{N_i})_{t_i}$  as the  $(k+1)$ -th occurrence.

Hence,  $\tau_{22}$  is free of synchronization primitives from  $bk(N)$ . By the premise  $I\text{Stable}(C_{t_j}, \overline{bk(N)})$ , the context  $C_{t_j}$  is therefore preserved through this interval, that is  $C_{t_j}$  holds before executing the path  $(\pi_N)_{t_i}$ .

We can now exploit the contexts for the commutativity movers over KAT. By the definition of the commutativity right mover,  $R \triangleright_{t_j} C$ , and the fact that  $\pi_R \in \llbracket R \rrbracket$ , we infer that:

$$\exists \tau'_1, \tau'_2 \in E\bar{v}^*. \tau \leq \tau'_1(\pi_{N_k})_{t_k} \tau'_2(\pi_R)_{t_j} (\pi_N)_{t_i} \tau_3 \quad (4)$$

Similarly, by the definition of  $C'_{t_j} \triangleleft L$  and the fact that  $\pi_L \in \llbracket L \rrbracket$ :

$$\exists \tau'_1, \tau'_2, \tau'_3 \in E\bar{v}^*. \tau \leq \tau'_1(\pi_{N_k})_{t_k} \tau'_2(\pi_R)_{t_j} (\pi_N)_{t_i} (\pi_L)_{t_j} \tau'_3 \quad (5)$$

Therefore we must have:

$$\forall t_j \in J. \exists \tau'_1, \tau'_2, \tau'_3 \in E\bar{v}^*. \tau \leq \tau'_1(\pi_{N_k})_{t_k} \tau'_2(\pi_R)_{t_j} (\pi_N)_{t_i} (\pi_L)_{t_j} \tau'_3$$

Assume that  $J = \{t_{j_1}, \dots, t_{j_m}\}$  such that  $t_{j_1} < \dots < t_{j_m}$ . The condition  $R \rightleftarrows R$  (resp.  $L \rightleftarrows L$ ) states that every path in  $R$  (resp.  $L$ ) consists of commutative actions, and consequently, we can rearrange them in increasing (resp. decreasing) order of their thread identifiers, it follows that:

$$\tau \leq \tau'_1(\pi_{N_k})_{t_k} \tau'_2(\pi_{R_1})_{t_{j_1}} \dots (\pi_{R_m})_{t_{j_m}} (\pi_N)_{t_i} (\pi_{L_m})_{t_{j_m}} \dots (\pi_{L_1})_{t_{j_1}} \tau'_3$$

for some  $\tau'_1, \tau'_2, \tau'_3$ . Let  $\tau'$  denote this reordered trace. For brevity of notation, we write:

$$\tau' = \tau'_1(\pi_{N_k})_{t_k} \tau'_2 \overrightarrow{(\pi_{R_j})_{t_j}} (\pi_{N_i})_{t_i} \overleftarrow{(\pi_{L_j})_{t_j}} \tau'_3$$

The last step of this inductive step is to show that  $\tau' \in T_{k+1}^\alpha$ .

By the fact that the bijections  $\forall t \in \mathcal{T}. \text{occ}_{\alpha^\cup}(\tau|_t) \leftrightarrow \text{occ}_N(\tau|_t)$  are also order-preserving, we infer that the first  $k$  occurrences of trace fragments  $\pi \in \llbracket N \rrbracket$  must occur in the prefix of  $\tau$  up to the occurrence of  $(\pi_{N_i})_{t_i}$ . That it,  $\tau$  admitted the factorization

$$\tau = \tau_{10} \tau_1 \tau_2 \tau_{12} \dots (\tau_k) \tau_{1k} (\pi_{N_i})_{t_i} \tau_3$$

where for any  $l \leq k$  there exists  $t_l$  and  $J_l \subseteq \mathcal{T} \setminus \{t_l\}$  such that:

$$\tau_l \in (\vec{\Pi}_{t_j \in J_l} \llbracket R \rrbracket_{t_j}) \llbracket N \rrbracket_{t_l} (\overleftarrow{\Pi}_{t_j \in J_l} \llbracket L \rrbracket_{t_j})$$

We show next that the first  $k$  canonical groups of events have not been affected by the construction of  $\tau'$ . The proof is by case analysis on whether there exists  $l \leq k$ , s.t.  $J_l \cap J \neq \emptyset$ . That is, whether there exists at least one thread  $t_j$  that contributes to both an early  $\tau_l$  and also to the cooperative block corresponding to  $(k+1)$ -th occurrence of  $\mathbf{a}$ .

*Case  $\nexists l \leq k$ .  $J_l \cap J \neq \emptyset$ .* Then all trace fragments  $(\tau_l)$  occur in the prefix  $\tau'_1(\pi_{N_k})_{t_k}\tau'_2$  of  $\tau'$ , since their actions were not actively involved during the rewritings that constructed  $\tau'$ . This concludes this case.

*Case  $\exists l \leq k$ .  $J_l \cap J \neq \emptyset$ .* Let  $t_\xi$  be some thread that is in both  $J$  and  $J_l$ . We prove by contradiction that the trace fragment  $\tau_l$  in  $\tau$  has not been affected by the construction. We focus on the restriction of the trace  $\tau'$  to  $t_\xi$ . Suppose it is affected, and let  $a$  be the first action in the path  $\pi_{R_a}\pi_{L_a}$  within the earlier  $\tau_l$  that occurs also in the path  $\pi_{R_b}\pi_{L_b}$  contributing to the  $(k+1)$ -th cooperative block. This implies that the suffix  $\pi_{R_a}\pi_{L_a}$  and prefix of  $\pi_{R_b}\pi_{L_b}$  overlap. However this contradicts the non-ambiguity condition of  $\alpha$ .

It follows that  $\tau'$  admits a factorization:

$$\tau' = \tau_{10} \tau_1 \tau_{11} \tau_2 \tau_{12} \cdots (\tau_k) \tau_{1k} (\pi_{N_i})_{t_i} \tau'_2 (\overrightarrow{\pi_{R_j}})_{t_j} (\pi_{N_i})_{t_i} (\overleftarrow{\pi_{L_j}})_{t_j} \tau'_3$$

where for all  $l \leq k$  it is the case that  $\tau_l \in \langle \llbracket R[N]L \rrbracket \rangle$ . This concludes the proof that  $\tau' \in T_{k+1}^\alpha$ .

*Step 2.* We prove that  $\inf_{k \geq 0} T_k^\alpha = \text{Traces}(O[\llbracket R[N]L \rrbracket]) = \llbracket \alpha \rrbracket(T)$ . The argument is analogous to that in the soundness proof of the AROM, mut. mut. The infimum  $\inf_{k \geq 0} T_k^\alpha$  is given by the limit of the chain  $T = T_0^\alpha \supseteq T_1^\alpha \rightarrow T_2^\alpha \dots$ , which is precisely  $\text{Trace}(O[\llbracket R[N]L \rrbracket])$ , and since the chain keeps the representatives of equivalent classes that are also  $\preceq_r$ -maximal, it follows that it also coincides with  $\llbracket \alpha \rrbracket(T)$ .

This concludes our proof.  $\square$

LEMMA A.6 (CSTRUCT SOUNDNESS). *Let  $\mathcal{A}$  be a collection of IRSspecs for a concurrent object  $O$ . For every cooperative IRSspec  $\alpha \in \mathcal{A}$ , the rule CSTRUCT is sound.*

PROOF. By Lemma 5.3

$$\langle \llbracket R[\sum_{N \in N_U} N]L \rrbracket \rangle = \sum_{N \in N_U} \langle \llbracket R[N]L \rrbracket \rangle.$$

Let  $\tau \in \text{Traces}(O[\sum_{N \in N_U} N])$ . Then  $\tau$  is centered around some  $N \in N_U$ .

By the premise  $[N] \stackrel{\alpha}{\supseteq} \langle \llbracket R[N]L \rrbracket \rangle$ ,  $\tau$  reduces to a trace in  $\text{Traces}(O[\langle \llbracket R[N]L \rrbracket \rangle])$ , hence to a trace in

$$\text{Traces}\left(O\left[\sum_{N \in N_U} \langle \llbracket R[N]L \rrbracket \rangle\right]\right) = \text{Traces}\left(O\left[\langle \llbracket R[\sum_{N \in N_U} N]L \rrbracket \rangle\right]\right).$$

Therefore the conclusion follows.  $\square$

LEMMA A.7 (CSEQ SOUNDNESS). *Let  $\mathcal{A}$  be a collection of IRSspecs for a concurrent object  $O$ . For every two cooperative IRSspec,  $\alpha_1, \alpha_2 \in \mathcal{A}$ , the rule CSEQ is sound*

PROOF. Let  $\tau \in \text{Traces}(O[N])$ . By the premises,

$$[N] \stackrel{\alpha_1}{\supseteq} \langle \llbracket R_1[N]L_1 \rrbracket \rangle \quad \text{and} \quad [N] \stackrel{\alpha_2}{\supseteq} \langle \llbracket R_2[N]L_2 \rrbracket \rangle,$$

the trace  $\tau$  can be reduced, respectively, to traces in

$$\text{Traces}(O[\langle \llbracket R_1[N]L_1 \rrbracket \rangle]) \quad \text{and} \quad \text{Traces}(O[\langle \llbracket R_2[N]L_2 \rrbracket \rangle]).$$

By orthogonality of  $R_1L_1$  and  $R_2L_2$ , the two reductions are non-interfering: the surrounding paths selected by one rule neither overlap with nor factor through those selected by the other. Hence both reductions may be soundly combined around the same anchor  $N$ .

Further, since  $R_1 \curvearrowright R_2$  and  $L_1 \curvearrowright L_2$ , the right surrounding paths commute with one another and the left surrounding paths commute with one another. Therefore the set of traces obtained by applying both reductions is exactly captured by allowing, independently on each side of the anchor, a choice between the corresponding components. That is, the resulting traces belong to

$$\text{Traces}(O[\langle\langle (R_1 + R_2)[N](L_1 + L_2) \rangle\rangle]).$$

Thus every trace in  $\text{Traces}(O[N])$  reduces, via  $\alpha_1; \alpha_2$ , to a trace in the above language, which proves

$$[N] \stackrel{\alpha_1; \alpha_2}{\sqsupseteq} \langle\langle (R_1 + R_2)[N](L_1 + L_2) \rangle\rangle.$$

□

**THEOREM A.8 (CLOSED FORM OF THE CANONICAL TRACES).** *Let  $O$  be a concurrent object, and  $\mathcal{A}$  a collection of sound IRSpecs that is covering and orthogonal for  $O$ . Let  $r_1, \dots, r_m$  be the reduction patterns of the IRSpecs in  $\mathcal{A}$  that are maximal w.r.t. the precedence order  $\prec$ . Then, for any admissible linear extension  $\prec_1$  of  $\prec$ , we have  $O_{can}^{\prec_1} \sqsubseteq O$  and  $(r_1 + \dots + r_m)^*$  is an abstraction of the  $O_{can}^{\prec_1}$ , that is:*

$$\text{Traces}(O_{can}^{\prec_1}) \subseteq (r_1 + \dots + r_m)^*.$$

**PROOF.** Let  $\prec_1 \triangleq \alpha_1; \dots; \alpha_n$  be an admissible linear extension of the precedence order on  $\mathcal{A}$ , and let

$$O \stackrel{\alpha_1}{\sqsupseteq} O_1 \stackrel{\alpha_2}{\sqsupseteq} \dots \stackrel{\alpha_n}{\sqsupseteq} O_{can}^{\prec_1}$$

be the corresponding canonicalization sequence.

Because  $\prec_1$  is an admissible linear extension of the precedence order, every cooperative IRSpec is applied only after the atomic IRSpecs on which its anchor depends. Thus the sequential canonicalization is well-defined, and each factor in the above decomposition is reduced to the canonical fragment determined by its IRSpec. The second clause of covering guarantees in particular that every cooperative anchor is a sum of canonical fragments coming from preceding atomic IRSpecs, so cooperative reductions are always performed over already-canonicalized atomic anchors.

Consider an arbitrary trace  $\tau \in O_{can}^{\prec_1}$ . It must be that  $\tau$  admits a factorization:

$$\tau = u_0 \tau_1 u_1 \tau_2 u_2 \dots$$

alternating trace fragments  $\tau_i \in \langle |r| \rangle$  in the language of canonical trace expressions  $r = \text{canon}(\alpha)$  of some  $\alpha \in \mathcal{A}$  with residual trace fragments  $u_i$  that are not grouped. We call *maximal canonical expressions* the  $r$ 's that survive the reductions sequence.

Since  $O_{can}^{\prec_1} \sqsubseteq O$ ,  $\tau$  arises from a trace of  $O$  by a sequence of sound reductions induced by the IRSpecs in  $\mathcal{A}$ . It must be that  $r$  can take either the shape  $[RNL]$  or  $\langle\langle (\sum R_i)[\sum N](\sum L_i) \rangle\rangle$ . By the soundness of rule CSTRUCT, we can further simplify the second shape to  $\langle\langle (\sum R_i)[N](\sum L_i) \rangle\rangle$ , where each  $R_iL_i$  be the reduction pattern of some cooperative  $\alpha \text{ in } \mathcal{A}$ .

We proceed in two step. First, we prove that there is no overlap between trace fragments isolated by  $r$ 's. Second, we show that there are no gaps between any two subsequent trace fragments in the language of  $r$ 's, that is, all  $u_i$  are empty.

*1. Canonical fragments do not overlap.* We prove that if  $\tau_i$  and  $\tau_j$  are two trace fragments isolated by maximal canonical expressions, then they are disjoint.

Suppose for contradiction that two such fragments overlap.

If both fragments are isolated by atomic expressions  $R_iN_iL_i$  and  $R_jN_jL_j$ , then a nonempty overlap induces either:

- (i) a nonempty suffix of  $R_i N_i L_i$  that is also a prefix of  $R_j N_j L_j$ , or vice versa, or
- (ii) one reduction pattern occurs as a factor of the other.

Both possibilities contradict orthogonality. If  $i = j$ , then the overlap is a self-overlap of the same reduction pattern, which is ruled out by the non-self-overlap premise used in the soundness of the rule **ATOM**.

If one or both fragments are isolated by cooperative expressions, project the overlap to the thread(s) participating in the corresponding canonical expression. Any nonempty overlap must then occur either inside the anchor component, between a surrounding component and an anchor, or between surrounding components on the same side of the anchor. In every case, the overlap induces either a prefix/suffix overlap or a factorization relation between the underlying reduction patterns. Again this contradicts orthogonality. Therefore no two maximal canonical fragments overlap. Hence the family of maximal canonical fragments in  $\tau$  is pairwise disjoint.

2. *There are no gaps between consecutive canonical fragments.* We prove that every residual fragment  $u_j$  is empty.

Assume, for contradiction, that some  $u_j$  contains an event  $a_t$  that does not belong to any maximal canonical fragment of  $\tau$ . Let  $k_{op}$  be the implementation of the operation in which  $a$  occurs. Since  $\mathcal{A}$  is covering for  $O$ , condition (i) of covering implies that every control path of  $k_{op}$  is contained in the starred union of **IRSpec** reduction patterns. Hence there exists some  $\alpha \in \mathcal{A}$  such that  $a$  occurs in the reduction pattern  $[\alpha]$ . Since the event  $a_t$  occurs in the trace, the corresponding occurrence of  $[\alpha]$  must also occur in the underlying execution. But  $\prec_1$  enforces every  $\alpha$ -reduction during the canonicalization sequence, and by soundness that occurrence must therefore be grouped into the canonical fragment  $canon(\alpha)$  in  $\tau$ . This contradicts the assumption that  $a_t$  lies outside all maximal canonical fragments. Therefore every  $u_j$  is empty.

Combining the two steps, we conclude that  $\tau$  admits a factorization

$$\tau = \tau_1 \tau_2 \cdots \tau_\ell$$

such that each  $\tau_k$  belongs to the language of a basic canonical expression  $r_i$  for some  $i \in \{1, \dots, m\}$ . Equivalently,

$$\tau \in (r_1 + \cdots + r_m)^*.$$

Since  $\tau$  was arbitrary, it follows that

$$Traces(O_{can}^{\prec_1}) \subseteq (r_1 + \cdots + r_m)^*.$$

This proves the theorem. □

## B Results of applying Q<sub>REF</sub> to the benchmarks

### B.1 counter

26 obligations across 7 specs: atom::aDecrS, atom::aDecrS.R, atom::aDecrZero, atom::aIncrS, coop::cDecrF, coop::cDecrF.R, coop::cIncrF.

#### B.1.1 Sources.

#### Implementation (.cl)

```

1 /* Concurrent counter */

3 int ctr = 0;

5 invariant ctr >= 0;

7 int incr() {
8   while (true) {
9     int c = ctr;
10    if CAS(ctr, c, c + 1) {
11      return c;
12    }
13  }
14}

16 int decr() {
17   while (true) {
18     int c = ctr;
19     if (c == 0) {
20       return 0;
21     }
22     else {
23       if CAS(ctr, c, c - 1) {
24         return c;
25       }
26     }
27   }
28}

```

#### Labeled program (.cll)

```

1 /*****
2 /* [generated] KAT-friendly program with label annotations */
3 /* --
4 /* counter.cll
5 /*****

7 int ctr = 0;

9 invariant ctr >= 0;

```



```

11 int incr() {
12   while (true) {
13     [@ia1] int c = ctr;
14     if * {
15       [@icas2t] CAS(ctr, c, c + 1);
16       [@ia3] return c;
17     }
18     else {
19       [@icas2f] assume(ctr ≠ c);
20     }
21   }
22 }

24 int decr() {
25   while (true) {
26     [@da1] int c = ctr;
27     if * {
28       [@db2t] assume(c == 0);
29       [@da3] return 0;
30     }
31     else {
32       [@db2f] assume(c ≠ 0);
33       if * {
34         [@dcas4t] CAS(ctr, c, c - 1);
35         [@da5] return c;
36       }
37       else {
38         [@dcas4f] assume(ctr ≠ c);
39       }
40     }
41   }
42 }

```

### B.1.2 *IRSpec.*

#### IR specifications (.irspecs)

```

1 /* OBJECT: Counter */
2 aIncrS = atom{
3   ind = icas2t,
4   red = ia1 |> icas2t <| ia3
5 }

7 aDecrS = atom{
8   ind = dcas4t,
9   red = [da1;db2f] |> dcas4t <| da5
10 }

12 cIncrF = coop{
13   ind = icas2t + dcas4t,
14   red = ia1 |> [aIncrS] + [aDecrS] <| icas2f
15 }

```

```

17 cDecrF = coop{
18   ind = icas2t + dcas4t,
19   red = [da1;db2f] |> [aIncrS] + [aDecrS] <| dcas4f
20 }

22 aDecrZero = atom{
23   ind = da3,
24   red = [da1; db2t; da3]
25 }

```

### B.1.3 Contextual Commutativity (16).

| #  | Kind | Spec                           | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Result |
|----|------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1  | Inv  |                                | $\text{ctr} \geq 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | OK     |
| 2  | RMov | $\text{atom}::\text{aIncrS}$   | $\forall t:\mathbb{N}. c[t] = \text{ctr} \triangleright \{(ctr == c[t] \wedge c[t] + 1 \geq 0)\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | OK     |
| 3  | HTrp | $\text{atom}::\text{aIncrS}$   | $\forall t:\mathbb{N}. \{(ctr == c[t] \wedge c[t] + 1 \geq 0)\} \text{CAS}(ctr, c[t], c[t] + 1) \{\exists \text{old\_ctr}. (\text{old\_ctr} == c[t] \wedge c[t] + 1 \geq 0 \wedge ctr == c[t] + 1)\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | OK     |
| 4  | LMov | $\text{atom}::\text{aIncrS}$   | $\forall t:\mathbb{N}. \{\exists \text{old\_ctr}. (\text{old\_ctr} == c[t] \wedge c[t] + 1 \geq 0 \wedge ctr == c[t] + 1)\} \triangleleft \text{return } c[t]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | OK     |
| 5  | Atom | $\text{atom}::\text{aDecrS.R}$ | $[\text{int } c = \text{ctr}; \text{assume}(c \neq 0)]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | OK     |
| 6  | RMov | $\text{atom}::\text{aDecrS}$   | $\forall t:\mathbb{N}. c[t] = \text{ctr}; \text{assume}(c[t] \neq 0) \triangleright \{(ctr == c[t] \wedge c[t] - 1 \geq 0)\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | OK     |
| 7  | HTrp | $\text{atom}::\text{aDecrS}$   | $\forall t:\mathbb{N}. \{(ctr == c[t] \wedge c[t] - 1 \geq 0)\} \text{CAS}(ctr, c[t], c[t] - 1) \{\exists \text{old\_ctr}. (\text{old\_ctr} == c[t] \wedge c[t] - 1 \geq 0 \wedge ctr == c[t] - 1)\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | OK     |
| 8  | LMov | $\text{atom}::\text{aDecrS}$   | $\forall t:\mathbb{N}. \{\exists \text{old\_ctr}. (\text{old\_ctr} == c[t] \wedge c[t] - 1 \geq 0 \wedge ctr == c[t] - 1)\} \triangleleft \text{return } c[t]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | OK     |
| 9  | RMov | $\text{coop}::\text{cIncrF}$   | $\forall tj:\mathbb{N}. c[tj] = \text{ctr} \triangleright \{(ctr \geq 0 \wedge c[tj] == \text{ctr})\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | OK     |
| 10 | HTrp | $\text{coop}::\text{cIncrF}$   | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{(ctr \geq 0 \wedge c[tj] == \text{ctr})\} ([c[ti] = \text{ctr}; \text{CAS}(ctr, c[ti], c[ti] + 1); \text{return } c[ti]] + [[c[ti] = \text{ctr}; \text{assume}(c[ti] \neq 0); \text{CAS}(ctr, c[ti], c[ti] - 1); \text{return } c[ti]]) \{\exists \text{old\_ctr}. (\text{old\_ctr} \geq 0 \wedge c[tj] == \text{old\_ctr} \wedge c[ti] == \text{old\_ctr} \wedge c[ti] \neq \text{old\_ctr} \wedge ctr == c[ti] + 1) \vee \exists \text{old\_ctr}. (\text{old\_ctr} \geq 0 \wedge c[tj] == \text{old\_ctr} \wedge c[ti] == \text{old\_ctr} \wedge c[ti] \neq 0 \wedge ctr == c[ti] - 1)\}) \Rightarrow \{ctr \neq c[tj]\}$ | OK     |
| 11 | Subs | $\text{coop}::\text{cIncrF}$   | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{(\exists \text{old\_ctr}. (\text{old\_ctr} \geq 0 \wedge c[tj] == \text{old\_ctr} \wedge c[ti] == \text{old\_ctr} \wedge ctr == c[ti] + 1) \vee \exists \text{old\_ctr}. (\text{old\_ctr} \geq 0 \wedge c[tj] == \text{old\_ctr} \wedge c[ti] == \text{old\_ctr} \wedge c[ti] \neq 0 \wedge ctr == c[ti] - 1))\} \Rightarrow \{ctr \neq c[tj]\}$                                                                                                                                                                                                                                                                            | OK     |
| 12 | Atom | $\text{coop}::\text{cDecrF.R}$ | $[\text{int } c = \text{ctr}; \text{assume}(c \neq 0)]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | OK     |
| 13 | RMov | $\text{coop}::\text{cDecrF}$   | $\forall tj:\mathbb{N}. c[tj] = \text{ctr}; \text{assume}(c[tj] \neq 0) \triangleright \{(ctr \geq 0 \wedge c[tj] == \text{ctr} \wedge c[tj] \neq 0)\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | OK     |

| #  | Kind | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Result |
|----|------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 14 | HTrp | coop::cDecrF    | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{(ctr \geq 0 \wedge c[tj] == ctr \wedge c[tj] \neq 0)\} \{[c[ti] = ctr; CAS(ctr, c[ti], c[ti] + 1); return c[ti]] + [[c[ti] = ctr; assume(c[ti] \neq 0)]; CAS(ctr, c[ti], c[ti] - 1); return c[ti]]\} \{(\exists old\_ctr. (old\_ctr \geq 0 \wedge c[tj] == old\_ctr \wedge c[tj] \neq 0 \wedge c[ti] == old\_ctr \wedge ctr == c[ti] + 1) \vee \exists old\_ctr. (old\_ctr \geq 0 \wedge c[tj] == old\_ctr \wedge c[tj] \neq 0 \wedge c[ti] == old\_ctr \wedge c[ti] \neq 0 \wedge ctr == c[ti] - 1))\}$ | OK     |
| 15 | Subs | coop::cDecrF    | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{(\exists old\_ctr. (old\_ctr \geq 0 \wedge c[tj] == old\_ctr \wedge c[tj] \neq 0 \wedge c[ti] == old\_ctr \wedge ctr == c[ti] + 1) \vee \exists old\_ctr. (old\_ctr \geq 0 \wedge c[tj] == old\_ctr \wedge c[tj] \neq 0 \wedge c[ti] == old\_ctr \wedge c[ti] \neq 0 \wedge ctr == c[ti] - 1))\} \Rightarrow \{ctr \neq c[tj]\}$                                                                                                                                                                         | OK     |
| 16 | Atom | atom::aDecrZero | $[int\ c = ctr; assume(c == 0); return\ 0]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | OK     |

#### B.1.4 Automata (40).

coverage.

| Check         | Result |
|---------------|--------|
| incr: COVERED | OK     |
| decr: COVERED | OK     |

non-overlapping.

1 all distinct pairs are non-overlapping

non-self-overlapping.

1 no spec self-overlaps

bijection part I: well-definedness.

| Check                                                        | Result |
|--------------------------------------------------------------|--------|
| aIncrS: induction_pattern $\subseteq$ exactly_one_ais: OK    | OK     |
| aDecrZero: induction_pattern $\subseteq$ exactly_one_ais: OK | OK     |
| aDecrS: induction_pattern $\subseteq$ exactly_one_ais: OK    | OK     |
| cIncrF: induction_pattern $\subseteq$ exactly_one_ais: OK    | OK     |
| cDecrF: induction_pattern $\subseteq$ exactly_one_ais: OK    | OK     |

bijection part II: surjectivity.

| Check                                                        | Result |
|--------------------------------------------------------------|--------|
| aDecrZero $\cap$ at_least_one_ais(aDecrS) = $\emptyset$ : OK | OK     |
| aIncrS $\cap$ at_least_one_ais(aDecrS) = $\emptyset$ : OK    | OK     |
| cDecrF $\subseteq$ at_most_one_ais(aDecrS): OK               | OK     |
| cIncrF $\subseteq$ at_most_one_ais(aDecrS): OK               | OK     |
| aDecrS $\cap$ at_least_one_ais(aDecrZero) = $\emptyset$ : OK | OK     |

| Check                                                         | Result |
|---------------------------------------------------------------|--------|
| $aIncrS \cap at\_least\_one\_ais(aDecrZero) = \emptyset$ : OK | OK     |
| $cDecrF \cap at\_least\_one\_ais(aDecrZero) = \emptyset$ : OK | OK     |
| $cIncrF \cap at\_least\_one\_ais(aDecrZero) = \emptyset$ : OK | OK     |
| $aDecrS \cap at\_least\_one\_ais(aIncrS) = \emptyset$ : OK    | OK     |
| $aDecrZero \cap at\_least\_one\_ais(aIncrS) = \emptyset$ : OK | OK     |
| $cDecrF \subseteq at\_most\_one\_ais(aIncrS)$ : OK            | OK     |
| $cIncrF \subseteq at\_most\_one\_ais(aIncrS)$ : OK            | OK     |
| $aDecrZero \cap at\_least\_one\_ais(cDecrF) = \emptyset$ : OK | OK     |
| $cIncrF \cap at\_least\_one\_ais(cDecrF) = \emptyset$ : OK    | OK     |
| $aDecrZero \cap at\_least\_one\_ais(cIncrF) = \emptyset$ : OK | OK     |
| $cDecrF \cap at\_least\_one\_ais(cIncrF) = \emptyset$ : OK    | OK     |

*bijjective support (IRSspecs).*

```

1aIncrS: {ia1, ia3, icas2t}
2aDecrZero: {da3, db2t}
3aDecrS: {da5, db2f, dcas4t}
4cIncrF: {da5, db2f, dcas4t, ia1, ia3, icas2t}
5cDecrF: {da5, db2f, dcas4t, ia1, ia3, icas2t}

```

### B.1.5 Interference Stability (10).

| #  | Kind  | Spec         | Obligation                                                                                                                                                                                | Result |
|----|-------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1  | IStbl | coop::cIncrF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(ctr \geq 0 \wedge c[tj] == ctr)\} c[tk] = ctr \{(ctr \geq 0 \wedge c[tj] == ctr)\}$                                                           | OK     |
| 2  | IStbl | coop::cIncrF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(ctr \geq 0 \wedge c[tj] == ctr)\} \text{return } 0 \{(ctr \geq 0 \wedge c[tj] == ctr)\}$                                                      | OK     |
| 3  | IStbl | coop::cIncrF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(ctr \geq 0 \wedge c[tj] == ctr)\} \text{assume}(c[tk] == 0) \{(ctr \geq 0 \wedge c[tj] == ctr)\}$                                             | OK     |
| 4  | IStbl | coop::cIncrF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(ctr \geq 0 \wedge c[tj] == ctr)\} \text{assume}(ctr \neq c[tk]) \{(ctr \geq 0 \wedge c[tj] == ctr)\}$                                         | OK     |
| 5  | IStbl | coop::cIncrF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(ctr \geq 0 \wedge c[tj] == ctr)\} \text{assume}(ctr \neq c[tk]) \{(ctr \geq 0 \wedge c[tj] == ctr)\}$                                         | OK     |
| 6  | IStbl | coop::cDecrF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(ctr \geq 0 \wedge c[tj] == ctr \wedge c[tj] \neq 0)\} c[tk] = ctr \{(ctr \geq 0 \wedge c[tj] == ctr \wedge c[tj] \neq 0)\}$                   | OK     |
| 7  | IStbl | coop::cDecrF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(ctr \geq 0 \wedge c[tj] == ctr \wedge c[tj] \neq 0)\} \text{return } 0 \{(ctr \geq 0 \wedge c[tj] == ctr \wedge c[tj] \neq 0)\}$              | OK     |
| 8  | IStbl | coop::cDecrF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(ctr \geq 0 \wedge c[tj] == ctr \wedge c[tj] \neq 0)\} \text{assume}(c[tk] == 0) \{(ctr \geq 0 \wedge c[tj] == ctr \wedge c[tj] \neq 0)\}$     | OK     |
| 9  | IStbl | coop::cDecrF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(ctr \geq 0 \wedge c[tj] == ctr \wedge c[tj] \neq 0)\} \text{assume}(ctr \neq c[tk]) \{(ctr \geq 0 \wedge c[tj] == ctr \wedge c[tj] \neq 0)\}$ | OK     |
| 10 | IStbl | coop::cDecrF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(ctr \geq 0 \wedge c[tj] == ctr \wedge c[tj] \neq 0)\} \text{assume}(ctr \neq c[tk]) \{(ctr \geq 0 \wedge c[tj] == ctr \wedge c[tj] \neq 0)\}$ | OK     |

**Veracity Program Examples (one per obligation kind)**

```

RMov [OK].
1 int ctr = 0;

3 int main() {
4   havoc ctr;
5   assume(ctr >= 0);
6   int c_size = 0;
7   havoc c_size;
8   int[] c = new int[c_size];
9   int t = 1;
10  havoc t;
11  commute_right_ctx (ctr >= 0) {
12    {
13      c[t] = ctr;
14    }
15    {
16      havoc ctr;
17    }

19    context: ctr >= 0 && ctr == c[t] && c[t] + 1 >= 0
20  }
21  return 0;
22}

```

```

HTrp [OK].
1 int ctr = 0;

3 int main() {
4   havoc ctr;
5   int c_size = 0;
6   havoc c_size;
7   int[] c = new int[c_size];
8   int t = 1;
9   havoc t;
10  assume(ctr == c[t] && c[t] + 1 >= 0);
11  assume(ctr == c[t]);
12  ctr = c[t] + 1;
13  assert(exists old_ctr : int . old_ctr == c[t] && c[t] + 1 >= 0 &&
14         ctr == c[t] + 1);
15  return 0;
16}

```

```

LMov [OK].
1 int ctr = 0;

3 int main() {
4   havoc ctr;
5   assume(ctr >= 0);
6   int c_size = 0;
7   havoc c_size;
8   int[] c = new int[c_size];

```

```

9  int t = 1;
10 havoc t;
11 commute_left (exists old__ctr : int . old__ctr == c[t] && c[t] + 1 >= 0 &&
12               ctr == c[t] + 1) {
13   {
14     int steps = 1;
15     havoc steps;
16     while (steps > 0) invariant ctr >= 0 {
17       steps = steps - 1;
18       int tk = 1;
19       havoc tk;
20       assume(tk ≠ t);
21       int nd = 1;
22       havoc nd;
23       if (nd == 1) {
24         havoc ctr;
25         assume(ctr == c[tk] + 1);
26       } else {
27         if (true) {
28           havoc ctr;
29           assume(ctr == c[tk] - 1);
30         }
31       }
32     }
33   }
34   {
35     return c[t];
36   }
37 }
38 }
39 return 0;
40}

```

Subs [OK].

```

1 int ctr = 0;

3 int main() {
4   havoc ctr;
5   int c_size = 0;
6   havoc c_size;
7   int[] c = new int[c_size];
8   int ti = 1;
9   havoc ti;
10  int tj = 1;
11  havoc tj;
12  assume(exists old__ctr : int . old__ctr >= 0 && c[tj] == old__ctr &&
13         c[ti] == old__ctr && ctr == c[ti] + 1
14         ||
15         exists old__ctr : int . old__ctr >= 0 && c[tj] == old__ctr &&
16         c[ti] == old__ctr && c[ti] ≠ 0 &&
17         ctr == c[ti] - 1);
18  assert(ctr ≠ c[tj]);

```

```

19 return 0;
20 }

```

## B.2 es

548 obligations across 31 specs: atom::fAPopEmp, atom::fAPopS, atom::fAPopS.R, atom::fAPushS, atom::fAPushS.R, atom::fAbadPartnerP, atom::fAbadPartnerP.R, atom::fAbadPartnerQ, atom::fAbadPartnerQ.R, atom::fAfallbackSP, atom::fAfallbackSQ, atom::fAhappySP, atom::fAhappySP.R, atom::fAhappySQ, atom::fAhappySQ.R, atom::publoop, atom::publoop.R, atom::publoop\_pop, atom::publoop\_pop.R, coop::fAfallbackFP, coop::fAfallbackFQ, coop::fAhappyFH, coop::fAhappyFI, coop::fCPopF, coop::fCPopF.R, coop::fCPushF, coop::fCPushF.R, coop::publoopC, coop::publoopC.R, coop::publoopC\_pop, coop::publoopC\_pop.

### B.2.1 Sources.

#### Implementation (.cl)

```

1 /* Elimination Stack: Extension of Treiber Stack with a collision mechanism for an
   immediate resolution of subsequent dual operations */

3 /* Each operation descriptor is stored in an array of locations.
4  Example: a push of value 10 issued by thread 42 can be represented as:
5  tloc[] op = new_arr();
6  op[0] = new {42 {} NULL}; // thread descriptor
7  op[1] = new {1 {} NULL}; // operation tag for push
8  op[2] = new {10 {} NULL}; // payload
9 */

11 readonly tloc stacktop = NULL;
12 tloc[][] location = new_arr();
13 int[] collision = new_arr();

15 int push(tloc[] p) {
16   while (true) {
17     tloc top = stacktop;
18     int val = p[2]→val;
19     tloc ntop = new {val {} top};

21     if CAS(stacktop, top, ntop) {
22       return 1;
23     }
24     else {
25       int mytid; /* variable "mytid" has uniqueness semantics */
26       location[mytid] = p;
27       int pos = nondet();

29       int him = collision[pos];
30       if(CAS(collision[pos], him, mytid)) {

32         if (him ≠ NULL && him ≠ mytid) {
33           tloc[] q = location[him];

```

```

34     if ((q ≠ NULL) && (q[0]→val == him) && (q[1]→val ≠ p[1]→val)) {
35         if CAS(location[mytid], p, NULL) {
36             if CAS(location[him], q, p) {
37                 return 1;
38             }
39             else {
40                 continue;
41             }
42         }
43         else {
44             location[mytid] = NULL;
45             return 1;
46         }
47     }
48 }
49 else {
50     if (!CAS(location[mytid], p, NULL)) {
51         location[mytid] = NULL;
52         return 1;
53     }
54 }
55 } else { }
56 }
57 }
58 }

60 int pop(tloc[] p) {
61     while (true) {
62         tloc top = stacktop;
63         if (top == NULL) {
64             return 0;
65         }
66         else {
67             int val = top→val;
68             tloc ntop = top→next;
69             if CAS(stacktop, top, ntop) {
70                 return val;
71             }
72             else {
73                 int mytid; /* variable "mytid" has uniqueness semantics */
74                 location[mytid] = p;
75                 int pos = nondet();

77                 int him = collision[pos];
78                 if(CAS(collision[pos], him, mytid)) {
79                     if (him ≠ NULL && him ≠ mytid) {
80                         tloc[] q = location[him];
81                         if ((q ≠ NULL) && (q[0]→val == him) && (q[1]→val ≠ p[1]→val)) {
82                             if CAS(location[mytid], p, NULL) {
83                                 if CAS(location[him], q, NULL) {
84                                     return q[2]→val;
85                                 }

```



```

86         else {
87             continue;
88         }
89     }
90     else {
91         int val = q[2]→val;
92         location[mytid] = NULL;
93         return val;
94     }
95 }
96 }
97 else {
98     if (!CAS(location[mytid], p, NULL)) {
99         int val = q[2]→val;
100         location[mytid] = NULL;
101         return val;
102     }
103 }
104 } else { }
105 }
106 }
107 }
108 }

```

### Labeled program (.cll)

```

1 /*****/
2 /* [generated] KAT-friendly program with label annotations */
3 /* -- */
4 /* es.cl */
5 /*****/

7 readonly tloc stacktop = NULL;
8 tloc[][] location = new_arr();
9 int[] collision = new_arr();

11 int push(tloc[] p) {
12     while (true) {
13         [@pua1] tloc top = stacktop;
14         [@pua2] int val = p[2]→val;
15         [@pua3] tloc ntop = new {val {} top};
16         if * {
17             [@pucas4t] CAS(stacktop, top, ntop);
18             [@pua5] return 1;
19         }
20     } else {
21         [@pucas4f] assume(stacktop ≠ top);
22         [@pua6] int mytid;
23         [@pua7] location[mytid] = p;
24         [@pua8] int pos = nondet();
25         [@pua9] int him = collision[pos];

```

```

26     if * {
27         [@pucas10t] CAS(collision[pos], him, mytid);
28         if * {
29             [@pub11t] assume(him ≠ NULL && him ≠ mytid);
30             [@pua12] tloc[] q = location[him];
31             if * {
32                 [@pub13t] assume(q ≠ NULL && q[0]→val == him && q[1]→val ≠ p[1]→val);
33                 if * {
34                     [@pucas14t] CAS(location[mytid], p, NULL);
35                     if * {
36                         [@pucas15t] CAS(location[him], q, p);
37                         [@pua16] return 1;
38                     }
39                     else {
40                         [@pucas15f] assume(location[him] ≠ q);
41                         [@pua17] continue;
42                     }
43                 }
44                 else {
45                     [@pucas14f] assume(location[mytid] ≠ p);
46                     [@pua18] location[mytid] = NULL;
47                     [@pua19] return 1;
48                 }
49             }
50             else {
51                 [@pub13f] assume(!(q ≠ NULL && q[0]→val == him && q[1]→val ≠ p[1]→val
52             ));
53         }
54         else {
55             [@pub11f] assume(!(him ≠ NULL && him ≠ mytid));
56             if * {
57                 [@pucas20f] assume(location[mytid] ≠ p);
58                 [@pua21] location[mytid] = NULL;
59                 [@pua22] return 1;
60             }
61             else {
62                 [@pucas20t] CAS(location[mytid], p, NULL);
63             }
64         }
65     }
66     else {
67         [@pucas10f] assume(collision[pos] ≠ him);
68     }
69 }
70 }
71}

73int pop(tloc[] p) {
74    while (true) {
75        [@poa1] tloc top = stacktop;
76        if * {

```

```

77     [@pob2t] assume(top == NULL);
78     [@poa3] return 0;
79 }
80 else {
81     [@pob2f] assume(top ≠ NULL);
82     [@poa4] int val = top→val;
83     [@poa5] tloc ntop = top→next;
84     if * {
85         [@pocas6t] CAS(stacktop, top, ntop);
86         [@poa7] return val;
87     }
88     else {
89         [@pocas6f] assume(stacktop ≠ top);
90         [@poa8] int mytid;
91         [@poa9] location[mytid] = p;
92         [@poa10] int pos = nondet();
93         [@poa11] int him = collision[pos];
94         if * {
95             [@pocas12t] CAS(collision[pos], him, mytid);
96             if * {
97                 [@pob13t] assume(him ≠ NULL && him ≠ mytid);
98                 [@poa14] tloc[] q = location[him];
99                 if * {
100                     [@pob15t] assume(q ≠ NULL && q[0]→val == him && q[1]→val ≠ p[1]→val
101 );
102                     if * {
103                         [@pocas16t] CAS(location[mytid], p, NULL);
104                         if * {
105                             [@pocas17t] CAS(location[him], q, NULL);
106                             [@poa18] return q[2]→val;
107                         }
108                         else {
109                             [@pocas17f] assume(location[him] ≠ q);
110                             [@poa19] continue;
111                         }
112                     }
113                     else {
114                         [@pocas16f] assume(location[mytid] ≠ p);
115                         [@poa20] int val = q[2]→val;
116                         [@poa21] location[mytid] = NULL;
117                         [@poa22] return val;
118                     }
119                 }
120                 else {
121                     [@pob15f] assume(!(q ≠ NULL && q[0]→val == him && q[1]→val ≠ p[1]→
122 val));
123                 }
124             }
125             else {
126                 [@pob13f] assume(!(him ≠ NULL && him ≠ mytid));
127                 if * {
128                     [@pocas23f] assume(location[mytid] ≠ p);

```

```

127         [@poa24] int val = q[2]→val;
128         [@poa25] location[mytid] = NULL;
129         [@poa26] return val;
130     }
131     else {
132         [@pocas23t] CAS(location[mytid], p, NULL);
133     }
134 }
135 }
136 else {
137     [@pocas12f] assume(collision[pos] ≠ him);
138 }
139 }
140 }
141 }
142 }

```

### B.2.2 *IRSpec.*

#### IR specifications (.irspecs)

```

1/* OBJECT: Elimination Stack */

3/* ===== */
4/* Elimination Stack (Hendler et al.) IRSpecs */
5/* */
6/* Abstract letter → CLL label mapping (regenerated es.cll): */
7/* push Treiber: a,b,c,d=pua1;pua2;pua3 e=pucas4t f=pucas4f */
8/* push Treiber: g=pua5 */
9/* pop Treiber: h=poa1 i=pob2t j=poa3 k=pob2f */
10/* pop Treiber: l=poa4;poa5 m=pocas6t n=poa7 o=pocas6f */
11/* push publish CAS: success=pucas10t fail=pucas10f */
12/* pop publish CAS: success=pocas12t fail=pocas12f */
13/* elim push: E=pub11t F=pua12 rst=pub13t H=pucas14t */
14/* I=pucas15t J=pua16 K=pucas15f L=pua17 */
15/* M=pucas14f NO=pua18 P=pua19 */
16/* elim pop: E=pob13t F=poa14 rst=pob15t H=pocas16t */
17/* I=pocas17t J=poa18 K=pocas17f L=poa19 */
18/* M=pocas16f N=poa20 O=poa21 P=poa22 */
19/* fallback push: R=pub13f S=pub11f U=pucas20f VW=pua21 */
20/* X=pua22 Y=pucas20t */
21/* fallback pop: R=pob15f S=pob13f U=pocas23f V=poa24 */
22/* W=poa25 X=poa26 Y=pocas23t */
23/* ===== */

25/* ---- Treiber-style CAS on Top ----- */

27/* Push succeeds via Treiber-style CAS */
28fAPushS = atom{
29 ind = pucas4t,
30 red = [pua1; pua2; pua3] |> pucas4t <| pua5
31}

```

```

33/* Pop succeeds via Treiber-style CAS */
34fAPopS = atom{
35  ind = pocas6t,
36  red = [poa1; pob2f; poa4; poa5] |> pocas6t <| poa7
37}

39/* A successful push/pop causes another push to fail its Top CAS */
40fCPushF = coop{
41  ind = pucas4t + pocas6t,
42  red = [poa1; pua2; pua3] |> [fAPushS] + [fAPopS] <| pucas4f
43}

45/* A successful push/pop causes another pop to fail its Top CAS */
46fCPopF = coop{
47  ind = pucas4t + pocas6t,
48  red = [poa1; pob2f; poa4; poa5] |> [fAPushS] + [fAPopS] <| pocas6f
49}

51/* Pop attempts on an empty stack */
52fAPopEmp = atom{
53  ind = poa3,
54  red = EPS |> poa1 <| pob2t; poa3
55}

57/* ---- Publication loop ----- */

59/* Push publication loop: CAS into the Collision array succeeds
60   (publishes this thread's descriptor). Anchor: pucas10t. */
61publoop = atom{
62  ind = pucas10t,
63  red = [pua6; pua7; pua8; pua9] |> pucas10t <| EPS
64}

66/* A successful publish causes a concurrent thread's publish CAS to fail
67   (it now observes Collision[pos] ≠ him). Left-mover: pucas10f. */
68publoopC = coop{
69  ind = pucas10t,
70  red = [pua6; pua7; pua8; pua9] |> [publoop] <| pucas10f
71}

73/* Push finds a partner but operation type mismatch – no exchange occurs */
74fAbadPartnerP = atom{
75  ind = pub13f,
76  red = [pub11t; pua12] |> pub13f <| EPS
77}

79/* Pop finds a partner but operation type mismatch – no exchange occurs */
80fAbadPartnerQ = atom{
81  ind = pob15f,
82  red = [pob13t; poa14] |> pob15f <| EPS
83}

```

```

85/* Pop publication loop: CAS into the Collision array succeeds.  Anchor: pocas12t. */
86publoop_pop = atom{
87  ind = pocas12t,
88  red = [poa8; poa9; poa10; poa11] |> pocas12t <| EPS
89}

91/* A successful pop publish causes a concurrent thread's publish CAS to fail
92   (Collision[pos] ≠ him).  Left-mover: pocas12f. */
93publoopC_pop = coop{
94  ind = pocas12t,
95  red = [poa8; poa9; poa10; poa11] |> [publoop_pop] <| pocas12f
96}

98/* ---- Elimination happy path ----- */

100/* Push thread finds pop partner and succeeds at CAS(Location[him], q, p) */
101fAhappySP = atom{
102  ind = pucas15t,
103  red = [pub11t; pua12; pub13t; pucas14t] |> pucas15t <| pua16
104}

107/* Pop thread finds push partner and succeeds at CAS(Location[him], q, NULL) */
108fAhappySQ = atom{
109  ind = pocas17t,
110  red = [pob13t; poa14; pob15t; pocas16t] |> pocas17t <| poa18
111}

113/* Happy path causes competitor to fail at CAS(Location[him]) (step I) */
114fAhappyFI = coop{
115  ind = pucas15t + pocas17t,
116  red = (pub11t; pua12; pub13t; pucas14t; pucas15f) +
117        (pob13t; poa14; pob15t; pocas16t; pocas17f)
118        |> [fAhappySP] + [fAhappySQ] <|
119        pua17 + poa19
120}

122/* Happy path causes competitor to fail at CAS(Location[mytid]) (step H) */
123fAhappyFH = coop{
124  ind = pucas15t + pocas17t,
125  red = (pub11t; pua12; pub13t; pucas14f) +
126        (pob13t; poa14; pob15t; pocas16f)
127        |> [fAhappySP] + [fAhappySQ] <|
128        (pua18; pua19) + (poa20; poa21; poa22)
129}

131/* ---- Fallback path ----- */

133/* Push fallback: no valid partner found; CAS(Location[mytid]) fails
134   (someone already cleared our descriptor for us) */
135fAfallbackSP = atom{

```

```

136  ind = pucas20f,
137  red = [(pub11f + [fAbadPartnerP])] |> pucas20f <| pua21; pua22
138}

140/* Pop fallback: no valid partner found; CAS(Location[mytid]) fails
141   (someone already cleared our descriptor for us) */
142fAfallbackSQ = atom{
143  ind = pocas23f,
144  red = [(pob13f + [fAbadPartnerQ])] |> pocas23f <| poa24; poa25; poa26
145}

147/* Anchor's fallback CAS fails because mover already cleared the descriptor (push) */
148fAfallbackFP = coop{
149  ind = pucas20f,
150  red = [(pub11f + [fAbadPartnerP]); pucas20t]
151        |> [fAfallbackSP] <| EPS
152}

154/* Anchor's fallback CAS fails because mover already cleared the descriptor (pop) */
155fAfallbackFQ = coop{
156  ind = pocas23f,
157  red = [(pob13f + [fAbadPartnerQ]); pocas23t]
158        |> [fAfallbackSQ] <| EPS
159}

```

### B.2.3 Contextual Commutativity (97).

| # | Kind | Spec            | Obligation                                                                                                                                                                                                                                         |
|---|------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Atom | atom::fAPushS.R | [tloc top = stacktop; int val = p[2]->val; tloc ntop = new {val {} top}]                                                                                                                                                                           |
| 2 | RMov | atom::fAPushS   | $\forall t:\mathbb{N}. \text{top}[t] = \text{stacktop}; \text{val}[t] = p[t][2]\text{->val}; \text{ntop}[t] = \text{tloc\_pua3\_t} \triangleright \{\text{stacktop} == \text{top}[t]\}$                                                            |
| 3 | HTrp | atom::fAPushS   | $\forall t:\mathbb{N}. \{\text{stacktop} == \text{top}[t]\} \text{CAS}(\text{stacktop}, \text{top}[t], \text{ntop}[t]) \{\exists \text{old\_stacktop}. (\text{old\_stacktop} == \text{top}[t] \wedge \text{stacktop} == \text{ntop}[t])\}$         |
| 4 | LMov | atom::fAPushS   | $\forall t:\mathbb{N}. \{\exists \text{old\_stacktop}. (\text{old\_stacktop} == \text{top}[t] \wedge \text{stacktop} == \text{ntop}[t])\} \triangleleft \text{return } 1$                                                                          |
| 5 | Atom | atom::fAPopS.R  | [tloc top = stacktop; assume(top $\neq$ NULL); int val = top->val; tloc ntop = top->next]                                                                                                                                                          |
| 6 | RMov | atom::fAPopS    | $\forall t:\mathbb{N}. \text{top}[t] = \text{stacktop}; \text{assume}(\text{top}[t] \neq \text{NULL}); \text{val}[t] = \text{top}[t]\text{->val}; \text{ntop}[t] = \text{top}[t]\text{->next} \triangleright \{\text{stacktop} == \text{top}[t]\}$ |
| 7 | HTrp | atom::fAPopS    | $\forall t:\mathbb{N}. \{\text{stacktop} == \text{top}[t]\} \text{CAS}(\text{stacktop}, \text{top}[t], \text{ntop}[t]) \{\exists \text{old\_stacktop}. (\text{old\_stacktop} == \text{top}[t] \wedge \text{stacktop} == \text{ntop}[t])\}$         |
| 8 | LMov | atom::fAPopS    | $\forall t:\mathbb{N}. \{\exists \text{old\_stacktop}. (\text{old\_stacktop} == \text{top}[t] \wedge \text{stacktop} == \text{ntop}[t])\} \triangleleft \text{return } \text{val}[t]$                                                              |
| 9 | Atom | coop::fCPushF.R | [tloc top = stacktop; int val = p[2]->val; tloc ntop = new {val {} top}]                                                                                                                                                                           |

| #  | Kind | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----|------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10 | RMov | coop::fCPushF   | $\forall tj:\mathbb{N}. \text{top}[tj] = \text{stacktop}; \text{val}[tj] = p[tj][2] \rightarrow \text{val}$<br>$\text{ntop}[tj] = \text{tloc\_pua3\_tj} \triangleright \{(\text{top}[tj] == \text{stacktop} \wedge$<br>$\text{val}[tj] == p[tj][2] \rightarrow \text{val} \wedge \text{ntop}[tj] == \text{tloc\_pua3\_tj})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 11 | HTrp | coop::fCPushF   | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{(\text{top}[tj] == \text{stacktop} \wedge \text{val}[tj]$<br>$p[tj][2] \rightarrow \text{val} \wedge \text{ntop}[tj] == \text{tloc\_pua3\_tj}) \} ([\text{top}[$<br>$= \text{stacktop}; \text{val}[ti] = p[ti][2] \rightarrow \text{val}; \text{ntop}[ti] =$<br>$\text{tloc\_pua3\_ti}; \text{CAS}(\text{stacktop}, \text{top}[ti], \text{ntop}[ti]); \text{return}$<br>$1) + [[\text{top}[ti] = \text{stacktop}; \text{assume}(\text{top}[ti] \neq \text{NULL});$<br>$\text{val}[ti] = \text{top}[ti] \rightarrow \text{val}; \text{ntop}[ti] = \text{top}[ti] \rightarrow \text{next};$<br>$\text{CAS}(\text{stacktop}, \text{top}[ti], \text{ntop}[ti]); \text{return val}[ti]]$<br>$\{(\exists \text{old\_stacktop}. (\text{top}[tj] == \text{old\_stacktop} \wedge \text{val}[$<br>$== p[tj][2] \rightarrow \text{val} \wedge \text{ntop}[tj] == \text{tloc\_pua3\_tj} \wedge \text{top}[$<br>$\text{old\_stacktop} \wedge \text{val}[ti] == p[ti][2] \rightarrow \text{val} \wedge \text{ntop}[ti]$<br>$\text{tloc\_pua3\_ti} \wedge \text{stacktop} == \text{ntop}[ti]) \vee \exists \text{old\_stack}$<br>$(\text{top}[tj] == \text{old\_stacktop} \wedge \text{val}[tj] == p[tj][2] \rightarrow \text{val}$<br>$\text{ntop}[tj] == \text{tloc\_pua3\_tj} \wedge \text{top}[ti] == \text{old\_stacktop}$<br>$\text{top}[ti] \neq \text{NULL} \wedge \text{val}[ti] == \text{top}[ti] \rightarrow \text{val} \wedge \text{ntop}[ti]$<br>$\text{top}[ti] \rightarrow \text{next} \wedge \text{stacktop} == \text{ntop}[ti]) \}$                                                                                                                                                                                      |
| 12 | LMov | coop::fCPushF   | $\forall ti, tj:\mathbb{N}, ti \neq tj. \text{assume}(\text{stacktop} \neq \text{top}[tj])$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 13 | Atom | coop::fCPopF.R  | $[\text{tloc top} = \text{stacktop}; \text{assume}(\text{top} \neq \text{NULL}); \text{int val} =$<br>$\text{top} \rightarrow \text{val}; \text{tloc ntop} = \text{top} \rightarrow \text{next}]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 14 | RMov | coop::fCPopF    | $\forall tj:\mathbb{N}. \text{top}[tj] = \text{stacktop}; \text{assume}(\text{top}[tj] \neq \text{NULL})$<br>$\text{val}[tj] = \text{top}[tj] \rightarrow \text{val}; \text{ntop}[tj] = \text{top}[tj] \rightarrow \text{next} \triangleright$<br>$\{(\text{top}[tj] == \text{stacktop} \wedge \text{top}[tj] \neq \text{NULL} \wedge \text{val}[tj] =$<br>$\text{top}[tj] \rightarrow \text{val} \wedge \text{ntop}[tj] == \text{top}[tj] \rightarrow \text{next}) \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 15 | HTrp | coop::fCPopF    | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{(\text{top}[tj] == \text{stacktop} \wedge \text{top}[tj]$<br>$\neq \text{NULL} \wedge \text{val}[tj] == \text{top}[tj] \rightarrow \text{val} \wedge \text{ntop}[tj] ==$<br>$\text{top}[tj] \rightarrow \text{next}) \} ([[\text{top}[ti] = \text{stacktop}; \text{val}[ti] =$<br>$p[ti][2] \rightarrow \text{val}; \text{ntop}[ti] = \text{tloc\_pua3\_ti}; \text{CAS}(\text{stacktop},$<br>$\text{top}[ti], \text{ntop}[ti]); \text{return 1} + [[\text{top}[ti] = \text{stacktop};$<br>$\text{assume}(\text{top}[ti] \neq \text{NULL}); \text{val}[ti] = \text{top}[ti] \rightarrow \text{val}; \text{ntop}[$<br>$= \text{top}[ti] \rightarrow \text{next}; \text{CAS}(\text{stacktop}, \text{top}[ti], \text{ntop}[ti]);$<br>$\text{return val}[ti]] \} \{(\exists \text{old\_stacktop}. (\text{top}[tj] ==$<br>$\text{old\_stacktop} \wedge \text{top}[tj] \neq \text{NULL} \wedge \text{val}[tj] == \text{top}[tj]$<br>$\wedge \text{ntop}[tj] == \text{top}[tj] \rightarrow \text{next} \wedge \text{top}[ti] == \text{old\_stacktop}$<br>$\wedge \text{val}[ti] == p[ti][2] \rightarrow \text{val} \wedge \text{ntop}[ti] == \text{tloc\_pua3\_ti}$<br>$\wedge \text{stacktop} == \text{ntop}[ti]) \vee \exists \text{old\_stacktop}. (\text{top}[tj]$<br>$\text{old\_stacktop} \wedge \text{top}[tj] \neq \text{NULL} \wedge \text{val}[tj] == \text{top}[tj]$<br>$\wedge \text{ntop}[tj] == \text{top}[tj] \rightarrow \text{next} \wedge \text{top}[ti] == \text{old\_stacktop}$<br>$\wedge \text{top}[ti] \neq \text{NULL} \wedge \text{val}[ti] == \text{top}[ti] \rightarrow \text{val} \wedge \text{ntop}[ti]$<br>$== \text{top}[ti] \rightarrow \text{next} \wedge \text{stacktop} == \text{ntop}[ti]) \}$ |
| 16 | LMov | coop::fCPopF    | $\forall ti, tj:\mathbb{N}, ti \neq tj. \text{assume}(\text{stacktop} \neq \text{top}[tj])$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 17 | HTrp | atom::fAPopEmp  | $\forall t:\mathbb{N}. \{\text{true}\} \text{top}[t] = \text{stacktop} \{\text{top}[t] == \text{stacktop}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 18 | LMov | atom::fAPopEmp  | $\forall t:\mathbb{N}. \text{assume}(\text{top}[t] == \text{NULL})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 19 | LMov | atom::fAPopEmp  | $\forall t:\mathbb{N}. \{(\text{top}[t] == \text{stacktop} \wedge \text{top}[t] == \text{NULL})\} \triangleleft n$<br>$0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 20 | Atom | atom::publoop.R | $[\text{int mytid}; \text{location}[\text{mytid}] = p; \text{int pos} = \text{nondet}()$<br>$\text{him} = \text{collision}[\text{pos}]]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



| #  | Kind | Spec                  | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----|------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21 | RMov | atom::publoop         | $\forall t:\mathbb{N}. \text{int mytid}; \text{location}[\text{mytid}[t]] = p[t]; \text{pos}[t] = \text{nondet}(); \text{him}[t] = \text{collision}[\text{pos}[t]] \triangleright \{\text{collision}[\text{pos}[t]] == \text{him}[t]\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 22 | HTrp | atom::publoop         | $\forall t:\mathbb{N}. \{\text{collision}[\text{pos}[t]] == \text{him}[t]\} \text{CAS}(\text{collision}[\text{pos}[t]], \text{him}[t], \text{mytid}[t]) \{\exists \text{old\_collision\_pos\_t\_}. (\text{old\_collision\_pos\_t\_} == \text{him}[t] \wedge \text{collision}[\text{pos}[t]] == \text{mytid}[t])\}$                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 23 | Atom | coop::publoopC.R      | $[\text{int mytid}; \text{location}[\text{mytid}] = p; \text{int pos} = \text{nondet}(); \text{him} = \text{collision}[\text{pos}]]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 24 | RMov | coop::publoopC        | $\forall tj:\mathbb{N}. \text{int mytid}; \text{location}[\text{mytid}[tj]] = p[tj]; \text{pos}[tj] = \text{nondet}(); \text{him}[tj] = \text{collision}[\text{pos}[tj]] \triangleright \{(\text{location}[\text{mytid}[tj]] == p[tj] \wedge \text{him}[tj] == \text{collision}[\text{pos}[tj]])\}$                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 25 | HTrp | coop::publoopC        | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{(\text{location}[\text{mytid}[tj]] == p[tj] \wedge \text{him}[tj] == \text{collision}[\text{pos}[tj]])\} [ [\text{int mytid}; \text{location}[\text{mytid}[ti]] = p[ti]; \text{pos}[ti] = \text{nondet}(); \text{him}[ti] = \text{collision}[\text{pos}[ti]]; \text{CAS}(\text{collision}[\text{pos}[ti]], \text{him}[ti], \text{mytid}[ti]); \text{skip} \{ \exists \text{old\_collision\_pos\_ti\_}. (\text{location}[\text{mytid}[tj]] == p[tj] \wedge \text{him}[tj] == \text{collision}[\text{pos}[tj]] \wedge \text{location}[\text{mytid}[ti]] == p[ti] \wedge \text{him}[ti] == \text{old\_collision\_pos\_ti\_} \wedge \text{collision}[\text{pos}[ti]] == \text{mytid}[ti]) \} ]$ |
| 26 | Subs | coop::publoopC        | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{ \exists \text{old\_collision\_pos\_ti\_}. (\text{location}[\text{mytid}[tj]] == p[tj] \wedge \text{him}[tj] == \text{collision}[\text{pos}[tj]] \wedge \text{location}[\text{mytid}[ti]] == p[ti] \wedge \text{him}[ti] == \text{old\_collision\_pos\_ti\_} \wedge \text{collision}[\text{pos}[ti]] == \text{mytid}[ti]) \} \Rightarrow \{ \text{collision}[\text{pos}[tj]] \neq \text{him}[tj] \}$                                                                                                                                                                                                                                                                                        |
| 27 | Atom | atom::fAbadPartnerP.R | $[\text{assume}(\text{him} \neq \text{NULL} \wedge \text{him} \neq \text{mytid}); \text{tloc}[] \text{ q} = \text{location}[\text{him}]]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 28 | RMov | atom::fAbadPartnerP   | $\forall t:\mathbb{N}. \text{assume}(\text{him}[t] \neq \text{NULL} \wedge \text{him}[t] \neq \text{mytid}[t]); = \text{location}[\text{him}[t]] \triangleright \{!(q[t] \neq \text{NULL} \wedge q[t][0] \rightarrow \text{val} \neq \text{him}[t] \wedge q[t][1] \rightarrow \text{val} \neq p[t][1] \rightarrow \text{val})\}$                                                                                                                                                                                                                                                                                                                                                                                                      |
| 29 | HTrp | atom::fAbadPartnerP   | $\forall t:\mathbb{N}. \{!(q[t] \neq \text{NULL} \wedge q[t][0] \rightarrow \text{val} == \text{him}[t] \wedge q[t][1] \rightarrow \text{val} \neq p[t][1] \rightarrow \text{val})\} \text{assume}(!(q[t] \neq \text{NULL} \wedge q[t][0] \rightarrow \text{val} == \text{him}[t] \wedge q[t][1] \rightarrow \text{val} \neq p[t][1] \rightarrow \text{val}) \wedge q[t][0] \rightarrow \text{val} == \text{him}[t] \wedge q[t][1] \rightarrow \text{val} \neq p[t][1] \rightarrow \text{val}) \neq p[t][1] \rightarrow \text{val})\}$                                                                                                                                                                                                |
| 30 | Atom | atom::fAbadPartnerQ.R | $[\text{assume}(\text{him} \neq \text{NULL} \wedge \text{him} \neq \text{mytid}); \text{tloc}[] \text{ q} = \text{location}[\text{him}]]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 31 | RMov | atom::fAbadPartnerQ   | $\forall t:\mathbb{N}. \text{assume}(\text{him}[t] \neq \text{NULL} \wedge \text{him}[t] \neq \text{mytid}[t]); = \text{location}[\text{him}[t]] \triangleright \{!(q[t] \neq \text{NULL} \wedge q[t][0] \rightarrow \text{val} \neq \text{him}[t] \wedge q[t][1] \rightarrow \text{val} \neq p[t][1] \rightarrow \text{val})\}$                                                                                                                                                                                                                                                                                                                                                                                                      |
| 32 | HTrp | atom::fAbadPartnerQ   | $\forall t:\mathbb{N}. \{!(q[t] \neq \text{NULL} \wedge q[t][0] \rightarrow \text{val} == \text{him}[t] \wedge q[t][1] \rightarrow \text{val} \neq p[t][1] \rightarrow \text{val})\} \text{assume}(!(q[t] \neq \text{NULL} \wedge q[t][0] \rightarrow \text{val} == \text{him}[t] \wedge q[t][1] \rightarrow \text{val} \neq p[t][1] \rightarrow \text{val}) \wedge q[t][0] \rightarrow \text{val} == \text{him}[t] \wedge q[t][1] \rightarrow \text{val} \neq p[t][1] \rightarrow \text{val}) \neq p[t][1] \rightarrow \text{val})\}$                                                                                                                                                                                                |
| 33 | Atom | atom::publoop_pop.R   | $[\text{int mytid}; \text{location}[\text{mytid}] = p; \text{int pos} = \text{nondet}(); \text{him} = \text{collision}[\text{pos}]]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| #  | Kind | Spec                 | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----|------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 34 | RMov | atom::publoop_pop    | $\forall t:\mathbb{N}. \text{int mytid}; \text{location}[\text{mytid}[t]] = p[t]; \text{pos}[t] = \text{nondet}(); \text{him}[t] = \text{collision}[\text{pos}[t]] \triangleright \{\text{collision}[\text{pos}[t]] == \text{him}[t]\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 35 | HTrp | atom::publoop_pop    | $\forall t:\mathbb{N}. \{\text{collision}[\text{pos}[t]] == \text{him}[t]\} \text{CAS}(\text{collision}[\text{pos}[t]], \text{him}[t], \text{mytid}[t]) \{\exists \text{old\_collision\_pos\_t\_}. (\text{old\_collision\_pos\_t\_} == \text{him}[t] \wedge \text{collision}[\text{pos}[t]] == \text{mytid}[t])\}$                                                                                                                                                                                                                                                                                                                                                                                                                |
| 36 | Atom | coop::publoopC_pop.R | $[\text{int mytid}; \text{location}[\text{mytid}] = p; \text{int pos} = \text{nondet}(); \text{him} = \text{collision}[\text{pos}]]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 37 | RMov | coop::publoopC_pop   | $\forall tj:\mathbb{N}. \text{int mytid}; \text{location}[\text{mytid}[tj]] = p[tj]; \text{pos}[tj] = \text{nondet}(); \text{him}[tj] = \text{collision}[\text{pos}[tj]] \triangleright \{(\text{location}[\text{mytid}[tj]] == p[tj] \wedge \text{him}[tj] == \text{collision}[\text{pos}[tj]])\}$                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 38 | HTrp | coop::publoopC_pop   | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{(\text{location}[\text{mytid}[tj]] == p[tj] \wedge \text{him}[tj] == \text{collision}[\text{pos}[tj]])\} [[\text{int mytid}; \text{location}[\text{mytid}[ti]] = p[ti]; \text{pos}[ti] = \text{nondet}(); \text{him}[ti] = \text{collision}[\text{pos}[ti]]; \text{CAS}(\text{collision}[\text{pos}[ti]], \text{him}[ti], \text{mytid}[ti]); \text{skip} \{\exists \text{old\_collision\_pos\_ti\_}. (\text{location}[\text{mytid}[tj]] == p[tj] \wedge \text{him}[tj] == \text{collision}[\text{pos}[tj]] \wedge \text{location}[\text{mytid}[ti]] == p[ti] \wedge \text{him}[ti] == \text{old\_collision\_pos\_ti\_} \wedge \text{collision}[\text{pos}[ti]] == \text{mytid}[ti])\}]$ |
| 39 | Subs | coop::publoopC_pop   | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{\exists \text{old\_collision\_pos\_ti\_}. (\text{location}[\text{mytid}[tj]] == p[tj] \wedge \text{him}[tj] == \text{collision}[\text{pos}[tj]] \wedge \text{location}[\text{mytid}[ti]] == p[ti] \wedge \text{him}[ti] == \text{old\_collision\_pos\_ti\_} \wedge \text{collision}[\text{pos}[ti]] == \text{mytid}[ti])\} \Rightarrow \{\text{collision}[\text{pos}[tj]] \neq \text{him}[tj]\}$                                                                                                                                                                                                                                                                                        |
| 40 | Atom | atom::fAhappySP.R    | $[\text{assume}(\text{him} \neq \text{NULL} \wedge \text{him} \neq \text{mytid}); \text{tloc}[] \text{ q} = \text{location}[\text{him}]; \text{assume}(\text{q} \neq \text{NULL} \wedge \text{q}[0] \rightarrow \text{val} == \text{him} \wedge \text{q}[1] \rightarrow \text{val} \neq p[1] \rightarrow \text{val}); \text{CAS}(\text{location}[\text{mytid}], p, \text{NULL})]$                                                                                                                                                                                                                                                                                                                                                 |
| 41 | RMov | atom::fAhappySP      | $\forall t:\mathbb{N}. \text{assume}(\text{him}[t] \neq \text{NULL} \wedge \text{him}[t] \neq \text{mytid}[t]); \text{q}[t] = \text{location}[\text{him}[t]]; \text{assume}(\text{q}[t] \neq \text{NULL} \wedge \text{q}[t][0] \rightarrow \text{val} == \text{him}[t] \wedge \text{q}[t][1] \rightarrow \text{val} \neq p[t][1] \rightarrow \text{val}); \text{CAS}(\text{location}[\text{mytid}[t]], p[t], \text{NULL}) \triangleright \{\text{location}[\text{him}[t]] == \text{q}[t]\}$                                                                                                                                                                                                                                       |
| 42 | HTrp | atom::fAhappySP      | $\forall t:\mathbb{N}. \{\text{location}[\text{him}[t]] == \text{q}[t]\} \text{CAS}(\text{location}[\text{him}[t]], \text{q}[t], p[t]) \{\exists \text{old\_location\_him\_t\_}. (\text{old\_location\_him\_t\_} == \text{q}[t] \wedge \text{location}[\text{him}[t]] == p[t])\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 43 | LMov | atom::fAhappySP      | $\forall t:\mathbb{N}. \{\exists \text{old\_location\_him\_t\_}. (\text{old\_location\_him\_t\_} == \text{q}[t] \wedge \text{location}[\text{him}[t]] == p[t])\} \triangleleft \text{return } 1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 44 | Atom | atom::fAhappySQ.R    | $[\text{assume}(\text{him} \neq \text{NULL} \wedge \text{him} \neq \text{mytid}); \text{tloc}[] \text{ q} = \text{location}[\text{him}]; \text{assume}(\text{q} \neq \text{NULL} \wedge \text{q}[0] \rightarrow \text{val} == \text{him} \wedge \text{q}[1] \rightarrow \text{val} \neq p[1] \rightarrow \text{val}); \text{CAS}(\text{location}[\text{mytid}], p, \text{NULL})]$                                                                                                                                                                                                                                                                                                                                                 |
| 45 | RMov | atom::fAhappySQ      | $\forall t:\mathbb{N}. \text{assume}(\text{him}[t] \neq \text{NULL} \wedge \text{him}[t] \neq \text{mytid}[t]); \text{q}[t] = \text{location}[\text{him}[t]]; \text{assume}(\text{q}[t] \neq \text{NULL} \wedge \text{q}[t][0] \rightarrow \text{val} == \text{him}[t] \wedge \text{q}[t][1] \rightarrow \text{val} \neq p[t][1] \rightarrow \text{val}); \text{CAS}(\text{location}[\text{mytid}[t]], p[t], \text{NULL}) \triangleright \{\text{location}[\text{him}[t]] == \text{q}[t]\}$                                                                                                                                                                                                                                       |

| #  | Kind | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----|------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 46 | HTrp | atom::fAhappySQ | $\forall t:\mathbb{N}. \{location[him[t]] == q[t]\} \text{ CAS}(location[him[t]], q[t], NULL) \{ \exists old\_location\_him\_t\_. (old\_location\_him\_t\_ == q[t] \wedge location[him[t]] == NULL) \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 47 | LMov | atom::fAhappySQ | $\forall t:\mathbb{N}. \{ \exists old\_location\_him\_t\_. (old\_location\_him\_t\_ == q[t] \wedge location[him[t]] == NULL) \} \triangleleft \text{return } q[t][2] \rightarrow val$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 48 | Subs | coop::fAhappyFI | $\forall tj:\mathbb{N}. \{ ((\exists old\_location\_mytid\_tj\_. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj][1] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] == q[tj]) \vee (\exists old\_location\_mytid\_tj\_. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj][1] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] == q[tj])) \} \Rightarrow \{ location[him[tj]] \neq q[tj] \}$                                |
| 49 | RMov | coop::fAhappyFI | $\forall tj:\mathbb{N}. \text{CAS}(location[mytid[tj]], p[tj], NULL) \triangleright \{ ((\exists old\_location\_mytid\_tj\_. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj][1] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj][1] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] == q[tj])) \}$ |
| 50 | RMov | coop::fAhappyFI | $\forall tj:\mathbb{N}. \text{assume}(q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 51 | RMov | coop::fAhappyFI | $\forall tj:\mathbb{N}. q[tj] = location[him[tj]] \triangleright \{ (location[mytid[tj]] == p[tj]) \wedge \exists old\_location\_mytid\_tj\_. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \wedge old\_location\_mytid\_tj\_ == p[tj]) \wedge location[him[tj]] \neq q[tj]) \}$                                                                                                                                                                                                                                                                                                                                                                                                         |
| 52 | RMov | coop::fAhappyFI | $\forall tj:\mathbb{N}. \text{assume}(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj])$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| #  | Kind | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----|------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 53 | Subs | coop::fAhappyFI | $\forall tj:\mathbb{N}. \{(\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])\} \Rightarrow \{location[him[tj]] \neq q[tj]\}$                         |
| 54 | RMov | coop::fAhappyFI | $\forall tj:\mathbb{N}. CAS(location[mytid[tj]], p[tj], NULL) \triangleright \{(\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])\}$ |
| 55 | RMov | coop::fAhappyFI | $\forall tj:\mathbb{N}. assume(q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 56 | RMov | coop::fAhappyFI | $\forall tj:\mathbb{N}. q[tj] = location[him[tj]] \triangleright \{(location[mytid[tj]] == p[tj] \wedge \exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj]) \wedge location[him[tj]] \neq q[tj])\}$                                                                                                                                                                                                                                                                                                                                                                              |
| 57 | RMov | coop::fAhappyFI | $\forall tj:\mathbb{N}. assume(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj])$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| #  | Kind | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----|------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 58 | HTrp | coop::fAhappyFI | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \} \{ [[assume(him[ti] \neq NULL \wedge him[ti] \neq mytid[ti]); q[ti] = location[him[ti]]; assume(q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val); CAS(location[mytid[ti]], p[ti], NULL)]; CAS(location[him[ti]], q[ti], p[ti]); return 1] + [[assume(him[ti] \neq NULL \wedge him[ti] \neq mytid[ti]); q[ti] = location[him[ti]]; assume(q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val); CAS(location[mytid[ti]], p[ti], NULL)]; CAS(location[him[ti]], q[ti], NULL); return q[ti][2] \rightarrow val] \} \{ (\exists old\_location\_him\_ti\_.. (\exists old\_location\_mytid\_ti\_.. (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj] \wedge him[ti] \neq mytid[ti] \wedge q[ti] == old\_location\_him\_tj\_ \wedge q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val \wedge old\_location\_mytid\_ti\_ == p[ti] \wedge location[mytid[ti]] == NULL) \wedge old\_location\_him\_tj\_ == q[ti] \wedge location[him[ti]] == p[ti]) \vee \exists old\_location\_him\_ti\_.. (\exists old\_location\_mytid\_ti\_.. (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj] \wedge him[ti] \neq NULL \wedge him[ti] \neq mytid[ti] \wedge q[ti] == old\_location\_him\_ti\_ \wedge q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val \wedge old\_location\_mytid\_ti\_ == p[ti] \wedge location[mytid[ti]] == NULL) \wedge old\_location\_him\_tj\_ == q[ti] \wedge location[him[ti]] == NULL) \} \}$ |

| #  | Kind | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----|------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 59 | LMov | coop::fAhappyFI | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ (\exists old\_location\_him\_ti\_.. (\exists old\_location\_mytid\_ti\_.. (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_.. == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj] \wedge him[ti] \neq mytid[ti] \wedge q[ti] == old\_location\_him\_ti\_.. q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val \wedge old\_location\_mytid\_ti\_.. == p[ti] \wedge location[mytid[ti]] == NULL) \wedge old\_location\_him\_ti\_.. == q[ti] \wedge location[him[ti]] == p[ti]) \vee \exists old\_location\_him\_ti\_.. (\exists old\_location\_mytid\_ti\_.. (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_.. == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj] \wedge him[ti] \neq NULL \wedge him[ti] \neq mytid[ti] \wedge q[ti] == old\_location\_him\_ti\_.. \wedge q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val \wedge old\_location\_mytid\_ti\_.. == p[ti] \wedge location[mytid[ti]] == NULL) \wedge old\_location\_him\_ti\_.. == q[ti] \wedge location[him[ti]] == NULL)) \}$ |
| 60 | LMov | coop::fAhappyFI | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ (\exists old\_location\_him\_ti\_.. (\exists old\_location\_mytid\_ti\_.. (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_.. == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj] \wedge him[ti] \neq mytid[ti] \wedge q[ti] == old\_location\_him\_ti\_.. q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val \wedge old\_location\_mytid\_ti\_.. == p[ti] \wedge location[mytid[ti]] == NULL) \wedge old\_location\_him\_ti\_.. == q[ti] \wedge location[him[ti]] == p[ti]) \vee \exists old\_location\_him\_ti\_.. (\exists old\_location\_mytid\_ti\_.. (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_.. == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj] \wedge him[ti] \neq NULL \wedge him[ti] \neq mytid[ti] \wedge q[ti] == old\_location\_him\_ti\_.. \wedge q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val \wedge old\_location\_mytid\_ti\_.. == p[ti] \wedge location[mytid[ti]] == NULL) \wedge old\_location\_him\_ti\_.. == q[ti] \wedge location[him[ti]] == NULL)) \}$ |

| #  | Kind | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----|------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 61 | Subs | coop::fAhappyFH | $\forall tj:\mathbb{N}. \{((him[tj] \neq \text{NULL} \wedge him[tj] \neq mytid[tj] \wedge q[tj] == \text{location}[him[tj]] \wedge q[tj] \neq \text{NULL} \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \wedge \text{location}[mytid[tj]] \neq p[tj]) \vee (him[tj] \neq \text{NULL} \wedge him[tj] \neq mytid[tj] \wedge q[tj] == \text{location}[him[tj]] \wedge q[tj] \neq \text{NULL} \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \neq p[tj][1] \rightarrow val \wedge \text{location}[mytid[tj]] \neq p[tj]))\} = \{\text{location}[mytid[tj]] \neq p[tj]\}$                     |
| 62 | RMov | coop::fAhappyFH | $\forall tj:\mathbb{N}. \text{assume}(q[tj] \neq \text{NULL} \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 63 | RMov | coop::fAhappyFH | $\forall tj:\mathbb{N}. q[tj] = \text{location}[him[tj]] \triangleright \{((him[tj] \neq \text{NULL} \wedge him[tj] \neq mytid[tj] \wedge q[tj] == \text{location}[him[tj]] \wedge q[tj] \neq \text{NULL} \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \neq p[tj][1] \rightarrow val \wedge \text{location}[mytid[tj]] \neq p[tj]) \vee (him[tj] \neq \text{NULL} \wedge him[tj] \neq mytid[tj] \wedge q[tj] == \text{location}[him[tj]] \wedge q[tj] \neq \text{NULL} \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge \text{location}[mytid[tj]] \neq p[tj]))\}$ |
| 64 | RMov | coop::fAhappyFH | $\forall tj:\mathbb{N}. \text{assume}(him[tj] \neq \text{NULL} \wedge him[tj] \neq mytid[tj])$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 65 | Subs | coop::fAhappyFH | $\forall tj:\mathbb{N}. \{((him[tj] \neq \text{NULL} \wedge him[tj] \neq mytid[tj] \wedge q[tj] == \text{location}[him[tj]] \wedge q[tj] \neq \text{NULL} \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \wedge \text{location}[mytid[tj]] \neq p[tj]) \vee (him[tj] \neq \text{NULL} \wedge him[tj] \neq mytid[tj] \wedge q[tj] == \text{location}[him[tj]] \wedge q[tj] \neq \text{NULL} \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \neq p[tj][1] \rightarrow val \wedge \text{location}[mytid[tj]] \neq p[tj]))\} = \{\text{location}[mytid[tj]] \neq p[tj]\}$                     |
| 66 | RMov | coop::fAhappyFH | $\forall tj:\mathbb{N}. \text{assume}(q[tj] \neq \text{NULL} \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 67 | RMov | coop::fAhappyFH | $\forall tj:\mathbb{N}. q[tj] = \text{location}[him[tj]] \triangleright \{((him[tj] \neq \text{NULL} \wedge him[tj] \neq mytid[tj] \wedge q[tj] == \text{location}[him[tj]] \wedge q[tj] \neq \text{NULL} \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \neq p[tj][1] \rightarrow val \wedge \text{location}[mytid[tj]] \neq p[tj]) \vee (him[tj] \neq \text{NULL} \wedge him[tj] \neq mytid[tj] \wedge q[tj] == \text{location}[him[tj]] \wedge q[tj] \neq \text{NULL} \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge \text{location}[mytid[tj]] \neq p[tj]))\}$ |
| 68 | RMov | coop::fAhappyFH | $\forall tj:\mathbb{N}. \text{assume}(him[tj] \neq \text{NULL} \wedge him[tj] \neq mytid[tj])$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| #  | Kind | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----|------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 69 | HTrp | coop::fAhappyFH | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ ((him[tj] \neq \text{NULL} \wedge him[tj] \neq mytid[tj] \wedge q[tj] == \text{location}[him[tj]] \wedge q[tj] \neq \text{NULL} \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge \text{location}[mytid[tj]] \neq p[tj]) \vee (him[tj] \neq \text{NULL} \wedge him[tj] \neq mytid[tj] \wedge q[tj] == \text{location}[him[tj]] \wedge q[tj] \neq \text{NULL} \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge \text{location}[mytid[tj]] \neq p[tj])) \} \langle [[\text{assume}(him[ti] \neq \text{NULL} \wedge him[ti] \neq mytid[ti]); q[ti] = \text{location}[him[ti]]; \text{assume}(q[ti] \neq \text{NULL} \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val); \text{CAS}(\text{location}[mytid[ti]], p[ti], \text{NULL}); \text{CAS}(\text{location}[him[ti]], q[ti], p[ti]); \text{return } 1] + [[\text{assume}(him[ti] \neq \text{NULL} \wedge him[ti] \neq mytid[ti]); q[ti] = \text{location}[him[ti]]; \text{assume}(q[ti] \neq \text{NULL} \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val); \text{CAS}(\text{location}[mytid[ti]], p[ti], \text{NULL}); \text{CAS}(\text{location}[him[ti]], q[ti], \text{NULL}); \text{return } q[ti][2] \rightarrow val] \} \{ (\exists old\_location\_him\_ti\_ \_. (\exists old\_location\_mytid\_ti\_ \_. (him[tj] \neq \text{NULL} \wedge him[tj] \neq mytid[tj] \wedge q[tj] == \text{location}[him[tj]] \wedge q[tj] \neq \text{NULL} \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge \text{location}[mytid[tj]] \neq p[tj] \wedge him[ti] \neq \text{NULL} \wedge him[ti] \neq mytid[ti] \wedge q[ti] == \text{old\_location\_him\_ti\_} \wedge q[ti] \neq \text{NULL} \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val \wedge \text{old\_location\_mytid\_ti\_} == p[ti] \wedge \text{location}[mytid[ti]] == \text{NULL}) \wedge \text{old\_location\_him\_ti\_} == q[ti] \wedge \text{location}[him[ti]] == p[ti]) \vee \exists old\_location\_him\_ti\_ \_. (\exists old\_location\_mytid\_ti\_ \_. (him[tj] \neq \text{NULL} \wedge him[tj] \neq mytid[tj] \wedge q[tj] == \text{location}[him[tj]] \wedge q[tj] \neq \text{NULL} \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge \text{location}[mytid[tj]] \neq p[tj] \wedge him[ti] \neq \text{NULL} \wedge him[ti] \neq mytid[ti] \wedge q[ti] == \text{old\_location\_him\_ti\_} \wedge q[ti] \neq \text{NULL} \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val \wedge \text{old\_location\_mytid\_ti\_} == p[ti] \wedge \text{location}[mytid[ti]] == \text{NULL}) \wedge \text{old\_location\_him\_ti\_} == q[ti] \wedge \text{location}[him[ti]] == \text{NULL})) \}$ |



| #  | Kind | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----|------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 70 | LMov | coop::fAhappyFH | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ (\exists old\_location\_him\_ti\_.. (\exists old\_location\_mytid\_ti\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge location[mytid[tj]] \neq p[tj] \wedge him[ti] \neq NULL \wedge him[ti] \neq mytid[ti] \wedge q[ti] == old\_location\_him\_ti\_ \wedge q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val \wedge old\_location\_mytid\_ti\_ == p[ti] \wedge location[mytid[ti]] == NULL) \wedge old\_location\_him\_ti\_ == q[ti] \wedge location[him[ti]] == p[ti]) \vee \exists old\_location\_him\_tj\_.. (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge location[mytid[tj]] \neq p[tj] \wedge him[ti] \neq NULL \wedge him[ti] \neq mytid[ti] \wedge q[ti] == old\_location\_him\_ti\_ \wedge q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val \wedge old\_location\_mytid\_ti\_ == p[ti] \wedge location[mytid[ti]] == NULL) \wedge old\_location\_him\_ti\_ == q[ti] \wedge location[him[ti]] == NULL)) \} \triangleleft location[mytid[tj]] == NULL \}$                                                                   |
| 71 | LMov | coop::fAhappyFH | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ \exists old\_location\_mytid\_tj\_.. ((\exists old\_location\_him\_ti\_.. (\exists old\_location\_mytid\_ti\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ \neq p[tj] \wedge him[ti] \neq NULL \wedge him[ti] \neq mytid[ti] \wedge q[ti] == old\_location\_him\_ti\_ \wedge q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val \wedge old\_location\_mytid\_ti\_ == p[ti] \wedge location[mytid[ti]] == NULL) \wedge old\_location\_him\_ti\_ == q[ti] \wedge location[him[ti]] == p[ti]) \vee \exists old\_location\_him\_tj\_.. (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ \neq p[tj] \wedge him[ti] \neq NULL \wedge him[ti] \neq mytid[ti] \wedge q[ti] == old\_location\_him\_ti\_ \wedge q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val \wedge old\_location\_mytid\_ti\_ == p[ti] \wedge location[mytid[ti]] == NULL) \wedge old\_location\_him\_ti\_ == q[ti] \wedge location[him[ti]] == p[ti]) \wedge location[mytid[tj]] == NULL) \} \triangleleft return 1$ |

| #  | Kind | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----|------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 72 | LMov | coop::fAhappyFH | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ (\exists old\_location\_him\_ti\_ \dots$ $(\exists old\_location\_mytid\_ti\_ \dots (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge location[mytid[tj]] \neq p[tj] \wedge him[ti] \neq NULL \wedge him[ti] \neq mytid[ti] \wedge q[ti] == old\_location\_him\_ti\_ \wedge q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val \wedge old\_location\_mytid\_ti\_ == p[ti] \wedge location[mytid[ti]] == NULL) \wedge old\_location\_him\_ti\_ == q[ti] \wedge location[him[ti]] == p[ti]) \vee \exists old\_location\_him\_tj \dots (\exists old\_location\_mytid\_tj \dots (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge location[mytid[tj]] \neq p[tj] \wedge him[ti] \neq NULL \wedge him[ti] \neq mytid[ti] \wedge q[ti] == old\_location\_him\_ti\_ \wedge q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val \wedge old\_location\_mytid\_ti\_ == p[ti] \wedge location[mytid[ti]] == NULL) \wedge old\_location\_him\_ti\_ == q[ti] \wedge location[him[ti]] == NULL) \} \triangleleft val[tj] = q[tj][2] \rightarrow val$                                     |
| 73 | LMov | coop::fAhappyFH | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ ((\exists old\_location\_him\_ti\_ \dots$ $(\exists old\_location\_mytid\_ti\_ \dots (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge location[mytid[tj]] \neq p[tj] \wedge him[ti] \neq NULL \wedge him[ti] \neq mytid[ti] \wedge q[ti] == old\_location\_him\_ti\_ \wedge q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val \wedge old\_location\_mytid\_ti\_ == p[ti] \wedge location[mytid[ti]] == NULL) \wedge old\_location\_him\_ti\_ == q[ti] \wedge location[him[ti]] == p[ti]) \vee \exists old\_location\_him\_tj \dots (\exists old\_location\_mytid\_tj \dots (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge location[mytid[tj]] \neq p[tj] \wedge him[ti] \neq NULL \wedge him[ti] \neq mytid[ti] \wedge q[ti] == old\_location\_him\_ti\_ \wedge q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val \wedge old\_location\_mytid\_ti\_ == p[ti] \wedge location[mytid[ti]] == NULL) \wedge old\_location\_him\_ti\_ == q[ti] \wedge location[him[ti]] == NULL) \wedge val[tj] = q[tj][2] \rightarrow val) \} \triangleleft location[mytid[tj]] = NULL$ |

| #  | Kind          | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----|---------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 74 | LMov          | coop::fAhappyFH    | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((\exists old\_location\_him\_ti\_ \dots (\exists old\_location\_mytid\_tj\_ \dots$ $(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] ==$ $location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val$ $== him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge$ $old\_location\_mytid\_tj\_ \neq p[tj] \wedge him[ti] \neq NULL$ $him[ti] \neq mytid[ti] \wedge q[ti] == old\_location\_him\_tj\_$ $q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1]$ $\neq p[ti][1] \rightarrow val \wedge old\_location\_mytid\_ti\_ == p[ti]$ $location[mytid[ti]] == NULL) \wedge old\_location\_him\_tj\_$ $== q[ti] \wedge location[him[ti]] == p[ti]) \vee$ $\exists old\_location\_him\_ti\_ \dots (\exists old\_location\_mytid\_tj\_ \dots$ $(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] ==$ $location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val$ $== him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge$ $old\_location\_mytid\_tj\_ \neq p[tj] \wedge him[ti] \neq NULL$ $him[ti] \neq mytid[ti] \wedge q[ti] == old\_location\_him\_tj\_$ $q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1]$ $\neq p[ti][1] \rightarrow val \wedge old\_location\_mytid\_ti\_ == p[ti]$ $location[mytid[ti]] == NULL) \wedge old\_location\_him\_tj\_$ $== q[ti] \wedge location[him[ti]] == NULL)) \wedge val[tj] =$ $q[tj][2] \rightarrow val \wedge location[mytid[tj]] == NULL) \} \triangleleft ret$ $val[tj]$ |
| 75 | IRSpecs Error | atom::fAfallbackSP | Atomic regex references atom [fAbadPartnerP] inside larger regex; use [fAbadPartnerP] as the whole atom regex                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 76 | RMov          | atom::fAfallbackSP | $\forall t: \mathbb{N}. \text{assume}(!(him[t] \neq NULL \wedge him[t] \neq mytid[t]))$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 77 | RMov          | atom::fAfallbackSP | $\forall t: \mathbb{N}. \text{assume}(him[t] \neq NULL \wedge him[t] \neq mytid[t]);$<br>$q[t] = location[him[t]]; \text{assume}(!(q[t] \neq NULL \wedge$<br>$q[t][0] \rightarrow val == him[t] \wedge q[t][1] \rightarrow val \neq p[t][1] \rightarrow val$<br>$\triangleright \{location[mytid[t]] \neq p[t]\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 78 | HTrp          | atom::fAfallbackSP | $\forall t: \mathbb{N}. \{location[mytid[t]] \neq p[t]\}$<br>$\text{assume}(location[mytid[t]] \neq p[t]) \{location[mytid[t]]$<br>$p[t]\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 79 | LMov          | atom::fAfallbackSP | $\forall t: \mathbb{N}. \{location[mytid[t]] \neq p[t]\} \triangleleft location[mytid[t]]$<br>$= NULL$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 80 | LMov          | atom::fAfallbackSP | $\forall t: \mathbb{N}. \{ \exists old\_location\_mytid\_tj\_ \dots$<br>$(old\_location\_mytid\_tj\_ \neq p[t] \wedge location[mytid[t]]$<br>$NULL) \} \triangleleft \text{return } 1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 81 | IRSpecs Error | atom::fAfallbackSQ | Atomic regex references atom [fAbadPartnerQ] inside larger regex; use [fAbadPartnerQ] as the whole atom regex                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 82 | RMov          | atom::fAfallbackSQ | $\forall t: \mathbb{N}. \text{assume}(!(him[t] \neq NULL \wedge him[t] \neq mytid[t]))$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 83 | RMov          | atom::fAfallbackSQ | $\forall t: \mathbb{N}. \text{assume}(him[t] \neq NULL \wedge him[t] \neq mytid[t]);$<br>$q[t] = location[him[t]]; \text{assume}(!(q[t] \neq NULL \wedge$<br>$q[t][0] \rightarrow val == him[t] \wedge q[t][1] \rightarrow val \neq p[t][1] \rightarrow val$<br>$\triangleright \{location[mytid[t]] \neq p[t]\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| #  | Kind          | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----|---------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 84 | HTrp          | atom::fAfallbackSQ | $\forall t:\mathbb{N}. \{location[mytid[t]] \neq p[t]\}$<br>assume( $location[mytid[t]] \neq p[t]$ ) $\{location[mytid[t]] \neq p[t]\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 85 | LMov          | atom::fAfallbackSQ | $\forall t:\mathbb{N}. \{location[mytid[t]] \neq p[t]\} \triangleleft val[t] = q[t][2] \rightarrow val$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 86 | LMov          | atom::fAfallbackSQ | $\forall t:\mathbb{N}. \{(location[mytid[t]] \neq p[t] \wedge val[t] == q[t][2] \rightarrow val)\} \triangleleft location[mytid[t]] = NULL$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 87 | LMov          | atom::fAfallbackSQ | $\forall t:\mathbb{N}. \{\exists old\_location\_mytid\_t\_.. (old\_location\_mytid\_t\_ \neq p[t] \wedge val[t] == q[t][2] \wedge location[mytid[t]] == NULL)\} \triangleleft return\ val[t]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 88 | IRSpecs Error | coop::fAfallbackFP | Atomic regex references atom [fAbadPartnerP] inside larger regex; use [fAbadPartnerP] as the whole atom regex                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 89 | RMov          | coop::fAfallbackFP | $\forall tj:\mathbb{N}. CAS(location[mytid[tj]], p[tj], NULL) \triangleright \{\exists old\_location\_mytid\_tj\_.. ((!(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj]) \vee (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge !(q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val)))) \wedge old\_location\_mytid\_tj\_ == p[tj], location[mytid[tj]] == NULL\}$                                                                                                                                                                                          |
| 90 | RMov          | coop::fAfallbackFP | $\forall tj:\mathbb{N}. assume(!(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj]))$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 91 | RMov          | coop::fAfallbackFP | $\forall tj:\mathbb{N}. assume(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj], q[tj] = location[him[tj]]); assume(!(q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val)) \triangleright \{(location[mytid[tj]] == p[tj] \wedge \exists old\_location\_mytid\_tj\_.. ((!(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj]) \vee (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge !(q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val)))) \wedge old\_location\_mytid\_tj\_ == p[tj],$ |

| #  | Kind          | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----|---------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 92 | HTrp          | coop::fAfallbackFP | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj]) \vee$ $(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] ==$ $location[him[tj]] \wedge !(q[tj] \neq NULL \wedge q[tj][0] \rightarrow val$ $== him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid$ $== NULL) \} [ [ (assume(!(him[ti] \neq NULL \wedge him[ti]$ $\neq mytid[ti])) + [ [ assume(him[ti] \neq NULL \wedge$ $him[ti] \neq mytid[ti]); q[ti] = location[him[ti]]];$ $assume(!(q[ti] \neq NULL \wedge q[ti][0] \rightarrow val == him[ti]$ $\wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val)); skip ] ] ;$ $assume(location[mytid[ti]] \neq p[ti]); location[mytid$ $= NULL; return 1 ] \{ \exists old\_location\_mytid\_ti\_..$ $((\exists old\_location\_mytid\_tj\_.. ((!(him[tj] \neq NULL \wedge$ $him[tj] \neq mytid[tj]) \vee (him[tj] \neq NULL \wedge him[tj] \neq$ $mytid[tj] \wedge q[tj] == location[him[tj]] \wedge !(q[tj] \neq$ $NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq$ $p[tj][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[tj]$ $\wedge location[mytid[tj]] == NULL) \wedge !(him[ti] \neq NULL$ $him[ti] \neq mytid[ti])) \vee (\exists old\_location\_mytid\_tj\_..$ $((!(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj]) \vee$ $(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] ==$ $location[him[tj]] \wedge !(q[tj] \neq NULL \wedge q[tj][0] \rightarrow val$ $== him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid$ $== NULL) \wedge him[ti] \neq NULL \wedge him[ti] \neq mytid[ti] \wedge$ $== location[him[ti]] \wedge !(q[ti] \neq NULL \wedge q[ti][0] \rightarrow$ $== him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_ti\_ \neq p[ti] \wedge location[mytid[$ $== NULL) \}$ |
| 93 | IRSpecs Error | coop::fAfallbackFQ | Atomic regex references atom [fAbadPartnerQ] inside larger regex; use [fAbadPartnerQ] as the whole atom regex                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 94 | RMov          | coop::fAfallbackFQ | $\forall tj: \mathbb{N}. CAS(location[mytid[tj]], p[tj], NULL) \triangleright$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[tj] \neq NULL \wedge$ $him[tj] \neq mytid[tj]) \vee (him[tj] \neq NULL \wedge him[tj] \neq$ $mytid[tj] \wedge q[tj] == location[him[tj]] \wedge !(q[tj] \neq$ $NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq$ $p[tj][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[tj]$ $\wedge location[mytid[tj]] == NULL) \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 95 | RMov          | coop::fAfallbackFQ | $\forall tj: \mathbb{N}. assume(!(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj]))$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 96 | RMov          | coop::fAfallbackFQ | $\forall tj: \mathbb{N}. assume(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj]);$ $q[tj] = location[him[tj]]; assume(!(q[tj] \neq NULL$ $\wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq$ $p[tj][1] \rightarrow val)) \triangleright \{ (location[mytid[tj]] == p[tj] \wedge$ $\exists old\_location\_mytid\_tj\_.. ((!(him[tj] \neq NULL \wedge$ $him[tj] \neq mytid[tj]) \vee (him[tj] \neq NULL \wedge him[tj] \neq$ $mytid[tj] \wedge q[tj] == location[him[tj]] \wedge !(q[tj] \neq$ $NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq$ $p[tj][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[tj]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| #  | Kind | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----|------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 97 | HTrp | coop::fAfallbackFQ | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj]) \vee$ $(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] ==$ $location[him[tj]] \wedge !(q[tj] \neq NULL \wedge q[tj][0] \rightarrow val$ $== him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid$ $== NULL) \} [ [ (assume(!(him[ti] \neq NULL \wedge him[ti] \neq$ $mytid[ti])) + [ [ (assume(him[ti] \neq NULL \wedge him[ti] \neq$ $mytid[ti]); q[ti] = location[him[ti]]; assume(!(q[ti]$ $\neq NULL \wedge q[ti][0] \rightarrow val == him[ti] \wedge q[ti][1] \rightarrow val$ $p[ti][1] \rightarrow val)); skip)]; assume(location[mytid[ti]]$ $\neq p[ti]); val[ti] = q[ti][2] \rightarrow val; location[mytid[ti]]$ $= NULL; return val[ti]] \{ \exists old\_location\_mytid\_ti\_ \dots$ $((\exists old\_location\_mytid\_tj\_ \dots ((!(him[tj] \neq NULL \wedge$ $him[tj] \neq mytid[tj]) \vee (him[tj] \neq NULL \wedge him[tj] \neq$ $mytid[tj] \wedge q[tj] == location[him[tj]] \wedge !(q[tj] \neq$ $NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq$ $p[tj][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[tj]$ $\wedge location[mytid[tj]] == NULL) \wedge !(him[ti] \neq NULL$ $him[ti] \neq mytid[ti])) \vee (\exists old\_location\_mytid\_tj\_ \dots$ $((!(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj]) \vee$ $(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] ==$ $location[him[tj]] \wedge !(q[tj] \neq NULL \wedge q[tj][0] \rightarrow val$ $== him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid$ $== NULL) \wedge him[ti] \neq NULL \wedge him[ti] \neq mytid[ti] \wedge$ $== location[him[ti]] \wedge !(q[ti] \neq NULL \wedge q[ti][0] \rightarrow$ $== him[ti] \wedge q[ti][1] \rightarrow val \neq p[ti][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_ti\_ \neq p[ti] \wedge val[ti] ==$ $q[ti][2] \rightarrow val \wedge location[mytid[ti]] == NULL) \}$ |

#### B.2.4 Automata (676).

coverage.

| Check         | Result |
|---------------|--------|
| push: COVERED | OK     |
| pop: COVERED  | OK     |

non-overlapping.

1 all distinct pairs are non-overlapping

non-self-overlapping.

1 no spec self-overlaps

bijection part I: well-definedness.

| Check                                                                              | Result |
|------------------------------------------------------------------------------------|--------|
| fAPopS: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK        | OK     |
| fAfallbackSQ: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK  | OK     |
| fAbadPartnerQ: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK | OK     |
| fAhappyFI: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK     | OK     |
| fCPushF: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK       | OK     |
| fAPopEmp: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK      | OK     |
| fAhappySP: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK     | OK     |
| fAhappyFH: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK     | OK     |
| fAfallbackFP: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK  | OK     |
| publoopC: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK      | OK     |
| publoopC_pop: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK  | OK     |
| fCPopF: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK        | OK     |
| fAfallbackSP: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK  | OK     |
| fAbadPartnerP: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK | OK     |
| publoop_pop: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK   | OK     |
| publoop: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK       | OK     |
| fAPushS: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK       | OK     |
| fAhappySQ: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK     | OK     |
| fAfallbackFQ: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK  | OK     |

*bijection part II: surjectivity.*

| Check                                                                             | Result |
|-----------------------------------------------------------------------------------|--------|
| fAPopS $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK        | OK     |
| fAPushS $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK       | OK     |
| fAbadPartnerP $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK | OK     |
| fAbadPartnerQ $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK | OK     |
| fAfallbackFP $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK  | OK     |
| fAfallbackFQ $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK  | OK     |
| fAfallbackSP $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK  | OK     |
| fAfallbackSQ $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK  | OK     |
| fAhappyFH $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK     | OK     |
| fAhappyFI $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK     | OK     |
| fAhappySP $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK     | OK     |
| fAhappySQ $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK     | OK     |
| fCPopF $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK        | OK     |
| fCPushF $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK       | OK     |
| publoop $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK       | OK     |
| publoopC $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK      | OK     |
| publoopC_pop $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK  | OK     |
| publoop_pop $\cap \text{at\_least\_one\_ais}(\text{fAPopEmp}) = \emptyset$ : OK   | OK     |
| fAPopEmp $\cap \text{at\_least\_one\_ais}(\text{fAPopS}) = \emptyset$ : OK        | OK     |
| fAPushS $\cap \text{at\_least\_one\_ais}(\text{fAPopS}) = \emptyset$ : OK         | OK     |
| fAbadPartnerP $\cap \text{at\_least\_one\_ais}(\text{fAPopS}) = \emptyset$ : OK   | OK     |
| fAbadPartnerQ $\cap \text{at\_least\_one\_ais}(\text{fAPopS}) = \emptyset$ : OK   | OK     |

| Check                                                                                               | Result |
|-----------------------------------------------------------------------------------------------------|--------|
| $fA_{\text{fallbackFP}} \cap \text{at\_least\_one\_ais}(fAPopS) = \emptyset$ : OK                   | OK     |
| $fA_{\text{fallbackFQ}} \cap \text{at\_least\_one\_ais}(fAPopS) = \emptyset$ : OK                   | OK     |
| $fA_{\text{fallbackSP}} \cap \text{at\_least\_one\_ais}(fAPopS) = \emptyset$ : OK                   | OK     |
| $fA_{\text{fallbackSQ}} \cap \text{at\_least\_one\_ais}(fAPopS) = \emptyset$ : OK                   | OK     |
| $fA_{\text{happyFH}} \cap \text{at\_least\_one\_ais}(fAPopS) = \emptyset$ : OK                      | OK     |
| $fA_{\text{happyFI}} \cap \text{at\_least\_one\_ais}(fAPopS) = \emptyset$ : OK                      | OK     |
| $fA_{\text{happySP}} \cap \text{at\_least\_one\_ais}(fAPopS) = \emptyset$ : OK                      | OK     |
| $fA_{\text{happySQ}} \cap \text{at\_least\_one\_ais}(fAPopS) = \emptyset$ : OK                      | OK     |
| $fCPopF \subseteq \text{at\_most\_one\_ais}(fAPopS)$ : OK                                           | OK     |
| $fCPushF \subseteq \text{at\_most\_one\_ais}(fAPopS)$ : OK                                          | OK     |
| $\text{publoop} \cap \text{at\_least\_one\_ais}(fAPopS) = \emptyset$ : OK                           | OK     |
| $\text{publoopC} \cap \text{at\_least\_one\_ais}(fAPopS) = \emptyset$ : OK                          | OK     |
| $\text{publoopC\_pop} \cap \text{at\_least\_one\_ais}(fAPopS) = \emptyset$ : OK                     | OK     |
| $\text{publoop\_pop} \cap \text{at\_least\_one\_ais}(fAPopS) = \emptyset$ : OK                      | OK     |
| $fAPopEmp \cap \text{at\_least\_one\_ais}(fAPushS) = \emptyset$ : OK                                | OK     |
| $fAPopS \cap \text{at\_least\_one\_ais}(fAPushS) = \emptyset$ : OK                                  | OK     |
| $fA_{\text{badPartnerP}} \cap \text{at\_least\_one\_ais}(fAPushS) = \emptyset$ : OK                 | OK     |
| $fA_{\text{badPartnerQ}} \cap \text{at\_least\_one\_ais}(fAPushS) = \emptyset$ : OK                 | OK     |
| $fA_{\text{fallbackFP}} \cap \text{at\_least\_one\_ais}(fAPushS) = \emptyset$ : OK                  | OK     |
| $fA_{\text{fallbackFQ}} \cap \text{at\_least\_one\_ais}(fAPushS) = \emptyset$ : OK                  | OK     |
| $fA_{\text{fallbackSP}} \cap \text{at\_least\_one\_ais}(fAPushS) = \emptyset$ : OK                  | OK     |
| $fA_{\text{fallbackSQ}} \cap \text{at\_least\_one\_ais}(fAPushS) = \emptyset$ : OK                  | OK     |
| $fA_{\text{happyFH}} \cap \text{at\_least\_one\_ais}(fAPushS) = \emptyset$ : OK                     | OK     |
| $fA_{\text{happyFI}} \cap \text{at\_least\_one\_ais}(fAPushS) = \emptyset$ : OK                     | OK     |
| $fA_{\text{happySP}} \cap \text{at\_least\_one\_ais}(fAPushS) = \emptyset$ : OK                     | OK     |
| $fA_{\text{happySQ}} \cap \text{at\_least\_one\_ais}(fAPushS) = \emptyset$ : OK                     | OK     |
| $fCPopF \subseteq \text{at\_most\_one\_ais}(fAPushS)$ : OK                                          | OK     |
| $fCPushF \subseteq \text{at\_most\_one\_ais}(fAPushS)$ : OK                                         | OK     |
| $\text{publoop} \cap \text{at\_least\_one\_ais}(fAPushS) = \emptyset$ : OK                          | OK     |
| $\text{publoopC} \cap \text{at\_least\_one\_ais}(fAPushS) = \emptyset$ : OK                         | OK     |
| $\text{publoopC\_pop} \cap \text{at\_least\_one\_ais}(fAPushS) = \emptyset$ : OK                    | OK     |
| $\text{publoop\_pop} \cap \text{at\_least\_one\_ais}(fAPushS) = \emptyset$ : OK                     | OK     |
| $fAPopEmp \cap \text{at\_least\_one\_ais}(fA_{\text{badPartnerP}}) = \emptyset$ : OK                | OK     |
| $fAPopS \cap \text{at\_least\_one\_ais}(fA_{\text{badPartnerP}}) = \emptyset$ : OK                  | OK     |
| $fAPushS \cap \text{at\_least\_one\_ais}(fA_{\text{badPartnerP}}) = \emptyset$ : OK                 | OK     |
| $fA_{\text{badPartnerQ}} \cap \text{at\_least\_one\_ais}(fA_{\text{badPartnerP}}) = \emptyset$ : OK | OK     |
| $fA_{\text{fallbackFP}} \subseteq \text{at\_most\_one\_ais}(fA_{\text{badPartnerP}})$ : OK          | OK     |
| $fA_{\text{fallbackFQ}} \cap \text{at\_least\_one\_ais}(fA_{\text{badPartnerP}}) = \emptyset$ : OK  | OK     |
| $fA_{\text{fallbackSP}} \subseteq \text{at\_most\_one\_ais}(fA_{\text{badPartnerP}})$ : OK          | OK     |
| $fA_{\text{fallbackSQ}} \cap \text{at\_least\_one\_ais}(fA_{\text{badPartnerP}}) = \emptyset$ : OK  | OK     |
| $fA_{\text{happyFH}} \cap \text{at\_least\_one\_ais}(fA_{\text{badPartnerP}}) = \emptyset$ : OK     | OK     |
| $fA_{\text{happyFI}} \cap \text{at\_least\_one\_ais}(fA_{\text{badPartnerP}}) = \emptyset$ : OK     | OK     |
| $fA_{\text{happySP}} \cap \text{at\_least\_one\_ais}(fA_{\text{badPartnerP}}) = \emptyset$ : OK     | OK     |
| $fA_{\text{happySQ}} \cap \text{at\_least\_one\_ais}(fA_{\text{badPartnerP}}) = \emptyset$ : OK     | OK     |
| $fCPopF \cap \text{at\_least\_one\_ais}(fA_{\text{badPartnerP}}) = \emptyset$ : OK                  | OK     |
| $fCPushF \cap \text{at\_least\_one\_ais}(fA_{\text{badPartnerP}}) = \emptyset$ : OK                 | OK     |



| Check                                                                                         | Result |
|-----------------------------------------------------------------------------------------------|--------|
| $\text{publoop} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerP}) = \emptyset$ : OK       | OK     |
| $\text{publoopC} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerP}) = \emptyset$ : OK      | OK     |
| $\text{publoopC\_pop} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerP}) = \emptyset$ : OK | OK     |
| $\text{publoop\_pop} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerP}) = \emptyset$ : OK  | OK     |
| $\text{fAPopEmp} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerQ}) = \emptyset$ : OK      | OK     |
| $\text{fAPoS} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerQ}) = \emptyset$ : OK         | OK     |
| $\text{fAPushS} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerQ}) = \emptyset$ : OK       | OK     |
| $\text{fAbadPartnerP} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerQ}) = \emptyset$ : OK | OK     |
| $\text{fAfallbackFP} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerQ}) = \emptyset$ : OK  | OK     |
| $\text{fAfallbackFQ} \subseteq \text{at\_most\_one\_ais}(\text{fAbadPartnerQ})$ : OK          | OK     |
| $\text{fAfallbackSP} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerQ}) = \emptyset$ : OK  | OK     |
| $\text{fAfallbackSQ} \subseteq \text{at\_most\_one\_ais}(\text{fAbadPartnerQ})$ : OK          | OK     |
| $\text{fAhappyFH} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerQ}) = \emptyset$ : OK     | OK     |
| $\text{fAhappyFI} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerQ}) = \emptyset$ : OK     | OK     |
| $\text{fAhappySP} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerQ}) = \emptyset$ : OK     | OK     |
| $\text{fAhappySQ} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerQ}) = \emptyset$ : OK     | OK     |
| $\text{fCPopF} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerQ}) = \emptyset$ : OK        | OK     |
| $\text{fCPushF} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerQ}) = \emptyset$ : OK       | OK     |
| $\text{publoop} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerQ}) = \emptyset$ : OK       | OK     |
| $\text{publoopC} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerQ}) = \emptyset$ : OK      | OK     |
| $\text{publoopC\_pop} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerQ}) = \emptyset$ : OK | OK     |
| $\text{publoop\_pop} \cap \text{at\_least\_one\_ais}(\text{fAbadPartnerQ}) = \emptyset$ : OK  | OK     |
| $\text{fAPopEmp} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFP}) = \emptyset$ : OK       | OK     |
| $\text{fAPoS} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFP}) = \emptyset$ : OK          | OK     |
| $\text{fAPushS} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFP}) = \emptyset$ : OK        | OK     |
| $\text{fAbadPartnerQ} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFP}) = \emptyset$ : OK  | OK     |
| $\text{fAfallbackFQ} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFP}) = \emptyset$ : OK   | OK     |
| $\text{fAfallbackSQ} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFP}) = \emptyset$ : OK   | OK     |
| $\text{fAhappyFH} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFP}) = \emptyset$ : OK      | OK     |
| $\text{fAhappyFI} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFP}) = \emptyset$ : OK      | OK     |
| $\text{fAhappySP} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFP}) = \emptyset$ : OK      | OK     |
| $\text{fAhappySQ} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFP}) = \emptyset$ : OK      | OK     |
| $\text{fCPopF} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFP}) = \emptyset$ : OK         | OK     |
| $\text{fCPushF} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFP}) = \emptyset$ : OK        | OK     |
| $\text{publoop} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFP}) = \emptyset$ : OK        | OK     |
| $\text{publoopC} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFP}) = \emptyset$ : OK       | OK     |
| $\text{publoopC\_pop} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFP}) = \emptyset$ : OK  | OK     |
| $\text{publoop\_pop} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFP}) = \emptyset$ : OK   | OK     |
| $\text{fAPopEmp} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFQ}) = \emptyset$ : OK       | OK     |
| $\text{fAPoS} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFQ}) = \emptyset$ : OK          | OK     |
| $\text{fAPushS} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFQ}) = \emptyset$ : OK        | OK     |
| $\text{fAbadPartnerP} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFQ}) = \emptyset$ : OK  | OK     |
| $\text{fAfallbackFP} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFQ}) = \emptyset$ : OK   | OK     |
| $\text{fAfallbackSP} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFQ}) = \emptyset$ : OK   | OK     |
| $\text{fAhappyFH} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFQ}) = \emptyset$ : OK      | OK     |
| $\text{fAhappyFI} \cap \text{at\_least\_one\_ais}(\text{fAfallbackFQ}) = \emptyset$ : OK      | OK     |

| Check                                                                         | Result |
|-------------------------------------------------------------------------------|--------|
| $fA_{happySP} \cap at\_least\_one\_ais(fA_{fallbackFQ}) = \emptyset$ : OK     | OK     |
| $fA_{happySQ} \cap at\_least\_one\_ais(fA_{fallbackFQ}) = \emptyset$ : OK     | OK     |
| $fCPopF \cap at\_least\_one\_ais(fA_{fallbackFQ}) = \emptyset$ : OK           | OK     |
| $fCPushF \cap at\_least\_one\_ais(fA_{fallbackFQ}) = \emptyset$ : OK          | OK     |
| $publoop \cap at\_least\_one\_ais(fA_{fallbackFQ}) = \emptyset$ : OK          | OK     |
| $publoopC \cap at\_least\_one\_ais(fA_{fallbackFQ}) = \emptyset$ : OK         | OK     |
| $publoopC\_pop \cap at\_least\_one\_ais(fA_{fallbackFQ}) = \emptyset$ : OK    | OK     |
| $publoop\_pop \cap at\_least\_one\_ais(fA_{fallbackFQ}) = \emptyset$ : OK     | OK     |
| $fAPopEmp \cap at\_least\_one\_ais(fA_{fallbackSP}) = \emptyset$ : OK         | OK     |
| $fAPoS \cap at\_least\_one\_ais(fA_{fallbackSP}) = \emptyset$ : OK            | OK     |
| $fAPushS \cap at\_least\_one\_ais(fA_{fallbackSP}) = \emptyset$ : OK          | OK     |
| $fA_{badPartnerQ} \cap at\_least\_one\_ais(fA_{fallbackSP}) = \emptyset$ : OK | OK     |
| $fA_{fallbackFP} \subseteq at\_most\_one\_ais(fA_{fallbackSP})$ : OK          | OK     |
| $fA_{fallbackFQ} \cap at\_least\_one\_ais(fA_{fallbackSP}) = \emptyset$ : OK  | OK     |
| $fA_{fallbackSQ} \cap at\_least\_one\_ais(fA_{fallbackSP}) = \emptyset$ : OK  | OK     |
| $fA_{happyFH} \cap at\_least\_one\_ais(fA_{fallbackSP}) = \emptyset$ : OK     | OK     |
| $fA_{happyFI} \cap at\_least\_one\_ais(fA_{fallbackSP}) = \emptyset$ : OK     | OK     |
| $fA_{happySP} \cap at\_least\_one\_ais(fA_{fallbackSP}) = \emptyset$ : OK     | OK     |
| $fA_{happySQ} \cap at\_least\_one\_ais(fA_{fallbackSP}) = \emptyset$ : OK     | OK     |
| $fCPopF \cap at\_least\_one\_ais(fA_{fallbackSP}) = \emptyset$ : OK           | OK     |
| $fCPushF \cap at\_least\_one\_ais(fA_{fallbackSP}) = \emptyset$ : OK          | OK     |
| $publoop \cap at\_least\_one\_ais(fA_{fallbackSP}) = \emptyset$ : OK          | OK     |
| $publoopC \cap at\_least\_one\_ais(fA_{fallbackSP}) = \emptyset$ : OK         | OK     |
| $publoopC\_pop \cap at\_least\_one\_ais(fA_{fallbackSP}) = \emptyset$ : OK    | OK     |
| $publoop\_pop \cap at\_least\_one\_ais(fA_{fallbackSP}) = \emptyset$ : OK     | OK     |
| $fAPopEmp \cap at\_least\_one\_ais(fA_{fallbackSQ}) = \emptyset$ : OK         | OK     |
| $fAPoS \cap at\_least\_one\_ais(fA_{fallbackSQ}) = \emptyset$ : OK            | OK     |
| $fAPushS \cap at\_least\_one\_ais(fA_{fallbackSQ}) = \emptyset$ : OK          | OK     |
| $fA_{badPartnerP} \cap at\_least\_one\_ais(fA_{fallbackSQ}) = \emptyset$ : OK | OK     |
| $fA_{fallbackFP} \cap at\_least\_one\_ais(fA_{fallbackSQ}) = \emptyset$ : OK  | OK     |
| $fA_{fallbackFQ} \subseteq at\_most\_one\_ais(fA_{fallbackSQ})$ : OK          | OK     |
| $fA_{fallbackSP} \cap at\_least\_one\_ais(fA_{fallbackSQ}) = \emptyset$ : OK  | OK     |
| $fA_{happyFH} \cap at\_least\_one\_ais(fA_{fallbackSQ}) = \emptyset$ : OK     | OK     |
| $fA_{happyFI} \cap at\_least\_one\_ais(fA_{fallbackSQ}) = \emptyset$ : OK     | OK     |
| $fA_{happySP} \cap at\_least\_one\_ais(fA_{fallbackSQ}) = \emptyset$ : OK     | OK     |
| $fA_{happySQ} \cap at\_least\_one\_ais(fA_{fallbackSQ}) = \emptyset$ : OK     | OK     |
| $fCPopF \cap at\_least\_one\_ais(fA_{fallbackSQ}) = \emptyset$ : OK           | OK     |
| $fCPushF \cap at\_least\_one\_ais(fA_{fallbackSQ}) = \emptyset$ : OK          | OK     |
| $publoop \cap at\_least\_one\_ais(fA_{fallbackSQ}) = \emptyset$ : OK          | OK     |
| $publoopC \cap at\_least\_one\_ais(fA_{fallbackSQ}) = \emptyset$ : OK         | OK     |
| $publoopC\_pop \cap at\_least\_one\_ais(fA_{fallbackSQ}) = \emptyset$ : OK    | OK     |
| $publoop\_pop \cap at\_least\_one\_ais(fA_{fallbackSQ}) = \emptyset$ : OK     | OK     |
| $fAPopEmp \cap at\_least\_one\_ais(fA_{happyFH}) = \emptyset$ : OK            | OK     |
| $fAPoS \cap at\_least\_one\_ais(fA_{happyFH}) = \emptyset$ : OK               | OK     |
| $fAPushS \cap at\_least\_one\_ais(fA_{happyFH}) = \emptyset$ : OK             | OK     |
| $fA_{badPartnerP} \cap at\_least\_one\_ais(fA_{happyFH}) = \emptyset$ : OK    | OK     |

| Check                                                                                           | Result |
|-------------------------------------------------------------------------------------------------|--------|
| $fA_{\text{badPartnerQ}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFH}}) = \emptyset$ : OK | OK     |
| $fA_{\text{fallbackFP}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFH}}) = \emptyset$ : OK  | OK     |
| $fA_{\text{fallbackFQ}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFH}}) = \emptyset$ : OK  | OK     |
| $fA_{\text{fallbackSP}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFH}}) = \emptyset$ : OK  | OK     |
| $fA_{\text{fallbackSQ}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFH}}) = \emptyset$ : OK  | OK     |
| $fA_{\text{happyFI}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFH}}) = \emptyset$ : OK     | OK     |
| $fCP_{\text{popF}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFH}}) = \emptyset$ : OK       | OK     |
| $fCP_{\text{pushF}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFH}}) = \emptyset$ : OK      | OK     |
| $\text{publoop} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFH}}) = \emptyset$ : OK          | OK     |
| $\text{publoopC} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFH}}) = \emptyset$ : OK         | OK     |
| $\text{publoopC\_pop} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFH}}) = \emptyset$ : OK    | OK     |
| $\text{publoop\_pop} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFH}}) = \emptyset$ : OK     | OK     |
| $fAP_{\text{popEmp}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFI}}) = \emptyset$ : OK     | OK     |
| $fAP_{\text{popS}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFI}}) = \emptyset$ : OK       | OK     |
| $fAP_{\text{pushS}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFI}}) = \emptyset$ : OK      | OK     |
| $fA_{\text{badPartnerP}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFI}}) = \emptyset$ : OK | OK     |
| $fA_{\text{badPartnerQ}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFI}}) = \emptyset$ : OK | OK     |
| $fA_{\text{fallbackFP}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFI}}) = \emptyset$ : OK  | OK     |
| $fA_{\text{fallbackFQ}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFI}}) = \emptyset$ : OK  | OK     |
| $fA_{\text{fallbackSP}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFI}}) = \emptyset$ : OK  | OK     |
| $fA_{\text{fallbackSQ}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFI}}) = \emptyset$ : OK  | OK     |
| $fA_{\text{happyFH}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFI}}) = \emptyset$ : OK     | OK     |
| $fCP_{\text{popF}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFI}}) = \emptyset$ : OK       | OK     |
| $fCP_{\text{pushF}} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFI}}) = \emptyset$ : OK      | OK     |
| $\text{publoop} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFI}}) = \emptyset$ : OK          | OK     |
| $\text{publoopC} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFI}}) = \emptyset$ : OK         | OK     |
| $\text{publoopC\_pop} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFI}}) = \emptyset$ : OK    | OK     |
| $\text{publoop\_pop} \cap \text{at\_least\_one\_ais}(fA_{\text{happyFI}}) = \emptyset$ : OK     | OK     |
| $fAP_{\text{popEmp}} \cap \text{at\_least\_one\_ais}(fA_{\text{happySP}}) = \emptyset$ : OK     | OK     |
| $fAP_{\text{popS}} \cap \text{at\_least\_one\_ais}(fA_{\text{happySP}}) = \emptyset$ : OK       | OK     |
| $fAP_{\text{pushS}} \cap \text{at\_least\_one\_ais}(fA_{\text{happySP}}) = \emptyset$ : OK      | OK     |
| $fA_{\text{badPartnerP}} \cap \text{at\_least\_one\_ais}(fA_{\text{happySP}}) = \emptyset$ : OK | OK     |
| $fA_{\text{badPartnerQ}} \cap \text{at\_least\_one\_ais}(fA_{\text{happySP}}) = \emptyset$ : OK | OK     |
| $fA_{\text{fallbackFP}} \cap \text{at\_least\_one\_ais}(fA_{\text{happySP}}) = \emptyset$ : OK  | OK     |
| $fA_{\text{fallbackFQ}} \cap \text{at\_least\_one\_ais}(fA_{\text{happySP}}) = \emptyset$ : OK  | OK     |
| $fA_{\text{fallbackSP}} \cap \text{at\_least\_one\_ais}(fA_{\text{happySP}}) = \emptyset$ : OK  | OK     |
| $fA_{\text{fallbackSQ}} \cap \text{at\_least\_one\_ais}(fA_{\text{happySP}}) = \emptyset$ : OK  | OK     |
| $fA_{\text{happyFH}} \subseteq \text{at\_most\_one\_ais}(fA_{\text{happySP}})$ : OK             | OK     |
| $fA_{\text{happyFI}} \subseteq \text{at\_most\_one\_ais}(fA_{\text{happySP}})$ : OK             | OK     |
| $fA_{\text{happySQ}} \cap \text{at\_least\_one\_ais}(fA_{\text{happySP}}) = \emptyset$ : OK     | OK     |
| $fCP_{\text{popF}} \cap \text{at\_least\_one\_ais}(fA_{\text{happySP}}) = \emptyset$ : OK       | OK     |
| $fCP_{\text{pushF}} \cap \text{at\_least\_one\_ais}(fA_{\text{happySP}}) = \emptyset$ : OK      | OK     |
| $\text{publoop} \cap \text{at\_least\_one\_ais}(fA_{\text{happySP}}) = \emptyset$ : OK          | OK     |
| $\text{publoopC} \cap \text{at\_least\_one\_ais}(fA_{\text{happySP}}) = \emptyset$ : OK         | OK     |
| $\text{publoopC\_pop} \cap \text{at\_least\_one\_ais}(fA_{\text{happySP}}) = \emptyset$ : OK    | OK     |
| $\text{publoop\_pop} \cap \text{at\_least\_one\_ais}(fA_{\text{happySP}}) = \emptyset$ : OK     | OK     |

| Check                                                                | Result |
|----------------------------------------------------------------------|--------|
| $fAPopEmp \cap at\_least\_one\_ais(fAHappySQ) = \emptyset$ : OK      | OK     |
| $fAPopS \cap at\_least\_one\_ais(fAHappySQ) = \emptyset$ : OK        | OK     |
| $fAPushS \cap at\_least\_one\_ais(fAHappySQ) = \emptyset$ : OK       | OK     |
| $fAbadPartnerP \cap at\_least\_one\_ais(fAHappySQ) = \emptyset$ : OK | OK     |
| $fAbadPartnerQ \cap at\_least\_one\_ais(fAHappySQ) = \emptyset$ : OK | OK     |
| $fAfallbackFP \cap at\_least\_one\_ais(fAHappySQ) = \emptyset$ : OK  | OK     |
| $fAfallbackFQ \cap at\_least\_one\_ais(fAHappySQ) = \emptyset$ : OK  | OK     |
| $fAfallbackSP \cap at\_least\_one\_ais(fAHappySQ) = \emptyset$ : OK  | OK     |
| $fAfallbackSQ \cap at\_least\_one\_ais(fAHappySQ) = \emptyset$ : OK  | OK     |
| $fAHappyFH \subseteq at\_most\_one\_ais(fAHappySQ)$ : OK             | OK     |
| $fAHappyFI \subseteq at\_most\_one\_ais(fAHappySQ)$ : OK             | OK     |
| $fAHappySP \cap at\_least\_one\_ais(fAHappySQ) = \emptyset$ : OK     | OK     |
| $fCPopF \cap at\_least\_one\_ais(fAHappySQ) = \emptyset$ : OK        | OK     |
| $fCPushF \cap at\_least\_one\_ais(fAHappySQ) = \emptyset$ : OK       | OK     |
| $publoop \cap at\_least\_one\_ais(fAHappySQ) = \emptyset$ : OK       | OK     |
| $publoopC \cap at\_least\_one\_ais(fAHappySQ) = \emptyset$ : OK      | OK     |
| $publoopC\_pop \cap at\_least\_one\_ais(fAHappySQ) = \emptyset$ : OK | OK     |
| $publoop\_pop \cap at\_least\_one\_ais(fAHappySQ) = \emptyset$ : OK  | OK     |
| $fAPopEmp \cap at\_least\_one\_ais(fCPopF) = \emptyset$ : OK         | OK     |
| $fAbadPartnerP \cap at\_least\_one\_ais(fCPopF) = \emptyset$ : OK    | OK     |
| $fAbadPartnerQ \cap at\_least\_one\_ais(fCPopF) = \emptyset$ : OK    | OK     |
| $fAfallbackFP \cap at\_least\_one\_ais(fCPopF) = \emptyset$ : OK     | OK     |
| $fAfallbackFQ \cap at\_least\_one\_ais(fCPopF) = \emptyset$ : OK     | OK     |
| $fAfallbackSP \cap at\_least\_one\_ais(fCPopF) = \emptyset$ : OK     | OK     |
| $fAfallbackSQ \cap at\_least\_one\_ais(fCPopF) = \emptyset$ : OK     | OK     |
| $fAHappyFH \cap at\_least\_one\_ais(fCPopF) = \emptyset$ : OK        | OK     |
| $fAHappyFI \cap at\_least\_one\_ais(fCPopF) = \emptyset$ : OK        | OK     |
| $fAHappySP \cap at\_least\_one\_ais(fCPopF) = \emptyset$ : OK        | OK     |
| $fAHappySQ \cap at\_least\_one\_ais(fCPopF) = \emptyset$ : OK        | OK     |
| $fCPushF \cap at\_least\_one\_ais(fCPopF) = \emptyset$ : OK          | OK     |
| $publoop \cap at\_least\_one\_ais(fCPopF) = \emptyset$ : OK          | OK     |
| $publoopC \cap at\_least\_one\_ais(fCPopF) = \emptyset$ : OK         | OK     |
| $publoopC\_pop \cap at\_least\_one\_ais(fCPopF) = \emptyset$ : OK    | OK     |
| $publoop\_pop \cap at\_least\_one\_ais(fCPopF) = \emptyset$ : OK     | OK     |
| $fAPopEmp \cap at\_least\_one\_ais(fCPushF) = \emptyset$ : OK        | OK     |
| $fAbadPartnerP \cap at\_least\_one\_ais(fCPushF) = \emptyset$ : OK   | OK     |
| $fAbadPartnerQ \cap at\_least\_one\_ais(fCPushF) = \emptyset$ : OK   | OK     |
| $fAfallbackFP \cap at\_least\_one\_ais(fCPushF) = \emptyset$ : OK    | OK     |
| $fAfallbackFQ \cap at\_least\_one\_ais(fCPushF) = \emptyset$ : OK    | OK     |
| $fAfallbackSP \cap at\_least\_one\_ais(fCPushF) = \emptyset$ : OK    | OK     |
| $fAfallbackSQ \cap at\_least\_one\_ais(fCPushF) = \emptyset$ : OK    | OK     |
| $fAHappyFH \cap at\_least\_one\_ais(fCPushF) = \emptyset$ : OK       | OK     |
| $fAHappyFI \cap at\_least\_one\_ais(fCPushF) = \emptyset$ : OK       | OK     |
| $fAHappySP \cap at\_least\_one\_ais(fCPushF) = \emptyset$ : OK       | OK     |
| $fAHappySQ \cap at\_least\_one\_ais(fCPushF) = \emptyset$ : OK       | OK     |
| $fCPopF \cap at\_least\_one\_ais(fCPushF) = \emptyset$ : OK          | OK     |

| Check                                                                                         | Result |
|-----------------------------------------------------------------------------------------------|--------|
| $\text{publoop} \cap \text{at\_least\_one\_ais}(\text{fCPushF}) = \emptyset$ : OK             | OK     |
| $\text{publoopC} \cap \text{at\_least\_one\_ais}(\text{fCPushF}) = \emptyset$ : OK            | OK     |
| $\text{publoopC\_pop} \cap \text{at\_least\_one\_ais}(\text{fCPushF}) = \emptyset$ : OK       | OK     |
| $\text{publoop\_pop} \cap \text{at\_least\_one\_ais}(\text{fCPushF}) = \emptyset$ : OK        | OK     |
| $\text{fAPopEmp} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK            | OK     |
| $\text{fAPopS} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK              | OK     |
| $\text{fAPushS} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK             | OK     |
| $\text{fAbadPartnerP} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK       | OK     |
| $\text{fAbadPartnerQ} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK       | OK     |
| $\text{fAfallbackFP} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK        | OK     |
| $\text{fAfallbackFQ} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK        | OK     |
| $\text{fAfallbackSP} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK        | OK     |
| $\text{fAfallbackSQ} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK        | OK     |
| $\text{fAhappyFH} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK           | OK     |
| $\text{fAhappyFI} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK           | OK     |
| $\text{fAhappySP} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK           | OK     |
| $\text{fAhappySQ} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK           | OK     |
| $\text{fCPopF} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK              | OK     |
| $\text{fCPushF} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK             | OK     |
| $\text{publoopC} \subseteq \text{at\_most\_one\_ais}(\text{publoop})$ : OK                    | OK     |
| $\text{publoopC\_pop} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK       | OK     |
| $\text{publoop\_pop} \cap \text{at\_least\_one\_ais}(\text{publoop}) = \emptyset$ : OK        | OK     |
| $\text{fAPopEmp} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK           | OK     |
| $\text{fAPopS} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK             | OK     |
| $\text{fAPushS} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK            | OK     |
| $\text{fAbadPartnerP} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK      | OK     |
| $\text{fAbadPartnerQ} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK      | OK     |
| $\text{fAfallbackFP} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK       | OK     |
| $\text{fAfallbackFQ} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK       | OK     |
| $\text{fAfallbackSP} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK       | OK     |
| $\text{fAfallbackSQ} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK       | OK     |
| $\text{fAhappyFH} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK          | OK     |
| $\text{fAhappyFI} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK          | OK     |
| $\text{fAhappySP} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK          | OK     |
| $\text{fAhappySQ} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK          | OK     |
| $\text{fCPopF} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK             | OK     |
| $\text{fCPushF} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK            | OK     |
| $\text{publoopC\_pop} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK      | OK     |
| $\text{publoop\_pop} \cap \text{at\_least\_one\_ais}(\text{publoopC}) = \emptyset$ : OK       | OK     |
| $\text{fAPopEmp} \cap \text{at\_least\_one\_ais}(\text{publoopC\_pop}) = \emptyset$ : OK      | OK     |
| $\text{fAPopS} \cap \text{at\_least\_one\_ais}(\text{publoopC\_pop}) = \emptyset$ : OK        | OK     |
| $\text{fAPushS} \cap \text{at\_least\_one\_ais}(\text{publoopC\_pop}) = \emptyset$ : OK       | OK     |
| $\text{fAbadPartnerP} \cap \text{at\_least\_one\_ais}(\text{publoopC\_pop}) = \emptyset$ : OK | OK     |
| $\text{fAbadPartnerQ} \cap \text{at\_least\_one\_ais}(\text{publoopC\_pop}) = \emptyset$ : OK | OK     |
| $\text{fAfallbackFP} \cap \text{at\_least\_one\_ais}(\text{publoopC\_pop}) = \emptyset$ : OK  | OK     |
| $\text{fAfallbackFQ} \cap \text{at\_least\_one\_ais}(\text{publoopC\_pop}) = \emptyset$ : OK  | OK     |

| Check                                                                   | Result |
|-------------------------------------------------------------------------|--------|
| $fAfallbackSP \cap at\_least\_one\_ais(publoopC\_pop) = \emptyset$ : OK | OK     |
| $fAfallbackSQ \cap at\_least\_one\_ais(publoopC\_pop) = \emptyset$ : OK | OK     |
| $fAhappyFH \cap at\_least\_one\_ais(publoopC\_pop) = \emptyset$ : OK    | OK     |
| $fAhappyFI \cap at\_least\_one\_ais(publoopC\_pop) = \emptyset$ : OK    | OK     |
| $fAhappySP \cap at\_least\_one\_ais(publoopC\_pop) = \emptyset$ : OK    | OK     |
| $fAhappySQ \cap at\_least\_one\_ais(publoopC\_pop) = \emptyset$ : OK    | OK     |
| $fCPopF \cap at\_least\_one\_ais(publoopC\_pop) = \emptyset$ : OK       | OK     |
| $fCPushF \cap at\_least\_one\_ais(publoopC\_pop) = \emptyset$ : OK      | OK     |
| $publoop \cap at\_least\_one\_ais(publoopC\_pop) = \emptyset$ : OK      | OK     |
| $publoopC \cap at\_least\_one\_ais(publoopC\_pop) = \emptyset$ : OK     | OK     |
| $fAPopEmp \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK      | OK     |
| $fAPoS \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK         | OK     |
| $fAPushS \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK       | OK     |
| $fAbadPartnerP \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK | OK     |
| $fAbadPartnerQ \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK | OK     |
| $fAfallbackFP \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK  | OK     |
| $fAfallbackFQ \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK  | OK     |
| $fAfallbackSP \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK  | OK     |
| $fAfallbackSQ \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK  | OK     |
| $fAhappyFH \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK     | OK     |
| $fAhappyFI \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK     | OK     |
| $fAhappySP \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK     | OK     |
| $fAhappySQ \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK     | OK     |
| $fCPopF \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK        | OK     |
| $fCPushF \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK       | OK     |
| $publoop \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK       | OK     |
| $publoopC \cap at\_least\_one\_ais(publoop\_pop) = \emptyset$ : OK      | OK     |
| $publoopC\_pop \subseteq at\_most\_one\_ais(publoop\_pop)$ : OK         | OK     |

*bijjective support (IRSspecs).*

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1fAPopS: {poa4, poa5, poa7, pob2f, pocas6t}
2fAfallbackSQ: {poa24, poa25, poa26, pocas23f}
3fAbadPartnerQ: {pob15f}
4fAPopEmp: {poa3, pob2t}
5fAhappySP: {pua16, pub13t, pucas14t, pucas15t}
6fAfallbackSP: {pua21, pua22, pucas20f}
7fAbadPartnerP: {pub13f}
8publoop_pop: {poa10, poa11, poa8, poa9, pocas12t}
9publoop: {pua6, pua7, pua8, pua9, pucas10t}
10fAPushS: {pua1, pua2, pua3, pua5, pucas4t}
11fAhappySQ: {poa18, pob15t, pocas16t, pocas17t}
12fCPushF: {poa4, poa5, poa7, pob2f, pocas6t, pua1, pua2, pua3, pua5, pucas4t}
13fCPopF: {poa4, poa5, poa7, pob2f, pocas6t, pua1, pua2, pua3, pua5, pucas4t}
14publoopC: {pua6, pua7, pua8, pua9, pucas10t}
15publoopC_pop: {poa10, poa11, poa8, poa9, pocas12t}
16fAhappyFI: {poa18, pob15t, pocas16t, pocas17t, pua16, pub13t, pucas14t, pucas15t}
17fAhappyFH: {poa18, pob15t, pocas16t, pocas17t, pua16, pub13t, pucas14t, pucas15t}

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18fAfallbackFP: {pua21, pua22, pucas20f}  
 19fAfallbackFQ: {poa24, poa25, poa26, pocas23f}

### B.2.5 Interference Stability (451).

| #  | Kind  | Spec          | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                  | Re |
|----|-------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1  | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{top}[tk] = \text{stacktop} \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                        | OK |
| 2  | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{pos}[tk] = \text{nondet}() \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                        | OK |
| 3  | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{him}[tk] = \text{collision}[\text{pos}[tk]] \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$       | OK |
| 4  | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{q}[tk] = \text{location}[\text{him}[tk]] \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$          | OK |
| 5  | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{return} \text{q}[tk][2] \rightarrow \text{val} \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$    | OK |
| 6  | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{continue} \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                         | OK |
| 7  | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{val}[tk] = \text{q}[tk][2] \rightarrow \text{val} \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ | OK |
| 8  | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{location}[\text{mytid}[tk]] = \text{NULL} \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$         | OK |
| 9  | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{return} \text{val}[tk] \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                            | OK |
| 10 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{val}[tk] = \text{q}[tk][2] \rightarrow \text{val} \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ | OK |
| 11 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{location}[\text{mytid}[tk]] = \text{NULL} \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$         | OK |



| #  | Kind  | Spec          | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Re |
|----|-------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 12 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ return $\text{val}[tk] \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                                                              | OK |
| 13 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ return 0 $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                                                                           | OK |
| 14 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ int mytid $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                                                                          | OK |
| 15 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ location[mytid[tk]] = p[tk] $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                                                        | OK |
| 16 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ assume(! (him[tk] $\neq$ NULL $\wedge$ him[tk] $\neq$ mytid[tk])) $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                  | OK |
| 17 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ assume(him[tk] $\neq$ NULL $\wedge$ him[tk] $\neq$ mytid[tk]) $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                      | OK |
| 18 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ assume(! (q[tk] $\neq$ NULL $\wedge$ q[tk][0] $\rightarrow$ val == him[tk] $\wedge$ q[tk][1] $\rightarrow$ val $\neq$ p[tk][1] $\rightarrow$ val)) $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ | OK |
| 19 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ assume(q[tk] $\neq$ NULL $\wedge$ q[tk][0] $\rightarrow$ val == him[tk] $\wedge$ q[tk][1] $\rightarrow$ val $\neq$ p[tk][1] $\rightarrow$ val) $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$     | OK |
| 20 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ assume(top[tk] == NULL) $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                                                            | OK |
| 21 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ assume(collision[pos[tk]] $\neq$ him[tk]) $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                                          | OK |



| #  | Kind  | Spec          | Obligation                                                                                                                                                                                                                                                                                                                                                                                             | Re |
|----|-------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 22 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>CAS(collision[pos[tk]], him[tk], mytid[tk]) $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ | OK |
| 23 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>assume(location[mytid[tk]] $\neq$ p[tk]) $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$    | OK |
| 24 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>CAS(location[mytid[tk]], p[tk], NULL) $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$       | OK |
| 25 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>assume(location[him[tk]] $\neq$ q[tk]) $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$      | OK |
| 26 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>CAS(location[him[tk]], q[tk], NULL) $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$         | OK |
| 27 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>assume(location[mytid[tk]] $\neq$ p[tk]) $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$    | OK |
| 28 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>CAS(location[mytid[tk]], p[tk], NULL) $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$       | OK |
| 29 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>assume(stacktop $\neq$ top[tk]) $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$             | OK |
| 30 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ q[tk] = location[him[tk]] $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                      | OK |
| 31 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ return 1 $\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                       | OK |

| #  | Kind  | Spec          | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Re |
|----|-------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 32 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{continue}$<br>$\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                           | OK |
| 33 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>$\text{location}[\text{mytid}[tk]] = \text{NULL} \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                           | OK |
| 34 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{return } 1$<br>$\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                          | OK |
| 35 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>$\text{location}[\text{mytid}[tk]] = \text{NULL} \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                           | OK |
| 36 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{return } 1$<br>$\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                          | OK |
| 37 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{int mytid}$<br>$\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                          | OK |
| 38 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>$\text{location}[\text{mytid}[tk]] = \text{p}[tk] \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                          | OK |
| 39 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{pos}[tk]$<br>$= \text{nondet}() \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                          | OK |
| 40 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\} \text{him}[tk] =$<br>$\text{collision}[\text{pos}[tk]] \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                         | OK |
| 41 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>$\text{assume}(\text{!}(\text{him}[tk] \neq \text{NULL} \wedge \text{him}[tk] \neq \text{mytid}[tk]))$<br>$\{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ | OK |
| 42 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>$\text{assume}(\text{him}[tk] \neq \text{NULL} \wedge \text{him}[tk] \neq \text{mytid}[tk]) \{(\text{top}[t_j] == \text{stacktop} \wedge \text{val}[t_j] == \text{p}[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                | OK |

| #  | Kind  | Spec          | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Re |
|----|-------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 43 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>$\text{assume}(!(\text{q}[tk] \neq \text{NULL} \wedge \text{q}[tk][0] \rightarrow \text{val} == \text{him}[tk] \wedge \text{q}[tk][1] \rightarrow \text{val} \neq p[tk][1] \rightarrow \text{val})) \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$ | OK |
| 44 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>$\text{assume}(\text{q}[tk] \neq \text{NULL} \wedge \text{q}[tk][0] \rightarrow \text{val} == \text{him}[tk] \wedge \text{q}[tk][1] \rightarrow \text{val} \neq p[tk][1] \rightarrow \text{val}) \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$    | OK |
| 45 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>$\text{assume}(\text{collision}[\text{pos}[tk]] \neq \text{him}[tk]) \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                                                                | OK |
| 46 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>$\text{CAS}(\text{collision}[\text{pos}[tk]], \text{him}[tk], \text{mytid}[tk]) \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                                                     | OK |
| 47 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>$\text{assume}(\text{location}[\text{mytid}[tk]] \neq p[tk]) \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                                                                        | OK |
| 48 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>$\text{CAS}(\text{location}[\text{mytid}[tk]], p[tk], \text{NULL}) \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                                                                  | OK |
| 49 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>$\text{assume}(\text{location}[\text{him}[tk]] \neq \text{q}[tk]) \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                                                                   | OK |
| 50 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>$\text{CAS}(\text{location}[\text{him}[tk]], \text{q}[tk], p[tk]) \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                                                                   | OK |
| 51 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>$\text{assume}(\text{location}[\text{mytid}[tk]] \neq p[tk]) \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                                                                        | OK |
| 52 | IStbl | coop::fCPushF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$<br>$\text{CAS}(\text{location}[\text{mytid}[tk]], p[tk], \text{NULL}) \{(\text{stacktop} \wedge \text{val}[t_j] == p[t_j][2] \rightarrow \text{val} \wedge \text{ntop}[t_j] == \text{tloc\_pua3\_tj})\}$                                                                                                                                  | OK |



| #  | Kind  | Spec         | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Result |
|----|-------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 63 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$ $val[tk] = q[tk][2] \rightarrow val \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$                                                                                                  | OK     |
| 64 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$ $location[mytid[tk]] = NULL \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$                                                                                                          | OK     |
| 65 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$ $return val[tk] \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$                                                                                                                      | OK     |
| 66 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$ $return \emptyset \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$                                                                                                                    | OK     |
| 67 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$ $int mytid \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$                                                                                                                           | OK     |
| 68 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$ $location[mytid[tk]] = p[tk] \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$                                                                                                         | OK     |
| 69 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$ $assume(!(him[tk] \neq NULL \wedge him[tk] \neq mytid[tk])) \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$                                                                          | OK     |
| 70 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$ $assume(him[tk] \neq NULL \wedge him[tk] \neq mytid[tk]) \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$                                                                             | OK     |
| 71 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$ $assume(!(q[tk] \neq NULL \wedge q[tk][0] \rightarrow val == him[tk] \wedge q[tk][1] \rightarrow val \neq p[tk][1] \rightarrow val)) \{ (top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next) \}$ | OK     |

| #  | Kind  | Spec         | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Re |
|----|-------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 72 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} \text{ assume}(q[tk] \neq NULL \wedge q[tk][0] \rightarrow val == him[tk] \wedge q[tk][1] \rightarrow val \neq p[tk][1] \rightarrow val) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$ | OK |
| 73 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} \text{ assume}(top[tk] == NULL) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$                                                                                                          | OK |
| 74 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} \text{ assume}(collision[pos[tk]] \neq him[tk]) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$                                                                                          | OK |
| 75 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} \text{ CAS}(collision[pos[tk]], him[tk], mytid[tk]) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$                                                                                      | OK |
| 76 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} \text{ assume}(\text{location}[mytid[tk]] \neq p[tk]) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$                                                                                    | OK |
| 77 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} \text{ CAS}(\text{location}[mytid[tk]], p[tk], NULL) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$                                                                                     | OK |
| 78 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} \text{ assume}(\text{location}[him[tk]] \neq q[tk]) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$                                                                                      | OK |
| 79 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} \text{ CAS}(\text{location}[him[tk]], q[tk], NULL) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$                                                                                       | OK |
| 80 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} \text{ assume}(\text{location}[mytid[tk]] \neq p[tk]) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$                                                                                    | OK |

| #  | Kind  | Spec         | Obligation                                                                                                                                                                                                                                                                                                                                                                      | Result |
|----|-------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 81 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} CAS(location[mytid[tk]], p[tk], NULL) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$ | OK     |
| 82 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} assume(stacktop \neq top[tk]) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$         | OK     |
| 83 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} q[tk] = location[him[tk]] \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$             | OK     |
| 84 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} return 1 \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$                              | OK     |
| 85 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} continue \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$                              | OK     |
| 86 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} location[mytid[tk]] = NULL \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$            | OK     |
| 87 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} return 1 \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$                              | OK     |
| 88 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} location[mytid[tk]] = NULL \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$            | OK     |
| 89 | IStbl | coop::fCPopF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} return 1 \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$                              | OK     |





| #   | Kind  | Spec           | Obligation                                                                                                                                                                                                                                                                                                                                                                            | Re |
|-----|-------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 99  | IStbl | coop::fCPopF   | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} CAS(collision[pos[tk]], him[tk], mytid[tk]) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$ | OK |
| 100 | IStbl | coop::fCPopF   | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} assume(location[mytid[tk]] \neq p[tk]) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$      | OK |
| 101 | IStbl | coop::fCPopF   | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} CAS(location[mytid[tk]], p[tk], NULL) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$       | OK |
| 102 | IStbl | coop::fCPopF   | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} assume(location[him[tk]] \neq q[tk]) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$        | OK |
| 103 | IStbl | coop::fCPopF   | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} CAS(location[him[tk]], q[tk], p[tk]) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$        | OK |
| 104 | IStbl | coop::fCPopF   | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} assume(location[mytid[tk]] \neq p[tk]) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$      | OK |
| 105 | IStbl | coop::fCPopF   | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} CAS(location[mytid[tk]], p[tk], NULL) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$       | OK |
| 106 | IStbl | coop::fCPopF   | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\} assume(stacktop \neq top[tk]) \{(top[t_j] == stacktop \wedge top[t_j] \neq NULL \wedge val[t_j] == top[t_j] \rightarrow val \wedge ntop[t_j] == top[t_j] \rightarrow next)\}$               | OK |
| 107 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]]\} top[tk] = stacktop \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]]\}$                                                                                                                                                                  | OK |
| 108 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]]\} pos[tk] = nondet() \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]]\}$                                                                                                                                                                  | OK |

| #   | Kind  | Spec           | Obligation                                                                                                                                                                                                                                                                                                                       | Re |
|-----|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 109 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{him}[tk] = \text{collision}[\text{pos}[tk]] \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$      | OK |
| 110 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} q[tk] = \text{location}[\text{him}[tk]] \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                | OK |
| 111 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{return } q[tk][2] \rightarrow \text{val} \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$         | OK |
| 112 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{continue} \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                        | OK |
| 113 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{val}[tk] = q[tk][2] \rightarrow \text{val} \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$       | OK |
| 114 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{location}[\text{mytid}[tk]] = \text{NULL} \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$        | OK |
| 115 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{return } \text{val}[tk] \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                          | OK |
| 116 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{val}[tk] = q[tk][2] \rightarrow \text{val} \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$       | OK |
| 117 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{location}[\text{mytid}[tk]] = \text{NULL} \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$        | OK |
| 118 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{return } \text{val}[tk] \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                          | OK |
| 119 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{return } 0 \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                       | OK |
| 120 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{val}[tk] = \text{top}[tk] \rightarrow \text{val} \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ | OK |

| #   | Kind  | Spec           | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Re |
|-----|-------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 121 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{ntop}[tk] = \text{top}[tk] \rightarrow \text{next} \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                                           | OK |
| 122 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{return val}[tk] \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                                                                              | OK |
| 123 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{int mytid} \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                                                                                   | OK |
| 124 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{location}[\text{mytid}[tk]] = p[tk] \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                                                          | OK |
| 125 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(!(\text{him}[tk] \neq \text{NULL} \wedge \text{him}[tk] \neq \text{mytid}[tk])) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                      | OK |
| 126 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{him}[tk] \neq \text{NULL} \wedge \text{him}[tk] \neq \text{mytid}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                         | OK |
| 127 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(!(\text{q}[tk] \neq \text{NULL} \wedge \text{q}[tk][0] \rightarrow \text{val} == \text{him}[tk] \wedge \text{q}[tk][1] \rightarrow \text{val} \neq p[tk][1] \rightarrow \text{val})) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ | OK |
| 128 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{q}[tk] \neq \text{NULL} \wedge \text{q}[tk][0] \rightarrow \text{val} == \text{him}[tk] \wedge \text{q}[tk][1] \rightarrow \text{val} \neq p[tk][1] \rightarrow \text{val})) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$   | OK |
| 129 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{top}[tk] \neq \text{NULL}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                                                     | OK |
| 130 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{top}[tk] == \text{NULL}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                                                       | OK |
| 131 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{collision}[\text{pos}[tk]] \neq \text{him}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                                | OK |

| #   | Kind  | Spec           | Obligation                                                                                                                                                                                                                                                                                                                                               | Re |
|-----|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 132 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{CAS}(\text{collision}[\text{pos}[tk]], \text{him}[tk], \text{mytid}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ | OK |
| 133 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{CAS}(\text{location}[\text{mytid}[tk]], p[tk], \text{NULL}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$              | OK |
| 134 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{CAS}(\text{location}[\text{mytid}[tk]], p[tk], \text{NULL}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$              | OK |
| 135 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{location}[\text{him}[tk]] \neq q[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                      | OK |
| 136 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{CAS}(\text{location}[\text{him}[tk]], q[tk], \text{NULL}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                | OK |
| 137 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{location}[\text{mytid}[tk]] \neq p[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                    | OK |
| 138 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{CAS}(\text{location}[\text{mytid}[tk]], p[tk], \text{NULL}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$              | OK |
| 139 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{stacktop} \neq \text{top}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                             | OK |
| 140 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{CAS}(\text{stacktop}, \text{top}[tk], \text{ntop}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                   | OK |
| 141 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{top}[tk] = \text{stacktop} \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                               | OK |
| 142 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} q[tk] = \text{location}[\text{him}[tk]] \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                        | OK |

| #   | Kind  | Spec           | Obligation                                                                                                                                                                                                                                                                                                                                                                              | Re |
|-----|-------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 143 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ return 1 $\{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                    | OK |
| 144 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ continue $\{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                    | OK |
| 145 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ location[mytid[tk]] = NULL $\{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                  | OK |
| 146 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ return 1 $\{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                    | OK |
| 147 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ val[tk] = p[tk][2]->val $\{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                     | OK |
| 148 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ location[mytid[tk]] = NULL $\{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                  | OK |
| 149 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ return 1 $\{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                    | OK |
| 150 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ ntop[tk] = tloc_pua3_tk $\{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                     | OK |
| 151 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ return 1 $\{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                    | OK |
| 152 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ assume(! (him[tk] $\neq$ NULL $\wedge$ him[tk] $\neq$ mytid[tk])) $\{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                           | OK |
| 153 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ assume(him[tk] $\neq$ NULL $\wedge$ him[tk] $\neq$ mytid[tk])) $\{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                              | OK |
| 154 | IStbl | coop::publoopC | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ assume(! (q[tk] $\neq$ NULL $\wedge$ q[tk][0]->val == him[tk] $\wedge$ q[tk][1]->val $\neq$ p[tk][1]->val)) $\{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                           | Re |
|-----|-------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 155 | IStbl | coop::publoopC     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(q[tk] \neq \text{NULL} \wedge q[tk][0] \rightarrow \text{val} == \text{him}[tk] \wedge q[tk][1] \rightarrow \text{val} \neq p[tk][1] \rightarrow \text{val}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ | OK |
| 156 | IStbl | coop::publoopC     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{collision}[\text{pos}[tk]] \neq \text{him}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                        | OK |
| 157 | IStbl | coop::publoopC     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{location}[\text{mytid}[tk]] \neq p[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                | OK |
| 158 | IStbl | coop::publoopC     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{CAS}(\text{location}[\text{mytid}[tk]], p[tk], \text{NULL}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                          | OK |
| 159 | IStbl | coop::publoopC     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{location}[\text{him}[tk]] \neq q[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                  | OK |
| 160 | IStbl | coop::publoopC     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{CAS}(\text{location}[\text{him}[tk]], q[tk], p[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                  | OK |
| 161 | IStbl | coop::publoopC     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{location}[\text{mytid}[tk]] \neq p[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                | OK |
| 162 | IStbl | coop::publoopC     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{CAS}(\text{location}[\text{mytid}[tk]], p[tk], \text{NULL}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                          | OK |
| 163 | IStbl | coop::publoopC     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{stacktop} \neq \text{top}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                         | OK |
| 164 | IStbl | coop::publoopC     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{CAS}(\text{stacktop}, \text{top}[tk], \text{ntop}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                               | OK |
| 165 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{top}[tk] = \text{stacktop} \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                                           | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                              | Re |
|-----|-------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 166 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} q[tk] = location[him[tk]] \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$           | OK |
| 167 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} return q[tk][2] \rightarrow val \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$     | OK |
| 168 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} continue \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$                            | OK |
| 169 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} val[tk] = q[tk][2] \rightarrow val \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$  | OK |
| 170 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} location[mytid[tk]] = NULL \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$          | OK |
| 171 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} return val[tk] \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$                      | OK |
| 172 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} val[tk] = q[tk][2] \rightarrow val \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$  | OK |
| 173 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} location[mytid[tk]] = NULL \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$          | OK |
| 174 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} return val[tk] \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$                      | OK |
| 175 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} return 0 \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$                            | OK |
| 176 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} val[tk] = top[tk] \rightarrow val \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$   | OK |
| 177 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} ntop[tk] = top[tk] \rightarrow next \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$ | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                            | Re |
|-----|-------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 178 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{return val}[tk]$<br>$\{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                                                  | OK |
| 179 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}!(\text{him}[tk] \neq \text{NULL} \wedge \text{him}[tk] \neq \text{mytid}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                 | OK |
| 180 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{him}[tk] \neq \text{NULL} \wedge \text{him}[tk] \neq \text{mytid}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                  | OK |
| 181 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}!(q[tk] \neq \text{NULL} \wedge q[tk][0] \rightarrow \text{val} == \text{him}[tk] \wedge q[tk][1] \rightarrow \text{val} \neq p[tk][1] \rightarrow \text{val}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ | OK |
| 182 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(q[tk] \neq \text{NULL} \wedge q[tk][0] \rightarrow \text{val} == \text{him}[tk] \wedge q[tk][1] \rightarrow \text{val} \neq p[tk][1] \rightarrow \text{val}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$  | OK |
| 183 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{top}[tk] \neq \text{NULL}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                              | OK |
| 184 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{top}[tk] == \text{NULL}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                                | OK |
| 185 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{collision}[\text{pos}[tk]] \neq \text{him}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                         | OK |
| 186 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{location}[\text{mytid}[tk]] \neq p[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                 | OK |
| 187 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{CAS}(\text{location}[\text{mytid}[tk]], p[tk], \text{NULL}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                           | OK |
| 188 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{location}[\text{him}[tk]] \neq q[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                   | OK |



| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                 | Re |
|-----|-------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 189 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} CAS(location[him[tk]], q[tk], NULL) \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$    | OK |
| 190 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} assume(location[mytid[tk]] \neq p[tk]) \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$ | OK |
| 191 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} CAS(location[mytid[tk]], p[tk], NULL) \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$  | OK |
| 192 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} assume(stacktop \neq top[tk]) \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$          | OK |
| 193 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} CAS(stacktop, top[tk], ntop[tk]) \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$       | OK |
| 194 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} top[tk] = stacktop \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$                     | OK |
| 195 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} q[tk] = location[him[tk]] \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$              | OK |
| 196 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} return 1 \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$                               | OK |
| 197 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} continue \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$                               | OK |
| 198 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} location[mytid[tk]] = NULL \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$             | OK |
| 199 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} return 1 \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$                               | OK |
| 200 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\} val[tk] = p[tk][2] \rightarrow val \{(location[mytid[t_j]] == p[t_j] \wedge him[t_j] == collision[pos[t_j]])\}$     | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                            | Re |
|-----|-------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 201 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{location}[\text{mytid}[tk]] = \text{NULL} \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                             | OK |
| 202 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{return } 1 \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                                                            | OK |
| 203 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{ntop}[tk] = \text{tloc\_pua3\_tk} \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                                     | OK |
| 204 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{return } 1 \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                                                            | OK |
| 205 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{int mytid} \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                                                            | OK |
| 206 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{location}[\text{mytid}[tk]] = p[tk] \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                                   | OK |
| 207 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{pos}[tk] = \text{nondet}() \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                                            | OK |
| 208 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{him}[tk] = \text{collision}[\text{pos}[tk]] \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                           | OK |
| 209 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}!(\text{him}[tk] \neq \text{NULL} \wedge \text{him}[tk] \neq \text{mytid}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                 | OK |
| 210 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{him}[tk] \neq \text{NULL} \wedge \text{him}[tk] \neq \text{mytid}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                  | OK |
| 211 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}!(q[tk] \neq \text{NULL} \wedge q[tk][0] \rightarrow \text{val} == \text{him}[tk] \wedge q[tk][1] \rightarrow \text{val} \neq p[tk][1] \rightarrow \text{val}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                           | Re |
|-----|-------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 212 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(q[tk] \neq \text{NULL} \wedge q[tk][0] \rightarrow \text{val} == \text{him}[tk] \wedge q[tk][1] \rightarrow \text{val} \neq p[tk][1] \rightarrow \text{val}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$ | OK |
| 213 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{collision}[\text{pos}[tk]] \neq \text{him}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                        | OK |
| 214 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{CAS}(\text{collision}[\text{pos}[tk]], \text{him}[tk], \text{mytid}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                             | OK |
| 215 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{location}[\text{mytid}[tk]] \neq p[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                | OK |
| 216 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{CAS}(\text{location}[\text{mytid}[tk]], p[tk], \text{NULL}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                          | OK |
| 217 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{location}[\text{him}[tk]] \neq q[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                  | OK |
| 218 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{CAS}(\text{location}[\text{him}[tk]], q[tk], p[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                  | OK |
| 219 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{location}[\text{mytid}[tk]] \neq p[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                | OK |
| 220 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{CAS}(\text{location}[\text{mytid}[tk]], p[tk], \text{NULL}) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                          | OK |
| 221 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{assume}(\text{stacktop} \neq \text{top}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                                         | OK |
| 222 | IStbl | coop::publoopC_pop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\} \text{CAS}(\text{stacktop}, \text{top}[tk], \text{ntop}[tk]) \{(\text{location}[\text{mytid}[t_j]] == p[t_j] \wedge \text{him}[t_j] == \text{collision}[\text{pos}[t_j]])\}$                                                                                                               | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Re |
|-----|-------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 223 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \{ (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \vee (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \} \} top[tk] = stacktop \{ \{ (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \vee (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \} \}$ | OK |
| 224 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \{ (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \vee (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \} \} pos[tk] = nondet() \{ \{ (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \vee (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \} \}$ | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Re |
|-----|-------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 225 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \{ (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \} \} \text{him}[tk] = \text{collision}[\text{pos}[tk]] \{ \{ (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \} \} \}$ | OK |
| 226 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \{ (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \} \} q[tk] = \text{location}[\text{him}[tk]] \{ \{ (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \} \} \}$           | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Re |
|-----|-------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 227 | ISTbl | coop::fAHappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ $\text{continue } \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$                                 | OK |
| 228 | ISTbl | coop::fAHappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ $\text{val}[tk] = q[tk][2] \rightarrow \text{val} \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Re |
|-----|-------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 229 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{location}[\text{mytid}[tk]] = \text{NULL} \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ | OK |
| 230 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{return val}[tk] \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$                           | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Re |
|-----|-------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 231 | ISTbl | coop::fAhappyFI | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \} val[tk] = q[tk][2] \rightarrow val \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$ | OK |
| 232 | ISTbl | coop::fAhappyFI | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \} location[mytid[tk]] = NULL \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$         | OK |



| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Re |
|-----|-------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 233 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{return val}[tk]$ $\{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$    | OK |
| 234 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{return } \emptyset$ $\{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Re |
|-----|-------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 235 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \vee (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j])) \} val[tk] = top[tk] \rightarrow val \{ ((\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \vee (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j])) \}$   | OK |
| 236 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \vee (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j])) \} ntop[tk] = top[tk] \rightarrow next \{ ((\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \vee (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j])) \}$ | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Re |
|-----|-------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 237 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{return val}[tk]$ $\{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ | OK |
| 238 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ $\text{int mytid} \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$      | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Re |
|-----|-------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 239 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{location}[\text{mytid}[tk]] = p[tk] \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \}$                                                  | OK |
| 240 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{assume}(!(\text{him}[tk] \neq \text{NULL} \wedge \text{him}[tk] \neq \text{mytid}[tk])) \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Re |
|-----|-------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 241 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \vee (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j])) \}$ $assume(him[tk] \neq NULL \wedge him[tk] \neq mytid[tk])$ $\{ ((\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \vee (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j])) \}$                                                                             | OK |
| 242 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \vee (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j])) \}$ $assume(!(q[tk] \neq NULL \wedge q[tk][0] \rightarrow val == him[tk] \wedge q[tk][1] \rightarrow val \neq p[tk][1] \rightarrow val))$ $\{ ((\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \vee (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j])) \}$ | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Re |
|-----|-------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 243 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{assume}(\text{top}[tk] \neq \text{NULL}) \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ | OK |
| 244 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{assume}(\text{top}[tk] == \text{NULL}) \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$   | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Re |
|-----|-------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 245 | IStbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \{ (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \} \} \text{assume}(\text{collision}[\text{pos}[tk]] \neq \text{him}[tk]) \{ \{ (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \} \} \}$            | OK |
| 246 | IStbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \{ (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \} \} \text{CAS}(\text{collision}[\text{pos}[tk]], \text{him}[tk], \text{mytid}[tk]) \{ \{ (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \} \} \}$ | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Re |
|-----|-------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 247 | ISTbl | coop::fAhappyFI | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee$ $(\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$ $assume(location[mytid[tk]] \neq p[tk])$ $\{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee$ $(\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$ | OK |
| 248 | ISTbl | coop::fAhappyFI | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee$ $(\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$ $assume(location[him[tk]] \neq q[tk])$ $\{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee$ $(\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$   | OK |



| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Re |
|-----|-------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 249 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \{ (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \} \} \text{assume}(\text{location}[\text{mytid}[tk]] \neq p[tk]) \{ \{ (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \} \} \}$       | OK |
| 250 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \{ (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \} \} \text{CAS}(\text{location}[\text{mytid}[tk]], p[tk], \text{NULL}) \{ \{ (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \} \} \}$ | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Re |
|-----|-------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 251 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{assume}(\text{stacktop} \neq \text{top}[tk]) \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$           | OK |
| 252 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{CAS}(\text{stacktop}, \text{top}[tk], \text{ntop}[tk]) \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Re |
|-----|-------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 253 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{top}[tk] = \text{stacktop} \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$        | OK |
| 254 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} q[tk] = \text{location}[\text{him}[tk]] \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Re |
|-----|-------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 255 | ISTbl | coop::fAhappyFI | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$ $continue \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$                   | OK |
| 256 | ISTbl | coop::fAhappyFI | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$ $location[mytid[tk]] = NULL \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$ | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Re |
|-----|-------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 257 | IStbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ $\text{return } 1 \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$                                 | OK |
| 258 | IStbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ $\text{val}[tk] = p[tk][2] \rightarrow \text{val} \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Re |
|-----|-------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 259 | ISTbl | coop::fAhappyFI | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \} location[mytid[tk]] = NULL \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$ | OK |
| 260 | ISTbl | coop::fAhappyFI | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \} return 1 \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$                   | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Re |
|-----|-------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 261 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{ntop}[tk] = \text{tloc\_pua3\_tk} \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ | OK |
| 262 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{return } 1 \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$                        | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Re |
|-----|-------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 263 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \int \text{mytid} \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$                         | OK |
| 264 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{location}[\text{mytid}[tk]] = p[tk] \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ | OK |



| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Re |
|-----|-------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 265 | IStbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{pos}[tk] = \text{nondet}()$ $\{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$                  | OK |
| 266 | IStbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{him}[tk] = \text{collision}[\text{pos}[tk]]$ $\{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Re |
|-----|-------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 267 | ISTbl | coop::fAhappyFI | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$ $assume(!(him[tk] \neq NULL \wedge him[tk] \neq mytid[tk]))$ $\{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$ | OK |
| 268 | ISTbl | coop::fAhappyFI | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$ $assume(him[tk] \neq NULL \wedge him[tk] \neq mytid[tk])$ $\{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$    | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Re |
|-----|-------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 269 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{assume}(! (q[tk] \neq \text{NULL} \wedge q[tk][0] \rightarrow \text{val} == \text{him}[tk] \wedge q[tk][1] \rightarrow \text{val} \neq p[tk][1] \rightarrow \text{val})) \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ | OK |
| 270 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{assume}(\text{collision}[\text{pos}[tk]] \neq \text{him}[tk]) \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$                                                                                                            | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Re |
|-----|-------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 271 | ISTbl | coop::fAhappyFI | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \} CAS(collision[pos[tk]], him[tk], mytid[tk]) \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$ | OK |
| 272 | ISTbl | coop::fAhappyFI | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj]) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \} assume(location[mytid[tk]] \neq p[tk]) \{ ((\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \vee (\exists old\_location\_mytid\_tj\_.. (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0] \rightarrow val == him[tj] \wedge q[tj][1] \rightarrow val \neq p[tj][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[tj] \wedge location[mytid[tj]] == NULL) \wedge location[him[tj]] \neq q[tj])) \}$      | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Re |
|-----|-------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 273 | IStbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \{ (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \} \} \text{assume}(\text{location}[\text{him}[tk]] \neq q[tk]) \{ \{ (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \} \} \}$   | OK |
| 274 | IStbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \{ (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \} \} \text{assume}(\text{location}[\text{mytid}[tk]] \neq p[tk]) \{ \{ (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \} \} \}$ | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Re |
|-----|-------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 275 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \{ (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \vee (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \} \} CAS(location[mytid[tk]], p[tk], NULL) \{ \{ (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \vee (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \} \}$ | OK |
| 276 | ISTbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \{ (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \vee (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \} \} assume(stacktop \neq top[tk]) \{ \{ (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \vee (\exists old\_location\_mytid\_tj\_.. (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]] == NULL) \wedge location[him[t_j]] \neq q[t_j]) \} \}$         | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Re |
|-----|-------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 277 | IStbl | coop::fAhappyFI | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \} \text{CAS}(\text{stacktop}, \text{top}[tk], \text{ntop}[tk])$ $\{ ((\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j]) \vee (\exists \text{old\_location\_mytid\_tj\_}. (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \wedge \text{location}[\text{him}[t_j]] \neq q[t_j])) \}$ | OK |
| 278 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{location}[\text{mytid}[t_j]] \neq p[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{location}[\text{mytid}[t_j]] \neq p[t_j])) \}$ $\text{top}[tk] = \text{stacktop} \{ ((\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{location}[\text{mytid}[t_j]] \neq p[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val} \wedge \text{location}[\text{mytid}[t_j]] \neq p[t_j])) \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | OK |





[illegible]

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Re |
|-----|-------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 285 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$ $return val[tk] \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$                     | OK |
| 286 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$ $val[tk] = q[tk][2] \rightarrow val \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$ | OK |
| 287 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$ $location[mytid[tk]] = NULL \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$         | OK |





| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Re |
|-----|-------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 294 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$ $location[mytid[tk]] = p[tk] \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$                                  | OK |
| 295 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$ $assume(!(him[tk] \neq NULL \wedge him[tk] \neq mytid[tk]))$ $\{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$ | OK |
| 296 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$ $assume(him[tk] \neq NULL \wedge him[tk] \neq mytid[tk]) \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$      | OK |









| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Re |
|-----|-------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 306 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$<br>assume(stacktop $\neq$ top[tk]) {((him[tj] $\neq$ NULL $\wedge$ him[tj] $\neq$ mytid[tj] $\wedge$ q[tj] == location[him[tj]] $\wedge$ q[tj] $\neq$ NULL $\wedge$ q[tj][0] $\rightarrow$ val == him[tj] $\wedge$ q[tj][1] $\rightarrow$ val $\neq$ p[tj][1] $\rightarrow$ val $\wedge$ location[mytid[tj]] $\neq$ p[tj]) $\vee$ (him[tj] $\neq$ NULL $\wedge$ him[tj] $\neq$ mytid[tj] $\wedge$ q[tj] == location[him[tj]] $\wedge$ q[tj] $\neq$ NULL $\wedge$ q[tj][0] $\rightarrow$ val == him[tj] $\wedge$ q[tj][1] $\rightarrow$ val $\neq$ p[tj][1] $\rightarrow$ val $\wedge$ location[mytid[tj]] $\neq$ p[tj]))}  | OK |
| 307 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$<br>CAS(stacktop, top[tk], ntop[tk]) {((him[tj] $\neq$ NULL $\wedge$ him[tj] $\neq$ mytid[tj] $\wedge$ q[tj] == location[him[tj]] $\wedge$ q[tj] $\neq$ NULL $\wedge$ q[tj][0] $\rightarrow$ val == him[tj] $\wedge$ q[tj][1] $\rightarrow$ val $\neq$ p[tj][1] $\rightarrow$ val $\wedge$ location[mytid[tj]] $\neq$ p[tj]) $\vee$ (him[tj] $\neq$ NULL $\wedge$ him[tj] $\neq$ mytid[tj] $\wedge$ q[tj] == location[him[tj]] $\wedge$ q[tj] $\neq$ NULL $\wedge$ q[tj][0] $\rightarrow$ val == him[tj] $\wedge$ q[tj][1] $\rightarrow$ val $\neq$ p[tj][1] $\rightarrow$ val $\wedge$ location[mytid[tj]] $\neq$ p[tj]))} | OK |
| 308 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$<br>top[tk] = stacktop {((him[tj] $\neq$ NULL $\wedge$ him[tj] $\neq$ mytid[tj] $\wedge$ q[tj] == location[him[tj]] $\wedge$ q[tj] $\neq$ NULL $\wedge$ q[tj][0] $\rightarrow$ val == him[tj] $\wedge$ q[tj][1] $\rightarrow$ val $\neq$ p[tj][1] $\rightarrow$ val $\wedge$ location[mytid[tj]] $\neq$ p[tj]) $\vee$ (him[tj] $\neq$ NULL $\wedge$ him[tj] $\neq$ mytid[tj] $\wedge$ q[tj] == location[him[tj]] $\wedge$ q[tj] $\neq$ NULL $\wedge$ q[tj][0] $\rightarrow$ val == him[tj] $\wedge$ q[tj][1] $\rightarrow$ val $\neq$ p[tj][1] $\rightarrow$ val $\wedge$ location[mytid[tj]] $\neq$ p[tj]))}               | OK |



| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Result |
|-----|-------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 312 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$<br>returnn 1 {((him[tj] ≠ NULL ∧ him[tj] ≠ mytid[tj] ∧ q[tj] == location[him[tj]] ∧ q[tj] ≠ NULL ∧ q[tj][0]→val == him[tj] ∧ q[tj][1]→val ≠ p[tj][1]→val ∧ location[mytid[tj]] ≠ p[tj]) ∨ (him[tj] ≠ NULL ∧ him[tj] ≠ mytid[tj] ∧ q[tj] == location[him[tj]] ∧ q[tj] ≠ NULL ∧ q[tj][0]→val == him[tj] ∧ q[tj][1]→val ≠ p[tj][1]→val ∧ location[mytid[tj]] ≠ p[tj]))}                  | OK     |
| 313 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$<br>val[tk] = p[tk][2]→val {((him[tj] ≠ NULL ∧ him[tj] ≠ mytid[tj] ∧ q[tj] == location[him[tj]] ∧ q[tj] ≠ NULL ∧ q[tj][0]→val == him[tj] ∧ q[tj][1]→val ≠ p[tj][1]→val ∧ location[mytid[tj]] ≠ p[tj]) ∨ (him[tj] ≠ NULL ∧ him[tj] ≠ mytid[tj] ∧ q[tj] == location[him[tj]] ∧ q[tj] ≠ NULL ∧ q[tj][0]→val == him[tj] ∧ q[tj][1]→val ≠ p[tj][1]→val ∧ location[mytid[tj]] ≠ p[tj]))}     | OK     |
| 314 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ ((him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge location[mytid[t_j]] \neq p[t_j])) \}$<br>location[mytid[tk]] = NULL {((him[tj] ≠ NULL ∧ him[tj] ≠ mytid[tj] ∧ q[tj] == location[him[tj]] ∧ q[tj] ≠ NULL ∧ q[tj][0]→val == him[tj] ∧ q[tj][1]→val ≠ p[tj][1]→val ∧ location[mytid[tj]] ≠ p[tj]) ∨ (him[tj] ≠ NULL ∧ him[tj] ≠ mytid[tj] ∧ q[tj] == location[him[tj]] ∧ q[tj] ≠ NULL ∧ q[tj][0]→val == him[tj] ∧ q[tj][1]→val ≠ p[tj][1]→val ∧ location[mytid[tj]] ≠ p[tj]))} | OK     |





| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Re |
|-----|-------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 321 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \{ (him[t_j] \neq \text{NULL} \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == \text{location}[him[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge \text{location}[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq \text{NULL} \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == \text{location}[him[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge \text{location}[mytid[t_j]] \neq p[t_j]) \} \} him[tk] = \text{collision}[\text{pos}[tk]] \{ \{ (him[t_j] \neq \text{NULL} \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == \text{location}[him[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge \text{location}[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq \text{NULL} \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == \text{location}[him[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge \text{location}[mytid[t_j]] \neq p[t_j]) \} \}$                        | OK |
| 322 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \{ (him[t_j] \neq \text{NULL} \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == \text{location}[him[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge \text{location}[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq \text{NULL} \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == \text{location}[him[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge \text{location}[mytid[t_j]] \neq p[t_j]) \} \} assume(!(him[tk] \neq \text{NULL} \wedge him[tk] \neq mytid[tk])) \{ \{ (him[t_j] \neq \text{NULL} \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == \text{location}[him[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge \text{location}[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq \text{NULL} \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == \text{location}[him[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge \text{location}[mytid[t_j]] \neq p[t_j]) \} \}$ | OK |
| 323 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \{ (him[t_j] \neq \text{NULL} \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == \text{location}[him[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge \text{location}[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq \text{NULL} \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == \text{location}[him[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge \text{location}[mytid[t_j]] \neq p[t_j]) \} \} assume(him[tk] \neq \text{NULL} \wedge him[tk] \neq mytid[tk]) \{ \{ (him[t_j] \neq \text{NULL} \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == \text{location}[him[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge \text{location}[mytid[t_j]] \neq p[t_j]) \vee (him[t_j] \neq \text{NULL} \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] == \text{location}[him[t_j]] \wedge q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val \wedge \text{location}[mytid[t_j]] \neq p[t_j]) \} \}$    | OK |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Result |
|-----|-------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 324 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, tj \neq tk. \{(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0]->val == him[tj] \wedge q[tj][1]->val \neq p[tj][1]->val \wedge location[mytid[tj]] \neq p[tj]) \vee (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0]->val == him[tj] \wedge q[tj][1]->val \neq p[tj][1]->val \wedge location[mytid[tj]] \neq p[tj]))\}$<br>assume(!(q[tk] ≠ NULL ∧ q[tk][0]→val == him[tk] ∧ q[tk][1]→val ≠ p[tk][1]→val)) {((him[tj] ≠ NULL ∧ him[tj] ≠ mytid[tj] ∧ q[tj] == location[him[tj]] ∧ q[tj] ≠ NULL ∧ q[tj][0]→val == him[tj] ∧ q[tj][1]→val ≠ p[tj][1]→val ∧ location[mytid[tj]] ≠ p[tj]) ∨ (him[tj] ≠ NULL ∧ him[tj] ≠ mytid[tj] ∧ q[tj] == location[him[tj]] ∧ q[tj] ≠ NULL ∧ q[tj][0]→val == him[tj] ∧ q[tj][1]→val ≠ p[tj][1]→val ∧ location[mytid[tj]] ≠ p[tj]))} | OK     |
| 325 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, tj \neq tk. \{(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0]->val == him[tj] \wedge q[tj][1]->val \neq p[tj][1]->val \wedge location[mytid[tj]] \neq p[tj]) \vee (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0]->val == him[tj] \wedge q[tj][1]->val \neq p[tj][1]->val \wedge location[mytid[tj]] \neq p[tj]))\}$<br>assume(collision[pos[tk]] ≠ him[tk]) {((him[tj] ≠ NULL ∧ him[tj] ≠ mytid[tj] ∧ q[tj] == location[him[tj]] ∧ q[tj] ≠ NULL ∧ q[tj][0]→val == him[tj] ∧ q[tj][1]→val ≠ p[tj][1]→val ∧ location[mytid[tj]] ≠ p[tj]) ∨ (him[tj] ≠ NULL ∧ him[tj] ≠ mytid[tj] ∧ q[tj] == location[him[tj]] ∧ q[tj] ≠ NULL ∧ q[tj][0]→val == him[tj] ∧ q[tj][1]→val ≠ p[tj][1]→val ∧ location[mytid[tj]] ≠ p[tj]))}                                            | OK     |
| 326 | IStbl | coop::fAhappyFH | $\forall t_j, tk: \mathbb{N}, tj \neq tk. \{(him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0]->val == him[tj] \wedge q[tj][1]->val \neq p[tj][1]->val \wedge location[mytid[tj]] \neq p[tj]) \vee (him[tj] \neq NULL \wedge him[tj] \neq mytid[tj] \wedge q[tj] == location[him[tj]] \wedge q[tj] \neq NULL \wedge q[tj][0]->val == him[tj] \wedge q[tj][1]->val \neq p[tj][1]->val \wedge location[mytid[tj]] \neq p[tj]))\}$<br>CAS(collision[pos[tk]], him[tk], mytid[tk]) {((him[tj] ≠ NULL ∧ him[tj] ≠ mytid[tj] ∧ q[tj] == location[him[tj]] ∧ q[tj] ≠ NULL ∧ q[tj][0]→val == him[tj] ∧ q[tj][1]→val ≠ p[tj][1]→val ∧ location[mytid[tj]] ≠ p[tj]) ∨ (him[tj] ≠ NULL ∧ him[tj] ≠ mytid[tj] ∧ q[tj] == location[him[tj]] ∧ q[tj] ≠ NULL ∧ q[tj][0]→val == him[tj] ∧ q[tj][1]→val ≠ p[tj][1]→val ∧ location[mytid[tj]] ≠ p[tj]))}                                     | OK     |







| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Re |
|-----|-------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 333 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{top}[tk] = \text{stacktop} \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$                  | OK |
| 334 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{pos}[tk] = \text{nondet}() \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$                  | OK |
| 335 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} ==$ $p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ $\text{him}[tk] = \text{collision}[\text{pos}[tk]] \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Re |
|-----|-------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 336 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \} q[tk] = \text{location}[\text{him}[tk]]$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge$ $\text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$        | OK |
| 337 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j]$ $\wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \} \text{return } q[tk][2] \rightarrow \text{val}$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge$ $\text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ | OK |
| 338 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \} \text{continue } \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$                                 | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Re |
|-----|-------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 339 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(\text{q}[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val})) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \} \text{val}[tk] = q[tk][2] \rightarrow \text{val}$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(\text{q}[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val})) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ | OK |
| 340 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(\text{q}[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val})) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j]$ $\wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \} \text{location}[\text{mytid}[tk]] =$ $\text{NULL} \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j]$ $\neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(\text{q}[t_j]$ $\neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val})) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$  | OK |
| 341 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(\text{q}[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val})) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \} \text{return val}[tk] \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(\text{q}[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val})) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$                            | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Re |
|-----|-------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 342 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL \} val[tk] = q[tk][2] \rightarrow val$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL \}$ | OK |
| 343 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j]$ $\wedge location[mytid[t_j]] == NULL \} location[mytid[tk]] =$ $NULL \{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL$ $\wedge him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j]$ $\neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j]$ $\neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL \}$         | OK |
| 344 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL \} return val[tk] \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL \}$                     | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Re |
|-----|-------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 345 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{return } 0 \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$                                           | OK |
| 346 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j]$ $\wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ $\text{val}[tk] = \text{top}[tk] \rightarrow \text{val}$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge$ $\text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$   | OK |
| 347 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ $\text{ntop}[tk] = \text{top}[tk] \rightarrow \text{next}$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge$ $\text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Re |
|-----|-------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 348 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $return val[tk] \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$              | OK |
| 349 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $int mytid \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$                   | OK |
| 350 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j]$ $\wedge location[mytid[t_j]] == NULL) \}$ $location[mytid[tk]] =$ $p[tk] \{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL$ $\wedge him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j]$ $\neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j]$ $\neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Re |
|-----|-------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 351 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{assume}(!(\text{him}[tk] \neq \text{NULL} \wedge \text{him}[tk] \neq$ $\text{mytid}[tk])) \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j]$ $\neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$                                                                                     | OK |
| 352 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{assume}(\text{him}[tk] \neq \text{NULL} \wedge \text{him}[tk] \neq \text{mytid}[tk])$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge$ $\text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$                                                                                      | OK |
| 353 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{assume}(!(q[tk] \neq \text{NULL} \wedge q[tk][0] \rightarrow \text{val}$ $== \text{him}[tk] \wedge q[tk][1] \rightarrow \text{val} \neq p[tk][1] \rightarrow \text{val}))$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge$ $\text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ | OK |



| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Re |
|-----|-------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 354 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(q[tk] \neq NULL \wedge q[tk][0] \rightarrow val$ $== him[tk] \wedge q[tk][1] \rightarrow val \neq p[tk][1] \rightarrow val)$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |
| 355 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j]$ $\wedge location[mytid[t_j]] == NULL) \}$ $assume(top[tk] \neq$ $NULL) \{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL$ $\wedge him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j]$ $\neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j]$ $\neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$                                                                                                          | OK |
| 356 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j]$ $\wedge location[mytid[t_j]] == NULL) \}$ $assume(top[tk] ==$ $NULL) \{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL$ $\wedge him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j]$ $\neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j]$ $\neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$                                                                                                            | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Re |
|-----|-------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 357 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}. \\ ((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee \\ (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \\ \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} \\ == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge \\ \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] \\ == \text{NULL}) \} \text{assume}(\text{collision}[\text{pos}[tk]] \neq \text{him}[tk]) \\ \{ \exists \text{old\_location\_mytid\_tj\_}. ((!(\text{him}[t_j] \neq \text{NULL} \wedge \\ \text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \\ \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \\ \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq \\ p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \\ \text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$            | OK |
| 358 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}. \\ ((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee \\ (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \\ \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} \\ == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge \\ \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] \\ == \text{NULL}) \} \text{CAS}(\text{collision}[\text{pos}[tk]], \text{him}[tk], \text{mytid}[tk]) \\ \{ \exists \text{old\_location\_mytid\_tj\_}. ((!(\text{him}[t_j] \neq \text{NULL} \wedge \\ \text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \\ \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \\ \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq \\ p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \\ \text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ | OK |
| 359 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}. \\ ((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee \\ (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \\ \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} \\ == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge \\ \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]] \\ == \text{NULL}) \} \text{assume}(\text{location}[\text{mytid}[tk]] \neq p[tk]) \\ \{ \exists \text{old\_location\_mytid\_tj\_}. ((!(\text{him}[t_j] \neq \text{NULL} \wedge \\ \text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \\ \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \\ \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq \\ p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \\ \text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$                    | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Re |
|-----|-------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 360 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \} CAS(location[mytid[tk]], p[tk], NULL)$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |
| 361 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \} assume(location[him[tk]] \neq q[tk])$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$  | OK |
| 362 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \} CAS(location[him[tk]], q[tk], NULL)$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$   | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Re |
|-----|-------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 363 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{assume}(\text{location}[\text{mytid}[tk]] \neq p[tk])$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge$ $\text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$       | OK |
| 364 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{CAS}(\text{location}[\text{mytid}[tk]], p[tk], \text{NULL})$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge$ $\text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ | OK |
| 365 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j]$ $\wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ $\text{assume}(\text{stacktop} \neq \text{top}[tk])$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL}$ $\wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j]$ $\neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j]$ $\neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$                | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Re |
|-----|-------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 366 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $CAS(stacktop, top[tk], ntop[tk])$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |
| 367 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $top[tk] = stacktop \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$               | OK |
| 368 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ $q[tk] = location[him[tk]]$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$        | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Re |
|-----|-------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 369 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $\text{return } 1 \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$           | OK |
| 370 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $\text{continue } \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$           | OK |
| 371 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j]$ $\wedge location[mytid[t_j]] == NULL) \}$ $location[mytid[tk]] =$ $NULL \{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL$ $\wedge him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j]$ $\neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j]$ $\neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Re |
|-----|-------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 372 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{return } 1 \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$                                   | OK |
| 373 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ $\text{val}[tk] = p[tk][2] \rightarrow \text{val}$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge$ $\text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ | OK |
| 374 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ $\text{ntop}[tk] = \text{tloc\_pua3\_tk}$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge$ $\text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$          | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Re |
|-----|-------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 375 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $\text{return } 1 \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$            | OK |
| 376 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $\text{int mytid} \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$            | OK |
| 377 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j]$ $\wedge location[mytid[t_j]] == NULL) \}$ $location[mytid[tk]] =$ $p[tk] \{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL$ $\wedge him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j]$ $\neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j]$ $\neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |



| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Re |
|-----|-------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 378 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $pos[tk] = nondet() \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$                                         | OK |
| 379 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ ==$ $p[t_j] \wedge location[mytid[t_j]] == NULL) \}$ $him[tk] = collision[pos[tk]] \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$                               | OK |
| 380 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(!(him[tk] \neq NULL \wedge him[tk] \neq$ $mytid[tk])) \{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq$ $NULL \wedge him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j]$ $\neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j]$ $\neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Re |
|-----|-------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 381 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(him[tk] \neq NULL \wedge him[tk] \neq mytid[tk])$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$                                                                               | OK |
| 382 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(!(q[tk] \neq NULL \wedge q[tk][0] \rightarrow val$ $== him[tk] \wedge q[tk][1] \rightarrow val \neq p[tk][1] \rightarrow val))$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |
| 383 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(q[tk] \neq NULL \wedge q[tk][0] \rightarrow val$ $== him[tk] \wedge q[tk][1] \rightarrow val \neq p[tk][1] \rightarrow val)$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$    | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Re |
|-----|-------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 384 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(collision[pos[tk]] \neq him[tk])$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$     | OK |
| 385 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $CAS(collision[pos[tk]], him[tk], mytid[tk])$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |
| 386 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(location[mytid[tk]] \neq p[tk])$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$      | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Re |
|-----|-------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 387 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{CAS}(\text{location}[\text{mytid}[tk]], p[tk], \text{NULL})$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ | OK |
| 388 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{assume}(\text{location}[\text{him}[tk]] \neq q[tk])$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$         | OK |
| 389 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{CAS}(\text{location}[\text{him}[tk]], q[tk], p[tk])$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$         | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Re |
|-----|-------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 390 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $CAS(location[mytid[tk]], p[tk], NULL)$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |
| 391 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j]$ $\wedge location[mytid[t_j]] == NULL) \}$ $assume(stacktop \neq top[tk]) \{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL$ $\wedge him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j]$ $\neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j]$ $\neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$           | OK |
| 392 | IStbl | coop::fAfallbackFP | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $CAS(stacktop, top[tk], ntop[tk])$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$      | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Re |
|-----|-------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 393 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{top}[tk] = \text{stacktop} \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$                  | OK |
| 394 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{pos}[tk] = \text{nondet}() \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$                  | OK |
| 395 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} ==$ $p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ $\text{him}[tk] = \text{collision}[\text{pos}[tk]] \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Re |
|-----|-------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 396 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \} q[tk] = \text{location}[\text{him}[tk]]$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge$ $\text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$        | OK |
| 397 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j]$ $\wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \} \text{return } q[tk][2] \rightarrow \text{val}$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge$ $\text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ | OK |
| 398 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \} \text{continue } \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$                                 | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Re |
|-----|-------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 399 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL \} val[tk] = q[tk][2] \rightarrow val$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL \}$ | OK |
| 400 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j]$ $\wedge location[mytid[t_j]] == NULL \} location[mytid[tk]] =$ $NULL \{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL$ $\wedge him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j]$ $\neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j]$ $\neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL \}$         | OK |
| 401 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL \} return val[tk] \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL \}$                     | OK |



| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Re |
|-----|-------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 402 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{return } 0 \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$                                           | OK |
| 403 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j]$ $\wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ $\text{val}[tk] = \text{top}[tk] \rightarrow \text{val}$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge$ $\text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$   | OK |
| 404 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ $\text{ntop}[tk] = \text{top}[tk] \rightarrow \text{next}$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge$ $\text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Re |
|-----|-------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 405 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $return val[tk] \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$              | OK |
| 406 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $int mytid \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$                   | OK |
| 407 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j]$ $\wedge location[mytid[t_j]] == NULL) \}$ $location[mytid[tk]] =$ $p[tk] \{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL$ $\wedge him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j]$ $\neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j]$ $\neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Re |
|-----|-------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 408 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(!(him[tk] \neq NULL \wedge him[tk] \neq$ $mytid[tk])) \{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq$ $NULL \wedge him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j]$ $\neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j]$ $\neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$                                                                            | OK |
| 409 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(him[tk] \neq NULL \wedge him[tk] \neq mytid[tk])$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$                                                                               | OK |
| 410 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(!(q[tk] \neq NULL \wedge q[tk][0] \rightarrow val$ $== him[tk] \wedge q[tk][1] \rightarrow val \neq p[tk][1] \rightarrow val))$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Re |
|-----|-------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 411 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(q[tk] \neq NULL \wedge q[tk][0] \rightarrow val$ $== him[tk] \wedge q[tk][1] \rightarrow val \neq p[tk][1] \rightarrow val)$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |
| 412 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j]$ $\wedge location[mytid[t_j]] == NULL) \}$ $assume(top[tk] \neq$ $NULL) \{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL$ $\wedge him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j]$ $\neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j]$ $\neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$                                                                                                          | OK |
| 413 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j]$ $\wedge location[mytid[t_j]] == NULL) \}$ $assume(top[tk] ==$ $NULL) \{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL$ $\wedge him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j]$ $\neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j]$ $\neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$                                                                                                            | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Re |
|-----|-------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 414 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(collision[pos[tk]] \neq him[tk])$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$     | OK |
| 415 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $CAS(collision[pos[tk]], him[tk], mytid[tk])$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |
| 416 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(location[mytid[tk]] \neq p[tk])$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$      | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Re |
|-----|-------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 417 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $CAS(location[mytid[tk]], p[tk], NULL)$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |
| 418 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(location[him[tk]] \neq q[tk])$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$  | OK |
| 419 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $CAS(location[him[tk]], q[tk], NULL)$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$   | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Re |
|-----|-------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 420 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $CAS(location[mytid[tk]], p[tk], NULL)$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |
| 421 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j]$ $\wedge location[mytid[t_j]] == NULL) \}$ $assume(stacktop \neq top[tk]) \{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL$ $\wedge him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j]$ $\neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j]$ $\neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$           | OK |
| 422 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $CAS(stacktop, top[tk], ntop[tk])$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$      | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Re |
|-----|-------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 423 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{top}[tk] = \text{stacktop} \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$          | OK |
| 424 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ $q[tk] = \text{location}[\text{him}[tk]]$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge$ $\text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ | OK |
| 425 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{return } 1 \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$                          | OK |



| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Re |
|-----|-------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 426 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{continue } \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$                                   | OK |
| 427 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j]$ $\wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ $\text{location}[\text{mytid}[tk]] =$ $\text{NULL } \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL}$ $\wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j]$ $\neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j]$ $\neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ | OK |
| 428 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{return } 1 \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$                                   | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Re |
|-----|-------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 429 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL \} val[tk] = p[tk][2] \rightarrow val$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL \}$ | OK |
| 430 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j])$ $\vee (him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge$ $q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq NULL$ $\wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j]$ $\wedge location[mytid[t_j]] == NULL \} location[mytid[tk]] =$ $NULL \{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL$ $\wedge him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j]$ $\neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j]$ $\neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL \}$         | OK |
| 431 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL \} return 1 \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL \}$                           | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Re |
|-----|-------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 432 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ $\text{ntop}[tk] = \text{tloc\_pua3\_tk}$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ | OK |
| 433 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{return } 1 \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$                        | OK |
| 434 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{int mytid } \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$                       | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Re |
|-----|-------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 435 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j]$ $\wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \} \text{location}[\text{mytid}[tk]] =$ $p[tk] \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j]$ $\neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j]$ $\neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$           | OK |
| 436 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \} \text{pos}[tk] = \text{nondet}() \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$                    | OK |
| 437 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} ==$ $p[t_j] \wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \} \text{him}[tk] =$ $\text{collision}[\text{pos}[tk]] \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Re |
|-----|-------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 438 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(!(him[tk] \neq NULL \wedge him[tk] \neq$ $mytid[tk])) \{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq$ $NULL \wedge him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j]$ $\neq mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j]$ $\neq NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$                                                                            | OK |
| 439 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(him[tk] \neq NULL \wedge him[tk] \neq mytid[tk])$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$                                                                               | OK |
| 440 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(!(q[tk] \neq NULL \wedge q[tk][0] \rightarrow val$ $== him[tk] \wedge q[tk][1] \rightarrow val \neq p[tk][1] \rightarrow val))$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Re |
|-----|-------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 441 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(q[tk] \neq NULL \wedge q[tk][0] \rightarrow val$ $== him[tk] \wedge q[tk][1] \rightarrow val \neq p[tk][1] \rightarrow val)$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |
| 442 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(collision[pos[tk]] \neq him[tk])$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$                                                                                            | OK |
| 443 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $CAS(collision[pos[tk]], him[tk], mytid[tk])$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$                                                                                        | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Re |
|-----|-------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 444 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(location[mytid[tk]] \neq p[tk])$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |
| 445 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $CAS(location[mytid[tk]], p[tk], NULL)$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$  | OK |
| 446 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_ \dots$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_ == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(location[him[tk]] \neq q[tk])$ $\{ \exists old\_location\_mytid\_tj\_ \dots ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_ == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$   | OK |

| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Re |
|-----|-------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 447 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $CAS(location[him[tk]], q[tk], p[tk])$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$   | OK |
| 448 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $assume(location[mytid[tk]] \neq p[tk])$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$ | OK |
| 449 | IStbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists old\_location\_mytid\_tj\_..$ $((!(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j]) \vee$ $(him[t_j] \neq NULL \wedge him[t_j] \neq mytid[t_j] \wedge q[t_j] ==$ $location[him[t_j]] \wedge !(q[t_j] \neq NULL \wedge q[t_j][0] \rightarrow val$ $== him[t_j] \wedge q[t_j][1] \rightarrow val \neq p[t_j][1] \rightarrow val))) \wedge$ $old\_location\_mytid\_tj\_.. == p[t_j] \wedge location[mytid[t_j]]$ $== NULL) \}$ $CAS(location[mytid[tk]], p[tk], NULL)$ $\{ \exists old\_location\_mytid\_tj\_.. ((!(him[t_j] \neq NULL \wedge$ $him[t_j] \neq mytid[t_j]) \vee (him[t_j] \neq NULL \wedge him[t_j] \neq$ $mytid[t_j] \wedge q[t_j] == location[him[t_j]] \wedge !(q[t_j] \neq$ $NULL \wedge q[t_j][0] \rightarrow val == him[t_j] \wedge q[t_j][1] \rightarrow val \neq$ $p[t_j][1] \rightarrow val))) \wedge old\_location\_mytid\_tj\_.. == p[t_j] \wedge$ $location[mytid[t_j]] == NULL) \}$  | OK |



| #   | Kind  | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Re |
|-----|-------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 450 | ISTbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j])$ $\vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge$ $q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL}$ $\wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j]$ $\wedge \text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ $\text{assume}(\text{stacktop} \neq \text{top}[tk]) \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j]$ $\neq \text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j]$ $\neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$             | OK |
| 451 | ISTbl | coop::fAfallbackFQ | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee$ $(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j] \wedge q[t_j] ==$ $\text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq \text{NULL} \wedge q[t_j][0] \rightarrow \text{val}$ $== \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq p[t_j][1] \rightarrow \text{val}))) \wedge$ $\text{old\_location\_mytid\_tj\_} == p[t_j] \wedge \text{location}[\text{mytid}[t_j]]$ $== \text{NULL}) \}$ $\text{CAS}(\text{stacktop}, \text{top}[tk], \text{ntop}[tk])$ $\{ \exists \text{old\_location\_mytid\_tj\_}.$ $((!(\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq \text{mytid}[t_j]) \vee (\text{him}[t_j] \neq \text{NULL} \wedge \text{him}[t_j] \neq$ $\text{mytid}[t_j] \wedge q[t_j] == \text{location}[\text{him}[t_j]] \wedge !(q[t_j] \neq$ $\text{NULL} \wedge q[t_j][0] \rightarrow \text{val} == \text{him}[t_j] \wedge q[t_j][1] \rightarrow \text{val} \neq$ $p[t_j][1] \rightarrow \text{val}))) \wedge \text{old\_location\_mytid\_tj\_} == p[t_j] \wedge$ $\text{location}[\text{mytid}[t_j]] == \text{NULL}) \}$ | OK |

### Veracity Program Examples (one per obligation kind)

```

RMov [FAIL].
1 tloc stacktop = null;

3 tloc[][] location = new tloc[][0];

5 int[] collision = new int[0];

7 int main() {
8   havoc stacktop;
9   havoc location;
10  havoc collision;
11  int mytid_size = 0;
12  havoc mytid_size;
13  int[] mytid = new int[mytid_size];
14  int p_size = 0;
15  havoc p_size;
16  tloc[][] p = new tloc[][p_size];
17  int t = 1;
18  havoc t;
19  int him_size = 0;
20  havoc him_size;

```

```

21 int[] him = new int[him_size];
22 int ntop_size = 0;
23 havoc ntop_size;
24 tloc[] ntop = new tloc[ntop_size];
25 int pos_size = 0;
26 havoc pos_size;
27 int[] pos = new int[pos_size];
28 int q_size = 0;
29 havoc q_size;
30 tloc[][] q = new tloc[][q_size];
31 int top_size = 0;
32 havoc top_size;
33 tloc[] top = new tloc[top_size];
34 commute_right_ctx (true && p[t][1] ≠ null && q[t][0] ≠ null &&
35                     q[t][1] ≠ null) {
36   {
37     assume(him[t] ≠ null && him[t] ≠ mytid[t]);
38     q[t] = location[him[t]];
39     assume(!(q[t] ≠ null && q[t][0]→value == him[t] &&
40             q[t][1]→value ≠ p[t][1]→value));
41   }
42   {
43     havoc collision;
44     havoc location;
45   }

47   context: location[mytid[t]] ≠ p[t]
48 }
49 return 0;
50}

```

```

HTrp [OK].
1 tloc stacktop = null;

3 tloc[][] location = new tloc[][0];

5 int[] collision = new int[0];

7 int main() {
8   havoc stacktop;
9   havoc location;
10  havoc collision;
11  int ntop_size = 0;
12  havoc ntop_size;
13  tloc[] ntop = new tloc[ntop_size];
14  int t = 1;
15  havoc t;
16  int top_size = 0;
17  havoc top_size;
18  tloc[] top = new tloc[top_size];
19  assume(stacktop == top[t]);
20  assume(stacktop == top[t]);

```

```

21 stacktop = ntop[t];
22 assert(exists old__stacktop : tloc . old__stacktop == top[t] &&
23         stacktop == ntop[t]);
24 return 0;
25}

```

```

  LMov  [OK].
1 tloc stacktop = null;

3 tloc[][] location = new tloc[][0];

5 int[] collision = new int[0];

7 int main() {
8   havoc stacktop;
9   havoc location;
10  havoc collision;
11  int ntop_size = 0;
12  havoc ntop_size;
13  tloc[] ntop = new tloc[ntop_size];
14  int t = 1;
15  havoc t;
16  int top_size = 0;
17  havoc top_size;
18  tloc[] top = new tloc[top_size];
19  int him_size = 0;
20  havoc him_size;
21  int[] him = new int[him_size];
22  int mytid_size = 0;
23  havoc mytid_size;
24  int[] mytid = new int[mytid_size];
25  int p_size = 0;
26  havoc p_size;
27  tloc[][] p = new tloc[][p_size];
28  int pos_size = 0;
29  havoc pos_size;
30  int[] pos = new int[pos_size];
31  int q_size = 0;
32  havoc q_size;
33  tloc[][] q = new tloc[][q_size];
34  commute_left (exists old__stacktop : tloc . old__stacktop == top[t] &&
35               stacktop == ntop[t]) {
36    {
37      int steps = 1;
38      havoc steps;
39      while (steps > 0) {
40        steps = steps - 1;
41        int tk = 1;
42        havoc tk;
43        assume(tk ≠ t);
44        int nd = 1;
45        havoc nd;

```

```

46     if (nd == 1) {
47         havoc stacktop;
48         assume(stacktop == ntop[tk]);
49     } else {
50         if (nd == 2) {
51             havoc collision[pos[tk]];
52             assume(collision[pos[tk]] == mytid[tk]);
53         } else {
54             if (nd == 3) {
55                 havoc location[mytid[tk]];
56                 assume(location[mytid[tk]] == null);
57             } else {
58                 if (nd == 4) {
59                     havoc location[him[tk]];
60                     assume(location[him[tk]] == p[tk]);
61                 } else {
62                     if (nd == 5) {
63                         havoc location[mytid[tk]];
64                         assume(location[mytid[tk]] == null);
65                     } else {
66                         if (nd == 6) {
67                             havoc stacktop;
68                             assume(stacktop == ntop[tk]);
69                         } else {
70                             if (nd == 7) {
71                                 havoc collision[pos[tk]];
72                                 assume(collision[pos[tk]] == mytid[tk]);
73                             } else {
74                                 if (nd == 8) {
75                                     havoc location[mytid[tk]];
76                                     assume(location[mytid[tk]] == null);
77                                 } else {
78                                     if (nd == 9) {
79                                         havoc location[him[tk]];
80                                         assume(location[him[tk]] == null);
81                                     } else {
82                                         if (true) {
83                                             havoc location[mytid[tk]];
84                                             assume(location[mytid[tk]] == null);
85                                         }
86                                     }
87                                 }
88                             }
89                         }
90                     }
91                 }
92             }
93         }
94     }
95 }
96 }
97 {

```

```

98     return 1;
99 }

101 }
102 return 0;
103 }

```

```

Subs [FAIL].
1 tloc stacktop = null;

3 tloc[][] location = new tloc[][0];

5 int[] collision = new int[0];

7 int main() {
8   havoc stacktop;
9   havoc location;
10  havoc collision;
11  int him_size = 0;
12  havoc him_size;
13  int[] him = new int[him_size];
14  int mytid_size = 0;
15  havoc mytid_size;
16  int[] mytid = new int[mytid_size];
17  int p_size = 0;
18  havoc p_size;
19  tloc[][] p = new tloc[][p_size];
20  int pos_size = 0;
21  havoc pos_size;
22  int[] pos = new int[pos_size];
23  int ti = 1;
24  havoc ti;
25  int tj = 1;
26  havoc tj;
27  assume(exists old__collision_pos_ti__ : int . location[mytid[tj]] == p[tj]
28          &&
29          him[tj] == collision[pos[tj]]
30          &&
31          location[mytid[ti]] == p[ti]
32          &&
33          him[ti] ==
34          old__collision_pos_ti__ &&
35          collision[pos[ti]] ==
36          mytid[ti]);
37  assert(collision[pos[tj]] != him[tj]);
38  return 0;
39 }

```

### B.3 msq

242 obligations across 28 specs: atom::aAdvPre, atom::aAdvPre.N, atom::aAdvRo, atom::aAdvRo.N, atom::aAdvS, atom::aDeqEmpS, atom::aDeqEmpS.N, atom::aDeqEmpS.R, atom::aDeqOneS,

atom::aDeqOneS.N, atom::aDeqOneS.R, atom::aDeqPre, atom::aDeqPre.N, atom::aDeqPre.R,  
 atom::aDeqS, atom::aEnqPre, atom::aEnqPre.N, atom::aEnqPre.R, atom::aEnqS, coop::cAdvF,  
 coop::cAdvFT, coop::cAdvFT.R, coop::cDeqF, coop::cDeqFH, coop::cDeqFH.R, coop::cEnqF,  
 coop::cEnqFStale, coop::cEnqFStale.R.

### B.3.1 Sources.

#### Implementation (.cl)

```

1 /* Michael & Scott Queue */

3 tloc Head = new_cell();
4 tloc Tail = new_cell();

6 invariant Head ≠ NULL && Tail ≠ NULL && (Head == Tail || Head→next ≠ null);

8 int enqueue(int val) {
9   while (true) {
10    tloc node = new_cell();
11    node→val = val;
12    tloc tail ∈ Tail;
13    tloc next = tail→next;

15    if (tail == Tail) {
16      if (next == NULL) {
17        if ICAS(node ≠ NULL && Tail == tail, tail→next, next, node) {
18          return 0;
19        }
20      }
21    }
22  }
23 }

25 int dequeue() {
26   while (true) {
27     int pval;
28     tloc head = Head;
29     tloc tail ∈ Tail;
30     tloc next = head→next;
31     if (Head == head) {
32       if (head == tail) {
33         if (next == NULL) {
34           return 0;
35         }
36       }
37       else {
38         pval = next→val;
39         if ICAS(next ≠ NULL, Head, head, next) {
40           return pval;
41         }
42       }

```

```

43     }
44 }
45}

47int tailAdvance() {
48   while (true) {
49       tloc tail ∈ Tail;
50       tloc next = tail→next;
51       if (next ≠ NULL) {
52           if ICAS(next ≠ NULL, Tail, tail, next) {
53               return 0;
54           }
55       }
56   }
57}

```

### Labeled program (.cll)

```

1/*****
2/* [generated] KAT-friendly program with label annotations */
3/* --
4/* msq.cll
5/*****

7tloc Head = new {0 {} NULL};
8tloc Tail = new {0 {} NULL};

10invariant Head ≠ NULL && Tail ≠ NULL && (Head == Tail || Head→next ≠ NULL);

12int enqueue(int val) {
13   while (true) {
14       [ea1] tloc node = new {0 {} NULL};
15       [ea2] node→val = val;
16       [ea3] tloc tail ∈ Tail;
17       [ea4] tloc next = tail→next;
18       if * {
19           [eb5t] assume(tail == Tail);
20           if * {
21               [eb6t] assume(next == NULL);
22               if * {
23                   [ecas7t] ICAS(node ≠ NULL && Tail == tail, tail→next, next, node);
24                   [ea8] return 0;
25               }
26           } else {
27               [ecas7f] assume(!(node ≠ NULL && Tail == tail && tail→next == next));
28           }
29       }
30       else {
31           [eb6f] assume(next ≠ NULL);
32       }
33   }

```

```

34     else {
35         [@eb5f] assume(tail ≠ Tail);
36     }
37 }
38}

40int dequeue() {
41     while (true) {
42         [@da1] int pval;
43         [@da2] tloc head = Head;
44         [@da3] tloc tail ∈ Tail;
45         [@da4] tloc next = head→next;
46         if * {
47             [@db5t] assume(Head == head);
48             if * {
49                 [@db6t] assume(head == tail);
50                 if * {
51                     [@db7t] assume(next == NULL);
52                     [@da8] return 0;
53                 }
54                 else {
55                     [@db7f] assume(next ≠ NULL);
56                 }
57             }
58             else {
59                 [@db6f] assume(head ≠ tail);
60                 [@da9] pval = next→val;
61                 if * {
62                     [@dcas10t] ICAS(next ≠ NULL, Head, head, next);
63                     [@da11] return pval;
64                 }
65                 else {
66                     [@dcas10f] assume(!(next ≠ NULL && Head == head));
67                 }
68             }
69         }
70         else {
71             [@db5f] assume(Head ≠ head);
72         }
73     }
74}

76int tailAdvance() {
77     while (true) {
78         [@ta1] tloc tail ∈ Tail;
79         [@ta2] tloc next = tail→next;
80         if * {
81             [@tb3t] assume(next ≠ NULL);
82             if * {
83                 [@tcas4t] ICAS(next ≠ NULL, Tail, tail, next);
84                 [@ta5] return 0;
85             }

```



```

86     else {
87         [etcas4f] assume(!(next ≠ NULL && Tail == tail));
88     }
89 }
90 else {
91     [etb3f] assume(next == NULL);
92 }
93 }
94 }

```

### B.3.2 *IRSpec.*

#### IR specifications (.irspecs)

```

1/* OBJECT: Michael-Scott Queue */

3/* Threads preparing to Enqueue that observe the end of the tail */
4aEnqPre = atom{
5    ind = eb6t,
6    red = [ea1; ea2; ea3]; ea4 |> [eb5t; eb6t] <| EPS
7}

9/* Enqueue happy path */
10aEnqS = atom{
11    ind = ecas7t,
12    red = [aEnqPre] |> ecas7t <| ea8
13}

15/* Two related paths: fail the CAS */
16cEnqF = coop{
17    ind = ecas7t,
18    red = [aEnqPre] |> [aEnqS] <| ecas7f
19}

21/* Two related paths: fail for observing a stale tail */
22cEnqFStale = coop{
23    ind = ecas7t,
24    red = [ea1; ea2; ea3] |> [aEnqS] <| ea4; eb5t; eb6f
25}

27aDeqPre = atom{
28    ind = db6f,
29    red = [da1; da2]; da3; da4 |> [db5t; db6f; da9] <| EPS
30}

32/* Dequeue happy path */
33aDeqS = atom{
34    ind = dcas10t,
35    red = [aDeqPre] |> dcas10t <| da11
36}

38/* Deq path that fails the CAS */

```

```

39 cDeqF = coop{
40   ind = dcas10t,
41   red = [aDeqPre] |> [aDeqS] <| dcas10f
42 }

44 /* Deq path that fails the agreement test between local head and global Head */
45 cDeqFH = coop{
46   ind = dcas10t,
47   red = [da1; da2] |> [aDeqS] <| da3; da4; db5f
48 }

50 /* Read-only path when queue is empty */
51 aDeqEmpS = atom{
52   ind = db7t,
53   red = [da1; da2]; da3; da4 |> [db5t; db6t; db7t] <| da8
54 }

56 /* Read-only path when queue has 1 element, but need advancer to act */
57 aDeqOneS = atom{
58   ind = db7f,
59   red = [da1; da2]; da3; da4 |> [db5t; db6t; db7f] <| EPS
60 }

62 /* Path preparing the tail-advance */
63 aAdvPre = atom{
64   ind = tb3t,
65   red = ta1 |> [ta2; tb3t] <| EPS
66 }

68 /* Tail-advance happy path */
69 aAdvS = atom{
70   ind = tcas4t,
71   red = [aAdvPre] |> tcas4t <| ta5
72 }

74 /* Tail-advance path failing at CAS */
75 cAdvF = coop{
76   ind = tcas4t,
77   red = [aAdvPre] |> [aAdvS] <| tcas4f
78 }

80 /* Enqueue path that fails due to advancer updating the tail */
81 cAdvFT = coop{
82   ind = tcas4t,
83   red = [ea1; ea2; ea3] |> [aAdvS] <| ea4; eb5f
84 }

86 /* Advancer read-only path when nothing to do */
87 aAdvRo = atom{
88   ind = tb3f,
89   red = ta1 |> [ta2; tb3f] <| EPS
90 }

```

## B.3.3 Contextual Commutativity (66).

| #  | Kind | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Result |
|----|------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1  | Inv  |                 | $\text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | OK     |
| 2  | Atom | atom::aEnqPre.R | $[\text{tloc node} = \text{new } \{\emptyset\} \text{ NULL}; \text{node} \rightarrow \text{val} = \text{val}; \text{tloc tail} \in \text{Tail}]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | OK     |
| 3  | Atom | atom::aEnqPre.N | $[\text{assume}(\text{tail} == \text{Tail}); \text{assume}(\text{next} == \text{NULL})]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | OK     |
| 4  | RMov | atom::aEnqPre   | $\forall t:\mathbb{N}. \text{next}[t] = \text{tail}[t] \rightarrow \text{next} \triangleright \{(\text{tail}[t] == \text{Tail} \wedge \text{next}[t] == \text{NULL} \wedge \text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL}))\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | OK     |
| 5  | RMov | atom::aEnqPre   | $\forall t:\mathbb{N}. \text{node}[t] = \text{tloc\_ea1\_t}; \text{node}[t] \rightarrow \text{val} = \text{val}[t]; \text{tail}[t] = \text{Tail} \triangleright \{(\text{tail}[t] == \text{Tail} \wedge \text{tail}[t] \rightarrow \text{next} == \text{NULL} \wedge \text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL}))\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | OK     |
| 6  | HTrp | atom::aEnqPre   | $\forall t:\mathbb{N}. \{(\text{tail}[t] == \text{Tail} \wedge \text{next}[t] == \text{NULL} \wedge \text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL}))\} [\text{assume}(\text{tail}[t] == \text{Tail}); \text{assume}(\text{next}[t] == \text{NULL})] \{(\text{tail}[t] == \text{Tail} \wedge \text{next}[t] == \text{NULL} \wedge \text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL}))\}$                                                                                                                                                                                                                                                                                        | OK     |
| 7  | RMov | atom::aEnqS     | $\forall t:\mathbb{N}. \text{node}[t] = \text{tloc\_ea1\_t}; \text{node}[t] \rightarrow \text{val} = \text{val}[t]; \text{tail}[t] = \text{Tail}; \text{next}[t] = \text{tail}[t] \rightarrow \text{next}; \text{assume}(\text{tail}[t] == \text{Tail}); \text{assume}(\text{next}[t] == \text{NULL}) \triangleright \{(\text{node}[t] \neq \text{NULL} \wedge \text{Tail} == \text{tail}[t] \wedge \text{tail}[t] \rightarrow \text{next} == \text{next}[t] \wedge \text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL}))\}$                                                                                                                                                                                                                                                                    | OK     |
| 8  | HTrp | atom::aEnqS     | $\forall t:\mathbb{N}. \{(\text{node}[t] \neq \text{NULL} \wedge \text{Tail} == \text{tail}[t] \wedge \text{tail}[t] \rightarrow \text{next} == \text{next}[t] \wedge \text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL}))\} \text{ICAS}(\text{node}[t] \neq \text{NULL} \wedge \text{Tail} == \text{tail}[t], \text{tail}[t] \rightarrow \text{next}, \text{next}[t], \text{node}[t]) \{\exists \text{old\_tail\_t\_next}. (\text{node}[t] \neq \text{NULL} \wedge \text{Tail} == \text{tail}[t] \wedge \text{old\_tail\_t\_next} == \text{next}[t] \wedge \text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL}) \wedge \text{tail}[t] \rightarrow \text{next} == \text{node}[t])\}$ | OK     |
| 9  | LMov | atom::aEnqS     | $\forall t:\mathbb{N}. \{\exists \text{old\_tail\_t\_next}. (\text{node}[t] \neq \text{NULL} \wedge \text{Tail} == \text{tail}[t] \wedge \text{old\_tail\_t\_next} == \text{next}[t] \wedge \text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL}) \wedge \text{tail}[t] \rightarrow \text{next} == \text{node}[t])\} \triangleleft \text{return } \emptyset$                                                                                                                                                                                                                                                                                                                                                                                                                                     | OK     |
| 10 | RMov | coop::cEnqF     | $\forall tj:\mathbb{N}. \text{node}[tj] = \text{tloc\_ea1\_tj}; \text{node}[tj] \rightarrow \text{val} = \text{val}[tj]; \text{tail}[tj] = \text{Tail}; \text{next}[tj] = \text{tail}[tj] \rightarrow \text{next}; \text{assume}(\text{tail}[tj] == \text{Tail}); \text{assume}(\text{next}[tj] == \text{NULL}) \triangleright \{(\text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL}) \wedge \text{node}[tj] == \text{tloc\_ea1\_tj} \wedge \text{node}[tj] \rightarrow \text{val} == \text{val}[tj] \wedge \text{tail}[tj] == \text{Tail} \wedge \text{next}[tj] == \text{tail}[tj] \rightarrow \text{next} \wedge \text{next}[tj] == \text{NULL})\}$                                                                                                                                         | OK     |

| #  | Kind | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Result |
|----|------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 11 | HTrp | coop::cEnqF        | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] == NULL) \}$ $[[[node[ti] = tloc\_ea1\_ti; node[ti] \rightarrow val = val[ti]; tail[ti] = Tail]; next[ti] = tail[ti] \rightarrow next; [assume(tail[ti] == Tail); assume(next[ti] == NULL)]; skip]; ICAS(node[ti] \neq NULL \wedge Tail == tail[ti], tail[ti] \rightarrow next, next[ti], node[ti]); return \emptyset]$ $\{ \exists old\_tail\_ti\_next. (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] == NULL \wedge node[ti] == tloc\_ea1\_ti \wedge node[ti] \rightarrow val == val[ti] \wedge tail[ti] == Tail \wedge next[ti] == old\_tail\_ti\_next \wedge next[ti] == NULL \wedge tail[ti] \rightarrow next == node[ti] \wedge node[ti] \neq NULL \wedge Tail == tail[ti]) \}$ | OK     |
| 12 | Subs | coop::cEnqF        | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ \exists old\_tail\_ti\_next. (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] == NULL \wedge node[ti] == tloc\_ea1\_ti \wedge node[ti] \rightarrow val == val[ti] \wedge tail[ti] == Tail \wedge next[ti] == old\_tail\_ti\_next \wedge next[ti] == NULL \wedge tail[ti] \rightarrow next == node[ti] \wedge node[ti] \neq NULL \wedge Tail == tail[ti]) \} \Rightarrow \{ !(node[tj] \neq NULL \wedge Tail == tail[tj] \wedge tail[tj] \rightarrow next == next[tj]) \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | OK     |
| 13 | Atom | coop::cEnqFStale.R | [tloc node = new {0 {} NULL}; node->val = val; tloc tail $\in$ Tail]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | OK     |
| 14 | RMov | coop::cEnqFStale   | $\forall tj: \mathbb{N}. node[tj] = tloc\_ea1\_tj; node[tj] \rightarrow val = val[tj]; tail[tj] = Tail \triangleright \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail) \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | OK     |
| 15 | HTrp | coop::cEnqFStale   | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail) \}$ $[[[node[ti] = tloc\_ea1\_ti; node[ti] \rightarrow val = val[ti]; tail[ti] = Tail]; next[ti] = tail[ti] \rightarrow next; [assume(tail[ti] == Tail); assume(next[ti] == NULL)]; skip]; ICAS(node[ti] \neq NULL \wedge Tail == tail[ti], tail[ti] \rightarrow next, next[ti], node[ti]); return \emptyset]$ $\{ \exists old\_tail\_ti\_next. (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail \wedge node[ti] == tloc\_ea1\_ti \wedge node[ti] \rightarrow val == val[ti] \wedge tail[ti] == Tail \wedge next[ti] == old\_tail\_ti\_next \wedge next[ti] == NULL \wedge tail[ti] \rightarrow next == node[ti] \wedge node[ti] \neq NULL \wedge Tail == tail[ti]) \}$                                                                                                                                           | OK     |

| #  | Kind | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Result |
|----|------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 16 | LMov | coop::cEnqFStale | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ \exists old\_tail\_ti\_next. (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail \wedge node[ti] == tloc\_ea1\_ti \wedge node[ti] \rightarrow val == val[ti] \wedge tail[ti] == Tail \wedge next[ti] == old\_tail\_ti\_next \wedge next[ti] == NULL \wedge tail[ti] \rightarrow next == node[ti] \wedge node[ti] \neq NULL \wedge Tail == tail[ti]) \} \wedge next[tj] = tail[tj] \rightarrow next$                                       | OK     |
| 17 | Subs | coop::cEnqFStale | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ (\exists old\_tail\_ti\_next. (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail \wedge node[ti] == tloc\_ea1\_ti \wedge node[ti] \rightarrow val == val[ti] \wedge tail[ti] == Tail \wedge next[ti] == old\_tail\_ti\_next \wedge next[ti] == NULL \wedge tail[ti] \rightarrow next == node[ti] \wedge node[ti] \neq NULL \wedge Tail == tail[ti]) \wedge next[tj] == tail[tj] \rightarrow next) \} \Rightarrow \{ tail[tj] == Tail \}$ | OK     |
| 18 | LMov | coop::cEnqFStale | $\forall ti, tj: \mathbb{N}, ti \neq tj. \text{assume}(next[tj] \neq NULL)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | OK     |
| 19 | Atom | atom::aDeqPre.R  | $[int \text{ pval}; tloc \text{ head} = Head]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | OK     |
| 20 | Atom | atom::aDeqPre.N  | $[assume(Head == head); assume(head \neq tail); pval = next \rightarrow val]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | OK     |
| 21 | RMov | atom::aDeqPre    | $\forall t: \mathbb{N}. next[t] = head[t] \rightarrow next \triangleright \{ (Head == head[t] \wedge head[t] \neq tail[t] \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL)) \}$                                                                                                                                                                                                                                                                                                                                                                                                  | OK     |
| 22 | RMov | atom::aDeqPre    | $\forall t: \mathbb{N}. tail[t] = Tail \triangleright \{ (Head == head[t] \wedge head[t] \neq tail[t] \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL)) \}$                                                                                                                                                                                                                                                                                                                                                                                                                      | FAIL   |
| 23 | RMov | atom::aDeqPre    | $\forall t: \mathbb{N}. int \text{ pval}; head[t] = Head \triangleright \{ (Head == head[t] \wedge head[t] \neq Tail \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL)) \}$                                                                                                                                                                                                                                                                                                                                                                                                       | OK     |
| 24 | HTrp | atom::aDeqPre    | $\forall t: \mathbb{N}. \{ (Head == head[t] \wedge head[t] \neq tail[t] \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL)) \} [assume(Head == head[t]); assume(head[t] \neq tail[t]); pval[t] = next[t] \rightarrow val] \{ (Head == head[t] \wedge head[t] \neq tail[t] \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge pval[t] == next[t] \rightarrow val) \}$                                                                                                                                                    | OK     |
| 25 | RMov | atom::aDeqS      | $\forall t: \mathbb{N}. int \text{ pval}; head[t] = Head; tail[t] = Tail; next[t] = head[t] \rightarrow next; assume(Head == head[t]); assume(head[t] \neq tail[t]); pval[t] = next[t] \rightarrow val \triangleright \{ (next[t] \neq NULL \wedge Head == head[t] \wedge next[t] \neq NULL \wedge Tail \neq NULL \wedge (next[t] == Tail \vee next[t] \rightarrow next \neq NULL)) \}$                                                                                                                                                                                                                                                | OK     |
| 26 | HTrp | atom::aDeqS      | $\forall t: \mathbb{N}. \{ (next[t] \neq NULL \wedge Head == head[t] \wedge next[t] \neq NULL \wedge Tail \neq NULL \wedge (next[t] == Tail \vee next[t] \rightarrow next \neq NULL)) \} ICAS(next[t] \neq NULL, Head, head[t], next[t]) \{ \exists old\_Head. (next[t] \neq NULL \wedge old\_Head == head[t] \wedge next[t] \neq NULL \wedge Tail \neq NULL \wedge (next[t] == Tail \vee next[t] \rightarrow next \neq NULL) \wedge Head == next[t]) \}$                                                                                                                                                                              | OK     |

| #  | Kind | Spec           | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Result |
|----|------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 27 | LMov | atom::aDeqS    | $\forall t:\mathbb{N}. \{\exists \text{old\_Head}. (\text{next}[t] \neq \text{NULL} \wedge \text{old\_Head} == \text{head}[t] \wedge \text{next}[t] \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{next}[t] == \text{Tail} \vee \text{next}[t] \rightarrow \text{next} \neq \text{NULL})) \wedge \text{Head} == \text{next}[t]\} \triangleleft$<br>return pval[t]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | OK     |
| 28 | RMov | coop::cDeqF    | $\forall tj:\mathbb{N}. \text{int pval}; \text{head}[tj] = \text{Head}; \text{tail}[tj] = \text{Tail};$<br>next[tj] = head[tj]→next; assume(Head == head[tj]);<br>assume(head[tj] ≠ tail[tj]); pval[tj] = next[tj]→val<br>▷ {(Head ≠ NULL ∧ Tail ≠ NULL ∧ (Head == Tail ∨<br>Head→next ≠ NULL) ∧ head[tj] == Head ∧ tail[tj] == Tail<br>∧ next[tj] == head[tj]→next ∧ head[tj] ≠ tail[tj] ∧<br>pval[tj] == next[tj]→val)}                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | OK     |
| 29 | HTrp | coop::cDeqF    | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{(\text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge$<br>(Head == Tail ∨ Head→next ≠ NULL) ∧ head[tj] == Head<br>∧ tail[tj] == Tail ∧ next[tj] == head[tj]→next ∧<br>head[tj] ≠ tail[tj] ∧ pval[tj] == next[tj]→val)}<br>[[[int pval; head[ti] = Head; tail[ti] = Tail;<br>next[ti] = head[ti]→next; [assume(Head == head[ti]);<br>assume(head[ti] ≠ tail[ti]); pval[ti] = next[ti]→val];<br>skip]; ICAS(next[ti] ≠ NULL, Head, head[ti], next[ti]);<br>return pval[ti]] {∃ old__Head. (old__Head ≠ NULL ∧ Tail<br>≠ NULL ∧ (old__Head == Tail ∨ old__Head→next ≠ NULL)<br>∧ head[tj] == old__Head ∧ tail[tj] == Tail ∧ next[tj]<br>== head[tj]→next ∧ head[tj] ≠ tail[tj] ∧ pval[tj] ==<br>next[tj]→val ∧ head[ti] == old__Head ∧ tail[ti] == Tail<br>∧ next[ti] == head[ti]→next ∧ head[ti] ≠ tail[ti] ∧<br>pval[ti] == next[ti]→val ∧ Head == next[ti] ∧ next[ti]<br>≠ NULL)} | OK     |
| 30 | Subs | coop::cDeqF    | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{\exists \text{old\_Head}. (\text{old\_Head} \neq \text{NULL} \wedge$<br>Tail ≠ NULL ∧ (old__Head == Tail ∨ old__Head→next<br>≠ NULL) ∧ head[tj] == old__Head ∧ tail[tj] == Tail<br>∧ next[tj] == head[tj]→next ∧ head[tj] ≠ tail[tj]<br>∧ pval[tj] == next[tj]→val ∧ head[ti] == old__Head<br>∧ tail[ti] == Tail ∧ next[ti] == head[ti]→next ∧<br>head[ti] ≠ tail[ti] ∧ pval[ti] == next[ti]→val ∧ Head<br>== next[ti] ∧ next[ti] ≠ NULL)} => {!(next[tj] ≠ NULL ∧<br>Head == head[tj])}                                                                                                                                                                                                                                                                                                                                                                                                             | OK     |
| 31 | Atom | coop::cDeqFH.R | [int pval; tloc head = Head]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | OK     |
| 32 | RMov | coop::cDeqFH   | $\forall tj:\mathbb{N}. \text{int pval}; \text{head}[tj] = \text{Head} \triangleright \{(\text{Head} \neq \text{NULL} \wedge$<br>Tail ≠ NULL ∧ (Head == Tail ∨ Head→next ≠ NULL) ∧<br>head[tj] == Head)}                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | OK     |

| #  | Kind | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Result |
|----|------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 33 | HTrp | coop::cDeqFH     | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{(\text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL}) \wedge \text{head}[tj] == \text{Head}) \} \llbracket [\text{int } pval; \text{head}[ti] = \text{Head}; \text{tail}[ti] = \text{Tail}; \text{next}[ti] = \text{head}[ti] \rightarrow \text{next}; [\text{assume}(\text{Head} == \text{head}[ti]); \text{assume}(\text{head}[ti] \neq \text{tail}[ti]); pval[ti] = \text{next}[ti] \rightarrow \text{val}; \text{skip}]; \text{ICAS}(\text{next}[ti] \neq \text{NULL}, \text{Head}, \text{head}[ti], \text{next}[ti]); \text{return } pval[ti]] \{ \exists \text{old\_Head}. (\text{old\_Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{old\_Head} == \text{Tail} \vee \text{old\_Head} \rightarrow \text{next} \neq \text{NULL}) \wedge \text{head}[tj] == \text{old\_Head} \wedge \text{head}[ti] == \text{old\_Head} \wedge \text{tail}[ti] == \text{Tail} \wedge \text{next}[ti] == \text{head}[ti] \rightarrow \text{next} \wedge \text{head}[ti] \neq \text{tail}[ti] \wedge pval[ti] == \text{next}[ti] \rightarrow \text{val} \wedge \text{Head} == \text{next}[ti] \wedge \text{next}[ti] \neq \text{NULL}) \}$ | OK     |
| 34 | LMov | coop::cDeqFH     | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ \exists \text{old\_Head}. (\text{old\_Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{old\_Head} == \text{Tail} \vee \text{old\_Head} \rightarrow \text{next} \neq \text{NULL}) \wedge \text{head}[tj] == \text{old\_Head} \wedge \text{head}[ti] == \text{old\_Head} \wedge \text{tail}[ti] == \text{Tail} \wedge \text{next}[ti] == \text{head}[ti] \rightarrow \text{next} \wedge \text{head}[ti] \neq \text{tail}[ti] \wedge pval[ti] == \text{next}[ti] \rightarrow \text{val} \wedge \text{Head} == \text{next}[ti] \wedge \text{next}[ti] \neq \text{NULL}) \} \triangleleft \text{tail}[tj] = \text{Tail}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | OK     |
| 35 | LMov | coop::cDeqFH     | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ (\exists \text{old\_Head}. (\text{old\_Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{old\_Head} == \text{Tail} \vee \text{old\_Head} \rightarrow \text{next} \neq \text{NULL}) \wedge \text{head}[tj] == \text{old\_Head} \wedge \text{head}[ti] == \text{old\_Head} \wedge \text{tail}[ti] == \text{Tail} \wedge \text{next}[ti] == \text{head}[ti] \rightarrow \text{next} \wedge \text{head}[ti] \neq \text{tail}[ti] \wedge pval[ti] == \text{next}[ti] \rightarrow \text{val} \wedge \text{Head} == \text{next}[ti] \wedge \text{next}[ti] \neq \text{NULL}) \wedge \text{tail}[tj] == \text{Tail}) \} \triangleleft \text{next}[tj] = \text{head}[tj] \rightarrow \text{next}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | OK     |
| 36 | Subs | coop::cDeqFH     | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ (\exists \text{old\_Head}. (\text{old\_Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{old\_Head} == \text{Tail} \vee \text{old\_Head} \rightarrow \text{next} \neq \text{NULL}) \wedge \text{head}[tj] == \text{old\_Head} \wedge \text{head}[ti] == \text{old\_Head} \wedge \text{tail}[ti] == \text{Tail} \wedge \text{next}[ti] == \text{head}[ti] \rightarrow \text{next} \wedge \text{head}[ti] \neq \text{tail}[ti] \wedge pval[ti] == \text{next}[ti] \rightarrow \text{val} \wedge \text{Head} == \text{next}[ti] \wedge \text{next}[ti] \neq \text{NULL}) \wedge \text{tail}[tj] == \text{Tail} \wedge \text{next}[tj] == \text{head}[tj] \rightarrow \text{next}) \} \Rightarrow \{ \text{Head} \neq \text{head}[tj] \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | OK     |
| 37 | Atom | atom::aDeqEmpS.R | $[\text{int } pval; \text{tloc head} = \text{Head}]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | OK     |
| 38 | Atom | atom::aDeqEmpS.N | $[\text{assume}(\text{Head} == \text{head}); \text{assume}(\text{head} == \text{tail}); \text{assume}(\text{next} == \text{NULL})]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | OK     |
| 39 | RMov | atom::aDeqEmpS   | $\forall t: \mathbb{N}. \text{next}[t] = \text{head}[t] \rightarrow \text{next} \triangleright \{ (\text{Head} == \text{head}[t] \wedge \text{head}[t] == \text{tail}[t] \wedge \text{next}[t] == \text{NULL} \wedge \text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL})) \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | OK     |
| 40 | RMov | atom::aDeqEmpS   | $\forall t: \mathbb{N}. \text{tail}[t] = \text{Tail} \triangleright \{ (\text{Head} == \text{head}[t] \wedge \text{head}[t] == \text{tail}[t] \wedge \text{head}[t] \rightarrow \text{next} == \text{NULL} \wedge \text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL})) \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | OK     |
| 41 | RMov | atom::aDeqEmpS   | $\forall t: \mathbb{N}. \text{int } pval; \text{head}[t] = \text{Head} \triangleright \{ (\text{Head} == \text{head}[t] \wedge \text{head}[t] == \text{Tail} \wedge \text{head}[t] \rightarrow \text{next} == \text{NULL} \wedge \text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL})) \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | OK     |

| #  | Kind | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Result |
|----|------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 42 | HTrp | atom::aDeqEmpS   | $\forall t:\mathbb{N}. \{(Head == head[t] \wedge head[t] == tail[t] \wedge next[t] == NULL \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL))\} [assume(Head == head[t]); assume(head[t] == tail[t]); assume(next[t] == NULL)] \{(Head == head[t] \wedge head[t] == tail[t] \wedge next[t] == NULL \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL))\}$       | OK     |
| 43 | LMov | atom::aDeqEmpS   | $\forall t:\mathbb{N}. \{(Head == head[t] \wedge head[t] == tail[t] \wedge next[t] == NULL \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL))\} \triangleleft return \emptyset$                                                                                                                                                                                                                                     | OK     |
| 44 | Atom | atom::aDeqOneS.R | [int pval; tloc head = Head]                                                                                                                                                                                                                                                                                                                                                                                                                                             | OK     |
| 45 | Atom | atom::aDeqOneS.N | [assume(Head == head); assume(head == tail); assume(next $\neq$ NULL)]                                                                                                                                                                                                                                                                                                                                                                                                   | OK     |
| 46 | RMov | atom::aDeqOneS   | $\forall t:\mathbb{N}. next[t] = head[t] \rightarrow next \triangleright \{(Head == head[t] \wedge head[t] == tail[t] \wedge next[t] \neq NULL \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL))\}$                                                                                                                                                                                                                | OK     |
| 47 | RMov | atom::aDeqOneS   | $\forall t:\mathbb{N}. tail[t] = Tail \triangleright \{(Head == head[t] \wedge head[t] == tail[t] \wedge head[t] \rightarrow next \neq NULL \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL))\}$                                                                                                                                                                                                                   | OK     |
| 48 | RMov | atom::aDeqOneS   | $\forall t:\mathbb{N}. int pval; head[t] = Head \triangleright \{(Head == head[t] \wedge head[t] == Tail \wedge head[t] \rightarrow next \neq NULL \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL))\}$                                                                                                                                                                                                            | OK     |
| 49 | HTrp | atom::aDeqOneS   | $\forall t:\mathbb{N}. \{(Head == head[t] \wedge head[t] == tail[t] \wedge next[t] \neq NULL \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL))\} [assume(Head == head[t]); assume(head[t] == tail[t]); assume(next[t] \neq NULL)] \{(Head == head[t] \wedge head[t] == tail[t] \wedge next[t] \neq NULL \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL))\}$ | OK     |
| 50 | Atom | atom::aAdvPre.N  | [tloc next = tail $\rightarrow$ next; assume(next $\neq$ NULL)]                                                                                                                                                                                                                                                                                                                                                                                                          | OK     |
| 51 | RMov | atom::aAdvPre    | $\forall t:\mathbb{N}. tail[t] = Tail \triangleright \{(tail[t] \rightarrow next \neq NULL \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL))\}$                                                                                                                                                                                                                                                                    | FAIL   |
| 52 | HTrp | atom::aAdvPre    | $\forall t:\mathbb{N}. \{(tail[t] \rightarrow next \neq NULL \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL))\} [next[t] = tail[t] \rightarrow next; assume(next[t] \neq NULL)] \{(tail[t] \rightarrow next \neq NULL \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge next[t] == tail[t] \rightarrow next \wedge next[t] \neq NULL)\}$              | OK     |
| 53 | RMov | atom::aAdvS      | $\forall t:\mathbb{N}. tail[t] = Tail; next[t] = tail[t] \rightarrow next; assume(next[t] \neq NULL) \triangleright \{(next[t] \neq NULL \wedge Tail == tail[t] \wedge Head \neq NULL \wedge next[t] \neq NULL \wedge (Head == next[t] \vee Head \rightarrow next \neq NULL))\}$                                                                                                                                                                                         | OK     |
| 54 | HTrp | atom::aAdvS      | $\forall t:\mathbb{N}. \{(next[t] \neq NULL \wedge Tail == tail[t] \wedge Head \neq NULL \wedge next[t] \neq NULL \wedge (Head == next[t] \vee Head \rightarrow next \neq NULL))\} ICAS(next[t] \neq NULL, Tail, tail[t], next[t]) \{\exists old\_Tail. (next[t] \neq NULL \wedge old\_Tail == tail[t] \wedge Head \neq NULL \wedge next[t] \neq NULL \wedge (Head == next[t] \vee Head \rightarrow next \neq NULL) \wedge Tail == next[t])\}$                           | OK     |



| #  | Kind | Spec           | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Result |
|----|------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 55 | LMov | atom::aAdvS    | $\forall t:\mathbb{N}. \{\exists \text{old\_Tail}. (\text{next}[t] \neq \text{NULL} \wedge \text{old\_Tail} == \text{tail}[t] \wedge \text{Head} \neq \text{NULL} \wedge \text{next}[t] \neq \text{NULL} \wedge (\text{Head} == \text{next}[t] \vee \text{Head} \rightarrow \text{next} \neq \text{NULL})) \wedge \text{Tail} == \text{next}[t]\} \triangleleft \text{return } 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | OK     |
| 56 | RMov | coop::cAdvF    | $\forall tj:\mathbb{N}. \text{tail}[tj] = \text{Tail}; \text{next}[tj] = \text{tail}[tj] \rightarrow \text{next}; \text{assume}(\text{next}[tj] \neq \text{NULL}) \triangleright \{(\text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL})) \wedge \text{tail}[tj] == \text{Tail} \wedge \text{next}[tj] == \text{tail}[tj] \rightarrow \text{next} \wedge \text{next}[tj] \neq \text{NULL}\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | OK     |
| 57 | HTrp | coop::cAdvF    | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{(\text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL})) \wedge \text{tail}[tj] == \text{Tail} \wedge \text{next}[tj] == \text{tail}[tj] \rightarrow \text{next} \wedge \text{next}[tj] \neq \text{NULL}\} \llbracket \text{tail}[ti] = \text{Tail}; [\text{next}[ti] = \text{tail}[ti] \rightarrow \text{next}; \text{assume}(\text{next}[ti] \neq \text{NULL})]; \text{skip}; \text{ICAS}(\text{next}[ti] \neq \text{NULL}, \text{Tail}, \text{tail}[ti], \text{next}[ti]); \text{return } 0 \rrbracket \{\exists \text{old\_Tail}. (\text{Head} \neq \text{NULL} \wedge \text{old\_Tail} \neq \text{NULL} \wedge (\text{Head} == \text{old\_Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL})) \wedge \text{tail}[tj] == \text{old\_Tail} \wedge \text{next}[tj] == \text{tail}[tj] \rightarrow \text{next} \wedge \text{next}[tj] \neq \text{NULL} \wedge \text{tail}[ti] == \text{old\_Tail} \wedge \text{next}[ti] == \text{tail}[ti] \rightarrow \text{next} \wedge \text{next}[ti] \neq \text{NULL} \wedge \text{Tail} == \text{next}[ti])\}$            | OK     |
| 58 | Subs | coop::cAdvF    | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{\exists \text{old\_Tail}. (\text{Head} \neq \text{NULL} \wedge \text{old\_Tail} \neq \text{NULL} \wedge (\text{Head} == \text{old\_Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL})) \wedge \text{tail}[tj] == \text{old\_Tail} \wedge \text{next}[tj] == \text{tail}[tj] \rightarrow \text{next} \wedge \text{next}[tj] \neq \text{NULL} \wedge \text{tail}[ti] == \text{old\_Tail} \wedge \text{next}[ti] == \text{tail}[ti] \rightarrow \text{next} \wedge \text{next}[ti] \neq \text{NULL} \wedge \text{Tail} == \text{next}[ti])\} \Rightarrow \{!(\text{next}[tj] \neq \text{NULL} \wedge \text{Tail} == \text{tail}[tj])\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | OK     |
| 59 | Atom | coop::cAdvFT.R | $[\text{tloc node} = \text{new } \{0\} \text{ NULL}; \text{node} \rightarrow \text{val} = \text{val}; \text{tloc tail} \in \text{Tail}]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | OK     |
| 60 | RMov | coop::cAdvFT   | $\forall tj:\mathbb{N}. \text{node}[tj] = \text{tloc\_ea1\_tj}; \text{node}[tj] \rightarrow \text{val} = \text{val}[tj]; \text{tail}[tj] = \text{Tail} \triangleright \{(\text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL})) \wedge \text{node}[tj] == \text{tloc\_ea1\_tj} \wedge \text{node}[tj] \rightarrow \text{val} == \text{val}[tj] \wedge \text{tail}[tj] == \text{Tail})\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | OK     |
| 61 | HTrp | coop::cAdvFT   | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{(\text{Head} \neq \text{NULL} \wedge \text{Tail} \neq \text{NULL} \wedge (\text{Head} == \text{Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL})) \wedge \text{node}[tj] == \text{tloc\_ea1\_tj} \wedge \text{node}[tj] \rightarrow \text{val} == \text{val}[tj] \wedge \text{tail}[tj] == \text{Tail})\} \llbracket \text{tail}[ti] = \text{Tail}; [\text{next}[ti] = \text{tail}[ti] \rightarrow \text{next}; \text{assume}(\text{next}[ti] \neq \text{NULL})]; \text{skip}; \text{ICAS}(\text{next}[ti] \neq \text{NULL}, \text{Tail}, \text{tail}[ti], \text{next}[ti]); \text{return } 0 \rrbracket \{\exists \text{old\_Tail}. (\text{Head} \neq \text{NULL} \wedge \text{old\_Tail} \neq \text{NULL} \wedge (\text{Head} == \text{old\_Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL})) \wedge \text{node}[tj] == \text{tloc\_ea1\_tj} \wedge \text{node}[tj] \rightarrow \text{val} == \text{val}[tj] \wedge \text{tail}[tj] == \text{old\_Tail} \wedge \text{tail}[ti] == \text{old\_Tail} \wedge \text{next}[ti] == \text{tail}[ti] \rightarrow \text{next} \wedge \text{next}[ti] \neq \text{NULL} \wedge \text{Tail} == \text{next}[ti])\}$ | OK     |
| 62 | LMov | coop::cAdvFT   | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{\exists \text{old\_Tail}. (\text{Head} \neq \text{NULL} \wedge \text{old\_Tail} \neq \text{NULL} \wedge (\text{Head} == \text{old\_Tail} \vee \text{Head} \rightarrow \text{next} \neq \text{NULL})) \wedge \text{node}[tj] == \text{tloc\_ea1\_tj} \wedge \text{node}[tj] \rightarrow \text{val} == \text{val}[tj] \wedge \text{tail}[tj] == \text{old\_Tail} \wedge \text{tail}[ti] == \text{old\_Tail} \wedge \text{next}[ti] == \text{tail}[ti] \rightarrow \text{next} \wedge \text{next}[ti] \neq \text{NULL} \wedge \text{Tail} == \text{next}[ti])\} \triangleleft \text{next}[tj] = \text{tail}[tj] \rightarrow \text{next}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | OK     |

| #  | Kind | Spec           | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Result |
|----|------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 63 | Subs | coop::cAdvFT   | $\forall ti, tj: \mathbb{N}. ti \neq tj. \{(\exists old\_Tail. (Head \neq NULL \wedge old\_Tail \neq NULL \wedge (Head == old\_Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == old\_Tail \wedge tail[ti] == old\_Tail \wedge next[ti] == tail[ti] \rightarrow next \wedge next[ti] \neq NULL \wedge Tail == next[ti]) \wedge next[tj] == tail[tj] \rightarrow next)\} \Rightarrow \{tail[tj] \neq Tail\}$ | OK     |
| 64 | Atom | atom::aAdvRo.N | $[tloc\ next = tail \rightarrow next; assume(next == NULL)]$                                                                                                                                                                                                                                                                                                                                                                                                                                        | OK     |
| 65 | RMov | atom::aAdvRo   | $\forall t: \mathbb{N}. tail[t] = Tail \triangleright \{(tail[t] \rightarrow next == NULL \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL))\}$                                                                                                                                                                                                                                                                                                | FAIL   |
| 66 | HTrp | atom::aAdvRo   | $\forall t: \mathbb{N}. \{(tail[t] \rightarrow next == NULL \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL))\} [next[t] = tail[t] \rightarrow next; assume(next[t] == NULL)] \{(tail[t] \rightarrow next == NULL \wedge Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge next[t] == tail[t] \rightarrow next \wedge next[t] == NULL)\}$                                                | OK     |

### B.3.4 Automata (408).

coverage.

| Check                | Result |
|----------------------|--------|
| enqueue: COVERED     | OK     |
| dequeue: COVERED     | OK     |
| tailAdvance: COVERED | OK     |

non-overlapping.

1 all distinct pairs are non-overlapping

non-self-overlapping.

1 no spec self-overlaps

bijection part I: well-definedness.

| Check                                                         | Result |
|---------------------------------------------------------------|--------|
| aDeqEmpS: induction_pattern $\subseteq$ exactly_one_ais: OK   | OK     |
| aAdvS: induction_pattern $\subseteq$ exactly_one_ais: OK      | OK     |
| cDeqFH: induction_pattern $\subseteq$ exactly_one_ais: OK     | OK     |
| aEnqS: induction_pattern $\subseteq$ exactly_one_ais: OK      | OK     |
| aAdvRo: induction_pattern $\subseteq$ exactly_one_ais: OK     | OK     |
| aEnqPre: induction_pattern $\subseteq$ exactly_one_ais: OK    | OK     |
| aDeqPre: induction_pattern $\subseteq$ exactly_one_ais: OK    | OK     |
| cDeqF: induction_pattern $\subseteq$ exactly_one_ais: OK      | OK     |
| cEnqFStale: induction_pattern $\subseteq$ exactly_one_ais: OK | OK     |
| aAdvPre: induction_pattern $\subseteq$ exactly_one_ais: OK    | OK     |
| aDeqS: induction_pattern $\subseteq$ exactly_one_ais: OK      | OK     |

| Check                                                       | Result |
|-------------------------------------------------------------|--------|
| aDeqOneS: induction_pattern $\subseteq$ exactly_one_ais: OK | OK     |
| cAdvF: induction_pattern $\subseteq$ exactly_one_ais: OK    | OK     |
| cAdvFT: induction_pattern $\subseteq$ exactly_one_ais: OK   | OK     |
| cEnqF: induction_pattern $\subseteq$ exactly_one_ais: OK    | OK     |

*bijection part II: surjectivity.*

| Check                                                          | Result |
|----------------------------------------------------------------|--------|
| aAdvRo $\cap$ at_least_one_ais(aAdvPre) = $\emptyset$ : OK     | OK     |
| aAdvS $\subseteq$ at_most_one_ais(aAdvPre): OK                 | OK     |
| aDeqEmpS $\cap$ at_least_one_ais(aAdvPre) = $\emptyset$ : OK   | OK     |
| aDeqOneS $\cap$ at_least_one_ais(aAdvPre) = $\emptyset$ : OK   | OK     |
| aDeqPre $\cap$ at_least_one_ais(aAdvPre) = $\emptyset$ : OK    | OK     |
| aDeqS $\cap$ at_least_one_ais(aAdvPre) = $\emptyset$ : OK      | OK     |
| aEnqPre $\cap$ at_least_one_ais(aAdvPre) = $\emptyset$ : OK    | OK     |
| aEnqS $\cap$ at_least_one_ais(aAdvPre) = $\emptyset$ : OK      | OK     |
| cAdvF $\subseteq$ at_most_one_ais(aAdvPre): OK                 | OK     |
| cAdvFT $\subseteq$ at_most_one_ais(aAdvPre): OK                | OK     |
| cDeqF $\cap$ at_least_one_ais(aAdvPre) = $\emptyset$ : OK      | OK     |
| cDeqFH $\cap$ at_least_one_ais(aAdvPre) = $\emptyset$ : OK     | OK     |
| cEnqF $\cap$ at_least_one_ais(aAdvPre) = $\emptyset$ : OK      | OK     |
| cEnqFStale $\cap$ at_least_one_ais(aAdvPre) = $\emptyset$ : OK | OK     |
| aAdvPre $\cap$ at_least_one_ais(aAdvRo) = $\emptyset$ : OK     | OK     |
| aAdvS $\cap$ at_least_one_ais(aAdvRo) = $\emptyset$ : OK       | OK     |
| aDeqEmpS $\cap$ at_least_one_ais(aAdvRo) = $\emptyset$ : OK    | OK     |
| aDeqOneS $\cap$ at_least_one_ais(aAdvRo) = $\emptyset$ : OK    | OK     |
| aDeqPre $\cap$ at_least_one_ais(aAdvRo) = $\emptyset$ : OK     | OK     |
| aDeqS $\cap$ at_least_one_ais(aAdvRo) = $\emptyset$ : OK       | OK     |
| aEnqPre $\cap$ at_least_one_ais(aAdvRo) = $\emptyset$ : OK     | OK     |
| aEnqS $\cap$ at_least_one_ais(aAdvRo) = $\emptyset$ : OK       | OK     |
| cAdvF $\cap$ at_least_one_ais(aAdvRo) = $\emptyset$ : OK       | OK     |
| cAdvFT $\cap$ at_least_one_ais(aAdvRo) = $\emptyset$ : OK      | OK     |
| cDeqF $\cap$ at_least_one_ais(aAdvRo) = $\emptyset$ : OK       | OK     |
| cDeqFH $\cap$ at_least_one_ais(aAdvRo) = $\emptyset$ : OK      | OK     |
| cEnqF $\cap$ at_least_one_ais(aAdvRo) = $\emptyset$ : OK       | OK     |
| cEnqFStale $\cap$ at_least_one_ais(aAdvRo) = $\emptyset$ : OK  | OK     |
| aAdvRo $\cap$ at_least_one_ais(aAdvS) = $\emptyset$ : OK       | OK     |
| aDeqEmpS $\cap$ at_least_one_ais(aAdvS) = $\emptyset$ : OK     | OK     |
| aDeqOneS $\cap$ at_least_one_ais(aAdvS) = $\emptyset$ : OK     | OK     |
| aDeqPre $\cap$ at_least_one_ais(aAdvS) = $\emptyset$ : OK      | OK     |
| aDeqS $\cap$ at_least_one_ais(aAdvS) = $\emptyset$ : OK        | OK     |
| aEnqPre $\cap$ at_least_one_ais(aAdvS) = $\emptyset$ : OK      | OK     |
| aEnqS $\cap$ at_least_one_ais(aAdvS) = $\emptyset$ : OK        | OK     |
| cAdvF $\subseteq$ at_most_one_ais(aAdvS): OK                   | OK     |
| cAdvFT $\subseteq$ at_most_one_ais(aAdvS): OK                  | OK     |

| Check                                                            | Result |
|------------------------------------------------------------------|--------|
| $cDeqF \cap at\_least\_one\_ais(aAdvS) = \emptyset$ : OK         | OK     |
| $cDeqFH \cap at\_least\_one\_ais(aAdvS) = \emptyset$ : OK        | OK     |
| $cEnqF \cap at\_least\_one\_ais(aAdvS) = \emptyset$ : OK         | OK     |
| $cEnqFStale \cap at\_least\_one\_ais(aAdvS) = \emptyset$ : OK    | OK     |
| $aAdvPre \cap at\_least\_one\_ais(aDeqEmpS) = \emptyset$ : OK    | OK     |
| $aAdvRo \cap at\_least\_one\_ais(aDeqEmpS) = \emptyset$ : OK     | OK     |
| $aAdvS \cap at\_least\_one\_ais(aDeqEmpS) = \emptyset$ : OK      | OK     |
| $aDeqOneS \cap at\_least\_one\_ais(aDeqEmpS) = \emptyset$ : OK   | OK     |
| $aDeqPre \cap at\_least\_one\_ais(aDeqEmpS) = \emptyset$ : OK    | OK     |
| $aDeqS \cap at\_least\_one\_ais(aDeqEmpS) = \emptyset$ : OK      | OK     |
| $aEnqPre \cap at\_least\_one\_ais(aDeqEmpS) = \emptyset$ : OK    | OK     |
| $aEnqS \cap at\_least\_one\_ais(aDeqEmpS) = \emptyset$ : OK      | OK     |
| $cAdvF \cap at\_least\_one\_ais(aDeqEmpS) = \emptyset$ : OK      | OK     |
| $cAdvFT \cap at\_least\_one\_ais(aDeqEmpS) = \emptyset$ : OK     | OK     |
| $cDeqF \cap at\_least\_one\_ais(aDeqEmpS) = \emptyset$ : OK      | OK     |
| $cDeqFH \cap at\_least\_one\_ais(aDeqEmpS) = \emptyset$ : OK     | OK     |
| $cEnqF \cap at\_least\_one\_ais(aDeqEmpS) = \emptyset$ : OK      | OK     |
| $cEnqFStale \cap at\_least\_one\_ais(aDeqEmpS) = \emptyset$ : OK | OK     |
| $aAdvPre \cap at\_least\_one\_ais(aDeqOneS) = \emptyset$ : OK    | OK     |
| $aAdvRo \cap at\_least\_one\_ais(aDeqOneS) = \emptyset$ : OK     | OK     |
| $aAdvS \cap at\_least\_one\_ais(aDeqOneS) = \emptyset$ : OK      | OK     |
| $aDeqEmpS \cap at\_least\_one\_ais(aDeqOneS) = \emptyset$ : OK   | OK     |
| $aDeqPre \cap at\_least\_one\_ais(aDeqOneS) = \emptyset$ : OK    | OK     |
| $aDeqS \cap at\_least\_one\_ais(aDeqOneS) = \emptyset$ : OK      | OK     |
| $aEnqPre \cap at\_least\_one\_ais(aDeqOneS) = \emptyset$ : OK    | OK     |
| $aEnqS \cap at\_least\_one\_ais(aDeqOneS) = \emptyset$ : OK      | OK     |
| $cAdvF \cap at\_least\_one\_ais(aDeqOneS) = \emptyset$ : OK      | OK     |
| $cAdvFT \cap at\_least\_one\_ais(aDeqOneS) = \emptyset$ : OK     | OK     |
| $cDeqF \cap at\_least\_one\_ais(aDeqOneS) = \emptyset$ : OK      | OK     |
| $cDeqFH \cap at\_least\_one\_ais(aDeqOneS) = \emptyset$ : OK     | OK     |
| $cEnqF \cap at\_least\_one\_ais(aDeqOneS) = \emptyset$ : OK      | OK     |
| $cEnqFStale \cap at\_least\_one\_ais(aDeqOneS) = \emptyset$ : OK | OK     |
| $aAdvPre \cap at\_least\_one\_ais(aDeqPre) = \emptyset$ : OK     | OK     |
| $aAdvRo \cap at\_least\_one\_ais(aDeqPre) = \emptyset$ : OK      | OK     |
| $aAdvS \cap at\_least\_one\_ais(aDeqPre) = \emptyset$ : OK       | OK     |
| $aDeqEmpS \cap at\_least\_one\_ais(aDeqPre) = \emptyset$ : OK    | OK     |
| $aDeqOneS \cap at\_least\_one\_ais(aDeqPre) = \emptyset$ : OK    | OK     |
| $aDeqS \subseteq at\_most\_one\_ais(aDeqPre)$ : OK               | OK     |
| $aEnqPre \cap at\_least\_one\_ais(aDeqPre) = \emptyset$ : OK     | OK     |
| $aEnqS \cap at\_least\_one\_ais(aDeqPre) = \emptyset$ : OK       | OK     |
| $cAdvF \cap at\_least\_one\_ais(aDeqPre) = \emptyset$ : OK       | OK     |
| $cAdvFT \cap at\_least\_one\_ais(aDeqPre) = \emptyset$ : OK      | OK     |
| $cDeqF \subseteq at\_most\_one\_ais(aDeqPre)$ : OK               | OK     |
| $cDeqFH \subseteq at\_most\_one\_ais(aDeqPre)$ : OK              | OK     |
| $cEnqF \cap at\_least\_one\_ais(aDeqPre) = \emptyset$ : OK       | OK     |
| $cEnqFStale \cap at\_least\_one\_ais(aDeqPre) = \emptyset$ : OK  | OK     |

| Check                                                         | Result |
|---------------------------------------------------------------|--------|
| $aAdvPre \cap at\_least\_one\_ais(aDeqS) = \emptyset$ : OK    | OK     |
| $aAdvRo \cap at\_least\_one\_ais(aDeqS) = \emptyset$ : OK     | OK     |
| $aAdvS \cap at\_least\_one\_ais(aDeqS) = \emptyset$ : OK      | OK     |
| $aDeqEmpS \cap at\_least\_one\_ais(aDeqS) = \emptyset$ : OK   | OK     |
| $aDeqOneS \cap at\_least\_one\_ais(aDeqS) = \emptyset$ : OK   | OK     |
| $aEnqPre \cap at\_least\_one\_ais(aDeqS) = \emptyset$ : OK    | OK     |
| $aEnqS \cap at\_least\_one\_ais(aDeqS) = \emptyset$ : OK      | OK     |
| $cAdvF \cap at\_least\_one\_ais(aDeqS) = \emptyset$ : OK      | OK     |
| $cAdvFT \cap at\_least\_one\_ais(aDeqS) = \emptyset$ : OK     | OK     |
| $cDeqF \subseteq at\_most\_one\_ais(aDeqS)$ : OK              | OK     |
| $cDeqFH \subseteq at\_most\_one\_ais(aDeqS)$ : OK             | OK     |
| $cEnqF \cap at\_least\_one\_ais(aDeqS) = \emptyset$ : OK      | OK     |
| $cEnqFStale \cap at\_least\_one\_ais(aDeqS) = \emptyset$ : OK | OK     |
| $aAdvPre \cap at\_least\_one\_ais(aEnqPre) = \emptyset$ : OK  | OK     |
| $aAdvRo \cap at\_least\_one\_ais(aEnqPre) = \emptyset$ : OK   | OK     |
| $aAdvS \cap at\_least\_one\_ais(aEnqPre) = \emptyset$ : OK    | OK     |
| $aDeqEmpS \cap at\_least\_one\_ais(aEnqPre) = \emptyset$ : OK | OK     |
| $aDeqOneS \cap at\_least\_one\_ais(aEnqPre) = \emptyset$ : OK | OK     |
| $aDeqPre \cap at\_least\_one\_ais(aEnqPre) = \emptyset$ : OK  | OK     |
| $aDeqS \cap at\_least\_one\_ais(aEnqPre) = \emptyset$ : OK    | OK     |
| $aEnqS \subseteq at\_most\_one\_ais(aEnqPre)$ : OK            | OK     |
| $cAdvF \cap at\_least\_one\_ais(aEnqPre) = \emptyset$ : OK    | OK     |
| $cAdvFT \cap at\_least\_one\_ais(aEnqPre) = \emptyset$ : OK   | OK     |
| $cDeqF \cap at\_least\_one\_ais(aEnqPre) = \emptyset$ : OK    | OK     |
| $cDeqFH \cap at\_least\_one\_ais(aEnqPre) = \emptyset$ : OK   | OK     |
| $cEnqF \subseteq at\_most\_one\_ais(aEnqPre)$ : OK            | OK     |
| $cEnqFStale \subseteq at\_most\_one\_ais(aEnqPre)$ : OK       | OK     |
| $aAdvPre \cap at\_least\_one\_ais(aEnqS) = \emptyset$ : OK    | OK     |
| $aAdvRo \cap at\_least\_one\_ais(aEnqS) = \emptyset$ : OK     | OK     |
| $aAdvS \cap at\_least\_one\_ais(aEnqS) = \emptyset$ : OK      | OK     |
| $aDeqEmpS \cap at\_least\_one\_ais(aEnqS) = \emptyset$ : OK   | OK     |
| $aDeqOneS \cap at\_least\_one\_ais(aEnqS) = \emptyset$ : OK   | OK     |
| $aDeqPre \cap at\_least\_one\_ais(aEnqS) = \emptyset$ : OK    | OK     |
| $aDeqS \cap at\_least\_one\_ais(aEnqS) = \emptyset$ : OK      | OK     |
| $cAdvF \cap at\_least\_one\_ais(aEnqS) = \emptyset$ : OK      | OK     |
| $cAdvFT \cap at\_least\_one\_ais(aEnqS) = \emptyset$ : OK     | OK     |
| $cDeqF \cap at\_least\_one\_ais(aEnqS) = \emptyset$ : OK      | OK     |
| $cDeqFH \cap at\_least\_one\_ais(aEnqS) = \emptyset$ : OK     | OK     |
| $cEnqF \subseteq at\_most\_one\_ais(aEnqS)$ : OK              | OK     |
| $cEnqFStale \subseteq at\_most\_one\_ais(aEnqS)$ : OK         | OK     |
| $aAdvRo \cap at\_least\_one\_ais(cAdvF) = \emptyset$ : OK     | OK     |
| $aDeqEmpS \cap at\_least\_one\_ais(cAdvF) = \emptyset$ : OK   | OK     |
| $aDeqOneS \cap at\_least\_one\_ais(cAdvF) = \emptyset$ : OK   | OK     |
| $aDeqPre \cap at\_least\_one\_ais(cAdvF) = \emptyset$ : OK    | OK     |
| $aDeqS \cap at\_least\_one\_ais(cAdvF) = \emptyset$ : OK      | OK     |
| $aEnqPre \cap at\_least\_one\_ais(cAdvF) = \emptyset$ : OK    | OK     |

| Check                                                          | Result |
|----------------------------------------------------------------|--------|
| $aEnqS \cap at\_least\_one\_ais(cAdvF) = \emptyset$ : OK       | OK     |
| $cAdvFT \cap at\_least\_one\_ais(cAdvF) = \emptyset$ : OK      | OK     |
| $cDeqF \cap at\_least\_one\_ais(cAdvF) = \emptyset$ : OK       | OK     |
| $cDeqFH \cap at\_least\_one\_ais(cAdvF) = \emptyset$ : OK      | OK     |
| $cEnqF \cap at\_least\_one\_ais(cAdvF) = \emptyset$ : OK       | OK     |
| $cEnqFStale \cap at\_least\_one\_ais(cAdvF) = \emptyset$ : OK  | OK     |
| $aAdvRo \cap at\_least\_one\_ais(cAdvFT) = \emptyset$ : OK     | OK     |
| $aDeqEmpS \cap at\_least\_one\_ais(cAdvFT) = \emptyset$ : OK   | OK     |
| $aDeqOneS \cap at\_least\_one\_ais(cAdvFT) = \emptyset$ : OK   | OK     |
| $aDeqPre \cap at\_least\_one\_ais(cAdvFT) = \emptyset$ : OK    | OK     |
| $aDeqS \cap at\_least\_one\_ais(cAdvFT) = \emptyset$ : OK      | OK     |
| $aEnqPre \cap at\_least\_one\_ais(cAdvFT) = \emptyset$ : OK    | OK     |
| $aEnqS \cap at\_least\_one\_ais(cAdvFT) = \emptyset$ : OK      | OK     |
| $cAdvF \cap at\_least\_one\_ais(cAdvFT) = \emptyset$ : OK      | OK     |
| $cDeqF \cap at\_least\_one\_ais(cAdvFT) = \emptyset$ : OK      | OK     |
| $cDeqFH \cap at\_least\_one\_ais(cAdvFT) = \emptyset$ : OK     | OK     |
| $cEnqF \cap at\_least\_one\_ais(cAdvFT) = \emptyset$ : OK      | OK     |
| $cEnqFStale \cap at\_least\_one\_ais(cAdvFT) = \emptyset$ : OK | OK     |
| $aAdvPre \cap at\_least\_one\_ais(cDeqF) = \emptyset$ : OK     | OK     |
| $aAdvRo \cap at\_least\_one\_ais(cDeqF) = \emptyset$ : OK      | OK     |
| $aAdvS \cap at\_least\_one\_ais(cDeqF) = \emptyset$ : OK       | OK     |
| $aDeqEmpS \cap at\_least\_one\_ais(cDeqF) = \emptyset$ : OK    | OK     |
| $aDeqOneS \cap at\_least\_one\_ais(cDeqF) = \emptyset$ : OK    | OK     |
| $aEnqPre \cap at\_least\_one\_ais(cDeqF) = \emptyset$ : OK     | OK     |
| $aEnqS \cap at\_least\_one\_ais(cDeqF) = \emptyset$ : OK       | OK     |
| $cAdvF \cap at\_least\_one\_ais(cDeqF) = \emptyset$ : OK       | OK     |
| $cAdvFT \cap at\_least\_one\_ais(cDeqF) = \emptyset$ : OK      | OK     |
| $cDeqFH \cap at\_least\_one\_ais(cDeqF) = \emptyset$ : OK      | OK     |
| $cEnqF \cap at\_least\_one\_ais(cDeqF) = \emptyset$ : OK       | OK     |
| $cEnqFStale \cap at\_least\_one\_ais(cDeqF) = \emptyset$ : OK  | OK     |
| $aAdvPre \cap at\_least\_one\_ais(cDeqFH) = \emptyset$ : OK    | OK     |
| $aAdvRo \cap at\_least\_one\_ais(cDeqFH) = \emptyset$ : OK     | OK     |
| $aAdvS \cap at\_least\_one\_ais(cDeqFH) = \emptyset$ : OK      | OK     |
| $aDeqEmpS \cap at\_least\_one\_ais(cDeqFH) = \emptyset$ : OK   | OK     |
| $aDeqOneS \cap at\_least\_one\_ais(cDeqFH) = \emptyset$ : OK   | OK     |
| $aEnqPre \cap at\_least\_one\_ais(cDeqFH) = \emptyset$ : OK    | OK     |
| $aEnqS \cap at\_least\_one\_ais(cDeqFH) = \emptyset$ : OK      | OK     |
| $cAdvF \cap at\_least\_one\_ais(cDeqFH) = \emptyset$ : OK      | OK     |
| $cAdvFT \cap at\_least\_one\_ais(cDeqFH) = \emptyset$ : OK     | OK     |
| $cDeqF \cap at\_least\_one\_ais(cDeqFH) = \emptyset$ : OK      | OK     |
| $cEnqF \cap at\_least\_one\_ais(cDeqFH) = \emptyset$ : OK      | OK     |
| $cEnqFStale \cap at\_least\_one\_ais(cDeqFH) = \emptyset$ : OK | OK     |
| $aAdvPre \cap at\_least\_one\_ais(cEnqF) = \emptyset$ : OK     | OK     |
| $aAdvRo \cap at\_least\_one\_ais(cEnqF) = \emptyset$ : OK      | OK     |
| $aAdvS \cap at\_least\_one\_ais(cEnqF) = \emptyset$ : OK       | OK     |
| $aDeqEmpS \cap at\_least\_one\_ais(cEnqF) = \emptyset$ : OK    | OK     |

| Check                                                            | Result |
|------------------------------------------------------------------|--------|
| $aDeqOneS \cap at\_least\_one\_ais(cEngF) = \emptyset$ : OK      | OK     |
| $aDeqPre \cap at\_least\_one\_ais(cEngF) = \emptyset$ : OK       | OK     |
| $aDeqS \cap at\_least\_one\_ais(cEngF) = \emptyset$ : OK         | OK     |
| $cAdvF \cap at\_least\_one\_ais(cEngF) = \emptyset$ : OK         | OK     |
| $cAdvFT \cap at\_least\_one\_ais(cEngF) = \emptyset$ : OK        | OK     |
| $cDeqF \cap at\_least\_one\_ais(cEngF) = \emptyset$ : OK         | OK     |
| $cDeqFH \cap at\_least\_one\_ais(cEngF) = \emptyset$ : OK        | OK     |
| $cEngFStale \cap at\_least\_one\_ais(cEngF) = \emptyset$ : OK    | OK     |
| $aAdvPre \cap at\_least\_one\_ais(cEngFStale) = \emptyset$ : OK  | OK     |
| $aAdvRo \cap at\_least\_one\_ais(cEngFStale) = \emptyset$ : OK   | OK     |
| $aAdvS \cap at\_least\_one\_ais(cEngFStale) = \emptyset$ : OK    | OK     |
| $aDeqEmpS \cap at\_least\_one\_ais(cEngFStale) = \emptyset$ : OK | OK     |
| $aDeqOneS \cap at\_least\_one\_ais(cEngFStale) = \emptyset$ : OK | OK     |
| $aDeqPre \cap at\_least\_one\_ais(cEngFStale) = \emptyset$ : OK  | OK     |
| $aDeqS \cap at\_least\_one\_ais(cEngFStale) = \emptyset$ : OK    | OK     |
| $cAdvF \cap at\_least\_one\_ais(cEngFStale) = \emptyset$ : OK    | OK     |
| $cAdvFT \cap at\_least\_one\_ais(cEngFStale) = \emptyset$ : OK   | OK     |
| $cDeqF \cap at\_least\_one\_ais(cEngFStale) = \emptyset$ : OK    | OK     |
| $cDeqFH \cap at\_least\_one\_ais(cEngFStale) = \emptyset$ : OK   | OK     |
| $cEngF \cap at\_least\_one\_ais(cEngFStale) = \emptyset$ : OK    | OK     |

*bijjective support (IRSspecs).*

```

1aDeqEmpS: {da8, db7t}
2aAdvS: {ta5, tb3t, tcas4t}
3aEngS: {ea8, eb5t, eb6t, ecas7t}
4aAdvRo: {tb3f}
5aEngPre: {eb5t, eb6t}
6aDeqPre: {da9, db6f}
7aAdvPre: {tb3t}
8aDeqS: {da11, da9, db6f, dcas10t}
9aDeqOneS: {db7f}
10cEngF: {ea8, eb5t, eb6t, ecas7t}
11cEngFStale: {ea8, eb5t, eb6t, ecas7t}
12cDeqF: {da11, da9, db6f, dcas10t}
13cDeqFH: {da11, da9, db6f, dcas10t}
14cAdvF: {ta5, tb3t, tcas4t}
15cAdvFT: {ta5, tb3t, tcas4t}

```

### B.3.5 Interference Stability (176).

| # | Kind  | Spec        | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Result |
|---|-------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1 | ISTbl | coop::cEnqF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$<br>$int\ pval\ \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$                            | OK     |
| 2 | ISTbl | coop::cEnqF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$<br>$return\ pval[tk]\ \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$                     | OK     |
| 3 | ISTbl | coop::cEnqF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$<br>$head[tk] = Head\ \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$                      | OK     |
| 4 | ISTbl | coop::cEnqF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$<br>$tail[tk] = Tail\ \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$                      | OK     |
| 5 | ISTbl | coop::cEnqF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$<br>$next[tk] = head[tk] \rightarrow next\ \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$ | OK     |
| 6 | ISTbl | coop::cEnqF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$<br>$return\ 0\ \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$                            | OK     |



| #  | Kind  | Spec        | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Result |
|----|-------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 7  | IStbl | coop::cEnqF | $\forall t_j, tk : \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tlloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL)\}$ <p>pval[tk] = next[tk] → val { (Head ≠ NULL ∧ Tail ≠ NULL ∧ (Head == Tail ∨ Head-&gt;next ≠ NULL) ∧ node[tj] == tlloc_ea1_tj ∧ node[tj]-&gt;val == val[tj] ∧ tail[tj] == Tail ∧ next[tj] == tail[tj]-&gt;next ∧ next[tj] == NULL)}</p>   | OK     |
| 8  | IStbl | coop::cEnqF | $\forall t_j, tk : \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tlloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL)\}$ <p>assume(Head ≠ head[tk]) {(Head ≠ NULL ∧ Tail ≠ NULL ∧ (Head == Tail ∨ Head-&gt;next ≠ NULL) ∧ node[tj] == tlloc_ea1_tj ∧ node[tj]-&gt;val == val[tj] ∧ tail[tj] == Tail ∧ next[tj] == tail[tj]-&gt;next ∧ next[tj] == NULL)}</p>      | OK     |
| 9  | IStbl | coop::cEnqF | $\forall t_j, tk : \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tlloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL)\}$ <p>assume(Head == head[tk]) {(Head ≠ NULL ∧ Tail ≠ NULL ∧ (Head == Tail ∨ Head-&gt;next ≠ NULL) ∧ node[tj] == tlloc_ea1_tj ∧ node[tj]-&gt;val == val[tj] ∧ tail[tj] == Tail ∧ next[tj] == tail[tj]-&gt;next ∧ next[tj] == NULL)}</p>     | OK     |
| 10 | IStbl | coop::cEnqF | $\forall t_j, tk : \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tlloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL)\}$ <p>assume(head[tk] ≠ tail[tk]) {(Head ≠ NULL ∧ Tail ≠ NULL ∧ (Head == Tail ∨ Head-&gt;next ≠ NULL) ∧ node[tj] == tlloc_ea1_tj ∧ node[tj]-&gt;val == val[tj] ∧ tail[tj] == Tail ∧ next[tj] == tail[tj]-&gt;next ∧ next[tj] == NULL)}</p>  | OK     |
| 11 | IStbl | coop::cEnqF | $\forall t_j, tk : \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tlloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL)\}$ <p>assume(head[tk] == tail[tk]) {(Head ≠ NULL ∧ Tail ≠ NULL ∧ (Head == Tail ∨ Head-&gt;next ≠ NULL) ∧ node[tj] == tlloc_ea1_tj ∧ node[tj]-&gt;val == val[tj] ∧ tail[tj] == Tail ∧ next[tj] == tail[tj]-&gt;next ∧ next[tj] == NULL)}</p> | OK     |
| 12 | IStbl | coop::cEnqF | $\forall t_j, tk : \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tlloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL)\}$ <p>assume(next[tk] ≠ NULL) {(Head ≠ NULL ∧ Tail ≠ NULL ∧ (Head == Tail ∨ Head-&gt;next ≠ NULL) ∧ node[tj] == tlloc_ea1_tj ∧ node[tj]-&gt;val == val[tj] ∧ tail[tj] == Tail ∧ next[tj] == tail[tj]-&gt;next ∧ next[tj] == NULL)}</p>      | OK     |

| #  | Kind  | Spec        | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Result |
|----|-------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 13 | ISTbl | coop::cEnqF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$<br>$assume(next[tk] == NULL) \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$                              | OK     |
| 14 | ISTbl | coop::cEnqF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$<br>$assume(!(next[tk] \neq NULL \wedge Head == head[tk])) \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$ | OK     |
| 15 | ISTbl | coop::cEnqF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$<br>$ICAS(next[tk] \neq NULL, Head, head[tk], next[tk]) \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$    | OK     |
| 16 | ISTbl | coop::cEnqF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$<br>$node[tk] = tloc\_ea1\_tk \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$                              | OK     |
| 17 | ISTbl | coop::cEnqF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$<br>$node[tk] \rightarrow val = val[tk] \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$                    | OK     |
| 18 | ISTbl | coop::cEnqF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$<br>$tail[tk] = Tail \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] == NULL) \}$                                       | OK     |

[illegible]



| #  | Kind  | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Result |
|----|-------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 31 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$ return pval[tk] $\{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$                      | OK     |
| 32 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$ head[tk] = Head $\{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$                      | OK     |
| 33 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$ tail[tk] = Tail $\{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$                      | OK     |
| 34 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$ next[tk] = head[tk] $\rightarrow next \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$ | OK     |
| 35 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$ return 0 $\{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$                             | OK     |
| 36 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$ pval[tk] = next[tk] $\rightarrow val \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$  | OK     |
| 37 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$ assume(Head $\neq$ head[tk]) $\{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$         | OK     |

| #  | Kind  | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Result |
|----|-------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 38 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_t_j \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ assume( $Head == head[tk]$ ) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_t_j \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$                              | OK     |
| 39 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_t_j \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ assume( $head[tk] \neq tail[tk]$ ) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_t_j \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$                        | OK     |
| 40 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_t_j \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ assume( $head[tk] == tail[tk]$ ) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_t_j \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$                          | OK     |
| 41 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_t_j \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ assume( $next[tk] \neq NULL$ ) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_t_j \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$                            | OK     |
| 42 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_t_j \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ assume( $next[tk] == NULL$ ) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_t_j \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$                              | OK     |
| 43 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_t_j \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ assume( $!(next[tk] \neq NULL \wedge Head == head[tk])$ ) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_t_j \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ | OK     |
| 44 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_t_j \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ ICAS( $next[tk] \neq NULL, Head, head[tk], next[tk]$ ) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_t_j \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$    | OK     |

| #  | Kind  | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Result |
|----|-------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 45 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ node[tk] = tloc_ea1_tk $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$                                                                                 | OK     |
| 46 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ node[tk] $\rightarrow val = val[tk] \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$                                                                     | OK     |
| 47 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ tail[tk] = Tail $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$                                                                                        | OK     |
| 48 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ next[tk] = tail[tk] $\rightarrow next \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$                                                                   | OK     |
| 49 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ assume(tail[tk] $\neq Tail$ ) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$                                                                          | OK     |
| 50 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ assume(next[tk] $\neq NULL$ ) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$                                                                          | OK     |
| 51 | IStbl | coop::cEnqFStale | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ assume(! (node[tk] $\neq NULL \wedge Tail == tail[tk] \wedge tail[tk] \rightarrow next == next[tk]$ )) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ | OK     |

| #  | Kind  | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Result |
|----|-------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 52 | IStbl | coop::cEnqFStale | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\} tail[tk] = Tail \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\}$                                       | OK     |
| 53 | IStbl | coop::cEnqFStale | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\} next[tk] = tail[tk] \rightarrow next \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\}$                  | OK     |
| 54 | IStbl | coop::cEnqFStale | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\} return \emptyset \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\}$                                      | OK     |
| 55 | IStbl | coop::cEnqFStale | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\} assume(next[tk] == NULL) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\}$                              | OK     |
| 56 | IStbl | coop::cEnqFStale | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\} assume(next[tk] \neq NULL) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\}$                            | OK     |
| 57 | IStbl | coop::cEnqFStale | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\} assume(!(next[tk] \neq NULL \wedge Tail == tail[tk])) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\}$ | OK     |
| 58 | IStbl | coop::cEnqFStale | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\} ICAS(next[tk] \neq NULL, Tail, tail[tk], next[tk]) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\}$    | OK     |



[illegible]



[illegible]



[illegible]

| #   | Kind  | Spec         | Obligation                                                                                                                                                                                                                                                                                                                                                       | Result |
|-----|-------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 90  | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$tail[tk] = Tail \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                                       | OK     |
| 91  | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$next[tk] = head[tk] \rightarrow next \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                  | OK     |
| 92  | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$return 0 \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                                              | OK     |
| 93  | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$assume(Head \neq head[tk]) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                            | OK     |
| 94  | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$assume(Head == head[tk]) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                              | OK     |
| 95  | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$assume(head[tk] == tail[tk]) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                          | OK     |
| 96  | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$assume(next[tk] \neq NULL) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                            | OK     |
| 97  | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$assume(next[tk] == NULL) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                              | OK     |
| 98  | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$assume(!(next[tk] \neq NULL \wedge Head == head[tk])) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$ | OK     |
| 99  | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$node[tk] = tloc\_ea1\_tk \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                              | OK     |
| 100 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$node[tk] \rightarrow val = val[tk] \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                    | OK     |

| #   | Kind  | Spec         | Obligation                                                                                                                                                                                                                                                                                                                                                                                                    | Result |
|-----|-------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 101 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$tail[tk] = Tail \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                                                                                    | OK     |
| 102 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$next[tk] = tail[tk] \rightarrow next \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                                                               | OK     |
| 103 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$return \emptyset \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                                                                                   | OK     |
| 104 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$assume(tail[tk] \neq Tail) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                                                                         | OK     |
| 105 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$assume(tail[tk] == Tail) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                                                                           | OK     |
| 106 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$assume(next[tk] \neq NULL) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                                                                         | OK     |
| 107 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$assume(next[tk] == NULL) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                                                                           | OK     |
| 108 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$assume(!(node[tk] \neq NULL \wedge Tail == tail[tk] \wedge tail[tk] \rightarrow next == next[tk])) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$ | OK     |
| 109 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$ICAS(node[tk] \neq NULL \wedge Tail == tail[tk], tail[tk] \rightarrow next, next[tk], node[tk]) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$    | OK     |
| 110 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$tail[tk] = Tail \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                                                                                    | OK     |
| 111 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>$next[tk] = tail[tk] \rightarrow next \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                                                               | OK     |

| #   | Kind  | Spec         | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Result |
|-----|-------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 112 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>return 0 $\{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                                                                                                                                                      | OK     |
| 113 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>assume(next[tk] == NULL) $\{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                                                                                                                                      | OK     |
| 114 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>assume(next[tk] $\neq$ NULL) $\{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                                                                                                                                  | OK     |
| 115 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>assume(! (next[tk] $\neq$ NULL $\wedge$ Tail == tail[tk])) $\{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                                                                                                    | OK     |
| 116 | IStbl | coop::cDeqFH | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$<br>ICAS(next[tk] $\neq$ NULL, Tail, tail[tk], next[tk]) $\{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge head[tj] == Head)\}$                                                                                                          | OK     |
| 117 | IStbl | coop::cAdvF  | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] \neq NULL)\}$ int<br>pval $\{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] \neq NULL)\}$        | OK     |
| 118 | IStbl | coop::cAdvF  | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] \neq NULL)\}$ return<br>pval[tk] $\{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] \neq NULL)\}$ | OK     |
| 119 | IStbl | coop::cAdvF  | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] \neq NULL)\}$ head[tk]<br>= Head $\{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] \neq NULL)\}$ | OK     |
| 120 | IStbl | coop::cAdvF  | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] \neq NULL)\}$ tail[tk]<br>= Tail $\{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] \neq NULL)\}$ | OK     |



[illegible]

| #   | Kind  | Spec        | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Result |
|-----|-------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 129 | IStbl | coop::cAdvF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] \neq NULL) \}$<br>assume( $next[tk] == NULL$ ) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] \neq NULL) \}$                              | OK     |
| 130 | IStbl | coop::cAdvF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] \neq NULL) \}$<br>assume( $!(next[tk] \neq NULL \wedge Head == head[tk])$ ) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] \neq NULL) \}$ | OK     |
| 131 | IStbl | coop::cAdvF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] \neq NULL) \}$<br>ICAS( $next[tk] \neq NULL, Head, head[tk], next[tk]$ ) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] \neq NULL) \}$    | OK     |
| 132 | IStbl | coop::cAdvF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] \neq NULL) \}$<br>node[tk] = tloc_ea1_tk $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] \neq NULL) \}$                                    | OK     |
| 133 | IStbl | coop::cAdvF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] \neq NULL) \}$<br>node[tk] $\rightarrow val = val[tk]$ $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] \neq NULL) \}$                      | OK     |
| 134 | IStbl | coop::cAdvF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] \neq NULL) \}$<br>tail[tk] = Tail $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] \neq NULL) \}$                                           | OK     |
| 135 | IStbl | coop::cAdvF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] \neq NULL) \}$<br>next[tk] = tail[tk] $\rightarrow next$ $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[t_j] == Tail \wedge next[t_j] == tail[t_j] \rightarrow next \wedge next[t_j] \neq NULL) \}$                    | OK     |

[illegible]

| #   | Kind  | Spec         | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Result |
|-----|-------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 143 | IStbl | coop::cAdvF  | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] \neq NULL)\} tail[tk] = Tail \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] \neq NULL)\}$                                       | OK     |
| 144 | IStbl | coop::cAdvF  | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] \neq NULL)\} next[tk] = tail[tk] \rightarrow next \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] \neq NULL)\}$                  | OK     |
| 145 | IStbl | coop::cAdvF  | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] \neq NULL)\} assume(next[tk] == NULL) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] \neq NULL)\}$                              | OK     |
| 146 | IStbl | coop::cAdvF  | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] \neq NULL)\} assume(!(next[tk] \neq NULL \wedge Tail == tail[tk])) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge tail[tj] == Tail \wedge next[tj] == tail[tj] \rightarrow next \wedge next[tj] \neq NULL)\}$ | OK     |
| 147 | IStbl | coop::cAdvFT | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\} int pval \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\}$                                    | OK     |
| 148 | IStbl | coop::cAdvFT | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\} return pval[tk] \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\}$                             | OK     |
| 149 | IStbl | coop::cAdvFT | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\} head[tk] = Head \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\}$                             | OK     |
| 150 | IStbl | coop::cAdvFT | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\} tail[tk] = Tail \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail)\}$                             | OK     |

| #   | Kind  | Spec         | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Result |
|-----|-------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 151 | IStbl | coop::cAdvFT | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ next[tk] = head[tk] $\rightarrow$ next $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ | OK     |
| 152 | IStbl | coop::cAdvFT | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ return 0 $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$                               | OK     |
| 153 | IStbl | coop::cAdvFT | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ pval[tk] = next[tk] $\rightarrow$ val $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$  | OK     |
| 154 | IStbl | coop::cAdvFT | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ assume(Head $\neq$ head[tk]) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$           | OK     |
| 155 | IStbl | coop::cAdvFT | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ assume(Head == head[tk]) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$               | OK     |
| 156 | IStbl | coop::cAdvFT | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ assume(head[tk] $\neq$ tail[tk]) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$       | OK     |
| 157 | IStbl | coop::cAdvFT | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$ assume(head[tk] == tail[tk]) $\{ (Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail) \}$           | OK     |



| #   | Kind  | Spec         | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Result |
|-----|-------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 165 | IStbl | coop::cAdvFT | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\} next[tk] = tail[tk] \rightarrow next \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$                                                               | OK     |
| 166 | IStbl | coop::cAdvFT | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\} return\ 0 \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$                                                                                          | OK     |
| 167 | IStbl | coop::cAdvFT | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\} assume(tail[tk] \neq Tail) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$                                                                         | OK     |
| 168 | IStbl | coop::cAdvFT | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\} assume(tail[tk] == Tail) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$                                                                           | OK     |
| 169 | IStbl | coop::cAdvFT | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\} assume(next[tk] \neq NULL) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$                                                                         | OK     |
| 170 | IStbl | coop::cAdvFT | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\} assume(next[tk] == NULL) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$                                                                           | OK     |
| 171 | IStbl | coop::cAdvFT | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\} assume(!(node[tk] \neq NULL \wedge Tail == tail[tk] \wedge tail[tk] \rightarrow next == next[tk])) \{(Head \neq NULL \wedge Tail \neq NULL \wedge (Head == Tail \vee Head \rightarrow next \neq NULL) \wedge node[t_j] == tloc\_ea1\_tj \wedge node[t_j] \rightarrow val == val[t_j] \wedge tail[t_j] == Tail)\}$ | OK     |

| #   | Kind  | Spec         | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Result |
|-----|-------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 172 | ISTbl | coop::cAdvFT | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ (Head \neq \text{NULL} \wedge Tail \neq \text{NULL} \wedge (Head == Tail \vee Head \rightarrow next \neq \text{NULL}) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail) \}$<br>$ICAS(node[tk] \neq \text{NULL} \wedge Tail == tail[tk], tail[tk] \rightarrow next, next[tk], node[tk]) \{ (Head \neq \text{NULL} \wedge Tail \neq \text{NULL} \wedge (Head == Tail \vee Head \rightarrow next \neq \text{NULL}) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail) \}$ | OK     |
| 173 | ISTbl | coop::cAdvFT | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ (Head \neq \text{NULL} \wedge Tail \neq \text{NULL} \wedge (Head == Tail \vee Head \rightarrow next \neq \text{NULL}) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail) \}$ $tail[tk] = Tail \{ (Head \neq \text{NULL} \wedge Tail \neq \text{NULL} \wedge (Head == Tail \vee Head \rightarrow next \neq \text{NULL}) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail) \}$                                                                                           | OK     |
| 174 | ISTbl | coop::cAdvFT | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ (Head \neq \text{NULL} \wedge Tail \neq \text{NULL} \wedge (Head == Tail \vee Head \rightarrow next \neq \text{NULL}) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail) \}$ $next[tk] = tail[tk] \rightarrow next \{ (Head \neq \text{NULL} \wedge Tail \neq \text{NULL} \wedge (Head == Tail \vee Head \rightarrow next \neq \text{NULL}) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail) \}$                                                                      | OK     |
| 175 | ISTbl | coop::cAdvFT | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ (Head \neq \text{NULL} \wedge Tail \neq \text{NULL} \wedge (Head == Tail \vee Head \rightarrow next \neq \text{NULL}) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail) \}$ $assume(next[tk] == \text{NULL}) \{ (Head \neq \text{NULL} \wedge Tail \neq \text{NULL} \wedge (Head == Tail \vee Head \rightarrow next \neq \text{NULL}) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail) \}$                                                                           | OK     |
| 176 | ISTbl | coop::cAdvFT | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ (Head \neq \text{NULL} \wedge Tail \neq \text{NULL} \wedge (Head == Tail \vee Head \rightarrow next \neq \text{NULL}) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail) \}$ $assume(!(next[tk] \neq \text{NULL} \wedge Tail == tail[tk])) \{ (Head \neq \text{NULL} \wedge Tail \neq \text{NULL} \wedge (Head == Tail \vee Head \rightarrow next \neq \text{NULL}) \wedge node[tj] == tloc\_ea1\_tj \wedge node[tj] \rightarrow val == val[tj] \wedge tail[tj] == Tail) \}$                                              | OK     |

### Veracity Program Examples (one per obligation kind)

```

RMov [FAIL].
1 tloc Head = null;

3 tloc Tail = null;

5 int main() {
6   havoc Head;
7   havoc Tail;
8   assume(Head != null && Tail != null && (Head == Tail || Head->next != null));
9   int tail_size = 0;
10  havoc tail_size;
11  tloc[] tail = new tloc[tail_size];
12  int head_size = 0;
13  havoc head_size;

```



```

14 tloc[] head = new tloc[head_size];
15 int t = 1;
16 havoc t;
17 int next_size = 0;
18 havoc next_size;
19 tloc[] next = new tloc[next_size];
20 int node_size = 0;
21 havoc node_size;
22 tloc[] node = new tloc[node_size];
23 commute_right_ctx (Head ≠ null && Tail ≠ null &&
24                   (Head == Tail || Head→next ≠ null) && Head ≠ null) {
25   {
26     havoc tail[t];
27     assume(tail[t] ≠ null);
28   }
29   {
30     havoc Head;
31     havoc Tail;
32   }

34   context: Head ≠ null && Tail ≠ null &&
35             (Head == Tail || Head→next ≠ null) &&
36             Head == head[t] && head[t] ≠ tail[t] &&
37             Head ≠ null && Tail ≠ null &&
38             (Head == Tail || Head→next ≠ null)
39 }
40 return 0;
41}

```

HTrp [OK].

```

1 tloc Head = null;

3 tloc Tail = null;

5 int main() {
6   havoc Head;
7   havoc Tail;
8   int next_size = 0;
9   havoc next_size;
10  tloc[] next = new tloc[next_size];
11  int t = 1;
12  havoc t;
13  int tail_size = 0;
14  havoc tail_size;
15  tloc[] tail = new tloc[tail_size];
16  assume(tail[t] == Tail && next[t] == null &&
17        Head ≠ null && Tail ≠ null && (Head == Tail || Head→next ≠ null));
18  assume(tail[t] == Tail);
19  assume(next[t] == null);
20  assert(tail[t] == Tail && next[t] == null &&
21        Head ≠ null && Tail ≠ null && (Head == Tail || Head→next ≠ null));
22  return 0;

```

23 }

```

    LMov  [OK].
1 tloc Head = null;

3 tloc Tail = null;

5 int main() {
6   havoc Head;
7   havoc Tail;
8   assume(Head ≠ null && Tail ≠ null && (Head == Tail || Head→next ≠ null));
9   int next_size = 0;
10  havoc next_size;
11  tloc[] next = new tloc[next_size];
12  int node_size = 0;
13  havoc node_size;
14  tloc[] node = new tloc[node_size];
15  int t = 1;
16  havoc t;
17  int tail_size = 0;
18  havoc tail_size;
19  tloc[] tail = new tloc[tail_size];
20  int head_size = 0;
21  havoc head_size;
22  tloc[] head = new tloc[head_size];
23  commute_left (exists old__tail_t__next : tloc . node[t] ≠ null &&
24                                     Tail == tail[t] &&
25                                     old__tail_t__next ==
26                                     next[t] &&
27                                     Head ≠ null &&
28                                     Tail ≠ null &&
29                                     (Head == Tail ||
30                                     Head→next ≠ null)
31                                     &&
32                                     tail[t]→next == node[t]
33                                     && Head ≠ null && tail[t] ≠ null) {
34  {
35    int steps = 1;
36    havoc steps;
37    while (steps > 0) invariant Head ≠ null && Tail ≠ null &&
38                                (Head == Tail || Head→next ≠ null) {
39      steps = steps - 1;
40      int tk = 1;
41      havoc tk;
42      assume(tk ≠ t);
43      int nd = 1;
44      havoc nd;
45      if (nd == 1) {
46        havoc tail[tk]→next;
47        assume(node[tk] ≠ null && Tail == tail[tk]);
48        assume(tail[tk]→next == node[tk]);
49      } else {

```

```

50         if (nd == 2) {
51             havoc Head;
52             assume(next[tk] ≠ null);
53             assume(Head == next[tk]);
54         } else {
55             if (true) {
56                 havoc Tail;
57                 assume(next[tk] ≠ null);
58                 assume(Tail == next[tk]);
59             }
60         }
61     }
62 }
63 }
64 {
65     return 0;
66 }

68 }
69 return 0;
70 }

```

Subs [OK].

```

1 tloc Head = null;

3 tloc Tail = null;

5 int main() {
6     havoc Head;
7     havoc Tail;
8     int next_size = 0;
9     havoc next_size;
10    tloc[] next = new tloc[next_size];
11    int node_size = 0;
12    havoc node_size;
13    tloc[] node = new tloc[node_size];
14    int tail_size = 0;
15    havoc tail_size;
16    tloc[] tail = new tloc[tail_size];
17    int ti = 1;
18    havoc ti;
19    int tj = 1;
20    havoc tj;
21    int val_size = 0;
22    havoc val_size;
23    int[] val = new int[val_size];
24    tloc tloc_ea1_ti = null;
25    havoc tloc_ea1_ti;
26    tloc tloc_ea1_tj = null;
27    havoc tloc_ea1_tj;
28    tloc_ea1_ti = new {0}null;
29    tloc_ea1_tj = new {0}null;

```

```

30  assume(exists old_tail_ti___next : tloc . Head ≠ null && Tail ≠ null &&
31      (Head == Tail ||
32      Head→next ≠ null)
33      && node[tj] == tloc_ea1_tj &&
34      node[tj]→value == val[tj] &&
35      tail[tj] == Tail &&
36      next[tj] == tail[tj]→next &&
37      next[tj] == null &&
38      node[ti] == tloc_ea1_ti &&
39      node[ti]→value == val[ti] &&
40      tail[ti] == Tail &&
41      next[ti] == old_tail_ti___next
42      && next[ti] == null &&
43      tail[ti]→next == node[ti] &&
44      node[ti] ≠ null &&
45      Tail == tail[ti]);
46  assert(!(node[tj] ≠ null && Tail == tail[tj] && tail[tj]→next == next[tj]));
47  return 0;
48 }

```

#### B.4 rdcss\_inlined

510 obligations across 38 specs: atom::aControlS, atom::aControlS.R, atom::aDataS, atom::aDataS.R, atom::aDataSLoop, atom::aDataSLoop.R, atom::aHelpS, atom::aHelpS.R, atom::aHelpSLoop, atom::aHelpSLoop.R, atom::aHelperRestoreS, atom::aHelperRestoreS.R, atom::aHelperRestoreSLoop, atom::aHelperRestoreSLoop.R, atom::aRestoreS, atom::aRestoreS.R, coop::cDataF, coop::cDataF.R, coop::cDataFLoop, coop::cDataFLoop.R, coop::cHelpF, coop::cHelpF.R, coop::cHelpFExit, coop::cHelpFExit.R, coop::cHelpFLoop, coop::cHelpFLoop.R, coop::cHelpFMain, coop::cHelpFMain.R, coop::cHelpFSkip, coop::cHelpFSkip.R, coop::cHelperRestoreF, coop::cHelperRestoreF.R, coop::cHelperRestoreFLoop, coop::cHelperRestoreFLoop.R, coop::cLoopSkip, coop::cLoopSkip.R, coop::cRestoreF, coop::cRestoreF.R.

##### B.4.1 Sources.

#### Implementation (.cl)

```

1/* RDCSS: Restricted form of Double Word CAS */
2/* See also https://arxiv.org/pdf/2301.05740, page 42*/
3/*
4  complete() inlined at each call site.
5  cptr1/cptr2/cptr3: local ptr for each complete() inline site.
6  ctrl1/ctrl2/ctrl3, cexp1/cexp2/cexp3: local scalars capturing
7  mem_control[cptr] and control_exp[r/tid] so the branch condition
8  is a Local (not GlobalRead) and acts as a right-mover.
9*/
10 readonly int MAX_REGS;
11 int mem_data[MAX_REGS];
12 readonly int mem_control[MAX_REGS];
14/* may need global invariant that all expected data is negative */

```

```

16 readonly int MAX_TID;
17 readonly int control_addr[MAX_TID];
18 readonly int control_exp[MAX_TID];
19 readonly int data_addr[MAX_TID];
20 readonly int data_exp[MAX_TID];
21 readonly int data_new[MAX_TID];

23 int RDCSS(int tid) {
24     int r;
25     int ptr = data_addr[tid];
26     r = RCAS(mem_data[ptr], data_exp[tid], tid);

28     /* inline: complete(r) */
29     if (r >= 0) {
30         int cptr1 = data_addr[r];
31         int ctrl1 = mem_control[cptr1];
32         int cexp1 = control_exp[r];
33         if (ctrl1 == cexp1) {
34             if CAS(mem_data[cptr1], r, data_new[r]) { } else { }
35         }
36         else {
37             if CAS(mem_data[cptr1], r, data_exp[r]) { } else { }
38         }
39     }

41     while (r != null) {
42         r = RCAS(mem_data[ptr], data_exp[tid], tid);

44         /* inline: complete(r) */
45         if (r >= 0) {
46             int cptr2 = data_addr[r];
47             int ctrl2 = mem_control[cptr2];
48             int cexp2 = control_exp[r];
49             if (ctrl2 == cexp2) {
50                 if CAS(mem_data[cptr2], r, data_new[r]) { } else { }
51             }
52             else {
53                 if CAS(mem_data[cptr2], r, data_exp[r]) { } else { }
54             }
55         }
56     }

58     /* inline: complete(tid) */
59     if (r == data_exp[tid]) {
60         int cptr3 = data_addr[tid];
61         int ctrl3 = mem_control[cptr3];
62         int cexp3 = control_exp[tid];
63         if (ctrl3 == cexp3) {
64             if CAS(mem_data[cptr3], tid, data_new[tid]) { } else { }
65         }
66         else {
67             if CAS(mem_data[cptr3], tid, data_exp[tid]) { } else { }

```

```

68     }
69 }

71 return r;
72 }

```

### Labeled program (.c11)

```

1 /*****
2 /* [generated] KAT-friendly program with label annotations */
3 /* --
4 /* rdcss_inlined.cl
5 *****/

7 readonly int MAX_REGS;
8 int mem_data[MAX_REGS];
9 readonly int mem_control[MAX_REGS];
10 readonly int MAX_TID;
11 readonly int control_addr[MAX_TID];
12 readonly int control_exp[MAX_TID];
13 readonly int data_addr[MAX_TID];
14 readonly int data_exp[MAX_TID];
15 readonly int data_new[MAX_TID];

17 int RDCSS(int tid) {
18     [@Ra1] int r;
19     [@Ra2] int ptr = data_addr[tid];
20     if * {
21         [@Rcas3t] RCAS(mem_data[ptr], data_exp[tid], tid, r);
22     }
23     else {
24         [@Rcas3f] assume(mem_data[ptr] ≠ data_exp[tid] && r == mem_data[ptr]);
25     }
26     if * {
27         [@Rb4t] assume(r ≥ 0);
28         [@Ra5] int cptr1 = data_addr[r];
29         [@Ra6] int ctrl1 = mem_control[cptr1];
30         [@Ra7] int cexp1 = control_exp[r];
31         if * {
32             [@Rb8t] assume(ctrl1 == cexp1);
33             if * {
34                 [@Rcas9t] CAS(mem_data[cptr1], r, data_new[r]);
35             }
36             else {
37                 [@Rcas9f] assume(mem_data[cptr1] ≠ r);
38             }
39         }
40         else {
41             [@Rb8f] assume(ctrl1 ≠ cexp1);
42             if * {
43                 [@Rcas10t] CAS(mem_data[cptr1], r, data_exp[r]);

```

```

44     }
45     else {
46         [@Rcas10f] assume(mem_data[cptr1] ≠ r);
47     }
48 }
49 }
50 else {
51     [@Rb4f] assume(r < 0);
52 }
53 while (true) {
54     [@Rb11t] assume(r ≠ NULL);
55     if * {
56         [@Rcas12t] RCAS(mem_data[ptr], data_exp[tid], tid, r);
57     }
58     else {
59         [@Rcas12f] assume(mem_data[ptr] ≠ data_exp[tid] && r == mem_data[ptr]);
60     }
61     if * {
62         [@Rb13t] assume(r ≥ 0);
63         [@Ra14] int cptr2 = data_addr[r];
64         [@Ra15] int ctrl2 = mem_control[cptr2];
65         [@Ra16] int cexp2 = control_exp[r];
66         if * {
67             [@Rb17t] assume(ctrl2 == cexp2);
68             if * {
69                 [@Rcas18t] CAS(mem_data[cptr2], r, data_new[r]);
70             }
71             else {
72                 [@Rcas18f] assume(mem_data[cptr2] ≠ r);
73             }
74         }
75         else {
76             [@Rb17f] assume(ctrl2 ≠ cexp2);
77             if * {
78                 [@Rcas19t] CAS(mem_data[cptr2], r, data_exp[r]);
79             }
80             else {
81                 [@Rcas19f] assume(mem_data[cptr2] ≠ r);
82             }
83         }
84     }
85     else {
86         [@Rb13f] assume(r < 0);
87     }
88 }
89 [@Rb11f] assume(r == NULL);
90 if * {
91     [@Rb20t] assume(r == data_exp[tid]);
92     [@Ra21] int cptr3 = data_addr[tid];
93     [@Ra22] int ctrl3 = mem_control[cptr3];
94     [@Ra23] int cexp3 = control_exp[tid];
95     if * {

```

```

96     [@Rb24t] assume(ctrl3 == cexp3);
97     if * {
98         [@Rcas25t] CAS(mem_data[cptr3], tid, data_new[tid]);
99     }
100    else {
101        [@Rcas25f] assume(mem_data[cptr3] ≠ tid);
102    }
103 }
104 else {
105     [@Rb24f] assume(ctrl3 ≠ cexp3);
106     if * {
107         [@Rcas26t] CAS(mem_data[cptr3], tid, data_exp[tid]);
108     }
109     else {
110         [@Rcas26f] assume(mem_data[cptr3] ≠ tid);
111     }
112 }
113 }
114 else {
115     [@Rb20f] assume(r ≠ data_exp[tid]);
116 }
117 [@Ra27] return r;
118 }

```

#### B.4.2 *IRSpec.*

##### IR specifications (.irspecs)

```

1 /* OBJECT: RDCSS (inlined) */
2 /* =====
3  RDCSS IRSpecs for the inlined program (benchmarks/rdcss_inlined.cll).

5  Translated from the manual example in irdefs.tex ("Harris et al. RDCSS
6  IRSpecs") + its companion program rdcss-rdcss.c. Manual single-letter
7  labels map onto the inlined CLL labels as:

9  S  init (int r; ptr = data_addr)      Ra1; Ra2
10  A  data RCAS succeeds                  Rcas3t  (loop copy: Rcas12t)
11  F  data RCAS fails (finds descriptor) Rcas3f  (loop copy: Rcas12f)
12  b  isDescriptor(r) holds              Rb4t    (loop copy: Rb13t)
13  c  control check passes (+ reads)    Ra5; Ra6; Ra7; Rb8t  (loop: Ra14;Ra15;
    Ra16;Rb17t)
14  d  help commit CAS succeeds          Rcas9t  (loop copy: Rcas18t)
15  u  help commit CAS fails             Rcas9f  (loop copy: Rcas18f)
16  e  control check fails (+ reads)    Ra5; Ra6; Ra7; Rb8f  (loop: Ra14;Ra15;
    Ra16;Rb17f)
17  f  help restore CAS succeeds         Rcas10t (loop copy: Rcas19t)
18  v  help restore CAS fails           Rcas10f (loop copy: Rcas19f)
19  g  !isDescriptor(r)                  Rb4f    (loop copy: Rb13f)
20  i  loop enters (isDescriptor(r))     Rb11t
21  j  loop exits (!isDescriptor(r))    Rb11f
22  k  r == data_exp                     Rb20t

```



```

23   l  main control check passes (+ reads) Ra21; Ra22; Ra23; Rb24t
24   m  main commit CAS succeeds           Rcas25t
25   s  main commit CAS fails              Rcas25f
26   n  main control check fails (+ reads) Ra21; Ra22; Ra23; Rb24f
27   o  main restore CAS succeeds          Rcas26t
28   t  main restore CAS fails             Rcas26f
29   p  r ≠ data_exp                       Rb20f
30   r  return                             Ra27

32  Every help / commit / restore CAS is wrapped in the CLL as
33    if CAS(...) { } else { }
34  so success (Rcas<N>t) and failure (Rcas<N>f) get distinct labels. A coop's
35  left-mover slot uses the CAS-FAIL label (Rcas<N>f) for the mover whose CAS is
36  turned into a no-op by the anchor's linearizing CAS.

38  DESIGN: LOCAL FRAGMENTS. The manual's specs are *local* label fragments
39  (e.g. controlS = jkl > m < r starts at the loop-exit j, not method entry),
40  not entire entry→return paths. The coverage query checks
41  L(method) ⊆ ( ∪ patterns )*, so local fragments tile the method language:
42  each pattern is one contiguous segment ending at a control-flow join, and
43  the Kleene star concatenates them (and repeats the loop body).

45  Each coop has a SINGLE-atom anchor. Every "+" branch of a manual coop's
46  right-moving (left prefix) or left-moving (right context) slot is its own
47  coop rule here, INCLUDING the observation-only branches (mover finds
48  !isDescriptor and skips = g,h; or exits the loop and returns = j,pqr) which
49  carry no CAS but tile the no-CAS "skip"/"give up" paths for coverage.

51  Manual order (dataS<dataF, controlS<controlF, helpS<helpF, restoreS<restoreF,
52  helperRestoreS<helperRestoreF) is encoded implicitly by definition order.
53  ===== */

56/* ---- data CAS install (dataS: (S+i) > A <) ----- */

58/* First-iteration install. Segment: Ra1;Ra2;Rcas3t. */
59aDataS = atom{
60  ind = Rcas3t,
61  red = [Ra1; Ra2] |> Rcas3t <| EPS
62}

64/* Loop-iteration install (the "i" alternative of (S+i)). Segment: Rb11t;Rcas12t. */
65aDataSLoop = atom{
66  ind = Rcas12t,
67  red = [Rb11t] |> Rcas12t <| EPS
68}

70/* dataF (S,F): install makes a concurrent thread's first data CAS fail.
71  Segment: Ra1;Ra2;Rcas3f. */
72cDataF = coop{
73  ind = Rcas3t,
74  red = [Ra1; Ra2] |> [aDataS] <| Rcas3f

```

```

75}

77/* dataF (i,F): install makes a concurrent thread's loop data CAS fail.
78   Segment: Rb11t;Rcas12f. */
79cDataFLoop = coop{
80   ind = Rcas12t,
81   red = [Rb11t] |> [aDataSLoop] <| Rcas12f
82}

85/* ---- help commit (helpS: bc > d <) ----- */

87/* First-iteration help-commit success.  Segment: Rb4t;Ra5;Ra6;Ra7;Rb8t;Rcas9t. */
88aHelpS = atom{
89   ind = Rcas9t,
90   red = [Rb4t; Ra5; Ra6; Ra7; Rb8t] |> Rcas9t <| EPS
91}

93/* Loop-iteration help-commit success.  Segment: Rb13t;Ra14;Ra15;Ra16;Rb17t;Rcas18t.
   */
94aHelpSLoop = atom{
95   ind = Rcas18t,
96   red = [Rb13t; Ra14; Ra15; Ra16; Rb17t] |> Rcas18t <| EPS
97}

100/* ---- main commit (controls: jkl > m < r) ----- */

102/* Segment: Rb11f;Rb20t;Ra21;Ra22;Ra23;Rb24t;Rcas25t;Ra27 (commit-success tail). */
103aControls = atom{
104   ind = Rcas25t,
105   red = [Rb11f; Rb20t; Ra21; Ra22; Ra23; Rb24t] |> Rcas25t <| Ra27
106}

108/* helpF (bc, u): helper's commit CAS fails due to either:
109   - main commit
110   - helper's commit
111   Segment: Rb4t;Ra5;Ra6;Ra7;Rb8t;Rcas9f.
112*/
113cHelpF = coop{
114   ind = Rcas25t + Rcas9t,
115   red = [Rb4t; Ra5; Ra6; Ra7; Rb8t] |> [aControls] + [aHelpS] <| Rcas9f
116}

118/* helpF (jkl,sr): a helper's commit makes the MAIN thread's commit CAS FAIL
119   (then it returns).  Segment: Rb11f..Rb24t;Rcas25f;Ra27 (commit-fail tail). */
120cHelpFMain = coop{
121   ind = Rcas9t,
122   red = [Rb11f; Rb20t; Ra21; Ra22; Ra23; Rb24t] |> [aHelpS] <| Rcas25f; Ra27
123}

125/* helpF loop copy (bc,u): a loop-iteration help-commit makes a concurrent

```

```

126  loop help-commit CAS FAIL.  Segment: Rb13t;Ra14;Ra15;Ra16;Rb17t;Rcas18f. */
127 cHelpFLoop = coop{
128   ind = Rcas18t,
129   red = [Rb13t; Ra14; Ra15; Ra16; Rb17t] |> [aHelpSLoop] <| Rcas18f
130 }

132 /* helpF (g,h) -- observation only: a thread observes !isDescriptor and skips
133    the help block.  Segment: Rb4f. */
134 cHelpFSkip = coop{
135   ind = Rcas9t,
136   red = [Rb4f] |> [aHelpS] <| EPS
137 }

139 /* Loop g -- observation only: mover observes !isDescriptor inside the loop and
140    re-checks.  Segment: Rb13f. */
141 cLoopSkip = coop{
142   ind = Rcas12t,
143   red = [Rb13f] |> [aDataSLoop] <| EPS
144 }

147 /* ---- help restore (helperRestoreS: be > f <) ----- */

149 /* First-iteration help-restore success.  Segment: Rb4t;Ra5;Ra6;Ra7;Rb8f;Rcas10t. */
150 aHelperRestoreS = atom{
151   ind = Rcas10t,
152   red = [Rb4t; Ra5; Ra6; Ra7; Rb8f] |> Rcas10t <| EPS
153 }

155 /* Loop-iteration help-restore success.  Segment: Rb13t;Ra14;Ra15;Ra16;Rb17f;Rcas19t.
    */
156 aHelperRestoreSLoop = atom{
157   ind = Rcas19t,
158   red = [Rb13t; Ra14; Ra15; Ra16; Rb17f] |> Rcas19t <| EPS
159 }

162 /* ---- main restore (restoreS: jkn > o < r) ----- */

164 /* Segment: Rb11f;Rb20t;Ra21;Ra22;Ra23;Rb24f;Rcas26t;Ra27 (restore-success tail). */
165 aRestoreS = atom{
166   ind = Rcas26t,
167   red = [Rb11f; Rb20t; Ra21; Ra22; Ra23; Rb24f] |> Rcas26t <| Ra27
168 }

170 /* restoreF (be,v): a main restore makes a concurrent helper's restore CAS FAIL.
171    Segment: Rb4t;Ra5;Ra6;Ra7;Rb8f;Rcas10f. */
172 cRestoreF = coop{
173   ind = Rcas26t,
174   red = [Rb4t; Ra5; Ra6; Ra7; Rb8f] |> [aRestoreS] <| Rcas10f
175 }

```

```

177/* helperRestoreF (jkn,tr): a helper's restore makes the MAIN thread's restore
178   CAS FAIL (then it returns). Segment: Rb11f..Rb24f;Rcas26f;Ra27 (restore-fail tail
   ). */
179cHelperRestoreF = coop{
180   ind = Rcas10t,
181   red = [Rb11f; Rb20t; Ra21; Ra22; Ra23; Rb24f] |> [aHelperRestoreS] <| Rcas26f; Ra27
182}

184/* helperRestoreF loop copy (be,v): a loop-iteration help-restore makes a
185   concurrent loop help-restore CAS FAIL. Segment: Rb13t;Ra14;Ra15;Ra16;Rb17f;
   Rcas19f. */
186cHelperRestoreFLoop = coop{
187   ind = Rcas19t,
188   red = [Rb13t; Ra14; Ra15; Ra16; Rb17f] |> [aHelperRestoreSLoop] <| Rcas19f
189}

191/* helpF (j,pqr) -- observation only: the mover exits the loop,
192   finds r ≠ data_exp, and returns. Same give-up tail in the case
193   of cooperating with a helper's restore
194   Segment: Rb11f;Rb20f;Ra27. */

196cHelpFExit = coop{
197   ind = Rcas9t + Rcas10t,
198   red = [Rb11f] |> [aHelpS] + [aHelperRestoreS] <| Rb20f; Ra27
199}

```

#### B.4.3 Contextual Commutativity (71).

| # | Kind | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---|------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Atom | atom::aDataS.R     | [int r; int ptr = data_addr[tid]]                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2 | RMov | atom::aDataS       | $\forall t:\mathbb{N}. \text{int } r; \text{ptr}[t] = \text{data\_addr}[\text{tid}[t]] \triangleright \{\text{mem\_data}[\text{ptr}[t]] == \text{data\_exp}[\text{tid}[t]]\}$                                                                                                                                                                                                                                                        |
| 3 | HTrp | atom::aDataS       | $\forall t:\mathbb{N}. \{\text{mem\_data}[\text{ptr}[t]] == \text{data\_exp}[\text{tid}[t]]\}$<br>$\text{RCAS}(\text{mem\_data}[\text{ptr}[t]], \text{data\_exp}[\text{tid}[t]], \text{tid}[t], r[t])$<br>$\{\exists \text{old\_mem\_data\_ptr\_t\_}. (\text{old\_mem\_data\_ptr\_t\_} == \text{data\_exp}[\text{tid}[t]] \wedge \text{mem\_data}[\text{ptr}[t]] == \text{tid}[t] \wedge r[t] == \text{old\_mem\_data\_ptr\_t\_})\}$ |
| 4 | Atom | atom::aDataSLoop.R | [assume(r ≠ NULL)]                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5 | RMov | atom::aDataSLoop   | $\forall t:\mathbb{N}. \text{assume}(r[t] \neq \text{NULL}) \triangleright \{\text{mem\_data}[\text{ptr}[t]] == \text{data\_exp}[\text{tid}[t]]\}$                                                                                                                                                                                                                                                                                   |
| 6 | HTrp | atom::aDataSLoop   | $\forall t:\mathbb{N}. \{\text{mem\_data}[\text{ptr}[t]] == \text{data\_exp}[\text{tid}[t]]\}$<br>$\text{RCAS}(\text{mem\_data}[\text{ptr}[t]], \text{data\_exp}[\text{tid}[t]], \text{tid}[t], r[t])$<br>$\{\exists \text{old\_mem\_data\_ptr\_t\_}. (\text{old\_mem\_data\_ptr\_t\_} == \text{data\_exp}[\text{tid}[t]] \wedge \text{mem\_data}[\text{ptr}[t]] == \text{tid}[t] \wedge r[t] == \text{old\_mem\_data\_ptr\_t\_})\}$ |
| 7 | Atom | coop::cDataF.R     | [int r; int ptr = data_addr[tid]]                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8 | RMov | coop::cDataF       | $\forall tj:\mathbb{N}. \text{int } r; \text{ptr}[tj] = \text{data\_addr}[\text{tid}[tj]] \triangleright \{\text{ptr}[tj] == \text{data\_addr}[\text{tid}[tj]]\}$                                                                                                                                                                                                                                                                    |

| #  | Kind | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----|------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9  | HTrp | coop::cDataF       | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ptr[tj] == data\_addr[tid[tj]]\} \llbracket [i$<br>$r; ptr[ti] = data\_addr[tid[ti]]\rrbracket; RCAS(mem\_data[ptr[ti],$<br>$data\_exp[tid[ti]], tid[ti], r[ti]); skip$<br>$\{\exists old\_mem\_data\_ptr\_ti\_.. (ptr[tj] == data\_addr[tid[tj]]$<br>$\wedge ptr[ti] == data\_addr[tid[ti]] \wedge old\_mem\_data\_ptr\_t$<br>$== data\_exp[tid[ti]] \wedge mem\_data[ptr[ti]] == tid[ti] \wedge$<br>$r[ti] == old\_mem\_data\_ptr\_ti\_)\}$ |
| 10 | Subs | coop::cDataF       | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{\exists old\_mem\_data\_ptr\_ti\_..$<br>$(ptr[tj] == data\_addr[tid[tj]] \wedge ptr[ti] ==$<br>$data\_addr[tid[ti]] \wedge old\_mem\_data\_ptr\_ti\_ ==$<br>$data\_exp[tid[ti]] \wedge mem\_data[ptr[ti]] == tid[ti] \wedge r[$<br>$== old\_mem\_data\_ptr\_ti\_)\} \Rightarrow \{mem\_data[ptr[tj]] \neq$<br>$data\_exp[tid[tj]] \wedge r[tj] == mem\_data[ptr[tj]]\}$                                                                       |
| 11 | Atom | coop::cDataFLoop.R | [assume( $r \neq NULL$ )]                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 12 | RMov | coop::cDataFLoop   | $\forall tj: \mathbb{N}. assume(r[tj] \neq NULL) \triangleright \{r[tj] \neq NULL\}$                                                                                                                                                                                                                                                                                                                                                                                                     |
| 13 | HTrp | coop::cDataFLoop   | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{r[tj] \neq NULL\} \llbracket [assume(r[ti] \neq$<br>$NULL)\rrbracket; RCAS(mem\_data[ptr[ti]], data\_exp[tid[ti]],$<br>$tid[ti], r[ti]); skip \{\exists old\_mem\_data\_ptr\_ti\_..$<br>$(r[tj] \neq NULL \wedge old\_mem\_data\_ptr\_ti\_ \neq NULL$<br>$\wedge old\_mem\_data\_ptr\_ti\_ == data\_exp[tid[ti]]$<br>$\wedge mem\_data[ptr[ti]] == tid[ti] \wedge r[ti] ==$<br>$old\_mem\_data\_ptr\_ti\_)\}$                                 |
| 14 | Subs | coop::cDataFLoop   | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{\exists old\_mem\_data\_ptr\_ti\_..$<br>$(r[tj] \neq NULL \wedge old\_mem\_data\_ptr\_ti\_ \neq NULL$<br>$\wedge old\_mem\_data\_ptr\_ti\_ == data\_exp[tid[ti]]$<br>$\wedge mem\_data[ptr[ti]] == tid[ti] \wedge r[ti] ==$<br>$old\_mem\_data\_ptr\_ti\_)\} \Rightarrow \{mem\_data[ptr[tj]] \neq$<br>$data\_exp[tid[tj]] \wedge r[tj] == mem\_data[ptr[tj]]\}$                                                                              |
| 15 | Atom | atom::aHelpS.R     | [assume( $r \geq 0$ ); int cptr1 = data_addr[r]; int ctrl1 = mem_control[cptr1]; int cexp1 = control_exp[r]; assume(ctrl1 == cexp1)]                                                                                                                                                                                                                                                                                                                                                     |
| 16 | RMov | atom::aHelpS       | $\forall t: \mathbb{N}. assume(r[t] \geq 0); cptr1[t] = data\_addr[r[t]];$<br>$ctrl1[t] = mem\_control[cptr1[t]]; cexp1[t] =$<br>$control\_exp[r[t]]; assume(ctrl1[t] == cexp1[t]) \triangleright$<br>$\{mem\_data[cptr1[t]] == r[t]\}$                                                                                                                                                                                                                                                  |
| 17 | HTrp | atom::aHelpS       | $\forall t: \mathbb{N}. \{mem\_data[cptr1[t]] == r[t]\}$<br>$CAS(mem\_data[cptr1[t]], r[t], data\_new[r[t]])$<br>$\{\exists old\_mem\_data\_cptr1\_t\_.. (old\_mem\_data\_cptr1\_t\_ ==$<br>$r[t] \wedge mem\_data[cptr1[t]] == data\_new[r[t]]\}$                                                                                                                                                                                                                                       |
| 18 | Atom | atom::aHelpSLoop.R | [assume( $r \geq 0$ ); int cptr2 = data_addr[r]; int ctrl2 = mem_control[cptr2]; int cexp2 = control_exp[r]; assume(ctrl2 == cexp2)]                                                                                                                                                                                                                                                                                                                                                     |
| 19 | RMov | atom::aHelpSLoop   | $\forall t: \mathbb{N}. assume(r[t] \geq 0); cptr2[t] = data\_addr[r[t]];$<br>$ctrl2[t] = mem\_control[cptr2[t]]; cexp2[t] =$<br>$control\_exp[r[t]]; assume(ctrl2[t] == cexp2[t]) \triangleright$<br>$\{mem\_data[cptr2[t]] == r[t]\}$                                                                                                                                                                                                                                                  |

| #  | Kind | Spec              | Obligation                                                                                                                                                                                                                                                                                                                                                                                                      |
|----|------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20 | HTrp | atom::aHelpSLoop  | $\forall t:\mathbb{N}. \{ \text{mem\_data}[\text{cptr2}[t]] == r[t] \}$<br>CAS(mem_data[cptr2[t]], r[t], data_new[r[t]])<br>$\{ \exists \text{old\_mem\_data\_cptr2\_t\_}. (\text{old\_mem\_data\_cptr2\_t\_} == r[t] \wedge \text{mem\_data}[\text{cptr2}[t]] == \text{data\_new}[r[t]]) \}$                                                                                                                   |
| 21 | Atom | atom::aControlS.R | [assume(r == NULL); assume(r == data_exp[tid]); int<br>cptr3 = data_addr[tid]; int ctrl3 = mem_control[cptr3];<br>int cexp3 = control_exp[tid]; assume(ctrl3 == cexp3)]                                                                                                                                                                                                                                         |
| 22 | RMov | atom::aControlS   | $\forall t:\mathbb{N}. \text{assume}(r[t] == \text{NULL}); \text{assume}(r[t] ==$<br>data_exp[tid[t]]); cptr3[t] = data_addr[tid[t]];<br>ctrl3[t] = mem_control[cptr3[t]]; cexp3[t] =<br>control_exp[tid[t]]; assume(ctrl3[t] == cexp3[t]) $\triangleright$<br>{mem_data[cptr3[t]] == tid[t]}                                                                                                                   |
| 23 | HTrp | atom::aControlS   | $\forall t:\mathbb{N}. \{ \text{mem\_data}[\text{cptr3}[t]] == \text{tid}[t] \}$<br>CAS(mem_data[cptr3[t]], tid[t], data_new[tid[t]])<br>$\{ \exists \text{old\_mem\_data\_cptr3\_t\_}. (\text{old\_mem\_data\_cptr3\_t\_} ==$<br>tid[t] $\wedge$ mem_data[cptr3[t]] == data_new[tid[t]]) \}                                                                                                                    |
| 24 | LMov | atom::aControlS   | $\forall t:\mathbb{N}. \{ \exists \text{old\_mem\_data\_cptr3\_t\_}. (\text{old\_mem\_data\_cptr3\_t\_} ==$<br>tid[t] $\wedge$ mem_data[cptr3[t]] == data_new[tid[t]]) \} $\triangleleft$ return r[t]                                                                                                                                                                                                           |
| 25 | Atom | coop::cHelpF.R    | [assume(r >= 0); int cptr1 = data_addr[r]; int ctrl1<br>= mem_control[cptr1]; int cexp1 = control_exp[r];<br>assume(ctrl1 == cexp1)]                                                                                                                                                                                                                                                                            |
| 26 | RMov | coop::cHelpF      | $\forall t_j:\mathbb{N}. \text{assume}(r[t_j] \geq 0); \text{cptr1}[t_j] =$<br>data_addr[r[tj]]; ctrl1[tj] = mem_control[cptr1[tj]];<br>cexp1[tj] = control_exp[r[tj]]; assume(ctrl1[tj]<br>== cexp1[tj]) $\triangleright$ {(r[tj] >= 0 $\wedge$ cptr1[tj] ==<br>data_addr[r[tj]] $\wedge$ ctrl1[tj] == mem_control[cptr1[tj]]<br>$\wedge$ cexp1[tj] == control_exp[r[tj]] $\wedge$ ctrl1[tj] ==<br>cexp1[tj])} |

| #  | Kind | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----|------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 27 | HTrp | coop::cHelpF       | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{(r[tj] \geq 0 \wedge cptr1[tj] == data\_addr[r[tj]] \wedge ctrl1[tj] == mem\_control[cptr1[tj]] \wedge cexp1[tj] == control\_exp[r[tj]] \wedge ctrl1[tj] == cexp1[tj]) \} ([[assume(r[ti] == NULL); assume(r[ti] == data\_exp[tid[ti]]); cptr3[ti] = data\_addr[tid[ti]]; ctrl3[ti] = mem\_control[cptr3[ti]]; cexp3[ti] = control\_exp[tid[ti]]; assume(ctrl3[ti] == cexp3[ti]); CAS(mem\_data[cptr3[ti]], tid[ti], data\_new[tid[ti]]); return r[ti]] + [[assume(r[ti] \geq 0); cptr1[ti] = data\_addr[r[ti]]; ctrl1[ti] = mem\_control[cptr1[ti]]; cexp1[ti] = control\_exp[r[ti]]; assume(ctrl1[ti] == cexp1[ti]); CAS(mem\_data[cptr1[ti]], r[ti], data\_new[r[ti]]); skip]) \{(\exists old\_mem\_data\_cptr3\_ti\_.. (r[tj] \geq 0 \wedge cptr1[tj] == data\_addr[r[tj]] \wedge ctrl1[tj] == mem\_control[cptr1[tj]] \wedge cexp1[tj] == control\_exp[r[tj]] \wedge ctrl1[tj] == cexp1[tj] \wedge r[ti] == NULL \wedge r[ti] == data\_exp[tid[ti]] \wedge cptr3[ti] == data\_addr[tid[ti]] \wedge ctrl3[ti] == mem\_control[cptr3[ti]] \wedge cexp3[ti] == control\_exp[tid[ti]] \wedge ctrl3[ti] == cexp3[ti] \wedge old\_mem\_data\_cptr3\_ti\_ == tid[ti] \wedge mem\_data[cptr3[ti]] == data\_new[tid[ti]]) \vee \exists old\_mem\_data\_cptr1\_ti\_.. (r[tj] \geq 0 \wedge cptr1[tj] == data\_addr[r[tj]] \wedge ctrl1[tj] == mem\_control[cptr1[tj]] \wedge cexp1[tj] == control\_exp[r[tj]] \wedge ctrl1[tj] == cexp1[tj] \wedge r[ti] \geq 0 \wedge cptr1[ti] == data\_addr[r[ti]] \wedge ctrl1[ti] == mem\_control[cptr1[ti]] \wedge cexp1[ti] == control\_exp[r[ti]] \wedge ctrl1[ti] == cexp1[ti] \wedge old\_mem\_data\_cptr1\_ti\_ == r[ti] \wedge mem\_data[cptr1[ti]] == data\_new[r[ti]])) \}$ |
| 28 | Subs | coop::cHelpF       | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{(\exists old\_mem\_data\_cptr3\_ti\_.. (r[tj] \geq 0 \wedge cptr1[tj] == data\_addr[r[tj]] \wedge ctrl1[tj] == mem\_control[cptr1[tj]] \wedge cexp1[tj] == control\_exp[r[tj]] \wedge ctrl1[tj] == cexp1[tj] \wedge r[ti] == NULL \wedge r[ti] == data\_exp[tid[ti]] \wedge cptr3[ti] == data\_addr[tid[ti]] \wedge ctrl3[ti] == mem\_control[cptr3[ti]] \wedge cexp3[ti] == control\_exp[tid[ti]] \wedge ctrl3[ti] == cexp3[ti] \wedge old\_mem\_data\_cptr3\_ti\_ == tid[ti] \wedge mem\_data[cptr3[ti]] == data\_new[tid[ti]]) \vee \exists old\_mem\_data\_cptr1\_ti\_.. (r[tj] \geq 0 \wedge cptr1[tj] == data\_addr[r[tj]] \wedge ctrl1[tj] == mem\_control[cptr1[tj]] \wedge cexp1[tj] == control\_exp[r[tj]] \wedge ctrl1[tj] == cexp1[tj] \wedge r[ti] \geq 0 \wedge cptr1[ti] == data\_addr[r[ti]] \wedge ctrl1[ti] == mem\_control[cptr1[ti]] \wedge cexp1[ti] == control\_exp[r[ti]] \wedge ctrl1[ti] == cexp1[ti] \wedge old\_mem\_data\_cptr1\_ti\_ == r[ti] \wedge mem\_data[cptr1[ti]] == data\_new[r[ti]])) \} \Rightarrow \{mem\_data[cptr1[tj]] \neq r[tj]\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 29 | Atom | coop::cHelpFMain.R | $[assume(r == NULL); assume(r == data\_exp[tid]); int cptr3 = data\_addr[tid]; int ctrl3 = mem\_control[cptr3]; int cexp3 = control\_exp[tid]; assume(ctrl3 == cexp3)]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| #  | Kind | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----|------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 30 | RMov | coop::cHelpFMain   | $\forall t_j:\mathbb{N}. \text{assume}(r[t_j] == \text{NULL}); \text{assume}(r[t_j] == \text{data\_exp}[\text{tid}[t_j]]); \text{cptr3}[t_j] = \text{data\_addr}[\text{tid}[t_j]];$<br>$\text{ctrl3}[t_j] = \text{mem\_control}[\text{cptr3}[t_j]]; \text{cexp3}[t_j] = \text{control\_exp}[\text{tid}[t_j]]; \text{assume}(\text{ctrl3}[t_j] == \text{cexp3}[t_j])$<br>$\triangleright \{(r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j])\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 31 | HTrp | coop::cHelpFMain   | $\forall ti, t_j:\mathbb{N}, ti \neq t_j. \{(r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j])\}$<br>$[[\text{assume}(r[ti] \geq 0); \text{cptr1}[ti] = \text{data\_addr}[r[ti]]; \text{ctrl1}[ti] = \text{mem\_control}[\text{cptr1}[ti]]; \text{cexp1}[ti] = \text{control\_exp}[r[ti]]; \text{assume}(\text{ctrl1}[ti] == \text{cexp1}[ti]);$<br>$\text{CAS}(\text{mem\_data}[\text{cptr1}[ti]], r[ti], \text{data\_new}[r[ti]]); \text{skip}];$<br>$\{\exists \text{old\_mem\_data\_cptr1\_ti\_}. (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j] \wedge r[ti] \geq 0 \wedge \text{cptr1}[ti] == \text{data\_addr}[r[ti]] \wedge \text{ctrl1}[ti] == \text{mem\_control}[\text{cptr1}[ti]] \wedge \text{cexp1}[ti] == \text{control\_exp}[r[ti]] \wedge \text{ctrl1}[ti] == \text{cexp1}[ti] \wedge \text{old\_mem\_data\_cptr1\_ti\_} = r[ti] \wedge \text{mem\_data}[\text{cptr1}[ti]] == \text{data\_new}[r[ti]]))\}$ |
| 32 | Subs | coop::cHelpFMain   | $\forall ti, t_j:\mathbb{N}, ti \neq t_j. \{\exists \text{old\_mem\_data\_cptr1\_ti\_}. (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j] \wedge r[ti] \geq 0 \wedge \text{cptr1}[ti] == \text{data\_addr}[r[ti]] \wedge \text{ctrl1}[ti] == \text{mem\_control}[\text{cptr1}[ti]] \wedge \text{cexp1}[ti] == \text{control\_exp}[r[ti]] \wedge \text{ctrl1}[ti] == \text{cexp1}[ti] \wedge \text{old\_mem\_data\_cptr1\_ti\_} = r[ti] \wedge \text{mem\_data}[\text{cptr1}[ti]] == \text{data\_new}[r[ti]]))\} \Rightarrow \{\text{mem\_data}[\text{cptr3}[t_j]] \neq \text{tid}[t_j]\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 33 | LMov | coop::cHelpFMain   | $\forall ti, t_j:\mathbb{N}, ti \neq t_j. \{(\exists \text{old\_mem\_data\_cptr1\_ti\_}. (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j] \wedge r[ti] \geq 0 \wedge \text{cptr1}[ti] == \text{data\_addr}[r[ti]] \wedge \text{ctrl1}[ti] == \text{mem\_control}[\text{cptr1}[ti]] \wedge \text{cexp1}[ti] == \text{control\_exp}[r[ti]] \wedge \text{ctrl1}[ti] == \text{cexp1}[ti] \wedge \text{old\_mem\_data\_cptr1\_ti\_} = r[ti] \wedge \text{mem\_data}[\text{cptr1}[ti]] == \text{data\_new}[r[ti]])) \wedge \text{mem\_data}[\text{cptr3}[t_j]] \neq \text{tid}[t_j])\}$<br>$\text{return } r[t_j]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 34 | Atom | coop::cHelpFLoop.R | $[\text{assume}(r \geq 0); \text{int } \text{cptr2} = \text{data\_addr}[r]; \text{int } \text{ctrl2} = \text{mem\_control}[\text{cptr2}]; \text{int } \text{cexp2} = \text{control\_exp}[r]; \text{assume}(\text{ctrl2} == \text{cexp2})]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |



| #  | Kind | Spec               | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----|------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 35 | RMov | coop::cHelpFLoop   | $\forall tj:\mathbb{N}. \text{assume}(r[tj] \geq 0); \text{cptr2}[tj] = \text{data\_addr}[r[tj]]; \text{ctrl2}[tj] = \text{mem\_control}[\text{cptr2}[tj]]; \text{cexp2}[tj] = \text{control\_exp}[r[tj]]; \text{assume}(\text{ctrl2}[tj] == \text{cexp2}[tj]) \triangleright \{(r[tj] \geq 0 \wedge \text{cptr2}[tj] == \text{data\_addr}[r[tj]] \wedge \text{ctrl2}[tj] == \text{mem\_control}[\text{cptr2}[tj]] \wedge \text{cexp2}[tj] == \text{control\_exp}[r[tj]] \wedge \text{ctrl2}[tj] == \text{cexp2}[tj])\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 36 | HTrp | coop::cHelpFLoop   | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{(r[tj] \geq 0 \wedge \text{cptr2}[tj] == \text{data\_addr}[r[tj]] \wedge \text{ctrl2}[tj] == \text{mem\_control}[\text{cptr2}[tj]] \wedge \text{cexp2}[tj] == \text{control\_exp}[r[tj]] \wedge \text{ctrl2}[tj] == \text{cexp2}[tj])\} \llbracket \text{assume}(r[ti] \geq 0); \text{cptr2}[ti] = \text{data\_addr}[r[ti]]; \text{ctrl2}[ti] = \text{mem\_control}[\text{cptr2}[ti]]; \text{cexp2}[ti] = \text{control\_exp}[r[ti]]; \text{assume}(\text{ctrl2}[ti] == \text{cexp2}[ti]); \text{CAS}(\text{mem\_data}[\text{cptr2}[ti]], r[ti], \text{data\_new}[r[ti]]); \text{skip} \{ \exists \text{old\_mem\_data\_cptr2\_ti\_}. (r[tj] \geq 0 \wedge \text{cptr2}[tj] == \text{data\_addr}[r[tj]] \wedge \text{ctrl2}[tj] == \text{mem\_control}[\text{cptr2}[tj]] \wedge \text{cexp2}[tj] == \text{control\_exp}[r[tj]] \wedge \text{ctrl2}[tj] == \text{cexp2}[tj] \wedge r[ti] \geq 0 \wedge \text{cptr2}[ti] == \text{data\_addr}[r[ti]] \wedge \text{ctrl2}[ti] == \text{mem\_control}[\text{cptr2}[ti]] \wedge \text{cexp2}[ti] == \text{control\_exp}[r[ti]] \wedge \text{ctrl2}[ti] == \text{cexp2}[ti] \wedge \text{old\_mem\_data\_cptr2\_ti\_} = r[ti] \wedge \text{mem\_data}[\text{cptr2}[ti]] == \text{data\_new}[r[ti]]) \}$ |
| 37 | Subs | coop::cHelpFLoop   | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{ \exists \text{old\_mem\_data\_cptr2\_ti\_}. (r[tj] \geq 0 \wedge \text{cptr2}[tj] == \text{data\_addr}[r[tj]] \wedge \text{ctrl2}[tj] == \text{mem\_control}[\text{cptr2}[tj]] \wedge \text{cexp2}[tj] == \text{control\_exp}[r[tj]] \wedge \text{ctrl2}[tj] == \text{cexp2}[tj] \wedge r[ti] \geq 0 \wedge \text{cptr2}[ti] == \text{data\_addr}[r[ti]] \wedge \text{ctrl2}[ti] == \text{mem\_control}[\text{cptr2}[ti]] \wedge \text{cexp2}[ti] == \text{control\_exp}[r[ti]] \wedge \text{ctrl2}[ti] == \text{cexp2}[ti] \wedge \text{old\_mem\_data\_cptr2\_ti\_} == r[ti] \wedge \text{mem\_data}[\text{cptr2}[ti]] == \text{data\_new}[r[ti]]) \} = \{ \text{mem\_data}[\text{cptr2}[tj]] \neq r[tj] \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 38 | Atom | coop::cHelpFSkip.R | $\llbracket \text{assume}(r < 0) \rrbracket$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 39 | RMov | coop::cHelpFSkip   | $\forall tj:\mathbb{N}. \text{assume}(r[tj] < 0) \triangleright \{r[tj] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 40 | HTrp | coop::cHelpFSkip   | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{r[tj] < 0\} \llbracket \text{assume}(r[ti] \geq 0); \text{cptr1}[ti] = \text{data\_addr}[r[ti]]; \text{ctrl1}[ti] = \text{mem\_control}[\text{cptr1}[ti]]; \text{cexp1}[ti] = \text{control\_exp}[r[ti]]; \text{assume}(\text{ctrl1}[ti] == \text{cexp1}[ti]); \text{CAS}(\text{mem\_data}[\text{cptr1}[ti]], r[ti], \text{data\_new}[r[ti]]); \text{skip} \{ \exists \text{old\_mem\_data\_cptr1\_ti\_}. (r[tj] < 0 \wedge r[ti] \geq 0 \wedge \text{cptr1}[ti] == \text{data\_addr}[r[ti]] \wedge \text{ctrl1}[ti] == \text{mem\_control}[\text{cptr1}[ti]] \wedge \text{cexp1}[ti] == \text{control\_exp}[r[ti]] \wedge \text{ctrl1}[ti] == \text{cexp1}[ti] \wedge \text{old\_mem\_data\_cptr1\_ti\_} = r[ti] \wedge \text{mem\_data}[\text{cptr1}[ti]] == \text{data\_new}[r[ti]]) \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 41 | Atom | coop::cLoopSkip.R  | $\llbracket \text{assume}(r < 0) \rrbracket$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 42 | RMov | coop::cLoopSkip    | $\forall tj:\mathbb{N}. \text{assume}(r[tj] < 0) \triangleright \{r[tj] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| #  | Kind | Spec                        | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----|------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 43 | HTrp | coop::cLoopSkip             | $\forall ti, tj: \mathbb{N}. ti \neq tj. \{r[tj] < 0\} \llbracket \text{assume}(r[ti] \neq \text{NULL}); \text{RCAS}(\text{mem\_data}[\text{ptr}[ti]], \text{data\_exp}[\text{tid}[ti]], \text{tid}[ti], r[ti]); \text{skip} \rrbracket \{\exists \text{old\_mem\_data\_ptr\_ti\_}. (r[tj] < 0 \wedge \text{old\_mem\_data\_ptr\_ti\_} \neq \text{NULL} \wedge \text{old\_mem\_data\_ptr\_ti\_} == \text{data\_exp}[\text{tid}[ti]] \wedge \text{mem\_data}[\text{ptr}[ti]] == \text{tid}[ti] \wedge r[ti] == \text{old\_mem\_data\_ptr\_ti\_})\}$ |
| 44 | Atom | atom::aHelperRestoreS.R     | $\llbracket \text{assume}(r \geq 0); \text{int cptr1} = \text{data\_addr}[r]; \text{int ctrl1} = \text{mem\_control}[\text{cptr1}]; \text{int cexp1} = \text{control\_exp}[r]; \text{assume}(\text{ctrl1} \neq \text{cexp1}) \rrbracket$                                                                                                                                                                                                                                                                                                           |
| 45 | RMov | atom::aHelperRestoreS       | $\forall t: \mathbb{N}. \text{assume}(r[t] \geq 0); \text{cptr1}[t] = \text{data\_addr}[r[t]]; \text{ctrl1}[t] = \text{mem\_control}[\text{cptr1}[t]]; \text{cexp1}[t] = \text{control\_exp}[r[t]]; \text{assume}(\text{ctrl1}[t] \neq \text{cexp1}[t]) \triangleright \{\text{mem\_data}[\text{cptr1}[t]] == r[t]\}$                                                                                                                                                                                                                              |
| 46 | HTrp | atom::aHelperRestoreS       | $\forall t: \mathbb{N}. \{\text{mem\_data}[\text{cptr1}[t]] == r[t]\} \llbracket \text{CAS}(\text{mem\_data}[\text{cptr1}[t]], r[t], \text{data\_exp}[r[t]]) \rrbracket \{\exists \text{old\_mem\_data\_cptr1\_t\_}. (\text{old\_mem\_data\_cptr1\_t\_} == r[t] \wedge \text{mem\_data}[\text{cptr1}[t]] == \text{data\_exp}[r[t]])\}$                                                                                                                                                                                                             |
| 47 | Atom | atom::aHelperRestoreSLoop.R | $\llbracket \text{assume}(r \geq 0); \text{int cptr2} = \text{data\_addr}[r]; \text{int ctrl2} = \text{mem\_control}[\text{cptr2}]; \text{int cexp2} = \text{control\_exp}[r]; \text{assume}(\text{ctrl2} \neq \text{cexp2}) \rrbracket$                                                                                                                                                                                                                                                                                                           |
| 48 | RMov | atom::aHelperRestoreSLoop   | $\forall t: \mathbb{N}. \text{assume}(r[t] \geq 0); \text{cptr2}[t] = \text{data\_addr}[r[t]]; \text{ctrl2}[t] = \text{mem\_control}[\text{cptr2}[t]]; \text{cexp2}[t] = \text{control\_exp}[r[t]]; \text{assume}(\text{ctrl2}[t] \neq \text{cexp2}[t]) \triangleright \{\text{mem\_data}[\text{cptr2}[t]] == r[t]\}$                                                                                                                                                                                                                              |
| 49 | HTrp | atom::aHelperRestoreSLoop   | $\forall t: \mathbb{N}. \{\text{mem\_data}[\text{cptr2}[t]] == r[t]\} \llbracket \text{CAS}(\text{mem\_data}[\text{cptr2}[t]], r[t], \text{data\_exp}[r[t]]) \rrbracket \{\exists \text{old\_mem\_data\_cptr2\_t\_}. (\text{old\_mem\_data\_cptr2\_t\_} == r[t] \wedge \text{mem\_data}[\text{cptr2}[t]] == \text{data\_exp}[r[t]])\}$                                                                                                                                                                                                             |
| 50 | Atom | atom::aRestoreS.R           | $\llbracket \text{assume}(r == \text{NULL}); \text{assume}(r == \text{data\_exp}[\text{tid}]); \text{int cptr3} = \text{data\_addr}[\text{tid}]; \text{int ctrl3} = \text{mem\_control}[\text{cptr3}]; \text{int cexp3} = \text{control\_exp}[\text{tid}]; \text{assume}(\text{ctrl3} \neq \text{cexp3}) \rrbracket$                                                                                                                                                                                                                               |
| 51 | RMov | atom::aRestoreS             | $\forall t: \mathbb{N}. \text{assume}(r[t] == \text{NULL}); \text{assume}(r[t] == \text{data\_exp}[\text{tid}[t]]); \text{cptr3}[t] = \text{data\_addr}[\text{tid}[t]]; \text{ctrl3}[t] = \text{mem\_control}[\text{cptr3}[t]]; \text{cexp3}[t] = \text{control\_exp}[\text{tid}[t]]; \text{assume}(\text{ctrl3}[t] \neq \text{cexp3}[t]) \triangleright \{\text{mem\_data}[\text{cptr3}[t]] == \text{tid}[t]\}$                                                                                                                                   |
| 52 | HTrp | atom::aRestoreS             | $\forall t: \mathbb{N}. \{\text{mem\_data}[\text{cptr3}[t]] == \text{tid}[t]\} \llbracket \text{CAS}(\text{mem\_data}[\text{cptr3}[t]], \text{tid}[t], \text{data\_exp}[\text{tid}[t]]) \rrbracket \{\exists \text{old\_mem\_data\_cptr3\_t\_}. (\text{old\_mem\_data\_cptr3\_t\_} == \text{tid}[t] \wedge \text{mem\_data}[\text{cptr3}[t]] == \text{data\_exp}[\text{tid}[t]])\}$                                                                                                                                                                |
| 53 | LMov | atom::aRestoreS             | $\forall t: \mathbb{N}. \{\exists \text{old\_mem\_data\_cptr3\_t\_}. (\text{old\_mem\_data\_cptr3\_t\_} == \text{tid}[t] \wedge \text{mem\_data}[\text{cptr3}[t]] == \text{data\_exp}[\text{tid}[t]])\} \triangleleft \text{return } r[t]$                                                                                                                                                                                                                                                                                                         |
| 54 | Atom | coop::cRestoreF.R           | $\llbracket \text{assume}(r \geq 0); \text{int cptr1} = \text{data\_addr}[r]; \text{int ctrl1} = \text{mem\_control}[\text{cptr1}]; \text{int cexp1} = \text{control\_exp}[r]; \text{assume}(\text{ctrl1} \neq \text{cexp1}) \rrbracket$                                                                                                                                                                                                                                                                                                           |

| #  | Kind | Spec                    | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----|------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 55 | RMov | coop::cRestoreF         | $\forall tj:\mathbb{N}. \text{assume}(r[tj] \geq 0); \text{cptr1}[tj] = \text{data\_addr}[r[tj]]; \text{ctrl1}[tj] = \text{mem\_control}[\text{cptr1}[tj]]; \text{cexp1}[tj] = \text{control\_exp}[r[tj]]; \text{assume}(\text{ctrl1}[tj] \neq \text{cexp1}[tj]) \triangleright \{(r[tj] \geq 0 \wedge \text{cptr1}[tj] == \text{data\_addr}[r[tj]] \wedge \text{ctrl1}[tj] == \text{mem\_control}[\text{cptr1}[tj]] \wedge \text{cexp1}[tj] == \text{control\_exp}[r[tj]] \wedge \text{ctrl1}[tj] \neq \text{cexp1}[tj])\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 56 | HTrp | coop::cRestoreF         | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{(r[tj] \geq 0 \wedge \text{cptr1}[tj] == \text{data\_addr}[r[tj]] \wedge \text{ctrl1}[tj] == \text{mem\_control}[\text{cptr1}[tj]] \wedge \text{cexp1}[tj] == \text{control\_exp}[r[tj]] \wedge \text{ctrl1}[tj] \neq \text{cexp1}[tj])\} \llbracket \text{assume}(r[ti] == \text{NULL}); \text{assume}(r[ti] == \text{data\_exp}[\text{tid}[ti]]); \text{cptr3}[ti] = \text{data\_addr}[\text{tid}[ti]]; \text{ctrl3}[ti] = \text{mem\_control}[\text{cptr3}[ti]]; \text{cexp3}[ti] = \text{control\_exp}[\text{tid}[ti]]; \text{assume}(\text{ctrl3}[ti] \neq \text{cexp3}[ti]); \text{CAS}(\text{mem\_data}[\text{cptr3}[ti]], \text{tid}[ti], \text{data\_exp}[\text{tid}[ti]]); \text{return } r[ti] \rrbracket \{\text{old\_mem\_data\_cptr3\_ti\_}.$<br>$(r[tj] \geq 0 \wedge \text{cptr1}[tj] == \text{data\_addr}[r[tj]] \wedge \text{ctrl1}[tj] == \text{mem\_control}[\text{cptr1}[tj]] \wedge \text{cexp1}[tj] == \text{control\_exp}[r[tj]] \wedge \text{ctrl1}[tj] \neq \text{cexp1}[tj] \wedge r[ti] == \text{NULL} \wedge r[ti] == \text{data\_exp}[\text{tid}[ti]] \wedge \text{cptr3}[ti] == \text{data\_addr}[\text{tid}[ti]] \wedge \text{ctrl3}[ti] == \text{mem\_control}[\text{cptr3}[ti]] \wedge \text{cexp3}[ti] == \text{control\_exp}[\text{tid}[ti]] \wedge \text{ctrl3}[ti] \neq \text{cexp3}[ti] \wedge \text{old\_mem\_data\_cptr3\_ti\_} == \text{tid}[ti] \wedge \text{mem\_data}[\text{cptr3}[ti]] == \text{data\_exp}[\text{tid}[ti]])\}$ |
| 57 | Subs | coop::cRestoreF         | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{\text{old\_mem\_data\_cptr3\_ti\_}.$<br>$(r[tj] \geq 0 \wedge \text{cptr1}[tj] == \text{data\_addr}[r[tj]] \wedge \text{ctrl1}[tj] == \text{mem\_control}[\text{cptr1}[tj]] \wedge \text{cexp1}[tj] == \text{control\_exp}[r[tj]] \wedge \text{ctrl1}[tj] \neq \text{cexp1}[tj] \wedge r[ti] == \text{NULL} \wedge r[ti] == \text{data\_exp}[\text{tid}[ti]] \wedge \text{cptr3}[ti] == \text{data\_addr}[\text{tid}[ti]] \wedge \text{ctrl3}[ti] == \text{mem\_control}[\text{cptr3}[ti]] \wedge \text{cexp3}[ti] == \text{control\_exp}[\text{tid}[ti]] \wedge \text{ctrl3}[ti] \neq \text{cexp3}[ti] \wedge \text{old\_mem\_data\_cptr3\_ti\_} == \text{tid}[ti] \wedge \text{mem\_data}[\text{cptr3}[ti]] == \text{data\_exp}[\text{tid}[ti]])\} \Rightarrow \{\text{mem\_data}[\text{cptr1}[tj]] \neq r[tj]\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 58 | Atom | coop::cHelperRestoreF.R | $\llbracket \text{assume}(r == \text{NULL}); \text{assume}(r == \text{data\_exp}[\text{tid}]); \text{int } \text{cptr3} = \text{data\_addr}[\text{tid}]; \text{int } \text{ctrl3} = \text{mem\_control}[\text{cptr3}]; \text{int } \text{cexp3} = \text{control\_exp}[\text{tid}]; \text{assume}(\text{ctrl3} \neq \text{cexp3}) \rrbracket$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 59 | RMov | coop::cHelperRestoreF   | $\forall tj:\mathbb{N}. \text{assume}(r[tj] == \text{NULL}); \text{assume}(r[tj] == \text{data\_exp}[\text{tid}[tj]]); \text{cptr3}[tj] = \text{data\_addr}[\text{tid}[tj]]; \text{ctrl3}[tj] = \text{mem\_control}[\text{cptr3}[tj]]; \text{cexp3}[tj] = \text{control\_exp}[\text{tid}[tj]]; \text{assume}(\text{ctrl3}[tj] \neq \text{cexp3}[tj]) \triangleright \{(r[tj] == \text{NULL} \wedge r[tj] == \text{data\_exp}[\text{tid}[tj]] \wedge \text{cptr3}[tj] == \text{data\_addr}[\text{tid}[tj]] \wedge \text{ctrl3}[tj] == \text{mem\_control}[\text{cptr3}[tj]] \wedge \text{cexp3}[tj] == \text{control\_exp}[\text{tid}[tj]] \wedge \text{ctrl3}[tj] \neq \text{cexp3}[tj])\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| #  | Kind | Spec                        | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----|------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 60 | HTrp | coop::cHelperRestoreF       | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{(r[tj] == \text{NULL} \wedge r[tj] == \text{data\_exp}[tid[tj]] \wedge \text{cptr3}[tj] == \text{data\_addr}[tid[tj]] \wedge \text{ctrl3}[tj] == \text{mem\_control}[\text{cptr3}[tj]] \wedge \text{cexp3}[tj] == \text{control\_exp}[tid[tj]] \wedge \text{ctrl3}[tj] \neq \text{cexp3}[tj])\}$<br>$[[\text{assume}(r[ti] \geq 0); \text{cptr1}[ti] = \text{data\_addr}[r[ti]]; \text{ctrl1}[ti] = \text{mem\_control}[\text{cptr1}[ti]]; \text{cexp1}[ti] = \text{control\_exp}[r[ti]]; \text{assume}(\text{ctrl1}[ti] \neq \text{cexp1}[ti]); \text{CAS}(\text{mem\_data}[\text{cptr1}[ti]], r[ti], \text{data\_exp}[r[ti]]); \text{skip}]]$<br>$\{\exists \text{old\_mem\_data\_cptr1\_ti\_}. (r[tj] == \text{NULL} \wedge r[tj] == \text{data\_exp}[tid[tj]] \wedge \text{cptr3}[tj] == \text{data\_addr}[tid[tj]] \wedge \text{ctrl3}[tj] == \text{mem\_control}[\text{cptr3}[tj]] \wedge \text{cexp3}[tj] == \text{control\_exp}[tid[tj]] \wedge \text{ctrl3}[tj] \neq \text{cexp3}[tj] \wedge r[ti] \geq 0 \wedge \text{cptr1}[ti] == \text{data\_addr}[r[ti]] \wedge \text{ctrl1}[ti] == \text{mem\_control}[\text{cptr1}[ti]] \wedge \text{cexp1}[ti] == \text{control\_exp}[r[ti]] \wedge \text{ctrl1}[ti] \neq \text{cexp1}[ti] \wedge \text{old\_mem\_data\_cptr1\_ti\_} == r[ti] \wedge \text{mem\_data}[\text{cptr1}[ti]] == \text{data\_exp}[r[ti]])\}$ |
| 61 | Subs | coop::cHelperRestoreF       | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{\exists \text{old\_mem\_data\_cptr1\_ti\_}. (r[tj] == \text{NULL} \wedge r[tj] == \text{data\_exp}[tid[tj]] \wedge \text{cptr3}[tj] == \text{data\_addr}[tid[tj]] \wedge \text{ctrl3}[tj] == \text{mem\_control}[\text{cptr3}[tj]] \wedge \text{cexp3}[tj] == \text{control\_exp}[tid[tj]] \wedge \text{ctrl3}[tj] \neq \text{cexp3}[tj] \wedge r[ti] \geq 0 \wedge \text{cptr1}[ti] == \text{data\_addr}[r[ti]] \wedge \text{ctrl1}[ti] == \text{mem\_control}[\text{cptr1}[ti]] \wedge \text{cexp1}[ti] == \text{control\_exp}[r[ti]] \wedge \text{ctrl1}[ti] \neq \text{cexp1}[ti] \wedge \text{old\_mem\_data\_cptr1\_ti\_} == r[ti] \wedge \text{mem\_data}[\text{cptr1}[ti]] == \text{data\_exp}[r[ti]])\} \Rightarrow \{\text{mem\_data}[\text{cptr3}[tj]] \neq \text{tid}[tj]\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 62 | LMov | coop::cHelperRestoreF       | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{(\exists \text{old\_mem\_data\_cptr1\_ti\_}. (r[tj] == \text{NULL} \wedge r[tj] == \text{data\_exp}[tid[tj]] \wedge \text{cptr3}[tj] == \text{data\_addr}[tid[tj]] \wedge \text{ctrl3}[tj] == \text{mem\_control}[\text{cptr3}[tj]] \wedge \text{cexp3}[tj] == \text{control\_exp}[tid[tj]] \wedge \text{ctrl3}[tj] \neq \text{cexp3}[tj] \wedge r[ti] \geq 0 \wedge \text{cptr1}[ti] == \text{data\_addr}[r[ti]] \wedge \text{ctrl1}[ti] == \text{mem\_control}[\text{cptr1}[ti]] \wedge \text{cexp1}[ti] == \text{control\_exp}[r[ti]] \wedge \text{ctrl1}[ti] \neq \text{cexp1}[ti] \wedge \text{old\_mem\_data\_cptr1\_ti\_} == r[ti] \wedge \text{mem\_data}[\text{cptr1}[ti]] == \text{data\_exp}[r[ti]]) \wedge \text{mem\_data}[\text{cptr3}[tj]] \neq \text{tid}[tj])\}$<br>$\text{return } r[tj]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 63 | Atom | coop::cHelperRestoreFLoop.R | $[\text{assume}(r \geq 0); \text{int } \text{cptr2} = \text{data\_addr}[r]; \text{int } \text{ctrl2} = \text{mem\_control}[\text{cptr2}]; \text{int } \text{cexp2} = \text{control\_exp}[r]; \text{assume}(\text{ctrl2} \neq \text{cexp2})]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 64 | RMov | coop::cHelperRestoreFLoop   | $\forall tj: \mathbb{N}. \text{assume}(r[tj] \geq 0); \text{cptr2}[tj] = \text{data\_addr}[r[tj]]; \text{ctrl2}[tj] = \text{mem\_control}[\text{cptr2}[tj]]; \text{cexp2}[tj] = \text{control\_exp}[r[tj]]; \text{assume}(\text{ctrl2}[tj] \neq \text{cexp2}[tj]) \triangleright \{(r[tj] \geq 0 \wedge \text{cptr2}[tj] == \text{data\_addr}[r[tj]] \wedge \text{ctrl2}[tj] == \text{mem\_control}[\text{cptr2}[tj]] \wedge \text{cexp2}[tj] == \text{control\_exp}[r[tj]] \wedge \text{ctrl2}[tj] \neq \text{cexp2}[tj])\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| #  | Kind | Spec                      | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----|------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 65 | HTrp | coop::cHelperRestoreFLoop | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ (r[tj] \geq 0 \wedge cptr2[tj] == data\_addr[r[tj]] \wedge ctrl2[tj] == mem\_control[cptr2[tj]] \wedge cexp2[tj] == control\_exp[r[tj]] \wedge ctrl2[tj] \neq cexp2[tj]) \} \llbracket assume(r[ti] \geq 0); cptr2[ti] = data\_addr[r[ti]]; ctrl2[ti] = mem\_control[cptr2[ti]]; cexp2[ti] = control\_exp[r[ti]]; assume(ctrl2[ti] \neq cexp2[ti]); CAS(mem\_data[cptr2[ti]], r[ti], data\_exp[r[ti]]); skip \} \{ \text{old\_mem\_data\_cptr2\_ti\_}.$ $(r[tj] \geq 0 \wedge cptr2[tj] == data\_addr[r[tj]] \wedge ctrl2[tj] == mem\_control[cptr2[tj]] \wedge cexp2[tj] == control\_exp[r[tj]] \wedge ctrl2[tj] \neq cexp2[tj] \wedge r[ti] \geq 0 \wedge cptr2[ti] == data\_addr[r[ti]] \wedge ctrl2[ti] == mem\_control[cptr2[ti]] \wedge cexp2[ti] == control\_exp[r[ti]] \wedge ctrl2[ti] \neq cexp2[ti] \wedge \text{old\_mem\_data\_cptr2\_ti\_} == r[ti] \wedge mem\_data[cptr2[ti]] == data\_exp[r[ti]]) \}$                                                                                                                                                                                                                                                              |
| 66 | Subs | coop::cHelperRestoreFLoop | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ \text{old\_mem\_data\_cptr2\_ti\_}.$ $(r[tj] \geq 0 \wedge cptr2[tj] == data\_addr[r[tj]] \wedge ctrl2[tj] == mem\_control[cptr2[tj]] \wedge cexp2[tj] == control\_exp[r[tj]] \wedge ctrl2[tj] \neq cexp2[tj] \wedge r[ti] \geq 0 \wedge cptr2[ti] == data\_addr[r[ti]] \wedge ctrl2[ti] == mem\_control[cptr2[ti]] \wedge cexp2[ti] == control\_exp[r[ti]] \wedge ctrl2[ti] \neq cexp2[ti] \wedge \text{old\_mem\_data\_cptr2\_ti\_} == r[ti] \wedge mem\_data[cptr2[ti]] == data\_exp[r[ti]]) \} \Rightarrow \{ mem\_data[cptr2[tj]] \neq r[tj] \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 67 | Atom | coop::cHelpFExit.R        | $\llbracket assume(r == NULL) \rrbracket$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 68 | RMov | coop::cHelpFExit          | $\forall tj: \mathbb{N}. assume(r[tj] == NULL) \triangleright \{ r[tj] == NULL \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 69 | HTrp | coop::cHelpFExit          | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ r[tj] == NULL \} \llbracket assume(r[ti] \geq 0); cptr1[ti] = data\_addr[r[ti]]; ctrl1[ti] = mem\_control[cptr1[ti]]; cexp1[ti] = control\_exp[r[ti]]; assume(ctrl1[ti] == cexp1[ti]); CAS(mem\_data[cptr1[ti]], r[ti], data\_new[r[ti]]); skip \} + \llbracket assume(r[ti] \geq 0); cptr1[ti] = data\_addr[r[ti]]; ctrl1[ti] = mem\_control[cptr1[ti]]; cexp1[ti] = control\_exp[r[ti]]; assume(ctrl1[ti] \neq cexp1[ti]); CAS(mem\_data[cptr1[ti]], r[ti], data\_exp[r[ti]]); skip \} \{ (\text{old\_mem\_data\_cptr1\_ti\_}.$ $(r[tj] == NULL \wedge r[ti] \geq 0 \wedge cptr1[ti] == data\_addr[r[ti]] \wedge ctrl1[ti] == mem\_control[cptr1[ti]] \wedge cexp1[ti] == control\_exp[r[ti]] \wedge ctrl1[ti] == cexp1[ti] \wedge \text{old\_mem\_data\_cptr1\_ti\_} == r[ti] \wedge mem\_data[cptr1[ti]] == data\_new[r[ti]])$ $\vee \text{old\_mem\_data\_cptr1\_ti\_}.$ $(r[tj] == NULL \wedge r[ti] \geq 0 \wedge cptr1[ti] == data\_addr[r[ti]] \wedge ctrl1[ti] == mem\_control[cptr1[ti]] \wedge cexp1[ti] == control\_exp[r[ti]] \wedge ctrl1[ti] \neq cexp1[ti] \wedge \text{old\_mem\_data\_cptr1\_ti\_} == r[ti] \wedge mem\_data[cptr1[ti]] == data\_exp[r[ti]]) \}$ |
| 70 | LMov | coop::cHelpFExit          | $\forall ti, tj: \mathbb{N}, ti \neq tj. assume(r[tj] \neq data\_exp[tid[tj]])$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| #  | Kind | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----|------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 71 | LMov | coop::cHelpFExit | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{((\exists old\_mem\_data\_cptr1\_ti\_ \_. (r[ti] == NULL \wedge r[ti] \geq 0 \wedge cptr1[ti] == data\_addr[r[ti]] \wedge ctrl1[ti] == mem\_control[cptr1[ti]] \wedge cexp1[ti] == control\_exp[r[ti]] \wedge ctrl1[ti] == cexp1[ti] \wedge old\_mem\_data\_cptr1\_ti\_ == r[ti] \wedge mem\_data[cptr1[ti]] == data\_new[r[ti]])) \vee \exists old\_mem\_data\_cptr1\_ti\_ \_. (r[ti] == NULL \wedge r[ti] \geq 0 \wedge cptr1[ti] == data\_addr[r[ti]] \wedge ctrl1[ti] == mem\_control[cptr1[ti]] \wedge cexp1[ti] == control\_exp[r[ti]] \wedge ctrl1[ti] \neq cexp1[ti] \wedge old\_mem\_data\_cptr1\_ti\_ == r[ti] \wedge mem\_data[cptr1[ti]] == data\_exp[r[ti]])) \wedge r[tj] \neq data\_exp[tid[tj]]))\}$ $\text{return } r[tj]$ |

#### B.4.4 Automata (684).

coverage.

| Check          | Result |
|----------------|--------|
| RDCSS: COVERED | OK     |

non-overlapping.

1 all distinct pairs are non-overlapping

non-self-overlapping.

1 no spec self-overlaps

bijection part I: well-definedness.

| Check                                                                  | Result |
|------------------------------------------------------------------------|--------|
| cHelperRestoreFLoop: induction_pattern $\subseteq$ exactly_one_ais: OK | OK     |
| cHelpFSkip: induction_pattern $\subseteq$ exactly_one_ais: OK          | OK     |
| aHelperRestoreSLoop: induction_pattern $\subseteq$ exactly_one_ais: OK | OK     |
| cHelpFExit: induction_pattern $\subseteq$ exactly_one_ais: OK          | OK     |
| cHelpFMain: induction_pattern $\subseteq$ exactly_one_ais: OK          | OK     |
| cLoopSkip: induction_pattern $\subseteq$ exactly_one_ais: OK           | OK     |
| cDataFLoop: induction_pattern $\subseteq$ exactly_one_ais: OK          | OK     |
| aHelpSLoop: induction_pattern $\subseteq$ exactly_one_ais: OK          | OK     |
| cRestoreF: induction_pattern $\subseteq$ exactly_one_ais: OK           | OK     |
| aRestoreS: induction_pattern $\subseteq$ exactly_one_ais: OK           | OK     |
| aHelperRestoreS: induction_pattern $\subseteq$ exactly_one_ais: OK     | OK     |
| aDataS: induction_pattern $\subseteq$ exactly_one_ais: OK              | OK     |
| aControls: induction_pattern $\subseteq$ exactly_one_ais: OK           | OK     |
| aHelpS: induction_pattern $\subseteq$ exactly_one_ais: OK              | OK     |
| cHelperRestoreF: induction_pattern $\subseteq$ exactly_one_ais: OK     | OK     |
| cHelpF: induction_pattern $\subseteq$ exactly_one_ais: OK              | OK     |

| Check                                                                           | Result |
|---------------------------------------------------------------------------------|--------|
| cHelpFLoop: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK | OK     |
| cDataF: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK     | OK     |
| aDataSLoop: $\text{induction\_pattern} \subseteq \text{exactly\_one\_ais}$ : OK | OK     |

*bijection part II: surjectivity.*

| Check                                                                                           | Result |
|-------------------------------------------------------------------------------------------------|--------|
| $\text{aDataS} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK              | OK     |
| $\text{aDataSLoop} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK          | OK     |
| $\text{aHelpS} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK              | OK     |
| $\text{aHelpSLoop} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK          | OK     |
| $\text{aHelperRestoreS} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK     | OK     |
| $\text{aHelperRestoreSLoop} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK | OK     |
| $\text{aRestoreS} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK           | OK     |
| $\text{cDataF} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK              | OK     |
| $\text{cDataFLoop} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK          | OK     |
| $\text{cHelpF} \subseteq \text{at\_most\_one\_ais}(\text{aControls})$ : OK                      | OK     |
| $\text{cHelpFExit} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK          | OK     |
| $\text{cHelpFLoop} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK          | OK     |
| $\text{cHelpFMain} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK          | OK     |
| $\text{cHelpFSkip} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK          | OK     |
| $\text{cHelperRestoreF} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK     | OK     |
| $\text{cHelperRestoreFLoop} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK | OK     |
| $\text{cLoopSkip} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK           | OK     |
| $\text{cRestoreF} \cap \text{at\_least\_one\_ais}(\text{aControls}) = \emptyset$ : OK           | OK     |
| $\text{aControls} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK              | OK     |
| $\text{aDataSLoop} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK             | OK     |
| $\text{aHelpS} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK                 | OK     |
| $\text{aHelpSLoop} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK             | OK     |
| $\text{aHelperRestoreS} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK        | OK     |
| $\text{aHelperRestoreSLoop} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK    | OK     |
| $\text{aRestoreS} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK              | OK     |
| $\text{cDataF} \subseteq \text{at\_most\_one\_ais}(\text{aDataS})$ : OK                         | OK     |
| $\text{cDataFLoop} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK             | OK     |
| $\text{cHelpF} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK                 | OK     |
| $\text{cHelpFExit} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK             | OK     |
| $\text{cHelpFLoop} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK             | OK     |
| $\text{cHelpFMain} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK             | OK     |
| $\text{cHelpFSkip} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK             | OK     |
| $\text{cHelperRestoreF} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK        | OK     |
| $\text{cHelperRestoreFLoop} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK    | OK     |
| $\text{cLoopSkip} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK              | OK     |
| $\text{cRestoreF} \cap \text{at\_least\_one\_ais}(\text{aDataS}) = \emptyset$ : OK              | OK     |
| $\text{aControls} \cap \text{at\_least\_one\_ais}(\text{aDataSLoop}) = \emptyset$ : OK          | OK     |

| Check                                                                       | Result |
|-----------------------------------------------------------------------------|--------|
| $aDataS \cap at\_least\_one\_ais(aDataSLoop) = \emptyset$ : OK              | OK     |
| $aHelpS \cap at\_least\_one\_ais(aDataSLoop) = \emptyset$ : OK              | OK     |
| $aHelpSLoop \cap at\_least\_one\_ais(aDataSLoop) = \emptyset$ : OK          | OK     |
| $aHelperRestoreS \cap at\_least\_one\_ais(aDataSLoop) = \emptyset$ : OK     | OK     |
| $aHelperRestoreSLoop \cap at\_least\_one\_ais(aDataSLoop) = \emptyset$ : OK | OK     |
| $aRestoreS \cap at\_least\_one\_ais(aDataSLoop) = \emptyset$ : OK           | OK     |
| $cDataF \cap at\_least\_one\_ais(aDataSLoop) = \emptyset$ : OK              | OK     |
| $cDataFLoop \subseteq at\_most\_one\_ais(aDataSLoop)$ : OK                  | OK     |
| $cHelpF \cap at\_least\_one\_ais(aDataSLoop) = \emptyset$ : OK              | OK     |
| $cHelpFExit \cap at\_least\_one\_ais(aDataSLoop) = \emptyset$ : OK          | OK     |
| $cHelpFLoop \cap at\_least\_one\_ais(aDataSLoop) = \emptyset$ : OK          | OK     |
| $cHelpFMain \cap at\_least\_one\_ais(aDataSLoop) = \emptyset$ : OK          | OK     |
| $cHelpFSkip \cap at\_least\_one\_ais(aDataSLoop) = \emptyset$ : OK          | OK     |
| $cHelperRestoreF \cap at\_least\_one\_ais(aDataSLoop) = \emptyset$ : OK     | OK     |
| $cHelperRestoreFLoop \cap at\_least\_one\_ais(aDataSLoop) = \emptyset$ : OK | OK     |
| $cLoopSkip \subseteq at\_most\_one\_ais(aDataSLoop)$ : OK                   | OK     |
| $cRestoreF \cap at\_least\_one\_ais(aDataSLoop) = \emptyset$ : OK           | OK     |
| $aControlS \cap at\_least\_one\_ais(aHelpS) = \emptyset$ : OK               | OK     |
| $aDataS \cap at\_least\_one\_ais(aHelpS) = \emptyset$ : OK                  | OK     |
| $aDataSLoop \cap at\_least\_one\_ais(aHelpS) = \emptyset$ : OK              | OK     |
| $aHelpSLoop \cap at\_least\_one\_ais(aHelpS) = \emptyset$ : OK              | OK     |
| $aHelperRestoreS \cap at\_least\_one\_ais(aHelpS) = \emptyset$ : OK         | OK     |
| $aHelperRestoreSLoop \cap at\_least\_one\_ais(aHelpS) = \emptyset$ : OK     | OK     |
| $aRestoreS \cap at\_least\_one\_ais(aHelpS) = \emptyset$ : OK               | OK     |
| $cDataF \cap at\_least\_one\_ais(aHelpS) = \emptyset$ : OK                  | OK     |
| $cDataFLoop \cap at\_least\_one\_ais(aHelpS) = \emptyset$ : OK              | OK     |
| $cHelpF \subseteq at\_most\_one\_ais(aHelpS)$ : OK                          | OK     |
| $cHelpFExit \subseteq at\_most\_one\_ais(aHelpS)$ : OK                      | OK     |
| $cHelpFLoop \cap at\_least\_one\_ais(aHelpS) = \emptyset$ : OK              | OK     |
| $cHelpFMain \subseteq at\_most\_one\_ais(aHelpS)$ : OK                      | OK     |
| $cHelpFSkip \subseteq at\_most\_one\_ais(aHelpS)$ : OK                      | OK     |
| $cHelperRestoreF \cap at\_least\_one\_ais(aHelpS) = \emptyset$ : OK         | OK     |
| $cHelperRestoreFLoop \cap at\_least\_one\_ais(aHelpS) = \emptyset$ : OK     | OK     |
| $cLoopSkip \cap at\_least\_one\_ais(aHelpS) = \emptyset$ : OK               | OK     |
| $cRestoreF \cap at\_least\_one\_ais(aHelpS) = \emptyset$ : OK               | OK     |
| $aControlS \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK           | OK     |
| $aDataS \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK              | OK     |
| $aDataSLoop \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK          | OK     |
| $aHelpS \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK              | OK     |
| $aHelperRestoreS \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK     | OK     |
| $aHelperRestoreSLoop \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK | OK     |
| $aRestoreS \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK           | OK     |
| $cDataF \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK              | OK     |



| Check                                                                            | Result |
|----------------------------------------------------------------------------------|--------|
| $cDataFLoop \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK               | OK     |
| $cHelpF \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK                   | OK     |
| $cHelpFExit \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK               | OK     |
| $cHelpFLoop \subseteq at\_most\_one\_ais(aHelpSLoop)$ : OK                       | OK     |
| $cHelpFMain \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK               | OK     |
| $cHelpFSkip \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK               | OK     |
| $cHelperRestoreF \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK          | OK     |
| $cHelperRestoreFLoop \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK      | OK     |
| $cLoopSkip \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK                | OK     |
| $cRestoreF \cap at\_least\_one\_ais(aHelpSLoop) = \emptyset$ : OK                | OK     |
| $aControlS \cap at\_least\_one\_ais(aHelperRestoreS) = \emptyset$ : OK           | OK     |
| $aDataS \cap at\_least\_one\_ais(aHelperRestoreS) = \emptyset$ : OK              | OK     |
| $aDataSLoop \cap at\_least\_one\_ais(aHelperRestoreS) = \emptyset$ : OK          | OK     |
| $aHelpS \cap at\_least\_one\_ais(aHelperRestoreS) = \emptyset$ : OK              | OK     |
| $aHelpSLoop \cap at\_least\_one\_ais(aHelperRestoreS) = \emptyset$ : OK          | OK     |
| $aHelperRestoreSLoop \cap at\_least\_one\_ais(aHelperRestoreS) = \emptyset$ : OK | OK     |
| $aRestoreS \cap at\_least\_one\_ais(aHelperRestoreS) = \emptyset$ : OK           | OK     |
| $cDataF \cap at\_least\_one\_ais(aHelperRestoreS) = \emptyset$ : OK              | OK     |
| $cDataFLoop \cap at\_least\_one\_ais(aHelperRestoreS) = \emptyset$ : OK          | OK     |
| $cHelpF \cap at\_least\_one\_ais(aHelperRestoreS) = \emptyset$ : OK              | OK     |
| $cHelpFExit \subseteq at\_most\_one\_ais(aHelperRestoreS)$ : OK                  | OK     |
| $cHelpFLoop \cap at\_least\_one\_ais(aHelperRestoreS) = \emptyset$ : OK          | OK     |
| $cHelpFMain \cap at\_least\_one\_ais(aHelperRestoreS) = \emptyset$ : OK          | OK     |
| $cHelpFSkip \cap at\_least\_one\_ais(aHelperRestoreS) = \emptyset$ : OK          | OK     |
| $cHelperRestoreF \subseteq at\_most\_one\_ais(aHelperRestoreS)$ : OK             | OK     |
| $cHelperRestoreFLoop \cap at\_least\_one\_ais(aHelperRestoreS) = \emptyset$ : OK | OK     |
| $cLoopSkip \cap at\_least\_one\_ais(aHelperRestoreS) = \emptyset$ : OK           | OK     |
| $cRestoreF \cap at\_least\_one\_ais(aHelperRestoreS) = \emptyset$ : OK           | OK     |
| $aControlS \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK       | OK     |
| $aDataS \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK          | OK     |
| $aDataSLoop \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK      | OK     |
| $aHelpS \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK          | OK     |
| $aHelpSLoop \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK      | OK     |
| $aHelperRestoreS \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK | OK     |
| $aRestoreS \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK       | OK     |
| $cDataF \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK          | OK     |
| $cDataFLoop \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK      | OK     |
| $cHelpF \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK          | OK     |

| Check                                                                            | Result |
|----------------------------------------------------------------------------------|--------|
| $cHelpFExit \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK      | OK     |
| $cHelpFLoop \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK      | OK     |
| $cHelpFMain \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK      | OK     |
| $cHelpFSkip \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK      | OK     |
| $cHelperRestoreF \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK | OK     |
| $cHelperRestoreFLoop \subseteq at\_most\_one\_ais(aHelperRestoreSLoop)$ : OK     | OK     |
| $cLoopSkip \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK       | OK     |
| $cRestoreF \cap at\_least\_one\_ais(aHelperRestoreSLoop) = \emptyset$ : OK       | OK     |
| $aControls \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK                 | OK     |
| $aDataS \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK                    | OK     |
| $aDataSLoop \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK                | OK     |
| $aHelpS \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK                    | OK     |
| $aHelpSLoop \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK                | OK     |
| $aHelperRestoreS \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK           | OK     |
| $aHelperRestoreSLoop \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK       | OK     |
| $cDataF \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK                    | OK     |
| $cDataFLoop \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK                | OK     |
| $cHelpF \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK                    | OK     |
| $cHelpFExit \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK                | OK     |
| $cHelpFLoop \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK                | OK     |
| $cHelpFMain \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK                | OK     |
| $cHelpFSkip \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK                | OK     |
| $cHelperRestoreF \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK           | OK     |
| $cHelperRestoreFLoop \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK       | OK     |
| $cLoopSkip \cap at\_least\_one\_ais(aRestoreS) = \emptyset$ : OK                 | OK     |
| $cRestoreF \subseteq at\_most\_one\_ais(aRestoreS)$ : OK                         | OK     |
| $aControls \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK                    | OK     |
| $aDataSLoop \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK                   | OK     |
| $aHelpS \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK                       | OK     |
| $aHelpSLoop \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK                   | OK     |
| $aHelperRestoreS \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK              | OK     |
| $aHelperRestoreSLoop \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK          | OK     |
| $aRestoreS \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK                    | OK     |
| $cDataFLoop \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK                   | OK     |
| $cHelpF \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK                       | OK     |
| $cHelpFExit \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK                   | OK     |
| $cHelpFLoop \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK                   | OK     |

| Check                                                                       | Result |
|-----------------------------------------------------------------------------|--------|
| $cHelpFMain \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK              | OK     |
| $cHelpFSkip \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK              | OK     |
| $cHelperRestoreF \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK         | OK     |
| $cHelperRestoreFLoop \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK     | OK     |
| $cLoopSkip \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK               | OK     |
| $cRestoreF \cap at\_least\_one\_ais(cDataF) = \emptyset$ : OK               | OK     |
| $aControls \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK           | OK     |
| $aDataS \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK              | OK     |
| $aHelpS \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK              | OK     |
| $aHelpSLoop \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK          | OK     |
| $aHelperRestoreS \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK     | OK     |
| $aHelperRestoreSLoop \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK | OK     |
| OK                                                                          |        |
| $aRestoreS \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK           | OK     |
| $cDataF \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK              | OK     |
| $cHelpF \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK              | OK     |
| $cHelpFExit \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK          | OK     |
| $cHelpFLoop \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK          | OK     |
| $cHelpFMain \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK          | OK     |
| $cHelpFSkip \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK          | OK     |
| $cHelperRestoreF \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK     | OK     |
| $cHelperRestoreFLoop \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK | OK     |
| OK                                                                          |        |
| $cLoopSkip \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK           | OK     |
| $cRestoreF \cap at\_least\_one\_ais(cDataFLoop) = \emptyset$ : OK           | OK     |
| $aDataS \cap at\_least\_one\_ais(cHelpF) = \emptyset$ : OK                  | OK     |
| $aDataSLoop \cap at\_least\_one\_ais(cHelpF) = \emptyset$ : OK              | OK     |
| $aHelpSLoop \cap at\_least\_one\_ais(cHelpF) = \emptyset$ : OK              | OK     |
| $aHelperRestoreS \cap at\_least\_one\_ais(cHelpF) = \emptyset$ : OK         | OK     |
| $aHelperRestoreSLoop \cap at\_least\_one\_ais(cHelpF) = \emptyset$ : OK     | OK     |
| $aRestoreS \cap at\_least\_one\_ais(cHelpF) = \emptyset$ : OK               | OK     |
| $cDataF \cap at\_least\_one\_ais(cHelpF) = \emptyset$ : OK                  | OK     |
| $cDataFLoop \cap at\_least\_one\_ais(cHelpF) = \emptyset$ : OK              | OK     |
| $cHelpFExit \cap at\_least\_one\_ais(cHelpF) = \emptyset$ : OK              | OK     |
| $cHelpFLoop \cap at\_least\_one\_ais(cHelpF) = \emptyset$ : OK              | OK     |
| $cHelpFMain \cap at\_least\_one\_ais(cHelpF) = \emptyset$ : OK              | OK     |
| $cHelpFSkip \cap at\_least\_one\_ais(cHelpF) = \emptyset$ : OK              | OK     |
| $cHelperRestoreF \cap at\_least\_one\_ais(cHelpF) = \emptyset$ : OK         | OK     |
| $cHelperRestoreFLoop \cap at\_least\_one\_ais(cHelpF) = \emptyset$ : OK     | OK     |
| $cLoopSkip \cap at\_least\_one\_ais(cHelpF) = \emptyset$ : OK               | OK     |
| $cRestoreF \cap at\_least\_one\_ais(cHelpF) = \emptyset$ : OK               | OK     |
| $aControls \cap at\_least\_one\_ais(cHelpFExit) = \emptyset$ : OK           | OK     |
| $aDataS \cap at\_least\_one\_ais(cHelpFExit) = \emptyset$ : OK              | OK     |
| $aDataSLoop \cap at\_least\_one\_ais(cHelpFExit) = \emptyset$ : OK          | OK     |
| $aHelpSLoop \cap at\_least\_one\_ais(cHelpFExit) = \emptyset$ : OK          | OK     |

| Check                                                                       | Result |
|-----------------------------------------------------------------------------|--------|
| $aHelperRestoreSLoop \cap at\_least\_one\_ais(cHelpFExit) = \emptyset$ : OK | OK     |
| $aRestoreS \cap at\_least\_one\_ais(cHelpFExit) = \emptyset$ : OK           | OK     |
| $cDataF \cap at\_least\_one\_ais(cHelpFExit) = \emptyset$ : OK              | OK     |
| $cDataFLoop \cap at\_least\_one\_ais(cHelpFExit) = \emptyset$ : OK          | OK     |
| $cHelpF \cap at\_least\_one\_ais(cHelpFExit) = \emptyset$ : OK              | OK     |
| $cHelpFLoop \cap at\_least\_one\_ais(cHelpFExit) = \emptyset$ : OK          | OK     |
| $cHelpFMain \cap at\_least\_one\_ais(cHelpFExit) = \emptyset$ : OK          | OK     |
| $cHelpFSkip \cap at\_least\_one\_ais(cHelpFExit) = \emptyset$ : OK          | OK     |
| $cHelperRestoreF \cap at\_least\_one\_ais(cHelpFExit) = \emptyset$ : OK     | OK     |
| $cHelperRestoreFLoop \cap at\_least\_one\_ais(cHelpFExit) = \emptyset$ : OK | OK     |
| $cLoopSkip \cap at\_least\_one\_ais(cHelpFExit) = \emptyset$ : OK           | OK     |
| $cRestoreF \cap at\_least\_one\_ais(cHelpFExit) = \emptyset$ : OK           | OK     |
| $aControls \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK           | OK     |
| $aDataS \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK              | OK     |
| $aDataSLoop \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK          | OK     |
| $aHelpS \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK              | OK     |
| $aHelperRestoreS \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK     | OK     |
| $aHelperRestoreSLoop \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK | OK     |
| $aRestoreS \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK           | OK     |
| $cDataF \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK              | OK     |
| $cDataFLoop \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK          | OK     |
| $cHelpF \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK              | OK     |
| $cHelpFExit \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK          | OK     |
| $cHelpFMain \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK          | OK     |
| $cHelpFSkip \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK          | OK     |
| $cHelperRestoreF \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK     | OK     |
| $cHelperRestoreFLoop \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK | OK     |
| $cLoopSkip \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK           | OK     |
| $cRestoreF \cap at\_least\_one\_ais(cHelpFLoop) = \emptyset$ : OK           | OK     |
| $aControls \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK           | OK     |
| $aDataS \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK              | OK     |
| $aDataSLoop \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK          | OK     |
| $aHelpSLoop \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK          | OK     |
| $aHelperRestoreS \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK     | OK     |
| $aHelperRestoreSLoop \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK | OK     |
| $aRestoreS \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK           | OK     |
| $cDataF \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK              | OK     |
| $cDataFLoop \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK          | OK     |
| $cHelpF \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK              | OK     |
| $cHelpFExit \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK          | OK     |
| $cHelpFLoop \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK          | OK     |

| Check                                                                            | Result |
|----------------------------------------------------------------------------------|--------|
| $cHelpFSkip \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK               | OK     |
| $cHelperRestoreF \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK          | OK     |
| $cHelperRestoreFLoop \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK      | OK     |
| $cLoopSkip \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK                | OK     |
| $cRestoreF \cap at\_least\_one\_ais(cHelpFMain) = \emptyset$ : OK                | OK     |
| $aControls \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK                | OK     |
| $aDataS \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK                   | OK     |
| $aDataSLoop \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK               | OK     |
| $aHelpSLoop \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK               | OK     |
| $aHelperRestoreS \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK          | OK     |
| $aHelperRestoreSLoop \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK      | OK     |
| $aRestoreS \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK                | OK     |
| $cDataF \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK                   | OK     |
| $cDataFLoop \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK               | OK     |
| $cHelpF \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK                   | OK     |
| $cHelpFExit \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK               | OK     |
| $cHelpFLoop \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK               | OK     |
| $cHelpFMain \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK               | OK     |
| $cHelperRestoreF \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK          | OK     |
| $cHelperRestoreFLoop \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK      | OK     |
| $cLoopSkip \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK                | OK     |
| $cRestoreF \cap at\_least\_one\_ais(cHelpFSkip) = \emptyset$ : OK                | OK     |
| $aControls \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK           | OK     |
| $aDataS \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK              | OK     |
| $aDataSLoop \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK          | OK     |
| $aHelpS \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK              | OK     |
| $aHelpSLoop \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK          | OK     |
| $aHelperRestoreSLoop \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK | OK     |
| $aRestoreS \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK           | OK     |
| $cDataF \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK              | OK     |
| $cDataFLoop \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK          | OK     |
| $cHelpF \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK              | OK     |
| $cHelpFExit \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK          | OK     |
| $cHelpFLoop \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK          | OK     |
| $cHelpFMain \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK          | OK     |
| $cHelpFSkip \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK          | OK     |
| $cHelperRestoreFLoop \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK | OK     |
| $cLoopSkip \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK           | OK     |
| $cRestoreF \cap at\_least\_one\_ais(cHelperRestoreF) = \emptyset$ : OK           | OK     |
| $aControls \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK       | OK     |

| Check                                                                            | Result |
|----------------------------------------------------------------------------------|--------|
| $aDataS \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK          | OK     |
| $aDataSLoop \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK      | OK     |
| $aHelpS \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK          | OK     |
| $aHelpSLoop \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK      | OK     |
| $aHelperRestoreS \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK | OK     |
| $aRestoreS \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK       | OK     |
| $cDataF \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK          | OK     |
| $cDataFLoop \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK      | OK     |
| $cHelpF \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK          | OK     |
| $cHelpFExit \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK      | OK     |
| $cHelpFLoop \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK      | OK     |
| $cHelpFMain \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK      | OK     |
| $cHelpFSkip \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK      | OK     |
| $cHelperRestoreF \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK | OK     |
| $cLoopSkip \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK       | OK     |
| $cRestoreF \cap at\_least\_one\_ais(cHelperRestoreFLoop) = \emptyset$ : OK       | OK     |
| $aControls \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK                 | OK     |
| $aDataS \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK                    | OK     |
| $aHelpS \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK                    | OK     |
| $aHelpSLoop \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK                | OK     |
| $aHelperRestoreS \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK           | OK     |
| $aHelperRestoreSLoop \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK       | OK     |
| $aRestoreS \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK                 | OK     |
| $cDataF \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK                    | OK     |
| $cDataFLoop \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK                | OK     |
| $cHelpF \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK                    | OK     |
| $cHelpFExit \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK                | OK     |
| $cHelpFLoop \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK                | OK     |
| $cHelpFMain \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK                | OK     |
| $cHelpFSkip \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK                | OK     |
| $cHelperRestoreF \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK           | OK     |
| $cHelperRestoreFLoop \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK       | OK     |
| $cRestoreF \cap at\_least\_one\_ais(cLoopSkip) = \emptyset$ : OK                 | OK     |

| Check                                                                      | Result |
|----------------------------------------------------------------------------|--------|
| $aControlS \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK           | OK     |
| $aDataS \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK              | OK     |
| $aDataSLoop \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK          | OK     |
| $aHelpS \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK              | OK     |
| $aHelpSLoop \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK          | OK     |
| $aHelperRestoreS \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK     | OK     |
| $aHelperRestoreSLoop \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK | OK     |
| OK                                                                         |        |
| $cDataF \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK              | OK     |
| $cDataFLoop \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK          | OK     |
| $cHelpF \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK              | OK     |
| $cHelpFExit \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK          | OK     |
| $cHelpFLoop \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK          | OK     |
| $cHelpFMain \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK          | OK     |
| $cHelpFSkip \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK          | OK     |
| $cHelperRestoreF \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK     | OK     |
| $cHelperRestoreFLoop \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK | OK     |
| OK                                                                         |        |
| $cLoopSkip \cap at\_least\_one\_ais(cRestoreF) = \emptyset$ : OK           | OK     |

*bijjective support (IRSpecs).*

```

1aHelperRestoreSLoop: {Rb17f, Rcas19t}
2aHelpSLoop: {Rb17t, Rcas18t}
3aRestoreS: {Rcas26t}
4aHelperRestoreS: {Rcas10t}
5aDataS: {Ra1, Ra2, Rcas3t}
6aControlS: {Rcas25t}
7aHelpS: {Rb8t, Rcas9t}
8aDataSLoop: {Rb11t, Rcas12t}
9cDataF: {Ra1, Ra2, Rcas3t}
10cDataFLoop: {Rb11t, Rcas12t}
11cHelpF: {Rb8t, Rcas25t, Rcas9t}
12cHelpFMain: {Rb8t, Rcas9t}
13cHelpFLoop: {Rb17t, Rcas18t}
14cHelpFSkip: {Rb8t, Rcas9t}
15cLoopSkip: {Rb11t, Rcas12t}
16cRestoreF: {Rcas26t}
17cHelperRestoreF: {Rcas10t}
18cHelperRestoreFLoop: {Rb17f, Rcas19t}
19cHelpFExit: {Rb8t, Rcas10t, Rcas9t}

```

#### B.4.5 Interference Stability (439).

| # | Kind  | Spec         | Obligation                                                                                                                                              |
|---|-------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | ISTbl | coop::cDataF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ptr[tj] == data\_addr[tid[tj]]\}$<br>$cptr2[tk] = data\_addr[r[tk]] \{ptr[tj] ==$<br>$data\_addr[tid[tj]]\}$ |

| #  | Kind  | Spec         | Obligation                                                                                                                                                       |
|----|-------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2  | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$ctrl2[tk] = mem\_control[cptr2[tk]] \{ptr[t_j] == data\_addr[tid[t_j]]\}$   |
| 3  | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$cexp2[tk] = control\_exp[r[tk]] \{ptr[t_j] == data\_addr[tid[t_j]]\}$       |
| 4  | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$cptr3[tk] = data\_addr[tid[tk]] \{ptr[t_j] == data\_addr[tid[t_j]]\}$       |
| 5  | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$ctrl3[tk] = mem\_control[cptr3[tk]] \{ptr[t_j] == data\_addr[tid[t_j]]\}$   |
| 6  | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$cexp3[tk] = control\_exp[tid[tk]] \{ptr[t_j] == data\_addr[tid[t_j]]\}$     |
| 7  | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$ ret<br>$r[tk] \{ptr[t_j] == data\_addr[tid[t_j]]\}$                             |
| 8  | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$cptr1[tk] = data\_addr[r[tk]] \{ptr[t_j] == data\_addr[tid[t_j]]\}$         |
| 9  | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$ctrl1[tk] = mem\_control[cptr1[tk]] \{ptr[t_j] == data\_addr[tid[t_j]]\}$   |
| 10 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$cexp1[tk] = control\_exp[r[tk]] \{ptr[t_j] == data\_addr[tid[t_j]]\}$       |
| 11 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$assume(r[tk] == NULL) \{ptr[t_j] == data\_addr[tid[t_j]]\}$                 |
| 12 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$assume(r[tk] \neq NULL) \{ptr[t_j] == data\_addr[tid[t_j]]\}$               |
| 13 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$assume(r[tk] < 0) \{ptr[t_j] == data\_addr[tid[t_j]]\}$                     |
| 14 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$assume(r[tk] \geq 0) \{ptr[t_j] == data\_addr[tid[t_j]]\}$                  |
| 15 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$assume(ctrl2[tk] \neq cexp2[tk]) \{ptr[t_j] == data\_addr[tid[t_j]]\}$      |
| 16 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$assume(ctrl2[tk] == cexp2[tk]) \{ptr[t_j] == data\_addr[tid[t_j]]\}$        |
| 17 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$assume(r[tk] \neq data\_exp[tid[tk]]) \{ptr[t_j] == data\_addr[tid[t_j]]\}$ |
| 18 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$assume(r[tk] == data\_exp[tid[tk]]) \{ptr[t_j] == data\_addr[tid[t_j]]\}$   |
| 19 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>$assume(ctrl3[tk] \neq cexp3[tk]) \{ptr[t_j] == data\_addr[tid[t_j]]\}$      |



| #  | Kind  | Spec         | Obligation                                                                                                                                                                                                 |
|----|-------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>assume(ctr13[tk] == cexp3[tk]) {ptr[tj] == data_addr[tid[tj]]}                                                         |
| 21 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>assume(r[tk] < 0) {ptr[tj] == data_addr[tid[tj]]}                                                                      |
| 22 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>assume(r[tk] >= 0) {ptr[tj] == data_addr[tid[tj]]}                                                                     |
| 23 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>assume(ctr11[tk] $\neq$ cexp1[tk]) {ptr[tj] == data_addr[tid[tj]]}                                                     |
| 24 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>assume(ctr11[tk] == cexp1[tk]) {ptr[tj] == data_addr[tid[tj]]}                                                         |
| 25 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>assume(mem_data[cptr1[tk]] $\neq$ r[tk]) {ptr[tj] == data_addr[tid[tj]]}                                               |
| 26 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>CAS(mem_data[cptr1[tk]], r[tk], data_exp[r[tk]])<br>{ptr[tj] == data_addr[tid[tj]]}                                    |
| 27 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>assume(mem_data[ptr[tk]] $\neq$ data_exp[tid[tk]] $\wedge$ r[tk] == mem_data[ptr[tk]]) {ptr[tj] == data_addr[tid[tj]]} |
| 28 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>RCAS(mem_data[ptr[tk]], data_exp[tid[tk]], tid[tk], r[tk]) {ptr[tj] == data_addr[tid[tj]]}                             |
| 29 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>assume(mem_data[cptr2[tk]] $\neq$ r[tk]) {ptr[tj] == data_addr[tid[tj]]}                                               |
| 30 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>CAS(mem_data[cptr2[tk]], r[tk], data_new[r[tk]])<br>{ptr[tj] == data_addr[tid[tj]]}                                    |
| 31 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>assume(mem_data[cptr2[tk]] $\neq$ r[tk]) {ptr[tj] == data_addr[tid[tj]]}                                               |
| 32 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>CAS(mem_data[cptr2[tk]], r[tk], data_exp[r[tk]])<br>{ptr[tj] == data_addr[tid[tj]]}                                    |
| 33 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>assume(mem_data[cptr3[tk]] $\neq$ tid[tk]) {ptr[tj] == data_addr[tid[tj]]}                                             |
| 34 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>CAS(mem_data[cptr3[tk]], tid[tk], data_new[tid[tk]])<br>{ptr[tj] == data_addr[tid[tj]]}                                |
| 35 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>assume(mem_data[cptr3[tk]] $\neq$ tid[tk]) {ptr[tj] == data_addr[tid[tj]]}                                             |
| 36 | IStbl | coop::cDataF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>CAS(mem_data[cptr3[tk]], tid[tk], data_exp[tid[tk]])<br>{ptr[tj] == data_addr[tid[tj]]}                                |

| #  | Kind  | Spec             | Obligation                                                                                                                                                                                                |
|----|-------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 37 | IStbl | coop::cDataF     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>assume(mem_data[ptr[tk]] $\neq$ data_exp[tid[tk]] $\wedge$ r[tk] = mem_data[ptr[tk]]) {ptr[tj] == data_addr[tid[tj]]} |
| 38 | IStbl | coop::cDataF     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>assume(mem_data[cptr1[tk]] $\neq$ r[tk]) {ptr[tj] == data_addr[tid[tj]]}                                              |
| 39 | IStbl | coop::cDataF     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ptr[t_j] == data\_addr[tid[t_j]]\}$<br>CAS(mem_data[cptr1[tk]], r[tk], data_new[r[tk]])<br>{ptr[tj] == data_addr[tid[tj]]}                                   |
| 40 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ int } r \{r[t_j] \neq \text{NULL}\}$                                                                                        |
| 41 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ cptr2[tk] = data\_addr[r[tk]] } \{r[t_j] \neq \text{NULL}\}$                                                                |
| 42 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ ctrl2[tk] = mem\_control[cptr2[tk]] } \{r[t_j] \neq \text{NULL}\}$                                                          |
| 43 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ cexp2[tk] = control\_exp[r[tk]] } \{r[t_j] \neq \text{NULL}\}$                                                              |
| 44 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ ptr[tk] = data\_addr[tid[tk]] } \{r[t_j] \neq \text{NULL}\}$                                                                |
| 45 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ cptr3[tk] = data\_addr[tid[tk]] } \{r[t_j] \neq \text{NULL}\}$                                                              |
| 46 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ ctrl3[tk] = mem\_control[cptr3[tk]] } \{r[t_j] \neq \text{NULL}\}$                                                          |
| 47 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ cexp3[tk] = control\_exp[tid[tk]] } \{r[t_j] \neq \text{NULL}\}$                                                            |
| 48 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ return } r[tk] \{r[t_j] \text{ NULL}\}$                                                                                     |
| 49 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ cptr1[tk] = data\_addr[r[tk]] } \{r[t_j] \neq \text{NULL}\}$                                                                |
| 50 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ ctrl1[tk] = mem\_control[cptr1[tk]] } \{r[t_j] \neq \text{NULL}\}$                                                          |
| 51 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ cexp1[tk] = control\_exp[r[tk]] } \{r[t_j] \neq \text{NULL}\}$                                                              |
| 52 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(r[tk] == \text{NULL}) \{r[t_j] \neq \text{NULL}\}$                                                                  |
| 53 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(r[tk] < 0) \{r[t_j] \neq \text{NULL}\}$                                                                             |
| 54 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(r[tk] \geq 0) \{r[t_j] \neq \text{NULL}\}$                                                                          |
| 55 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(ctrl2[tk] \neq \text{cexp2[tk]}) \{r[t_j] \neq \text{NULL}\}$                                                       |
| 56 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(ctrl2[tk] == \text{cexp2[tk]}) \{r[t_j] \neq \text{NULL}\}$                                                         |
| 57 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(r[tk] \neq \text{data\_exp[tid[tk]]}) \{r[t_j] \neq \text{NULL}\}$                                                  |
| 58 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(r[tk] == \text{data\_exp[tid[tk]]}) \{r[t_j] \neq \text{NULL}\}$                                                    |
| 59 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(ctrl3[tk] \neq \text{cexp3[tk]}) \{r[t_j] \neq \text{NULL}\}$                                                       |

| #  | Kind  | Spec             | Obligation                                                                                                                                                                                                                                  |
|----|-------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 60 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(\text{ctrl3}[tk] == \text{cexp3}[tk]) \{r[t_j] \neq \text{NULL}\}$                                                                                    |
| 61 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(r[tk] < 0) \{r[t_j] \neq \text{NULL}\}$                                                                                                               |
| 62 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(r[tk] \geq 0) \{r[t_j] \neq \text{NULL}\}$                                                                                                            |
| 63 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(\text{ctrl1}[tk] \neq \text{cexp1}[tk]) \{r[t_j] \neq \text{NULL}\}$                                                                                  |
| 64 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(\text{ctrl1}[tk] == \text{cexp1}[tk]) \{r[t_j] \neq \text{NULL}\}$                                                                                    |
| 65 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(\text{mem\_data}[\text{cptr1}[tk]] \neq r[tk]) \{r[t_j] \neq \text{NULL}\}$                                                                           |
| 66 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ CAS}(\text{mem\_data}[\text{cptr1}[tk]], r[tk], \text{data\_exp}[r[tk]]) \{r[t_j] \neq \text{NULL}\}$                                                         |
| 67 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(\text{mem\_data}[\text{ptr}[tk]] \neq \text{data\_exp}[\text{tid}[tk]] \wedge r[tk] == \text{mem\_data}[\text{ptr}[tk]]) \{r[t_j] \neq \text{NULL}\}$ |
| 68 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(\text{mem\_data}[\text{cptr2}[tk]] \neq r[tk]) \{r[t_j] \neq \text{NULL}\}$                                                                           |
| 69 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ CAS}(\text{mem\_data}[\text{cptr2}[tk]], r[tk], \text{data\_new}[r[tk]]) \{r[t_j] \neq \text{NULL}\}$                                                         |
| 70 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(\text{mem\_data}[\text{cptr2}[tk]] \neq r[tk]) \{r[t_j] \neq \text{NULL}\}$                                                                           |
| 71 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ CAS}(\text{mem\_data}[\text{cptr2}[tk]], r[tk], \text{data\_exp}[r[tk]]) \{r[t_j] \neq \text{NULL}\}$                                                         |
| 72 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(\text{mem\_data}[\text{cptr3}[tk]] \neq \text{tid}[tk]) \{r[t_j] \neq \text{NULL}\}$                                                                  |
| 73 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ CAS}(\text{mem\_data}[\text{cptr3}[tk]], \text{tid}[tk], \text{data\_new}[\text{tid}[tk]]) \{r[t_j] \neq \text{NULL}\}$                                       |
| 74 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(\text{mem\_data}[\text{cptr3}[tk]] \neq \text{tid}[tk]) \{r[t_j] \neq \text{NULL}\}$                                                                  |
| 75 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ CAS}(\text{mem\_data}[\text{cptr3}[tk]], \text{tid}[tk], \text{data\_exp}[\text{tid}[tk]]) \{r[t_j] \neq \text{NULL}\}$                                       |
| 76 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(\text{mem\_data}[\text{ptr}[tk]] \neq \text{data\_exp}[\text{tid}[tk]] \wedge r[tk] == \text{mem\_data}[\text{ptr}[tk]]) \{r[t_j] \neq \text{NULL}\}$ |
| 77 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ RCAS}(\text{mem\_data}[\text{ptr}[tk]], \text{data\_exp}[\text{tid}[tk]], \text{tid}[tk], r[tk]) \{r[t_j] \neq \text{NULL}\}$                                 |
| 78 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\} \text{ assume}(\text{mem\_data}[\text{cptr1}[tk]] \neq r[tk]) \{r[t_j] \neq \text{NULL}\}$                                                                           |

| #  | Kind  | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----|-------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 79 | IStbl | coop::cDataFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] \neq \text{NULL}\}$<br>$\text{CAS}(\text{mem\_data}[\text{cptr1}[tk]], r[tk], \text{data\_new}[r[tk]])$<br>$\{r[t_j] \neq \text{NULL}\}$                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 80 | IStbl | coop::cHelpF     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge \text{cptr1}[t_j] ==$<br>$\text{data\_addr}[r[t_j]] \wedge \text{ctrl1}[t_j] == \text{mem\_control}[\text{cptr1}[t_j]]$<br>$\wedge \text{cexp1}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl1}[t_j]$<br>$== \text{cexp1}[t_j])\}$ int $r \{(r[t_j] \geq 0 \wedge \text{cptr1}[t_j] ==$<br>$\text{data\_addr}[r[t_j]] \wedge \text{ctrl1}[t_j] == \text{mem\_control}[\text{cptr1}[t_j]]$<br>$\wedge \text{cexp1}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl1}[t_j] ==$<br>$\text{cexp1}[t_j])\}$                                                   |
| 81 | IStbl | coop::cHelpF     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge \text{cptr1}[t_j] ==$<br>$\text{data\_addr}[r[t_j]] \wedge \text{ctrl1}[t_j] == \text{mem\_control}[\text{cptr1}[t_j]]$<br>$\wedge \text{cexp1}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl1}[t_j] ==$<br>$\text{cexp1}[t_j])\}$ $\text{cptr2}[tk] = \text{data\_addr}[r[tk]] \{(r[t_j] \geq$<br>$0 \wedge \text{cptr1}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl1}[t_j] ==$<br>$\text{mem\_control}[\text{cptr1}[t_j]] \wedge \text{cexp1}[t_j] == \text{control\_exp}[r[t_j]$<br>$\wedge \text{ctrl1}[t_j] == \text{cexp1}[t_j])\}$              |
| 82 | IStbl | coop::cHelpF     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge \text{cptr1}[t_j] ==$<br>$\text{data\_addr}[r[t_j]] \wedge \text{ctrl1}[t_j] == \text{mem\_control}[\text{cptr1}[t_j]]$<br>$\wedge \text{cexp1}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl1}[t_j] ==$<br>$\text{cexp1}[t_j])\}$ $\text{ctrl2}[tk] = \text{mem\_control}[\text{cptr2}[tk]] \{(r[t_j]$<br>$\geq 0 \wedge \text{cptr1}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl1}[t_j] ==$<br>$\text{mem\_control}[\text{cptr1}[t_j]] \wedge \text{cexp1}[t_j] == \text{control\_exp}[r[t_j]$<br>$\wedge \text{ctrl1}[t_j] == \text{cexp1}[t_j])\}$ |
| 83 | IStbl | coop::cHelpF     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge \text{cptr1}[t_j] ==$<br>$\text{data\_addr}[r[t_j]] \wedge \text{ctrl1}[t_j] == \text{mem\_control}[\text{cptr1}[t_j]]$<br>$\wedge \text{cexp1}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl1}[t_j] ==$<br>$\text{cexp1}[t_j])\}$ $\text{cexp2}[tk] = \text{control\_exp}[r[tk]] \{(r[t_j]$<br>$\geq 0 \wedge \text{cptr1}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl1}[t_j] ==$<br>$\text{mem\_control}[\text{cptr1}[t_j]] \wedge \text{cexp1}[t_j] == \text{control\_exp}[r[t_j]$<br>$\wedge \text{ctrl1}[t_j] == \text{cexp1}[t_j])\}$            |
| 84 | IStbl | coop::cHelpF     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge \text{cptr1}[t_j] ==$<br>$\text{data\_addr}[r[t_j]] \wedge \text{ctrl1}[t_j] == \text{mem\_control}[\text{cptr1}[t_j]]$<br>$\wedge \text{cexp1}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl1}[t_j] ==$<br>$\text{cexp1}[t_j])\}$ $\text{ptr}[tk] = \text{data\_addr}[\text{tid}[tk]] \{(r[t_j] \geq$<br>$0 \wedge \text{cptr1}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl1}[t_j] ==$<br>$\text{mem\_control}[\text{cptr1}[t_j]] \wedge \text{cexp1}[t_j] == \text{control\_exp}[r[t_j]$<br>$\wedge \text{ctrl1}[t_j] == \text{cexp1}[t_j])\}$       |
| 85 | IStbl | coop::cHelpF     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge \text{cptr1}[t_j] ==$<br>$\text{data\_addr}[r[t_j]] \wedge \text{ctrl1}[t_j] == \text{mem\_control}[\text{cptr1}[t_j]]$<br>$\wedge \text{cexp1}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl1}[t_j] ==$<br>$\text{cexp1}[t_j])\}$ $\text{cptr3}[tk] = \text{data\_addr}[\text{tid}[tk]] \{(r[t_j]$<br>$\geq 0 \wedge \text{cptr1}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl1}[t_j] ==$<br>$\text{mem\_control}[\text{cptr1}[t_j]] \wedge \text{cexp1}[t_j] == \text{control\_exp}[r[t_j]$<br>$\wedge \text{ctrl1}[t_j] == \text{cexp1}[t_j])\}$     |

| #  | Kind  | Spec         | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----|-------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 86 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ $ctrl13[tk] = mem\_control[cptr3[tk]] \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ |
| 87 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ $cexp3[tk] = control\_exp[tid[tk]] \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$    |
| 88 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ $return\ r[tk] \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$                        |
| 89 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ $cptr1[tk] = data\_addr[r[tk]] \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$        |
| 90 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ $ctrl1[tk] = mem\_control[cptr1[tk]] \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$  |
| 91 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ $cexp1[tk] = control\_exp[r[tk]] \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$      |
| 92 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ $assume(r[tk] == NULL) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$                |

| #  | Kind  | Spec         | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----|-------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 93 | ISTbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\} \text{ assume}(r[tk] \neq \text{NULL}) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$        |
| 94 | ISTbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\} \text{ assume}(r[tk] < 0) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$                     |
| 95 | ISTbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\} \text{ assume}(r[tk] \geq 0) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$                  |
| 96 | ISTbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\} \text{ assume}(ctrl2[tk] \neq cexp2[tk]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$      |
| 97 | ISTbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\} \text{ assume}(ctrl2[tk] == cexp2[tk]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$        |
| 98 | ISTbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\} \text{ assume}(r[tk] \neq data\_exp[tid[tk]]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ |
| 99 | ISTbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\} \text{ assume}(r[tk] == data\_exp[tid[tk]]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$   |

| #   | Kind  | Spec         | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 100 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\} \text{ assume}(ctrl13[tk] \neq cexp3[tk]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$                  |
| 101 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\} \text{ assume}(ctrl13[tk] == cexp3[tk]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$                    |
| 102 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\} \text{ assume}(r[tk] < 0) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$                                  |
| 103 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\} \text{ assume}(r[tk] \geq 0) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$                               |
| 104 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\} \text{ assume}(ctrl11[tk] \neq cexp1[tk]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$                  |
| 105 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\} \text{ assume}(mem\_data[cptr1[tk]] \neq r[tk]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$            |
| 106 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\} \text{ CAS}(mem\_data[cptr1[tk]], r[tk], data\_exp[r[tk]]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ |

| #   | Kind  | Spec         | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----|-------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 107 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ $assume(mem\_data[ptr[tk]] \neq data\_exp[tid[tk]] \wedge r[tk] == mem\_data[ptr[tk]]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ |
| 108 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ $RCAS(mem\_data[ptr[tk]], data\_exp[tid[tk]], tid[tk], r[tk]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$                          |
| 109 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ $assume(mem\_data[cptr2[tk]] \neq r[tk]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$                                               |
| 110 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ $CAS(mem\_data[cptr2[tk]], r[tk], data\_new[r[tk]]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$                                    |
| 111 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ $assume(mem\_data[cptr2[tk]] \neq r[tk]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$                                               |
| 112 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$ $CAS(mem\_data[cptr2[tk]], r[tk], data\_exp[r[tk]]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j])\}$                                    |



| #   | Kind  | Spec         | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----|-------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 113 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j]) \}$ $\text{assume}(mem\_data[cptr3[tk]] \neq tid[tk])$ $\{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j]) \}$                                             |
| 114 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j]) \}$ $\text{assume}(mem\_data[cptr3[tk]] \neq tid[tk])$ $\{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j]) \}$                                             |
| 115 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j]) \}$ $\text{CAS}(mem\_data[cptr3[tk]], tid[tk], data\_exp[tid[tk]])$ $\{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j]) \}$                                |
| 116 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j]) \}$ $\text{assume}(mem\_data[ptr[tk]] \neq data\_exp[tid[tk]] \wedge r[tk] == mem\_data[ptr[tk]])$ $\{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j]) \}$ |
| 117 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j]) \}$ $\text{RCAS}(mem\_data[ptr[tk]], data\_exp[tid[tk]], tid[tk], r[tk])$ $\{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j]) \}$                          |
| 118 | IStbl | coop::cHelpF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j]) \}$ $\text{assume}(mem\_data[cptr1[tk]] \neq r[tk])$ $\{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] == cexp1[t_j]) \}$                                               |



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| #   | Kind  | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----|-------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 143 | ISTbl | coop::cHelpFMain | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j])\}$<br>$\text{assume}(\text{ctrl1}[tk] \neq \text{cexp1}[tk]) \{(r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j])\}$                                                                                  |
| 144 | ISTbl | coop::cHelpFMain | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j])\}$<br>$\text{assume}(\text{mem\_data}[\text{cptr1}[tk]] \neq r[tk]) \{(r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j])\}$                                                                           |
| 145 | ISTbl | coop::cHelpFMain | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j])\}$<br>$\text{CAS}(\text{mem\_data}[\text{cptr1}[tk]], r[tk], \text{data\_exp}[r[tk]]) \{(r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j])\}$                                                         |
| 146 | ISTbl | coop::cHelpFMain | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j])\}$<br>$\text{assume}(\text{mem\_data}[\text{ptr}[tk]] \neq \text{data\_exp}[\text{tid}[tk]] \wedge r[tk] == \text{mem\_data}[\text{ptr}[tk]]) \{(r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j])\}$ |
| 147 | ISTbl | coop::cHelpFMain | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j])\}$<br>$\text{RCAS}(\text{mem\_data}[\text{ptr}[tk]], \text{data\_exp}[\text{tid}[tk]], \text{tid}[tk], r[tk]) \{(r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j])\}$                                 |

| #   | Kind  | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----|-------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 148 | ISTbl | coop::cHelpFMain | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$ $\text{assume}(\text{mem\_data}[\text{cptr2}[tk]] \neq r[tk]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$                   |
| 149 | ISTbl | coop::cHelpFMain | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$ $\text{CAS}(\text{mem\_data}[\text{cptr2}[tk]], r[tk], \text{data\_new}[r[tk]]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$ |
| 150 | ISTbl | coop::cHelpFMain | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$ $\text{assume}(\text{mem\_data}[\text{cptr2}[tk]] \neq r[tk]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$                   |
| 151 | ISTbl | coop::cHelpFMain | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$ $\text{CAS}(\text{mem\_data}[\text{cptr2}[tk]], r[tk], \text{data\_exp}[r[tk]]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$ |
| 152 | ISTbl | coop::cHelpFMain | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$ $\text{assume}(\text{mem\_data}[\text{cptr3}[tk]] \neq tid[tk]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$                 |

| #   | Kind  | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----|-------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 153 | ISTbl | coop::cHelpFMain | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$ $\text{CAS}(\text{mem\_data}[\text{cptr3}[tk]], \text{tid}[tk], \text{data\_new}[\text{tid}[tk]])$ $\{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$                                     |
| 154 | ISTbl | coop::cHelpFMain | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$ $\text{assume}(\text{mem\_data}[\text{cptr3}[tk]] \neq \text{tid}[tk]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$                                                                  |
| 155 | ISTbl | coop::cHelpFMain | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$ $\text{CAS}(\text{mem\_data}[\text{cptr3}[tk]], \text{tid}[tk], \text{data\_exp}[\text{tid}[tk]])$ $\{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$                                     |
| 156 | ISTbl | coop::cHelpFMain | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$ $\text{assume}(\text{mem\_data}[\text{ptr}[tk]] \neq \text{data\_exp}[\text{tid}[tk]] \wedge r[tk] == \text{mem\_data}[\text{ptr}[tk]]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$ |
| 157 | ISTbl | coop::cHelpFMain | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$ $\text{RCAS}(\text{mem\_data}[\text{ptr}[tk]], \text{data\_exp}[\text{tid}[tk]], \text{tid}[tk], r[tk]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[\text{tid}[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[\text{tid}[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$                                 |



| #   | Kind  | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----|-------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 158 | IStbl | coop::cHelpFMain | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$ $\text{assume}(\text{mem\_data}[\text{cptr1}[tk]] \neq r[tk]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{cexp3}[t_j]) \}$ |
| 159 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{cexp2}[t_j]) \}$ $\text{int } r \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{cexp2}[t_j]) \}$                                                                                                                                                                |
| 160 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{cexp2}[t_j]) \}$ $\text{cptr2}[tk] = \text{data\_addr}[r[tk]] \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{cexp2}[t_j]) \}$                                                                                                                                  |
| 161 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{cexp2}[t_j]) \}$ $\text{ctrl2}[tk] = \text{mem\_control}[\text{cptr2}[tk]] \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{cexp2}[t_j]) \}$                                                                                                                     |
| 162 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{cexp2}[t_j]) \}$ $\text{cexp2}[tk] = \text{control\_exp}[r[tk]] \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{cexp2}[t_j]) \}$                                                                                                                                |
| 163 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{cexp2}[t_j]) \}$ $\text{ptr}[tk] = \text{data\_addr}[tid[tk]] \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{cexp2}[t_j]) \}$                                                                                                                                  |

| #   | Kind  | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----|-------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 164 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} cptr3[tk] = data\_addr[tid[tk]] \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$     |
| 165 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} ctrl3[tk] = mem\_control[cptr3[tk]] \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$ |
| 166 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} cexp3[tk] = control\_exp[tid[tk]] \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$   |
| 167 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} return r[tk] \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$                        |
| 168 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} cptr1[tk] = data\_addr[r[tk]] \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$       |
| 169 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} ctrl1[tk] = mem\_control[cptr1[tk]] \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$ |
| 170 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} cexp1[tk] = control\_exp[r[tk]] \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$     |

| #   | Kind  | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|-------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 171 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} \text{ assume}(r[tk] == \text{NULL}) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$          |
| 172 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} \text{ assume}(r[tk] \neq \text{NULL}) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$        |
| 173 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} \text{ assume}(r[tk] < 0) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$                     |
| 174 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} \text{ assume}(r[tk] \geq 0) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$                  |
| 175 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} \text{ assume}(ctrl2[tk] \neq cexp2[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$      |
| 176 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} \text{ assume}(r[tk] \neq data\_exp[tid[tk]]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$ |
| 177 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} \text{ assume}(r[tk] == data\_exp[tid[tk]]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$   |

| #   | Kind  | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----|-------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 178 | ISTbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} \text{ assume}(ctrl13[tk] \neq cexp3[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$       |
| 179 | ISTbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} \text{ assume}(ctrl13[tk] == cexp3[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$         |
| 180 | ISTbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} \text{ assume}(r[tk] < 0) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$                       |
| 181 | ISTbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} \text{ assume}(r[tk] \geq 0) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$                    |
| 182 | ISTbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} \text{ assume}(ctrl11[tk] \neq cexp1[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$       |
| 183 | ISTbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} \text{ assume}(ctrl11[tk] == cexp1[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$         |
| 184 | ISTbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} \text{ assume}(mem\_data[cptr1[tk]] \neq r[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$ |

| #   | Kind  | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----|-------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 185 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$ $CAS(mem\_data[cptr1[tk]], r[tk], data\_exp[r[tk]]) \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$                                    |
| 186 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$ $assume(mem\_data[ptr[tk]] \neq data\_exp[tid[tk]] \wedge r[tk] == mem\_data[ptr[tk]]) \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$ |
| 187 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$ $RCAS(mem\_data[ptr[tk]], data\_exp[tid[tk]], tid[tk], r[tk]) \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$                          |
| 188 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$ $assume(mem\_data[cptr2[tk]] \neq r[tk]) \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$                                               |
| 189 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$ $assume(mem\_data[cptr2[tk]] \neq r[tk]) \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$                                               |
| 190 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$ $CAS(mem\_data[cptr2[tk]], r[tk], data\_exp[r[tk]]) \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$                                    |

| #   | Kind  | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----|-------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 191 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$ $\text{assume}(mem\_data[cptr3[tk]] \neq tid[tk])$ $\{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$                                             |
| 192 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$ $\text{CAS}(mem\_data[cptr3[tk]], tid[tk], data\_new[tid[tk]])$ $\{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$                                |
| 193 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$ $\text{assume}(mem\_data[cptr3[tk]] \neq tid[tk])$ $\{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$                                             |
| 194 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$ $\text{CAS}(mem\_data[cptr3[tk]], tid[tk], data\_exp[tid[tk]])$ $\{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$                                |
| 195 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$ $\text{assume}(mem\_data[ptr[tk]] \neq data\_exp[tid[tk]] \wedge r[tk] == mem\_data[ptr[tk]])$ $\{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$ |
| 196 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$ $\text{RCAS}(mem\_data[ptr[tk]], data\_exp[tid[tk]], tid[tk], r[tk])$ $\{ (r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j]) \}$                          |

| #   | Kind  | Spec             | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 197 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} \text{ assume}(mem\_data[cptr1[tk]] \neq r[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$            |
| 198 | IStbl | coop::cHelpFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\} \text{ CAS}(mem\_data[cptr1[tk]], r[tk], data\_new[r[tk]]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] == cexp2[t_j])\}$ |
| 199 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ int } r \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                       |
| 200 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} cptr2[tk] = data\_addr[r[tk]] \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                        |
| 201 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} ctrl2[tk] = mem\_control[cptr2[tk]] \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                  |
| 202 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} cexp2[tk] = control\_exp[r[tk]] \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                      |
| 203 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} ptr[tk] = data\_addr[tid[tk]] \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                        |
| 204 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} cptr3[tk] = data\_addr[tid[tk]] \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                      |
| 205 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} ctrl3[tk] = mem\_control[cptr3[tk]] \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                  |
| 206 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} cexp3[tk] = control\_exp[tid[tk]] \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                    |
| 207 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ return } r[tk] \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                |
| 208 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} cptr1[tk] = data\_addr[r[tk]] \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                        |
| 209 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} ctrl1[tk] = mem\_control[cptr1[tk]] \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                  |
| 210 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} cexp1[tk] = control\_exp[r[tk]] \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                      |
| 211 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(r[tk] == \text{NULL}) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                 |
| 212 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(r[tk] \neq \text{NULL}) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                               |
| 213 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(r[tk] < 0) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                            |
| 214 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(r[tk] \geq 0) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                         |
| 215 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(ctrl2[tk] \neq cexp2[tk]) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                             |

| #   | Kind  | Spec             | Obligation                                                                                                                                                                                                        |
|-----|-------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 216 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{ctr12}[tk] == \text{cexp2}[tk]) \{r[t_j] < 0\}$                                                                                    |
| 217 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(r[tk] \neq \text{data\_exp}[\text{tid}[tk]]) \{r[t_j] < 0\}$                                                                             |
| 218 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(r[tk] == \text{data\_exp}[\text{tid}[tk]]) \{r[t_j] < 0\}$                                                                               |
| 219 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{ctr13}[tk] \neq \text{cexp3}[tk]) \{r[t_j] < 0\}$                                                                                  |
| 220 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{ctr13}[tk] == \text{cexp3}[tk]) \{r[t_j] < 0\}$                                                                                    |
| 221 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(r[tk] < 0) \{r[t_j] < 0\}$                                                                                                               |
| 222 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(r[tk] \geq 0) \{r[t_j] < 0\}$                                                                                                            |
| 223 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{ctr11}[tk] \neq \text{cexp1}[tk]) \{r[t_j] < 0\}$                                                                                  |
| 224 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{mem\_data}[\text{cptr1}[tk]] \neq r[tk]) \{r[t_j] < 0\}$                                                                           |
| 225 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ CAS}(\text{mem\_data}[\text{cptr1}[tk]], r[tk], \text{data\_exp}[r[tk]]) \{r[t_j] < 0\}$                                                         |
| 226 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{mem\_data}[\text{ptr}[tk]] \neq \text{data\_exp}[\text{tid}[tk]] \wedge r[tk] == \text{mem\_data}[\text{ptr}[tk]]) \{r[t_j] < 0\}$ |
| 227 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ RCAS}(\text{mem\_data}[\text{ptr}[tk]], \text{data\_exp}[\text{tid}[tk]], \text{tid}[tk], r[tk]) \{r[t_j] < 0\}$                                 |
| 228 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{mem\_data}[\text{cptr2}[tk]] \neq r[tk]) \{r[t_j] < 0\}$                                                                           |
| 229 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ CAS}(\text{mem\_data}[\text{cptr2}[tk]], r[tk], \text{data\_new}[r[tk]]) \{r[t_j] < 0\}$                                                         |
| 230 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{mem\_data}[\text{cptr2}[tk]] \neq r[tk]) \{r[t_j] < 0\}$                                                                           |
| 231 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ CAS}(\text{mem\_data}[\text{cptr2}[tk]], r[tk], \text{data\_exp}[r[tk]]) \{r[t_j] < 0\}$                                                         |
| 232 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{mem\_data}[\text{cptr3}[tk]] \neq \text{tid}[tk]) \{r[t_j] < 0\}$                                                                  |
| 233 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ CAS}(\text{mem\_data}[\text{cptr3}[tk]], \text{tid}[tk], \text{data\_new}[\text{tid}[tk]]) \{r[t_j] < 0\}$                                       |
| 234 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{mem\_data}[\text{cptr3}[tk]] \neq \text{tid}[tk]) \{r[t_j] < 0\}$                                                                  |
| 235 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ CAS}(\text{mem\_data}[\text{cptr3}[tk]], \text{tid}[tk], \text{data\_exp}[\text{tid}[tk]]) \{r[t_j] < 0\}$                                       |
| 236 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{mem\_data}[\text{ptr}[tk]] \neq \text{data\_exp}[\text{tid}[tk]] \wedge r[tk] == \text{mem\_data}[\text{ptr}[tk]]) \{r[t_j] < 0\}$ |
| 237 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ RCAS}(\text{mem\_data}[\text{ptr}[tk]], \text{data\_exp}[\text{tid}[tk]], \text{tid}[tk], r[tk]) \{r[t_j] < 0\}$                                 |
| 238 | IStbl | coop::cHelpFSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{mem\_data}[\text{cptr1}[tk]] \neq r[tk]) \{r[t_j] < 0\}$                                                                           |



| #   | Kind  | Spec            | Obligation                                                                                                                            |
|-----|-------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 239 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ int } r \{r[t_j] < 0\}$                                              |
| 240 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ cptr2}[tk] = \text{data\_addr}[r[tk]] \{r[t_j] < 0\}$                |
| 241 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ ctrl2}[tk] = \text{mem\_control}[\text{cptr2}[tk]] \{r[t_j] < 0\}$   |
| 242 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ cexp2}[tk] = \text{control\_exp}[r[tk]] \{r[t_j] < 0\}$              |
| 243 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ ptr}[tk] = \text{data\_addr}[\text{tid}[tk]] \{r[t_j] < 0\}$         |
| 244 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ cptr3}[tk] = \text{data\_addr}[\text{tid}[tk]] \{r[t_j] < 0\}$       |
| 245 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ ctrl3}[tk] = \text{mem\_control}[\text{cptr3}[tk]] \{r[t_j] < 0\}$   |
| 246 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ cexp3}[tk] = \text{control\_exp}[\text{tid}[tk]] \{r[t_j] < 0\}$     |
| 247 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ return } r[tk] \{r[t_j] < 0\}$                                       |
| 248 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ cptr1}[tk] = \text{data\_addr}[r[tk]] \{r[t_j] < 0\}$                |
| 249 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ ctrl1}[tk] = \text{mem\_control}[\text{cptr1}[tk]] \{r[t_j] < 0\}$   |
| 250 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ cexp1}[tk] = \text{control\_exp}[r[tk]] \{r[t_j] < 0\}$              |
| 251 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(r[tk] == \text{NULL}) \{r[t_j] < 0\}$                        |
| 252 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(r[tk] < 0) \{r[t_j] < 0\}$                                   |
| 253 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(r[tk] \geq 0) \{r[t_j] < 0\}$                                |
| 254 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{ctrl2}[tk] \neq \text{cexp2}[tk]) \{r[t_j] < 0\}$      |
| 255 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{ctrl2}[tk] == \text{cexp2}[tk]) \{r[t_j] < 0\}$        |
| 256 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(r[tk] \neq \text{data\_exp}[\text{tid}[tk]]) \{r[t_j] < 0\}$ |
| 257 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(r[tk] == \text{data\_exp}[\text{tid}[tk]]) \{r[t_j] < 0\}$   |
| 258 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{ctrl3}[tk] \neq \text{cexp3}[tk]) \{r[t_j] < 0\}$      |
| 259 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{ctrl3}[tk] == \text{cexp3}[tk]) \{r[t_j] < 0\}$        |
| 260 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(r[tk] < 0) \{r[t_j] < 0\}$                                   |
| 261 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(r[tk] \geq 0) \{r[t_j] < 0\}$                                |
| 262 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{ctrl1}[tk] \neq \text{cexp1}[tk]) \{r[t_j] < 0\}$      |
| 263 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{ctrl1}[tk] == \text{cexp1}[tk]) \{r[t_j] < 0\}$        |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----|-------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 264 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{mem\_data}[\text{cptr1}[t_j]] \neq r[tk]) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 265 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ CAS}(\text{mem\_data}[\text{cptr1}[t_j]], r[tk], \text{data\_exp}[r[tk]]) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 266 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{mem\_data}[\text{ptr}[t_j]] \neq \text{data\_exp}[\text{tid}[tk]] \wedge r[tk] == \text{mem\_data}[\text{ptr}[tk]]) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                   |
| 267 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{mem\_data}[\text{cptr2}[t_j]] \neq r[tk]) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 268 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ CAS}(\text{mem\_data}[\text{cptr2}[t_j]], r[tk], \text{data\_new}[r[tk]]) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 269 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{mem\_data}[\text{cptr2}[t_j]] \neq r[tk]) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 270 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ CAS}(\text{mem\_data}[\text{cptr2}[t_j]], r[tk], \text{data\_exp}[r[tk]]) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 271 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{mem\_data}[\text{cptr3}[t_j]] \neq \text{tid}[tk]) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 272 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ CAS}(\text{mem\_data}[\text{cptr3}[t_j]], \text{tid}[tk], \text{data\_new}[\text{tid}[tk]]) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 273 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{mem\_data}[\text{cptr3}[t_j]] \neq \text{tid}[tk]) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 274 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ CAS}(\text{mem\_data}[\text{cptr3}[t_j]], \text{tid}[tk], \text{data\_exp}[\text{tid}[tk]]) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 275 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{mem\_data}[\text{ptr}[t_j]] \neq \text{data\_exp}[\text{tid}[tk]] \wedge r[tk] == \text{mem\_data}[\text{ptr}[tk]]) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                   |
| 276 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ RCAS}(\text{mem\_data}[\text{ptr}[t_j]], \text{data\_exp}[\text{tid}[tk]], \text{tid}[tk], r[tk]) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 277 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ assume}(\text{mem\_data}[\text{cptr1}[t_j]] \neq r[tk]) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 278 | IStbl | coop::cLoopSkip | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] < 0\} \text{ CAS}(\text{mem\_data}[\text{cptr1}[t_j]], r[tk], \text{data\_new}[r[tk]]) \{r[t_j] < 0\}$                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 279 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge \text{cptr1}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl1}[t_j] == \text{mem\_control}[\text{cptr1}[t_j]] \wedge \text{cexp1}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl1}[t_j] \neq \text{cexp1}[t_j]) \} \text{ int } r \{(r[t_j] \geq 0 \wedge \text{cptr1}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl1}[t_j] == \text{mem\_control}[\text{cptr1}[t_j]] \wedge \text{cexp1}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl1}[t_j] \neq \text{cexp1}[t_j]) \}$                               |
| 280 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge \text{cptr1}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl1}[t_j] == \text{mem\_control}[\text{cptr1}[t_j]] \wedge \text{cexp1}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl1}[t_j] \neq \text{cexp1}[t_j]) \} \text{ cptr2}[tk] = \text{data\_addr}[r[tk]] \{(r[t_j] \geq 0 \wedge \text{cptr1}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl1}[t_j] == \text{mem\_control}[\text{cptr1}[t_j]] \wedge \text{cexp1}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl1}[t_j] \neq \text{cexp1}[t_j]) \}$ |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|-------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 281 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $ctrl2[tk] = mem\_control[cptr2[tk]] \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ |
| 282 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $cexp2[tk] = control\_exp[r[tk]] \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$     |
| 283 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $ptr[tk] = data\_addr[tid[tk]] \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$       |
| 284 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $cptr3[tk] = data\_addr[tid[tk]] \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$     |
| 285 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $ctrl3[tk] = mem\_control[cptr3[tk]] \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ |
| 286 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $cexp3[tk] = control\_exp[tid[tk]] \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$   |
| 287 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $return r[tk] \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$                        |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|-------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 288 | ISTbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $cptr1[tk] = data\_addr[r[tk]] \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$       |
| 289 | ISTbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $ctrl1[tk] = mem\_control[cptr1[tk]] \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ |
| 290 | ISTbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $cexp1[tk] = control\_exp[r[tk]] \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$     |
| 291 | ISTbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $assume(r[tk] == NULL) \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$               |
| 292 | ISTbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $assume(r[tk] \neq NULL) \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$             |
| 293 | ISTbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $assume(r[tk] < 0) \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$                   |
| 294 | ISTbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $assume(r[tk] \geq 0) \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$                |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|-------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 295 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\} \text{ assume}(ctrl2[tk] \neq cexp2[tk]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\}$      |
| 296 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\} \text{ assume}(ctrl2[tk] == cexp2[tk]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\}$        |
| 297 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\} \text{ assume}(r[tk] \neq data\_exp[tid[tk]]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\}$ |
| 298 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\} \text{ assume}(r[tk] == data\_exp[tid[tk]]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\}$   |
| 299 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\} \text{ assume}(ctrl3[tk] \neq cexp3[tk]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\}$      |
| 300 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\} \text{ assume}(ctrl3[tk] == cexp3[tk]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\}$        |
| 301 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\} \text{ assume}(r[tk] < 0) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\}$                     |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|-------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 302 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\} \text{ assume}(r[tk] \geq 0) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\}$                                                                  |
| 303 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\} \text{ assume}(ctrl1[tk] \neq cexp1[tk]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\}$                                                      |
| 304 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\} \text{ assume}(ctrl1[tk] == cexp1[tk]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\}$                                                        |
| 305 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\} \text{ assume}(mem\_data[cptr1[tk]] \neq r[tk]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\}$                                               |
| 306 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\} \text{ CAS}(mem\_data[cptr1[tk]], r[tk], data\_exp[r[tk]]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\}$                                    |
| 307 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\} \text{ assume}(mem\_data[ptr[tk]] \neq data\_exp[tid[tk]] \wedge r[tk] == mem\_data[ptr[tk]]) \{(r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j])\}$ |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----|-------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 308 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $RCAS(mem\_data[ptr[tk]], data\_exp[tid[tk]], tid[tk], r[tk]) \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ |
| 309 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $assume(mem\_data[cptr2[tk]] \neq r[tk]) \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$                      |
| 310 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $CAS(mem\_data[cptr2[tk]], r[tk], data\_new[r[tk]]) \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$           |
| 311 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $assume(mem\_data[cptr2[tk]] \neq r[tk]) \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$                      |
| 312 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $CAS(mem\_data[cptr2[tk]], r[tk], data\_exp[r[tk]]) \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$           |
| 313 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $assume(mem\_data[cptr3[tk]] \neq tid[tk]) \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$                    |

| #   | Kind  | Spec            | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----|-------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 314 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $CAS(mem\_data[cptr3[tk]], tid[tk], data\_new[tid[tk]]) \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$                                |
| 315 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $assume(mem\_data[cptr3[tk]] \neq tid[tk]) \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$                                             |
| 316 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $assume(mem\_data[ptr[tk]] \neq data\_exp[tid[tk]] \wedge r[tk] == mem\_data[ptr[tk]]) \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ |
| 317 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $RCAS(mem\_data[ptra[tk]], data\_exp[tid[tk]], tid[tk], r[tk]) \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$                         |
| 318 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $assume(mem\_data[cptr1[tk]] \neq r[tk]) \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$                                               |
| 319 | IStbl | coop::cRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$ $CAS(mem\_data[cptr1[tk]], r[tk], data\_new[r[tk]]) \{ (r[t_j] \geq 0 \wedge cptr1[t_j] == data\_addr[r[t_j]] \wedge ctrl1[t_j] == mem\_control[cptr1[t_j]] \wedge cexp1[t_j] == control\_exp[r[t_j]] \wedge ctrl1[t_j] \neq cexp1[t_j]) \}$                                    |



| #   | Kind  | Spec                  | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----|-------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 320 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ $\text{int } r \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$                                            |
| 321 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ $\text{cptr2}[tk] = \text{data\_addr}[r[tk]] \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$              |
| 322 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ $\text{ctrl2}[tk] = \text{mem\_control}[\text{cptr2}[tk]] \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ |
| 323 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ $\text{cexp2}[tk] = \text{control\_exp}[r[tk]] \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$            |
| 324 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ $\text{ptr}[tk] = \text{data\_addr}[tid[tk]] \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$              |
| 325 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ $\text{cptr3}[tk] = \text{data\_addr}[tid[tk]] \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$            |



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| #   | Kind  | Spec                  | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 344 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$<br>$\text{assume}(\text{ctrl1}[tk] \neq \text{cexp1}[tk]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$                                                                           |
| 345 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$<br>$\text{assume}(\text{ctrl1}[tk] == \text{cexp1}[tk]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$                                                                             |
| 346 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$<br>$\text{assume}(\text{mem\_data}[\text{cptr1}[tk]] \neq r[tk]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$                                                                    |
| 347 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$<br>$\text{assume}(\text{mem\_data}[\text{ptr}[tk]] \neq \text{data\_exp}[tid[tk]] \wedge r[tk] == \text{mem\_data}[\text{ptr}[tk]]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ |
| 348 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$<br>$\text{RCAS}(\text{mem\_data}[\text{ptr}[tk]], \text{data\_exp}[tid[tk]], \text{tid}[tk], r[tk]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$                                 |

| #   | Kind  | Spec                  | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|-------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 349 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$<br>$\text{assume}(\text{mem\_data}[\text{cptr2}[tk]] \neq r[tk]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$                   |
| 350 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$<br>$\text{CAS}(\text{mem\_data}[\text{cptr2}[tk]], r[tk], \text{data\_new}[r[tk]]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ |
| 351 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$<br>$\text{assume}(\text{mem\_data}[\text{cptr2}[tk]] \neq r[tk]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$                   |
| 352 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$<br>$\text{CAS}(\text{mem\_data}[\text{cptr2}[tk]], r[tk], \text{data\_exp}[r[tk]]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ |
| 353 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$<br>$\text{assume}(\text{mem\_data}[\text{cptr3}[tk]] \neq tid[tk]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$                 |

| #   | Kind  | Spec                  | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----|-------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 354 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ $\text{CAS}(\text{mem\_data}[\text{cptr3}[tk]], tid[tk], \text{data\_new}[tid[tk]])$ $\{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$                                            |
| 355 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ $\text{assume}(\text{mem\_data}[\text{cptr3}[tk]] \neq tid[tk]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$                                                                  |
| 356 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ $\text{CAS}(\text{mem\_data}[\text{cptr3}[tk]], tid[tk], \text{data\_exp}[tid[tk]])$ $\{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$                                            |
| 357 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ $\text{assume}(\text{mem\_data}[\text{ptr}[tk]] \neq \text{data\_exp}[tid[tk]] \wedge r[tk] == \text{mem\_data}[\text{ptr}[tk]]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ |
| 358 | IStbl | coop::cHelperRestoreF | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ $\text{RCAS}(\text{mem\_data}[\text{ptr}[tk]], \text{data\_exp}[tid[tk]], tid[tk], r[tk]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$                                        |

| #   | Kind  | Spec                      | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|-------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 359 | IStbl | coop::cHelperRestoreF     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$<br>$\text{assume}(\text{mem\_data}[\text{cptr1}[tk]] \neq r[tk]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$                   |
| 360 | IStbl | coop::cHelperRestoreF     | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$<br>$\text{CAS}(\text{mem\_data}[\text{cptr1}[tk]], r[tk], \text{data\_new}[r[tk]]) \{ (r[t_j] == \text{NULL} \wedge r[t_j] == \text{data\_exp}[tid[t_j]] \wedge \text{cptr3}[t_j] == \text{data\_addr}[tid[t_j]] \wedge \text{ctrl3}[t_j] == \text{mem\_control}[\text{cptr3}[t_j]] \wedge \text{cexp3}[t_j] == \text{control\_exp}[tid[t_j]] \wedge \text{ctrl3}[t_j] \neq \text{cexp3}[t_j]) \}$ |
| 361 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] \neq \text{cexp2}[t_j]) \}$<br>$\text{int } r \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] \neq \text{cexp2}[t_j]) \}$                                                                                                                                                                                  |
| 362 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] \neq \text{cexp2}[t_j]) \}$<br>$\text{cptr2}[tk] = \text{data\_addr}[r[tk]] \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] \neq \text{cexp2}[t_j]) \}$                                                                                                                                                    |
| 363 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] \neq \text{cexp2}[t_j]) \}$<br>$\text{ctrl2}[tk] = \text{mem\_control}[\text{cptr2}[tk]] \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] \neq \text{cexp2}[t_j]) \}$                                                                                                                                       |
| 364 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] \neq \text{cexp2}[t_j]) \}$<br>$\text{cexp2}[tk] = \text{control\_exp}[r[tk]] \{ (r[t_j] \geq 0 \wedge \text{cptr2}[t_j] == \text{data\_addr}[r[t_j]] \wedge \text{ctrl2}[t_j] == \text{mem\_control}[\text{cptr2}[t_j]] \wedge \text{cexp2}[t_j] == \text{control\_exp}[r[t_j]] \wedge \text{ctrl2}[t_j] \neq \text{cexp2}[t_j]) \}$                                                                                                                                                  |



| #   | Kind  | Spec                      | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----|-------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 365 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ ptr[tk] = data_addr[tid[tk]] $\{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$       |
| 366 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ cptr3[tk] = data_addr[tid[tk]] $\{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$     |
| 367 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ ctrl3[tk] = mem_control[cptr3[tk]] $\{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ |
| 368 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ cexp3[tk] = control_exp[tid[tk]] $\{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$   |
| 369 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ return r[tk] $\{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                       |
| 370 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ cptr1[tk] = data_addr[r[tk]] $\{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$       |
| 371 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ ctrl1[tk] = mem_control[cptr1[tk]] $\{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ |

| #   | Kind  | Spec                      | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|-------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 372 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} cexp1[tk] = control\_exp[r[tk]] \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$       |
| 373 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} assume(r[tk] == NULL) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                 |
| 374 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} assume(r[tk] \neq NULL) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$               |
| 375 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} assume(r[tk] < 0) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                     |
| 376 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} assume(r[tk] \geq 0) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                  |
| 377 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} assume(ctrl2[tk] == cexp2[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$        |
| 378 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} assume(r[tk] \neq data\_exp[tid[tk]]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ |

| #   | Kind  | Spec                      | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----|-------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 379 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} \text{ assume}(r[tk] == data\_exp[tid[tk]]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ |
| 380 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} \text{ assume}(ctrl13[tk] \neq cexp3[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$   |
| 381 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} \text{ assume}(ctrl13[tk] == cexp3[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$     |
| 382 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} \text{ assume}(r[tk] < 0) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                   |
| 383 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} \text{ assume}(r[tk] \geq 0) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                |
| 384 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} \text{ assume}(ctrl11[tk] \neq cexp1[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$   |
| 385 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} \text{ assume}(ctrl11[tk] == cexp1[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$     |

| #   | Kind  | Spec                      | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|-------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 386 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} \text{ assume}(mem\_data[cptr1[tk]] \neq r[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                                               |
| 387 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} \text{ CAS}(mem\_data[cptr1[tk]], r[tk], data\_exp[r[tk]]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                                    |
| 388 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} \text{ assume}(mem\_data[ptr[tk]] \neq data\_exp[tid[tk]] \wedge r[tk] == mem\_data[ptr[tk]]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ |
| 389 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} \text{ RCAS}(mem\_data[ptr[tk]], data\_exp[tid[tk]], tid[tk], r[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                          |
| 390 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} \text{ assume}(mem\_data[cptr2[tk]] \neq r[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                                               |
| 391 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} \text{ CAS}(mem\_data[cptr2[tk]], r[tk], data\_new[r[tk]]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                                    |

| #   | Kind  | Spec                      | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| 392 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ assume( $mem\_data[cptr2[tk]] \neq r[tk]$ ) $\{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                                               |
| 393 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ assume( $mem\_data[cptr3[tk]] \neq tid[tk]$ ) $\{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                                             |
| 394 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ CAS( $mem\_data[cptr3[tk]], tid[tk], data\_new[tid[tk]]$ ) $\{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                                |
| 395 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ assume( $mem\_data[cptr3[tk]] \neq tid[tk]$ ) $\{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                                             |
| 396 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ CAS( $mem\_data[cptr3[tk]], tid[tk], data\_exp[tid[tk]]$ ) $\{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                                |
| 397 | IStbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ assume( $mem\_data[ptr[tk]] \neq data\_exp[tid[tk]] \wedge r[tk] == mem\_data[ptr[tk]]$ ) $\{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ |

| #   | Kind  | Spec                      | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----|-------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 398 | ISTbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} RCAS(mem\_data[ptr[tk]], data\_exp[tid[tk]], tid[tk], r[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$ |
| 399 | ISTbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} assume(mem\_data[cptr1[tk]] \neq r[tk]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$                      |
| 400 | ISTbl | coop::cHelperRestoreFLoop | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\} CAS(mem\_data[cptr1[tk]], r[tk], data\_new[r[tk]]) \{(r[t_j] \geq 0 \wedge cptr2[t_j] == data\_addr[r[t_j]] \wedge ctrl2[t_j] == mem\_control[cptr2[t_j]] \wedge cexp2[t_j] == control\_exp[r[t_j]] \wedge ctrl2[t_j] \neq cexp2[t_j])\}$           |
| 401 | ISTbl | coop::cHelpFExit          | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == NULL\} int r \{r[t_j] == NULL\}$                                                                                                                                                                                                                                                                                                                                                                                              |
| 402 | ISTbl | coop::cHelpFExit          | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == NULL\} cptr2[tk] = data\_addr[r[tk]] \{r[t_j] == NULL\}$                                                                                                                                                                                                                                                                                                                                                                      |
| 403 | ISTbl | coop::cHelpFExit          | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == NULL\} ctrl2[tk] = mem\_control[cptr2[tk]] \{r[t_j] == NULL\}$                                                                                                                                                                                                                                                                                                                                                                |
| 404 | ISTbl | coop::cHelpFExit          | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == NULL\} cexp2[tk] = control\_exp[r[tk]] \{r[t_j] == NULL\}$                                                                                                                                                                                                                                                                                                                                                                    |
| 405 | ISTbl | coop::cHelpFExit          | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == NULL\} ptr[tk] = data\_addr[tid[tk]] \{r[t_j] == NULL\}$                                                                                                                                                                                                                                                                                                                                                                      |
| 406 | ISTbl | coop::cHelpFExit          | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == NULL\} cptr3[tk] = data\_addr[tid[tk]] \{r[t_j] == NULL\}$                                                                                                                                                                                                                                                                                                                                                                    |
| 407 | ISTbl | coop::cHelpFExit          | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == NULL\} ctrl3[tk] = mem\_control[cptr3[tk]] \{r[t_j] == NULL\}$                                                                                                                                                                                                                                                                                                                                                                |
| 408 | ISTbl | coop::cHelpFExit          | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == NULL\} cexp3[tk] = control\_exp[tid[tk]] \{r[t_j] == NULL\}$                                                                                                                                                                                                                                                                                                                                                                  |
| 409 | ISTbl | coop::cHelpFExit          | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == NULL\} return r[tk] \{r[t_j] == NULL\}$                                                                                                                                                                                                                                                                                                                                                                                       |
| 410 | ISTbl | coop::cHelpFExit          | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == NULL\} cptr1[tk] = data\_addr[r[tk]] \{r[t_j] == NULL\}$                                                                                                                                                                                                                                                                                                                                                                      |
| 411 | ISTbl | coop::cHelpFExit          | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == NULL\} ctrl1[tk] = mem\_control[cptr1[tk]] \{r[t_j] == NULL\}$                                                                                                                                                                                                                                                                                                                                                                |
| 412 | ISTbl | coop::cHelpFExit          | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == NULL\} cexp1[tk] = control\_exp[r[tk]] \{r[t_j] == NULL\}$                                                                                                                                                                                                                                                                                                                                                                    |
| 413 | ISTbl | coop::cHelpFExit          | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == NULL\} assume(r[tk] == NULL) \{r[t_j] == NULL\}$                                                                                                                                                                                                                                                                                                                                                                              |

| #   | Kind  | Spec             | Obligation                                                                                                                                                                                                                                  |
|-----|-------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 414 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\} \text{ assume}(r[tk] \neq \text{NULL})$<br>$\{r[t_j] == \text{NULL}\}$                                                                                                 |
| 415 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\} \text{ assume}(r[tk] < 0)$<br>$\{r[t_j] == \text{NULL}\}$                                                                                                              |
| 416 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\} \text{ assume}(r[tk] \geq 0)$<br>$\{r[t_j] == \text{NULL}\}$                                                                                                           |
| 417 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\} \text{ assume}(\text{ctrl2}[tk] \neq \text{cexp2}[tk]) \{r[t_j] == \text{NULL}\}$                                                                                      |
| 418 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\} \text{ assume}(\text{ctrl2}[tk] == \text{cexp2}[tk]) \{r[t_j] == \text{NULL}\}$                                                                                        |
| 419 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\} \text{ assume}(r[tk] \neq \text{data\_exp}[\text{tid}[tk]]) \{r[t_j] == \text{NULL}\}$                                                                                 |
| 420 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\} \text{ assume}(r[tk] == \text{data\_exp}[\text{tid}[tk]]) \{r[t_j] == \text{NULL}\}$                                                                                   |
| 421 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\} \text{ assume}(\text{ctrl3}[tk] \neq \text{cexp3}[tk]) \{r[t_j] == \text{NULL}\}$                                                                                      |
| 422 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\} \text{ assume}(\text{ctrl3}[tk] == \text{cexp3}[tk]) \{r[t_j] == \text{NULL}\}$                                                                                        |
| 423 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\} \text{ assume}(r[tk] < 0)$<br>$\{r[t_j] == \text{NULL}\}$                                                                                                              |
| 424 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\} \text{ assume}(r[tk] \geq 0)$<br>$\{r[t_j] == \text{NULL}\}$                                                                                                           |
| 425 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\} \text{ assume}(\text{ctrl1}[tk] \neq \text{cexp1}[tk]) \{r[t_j] == \text{NULL}\}$                                                                                      |
| 426 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\}$<br>$\text{assume}(\text{mem\_data}[\text{cptr1}[tk]] \neq r[tk]) \{r[t_j] == \text{NULL}\}$                                                                           |
| 427 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\}$<br>$\text{assume}(\text{mem\_data}[\text{ptr}[tk]] \neq \text{data\_exp}[\text{tid}[tk]] \wedge r[tk] == \text{mem\_data}[\text{ptr}[tk]]) \{r[t_j] == \text{NULL}\}$ |
| 428 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\}$<br>$\text{RCAS}(\text{mem\_data}[\text{ptr}[tk]], \text{data\_exp}[\text{tid}[tk]], \text{tid}[tk], r[tk]) \{r[t_j] == \text{NULL}\}$                                 |
| 429 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\}$<br>$\text{assume}(\text{mem\_data}[\text{cptr2}[tk]] \neq r[tk]) \{r[t_j] == \text{NULL}\}$                                                                           |
| 430 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\}$<br>$\text{CAS}(\text{mem\_data}[\text{cptr2}[tk]], r[tk], \text{data\_new}[r[tk]]) \{r[t_j] == \text{NULL}\}$                                                         |
| 431 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\}$<br>$\text{assume}(\text{mem\_data}[\text{cptr2}[tk]] \neq r[tk]) \{r[t_j] == \text{NULL}\}$                                                                           |
| 432 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\}$<br>$\text{CAS}(\text{mem\_data}[\text{cptr2}[tk]], r[tk], \text{data\_exp}[r[tk]]) \{r[t_j] == \text{NULL}\}$                                                         |
| 433 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\}$<br>$\text{assume}(\text{mem\_data}[\text{cptr3}[tk]] \neq \text{tid}[tk]) \{r[t_j] == \text{NULL}\}$                                                                  |
| 434 | IStbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\}$<br>$\text{CAS}(\text{mem\_data}[\text{cptr3}[tk]], \text{tid}[tk], \text{data\_new}[\text{tid}[tk]]) \{r[t_j] == \text{NULL}\}$                                       |

| #   | Kind  | Spec             | Obligation                                                                                                                                                                         |
|-----|-------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 435 | ISTbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\}$<br>assume(mem_data[cptr3[tk]] $\neq$ tid[tk]) {r[tj] == NULL}                                                |
| 436 | ISTbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\}$<br>CAS(mem_data[cptr3[tk]], tid[tk], data_exp[tid[tk]])<br>{r[tj] == NULL}                                   |
| 437 | ISTbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\}$<br>assume(mem_data[ptr[tk]] $\neq$ data_exp[tid[tk]] $\wedge$<br>r[tk] == mem_data[ptr[tk]]) {r[tj] == NULL} |
| 438 | ISTbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\}$<br>RCAS(mem_data[ptr[tk]], data_exp[tid[tk]], tid[tk],<br>r[tk]) {r[tj] == NULL}                             |
| 439 | ISTbl | coop::cHelpFExit | $\forall t_j, tk: \mathbb{N}, t_j \neq tk. \{r[t_j] == \text{NULL}\}$<br>assume(mem_data[cptr1[tk]] $\neq$ r[tk]) {r[tj] == NULL}                                                  |

### Veracity Program Examples (one per obligation kind)

```

RMov [OK].
1 int MAX_REGS = 0;

3 int[] mem_data = new int[0];

5 int[] mem_control = new int[0];

7 int MAX_TID = 0;

9 int[] control_addr = new int[0];

11 int[] control_exp = new int[0];

13 int[] data_addr = new int[0];

15 int[] data_exp = new int[0];

17 int[] data_new = new int[0];

19 int main() {
20   havoc MAX_REGS;
21   havoc mem_data;
22   havoc mem_control;
23   havoc MAX_TID;
24   havoc control_addr;
25   havoc control_exp;
26   havoc data_addr;
27   havoc data_exp;
28   havoc data_new;
29   int r_size = 0;
30   havoc r_size;
31   int[] r = new int[r_size];
32   int ptr_size = 0;

```



```

33 havoc ptr_size;
34 int[] ptr = new int[ptr_size];
35 int t = 1;
36 havoc t;
37 int tid_size = 0;
38 havoc tid_size;
39 int[] tid = new int[tid_size];
40 int cptr1_size = 0;
41 havoc cptr1_size;
42 int[] cptr1 = new int[cptr1_size];
43 int cptr2_size = 0;
44 havoc cptr2_size;
45 int[] cptr2 = new int[cptr2_size];
46 int cptr3_size = 0;
47 havoc cptr3_size;
48 int[] cptr3 = new int[cptr3_size];
49 commute_right_ctx (true) {
50   {
51     ptr[t] = data_addr[tid[t]];
52   }
53   {
54     havoc mem_data;
55   }

57   context: mem_data[ptr[t]] == data_exp[tid[t]]
58 }
59 return 0;
60}

```

```

HTrp [OK].
1 int MAX_REGS = 0;

3 int[] mem_data = new int[0];

5 int[] mem_control = new int[0];

7 int MAX_TID = 0;

9 int[] control_addr = new int[0];

11 int[] control_exp = new int[0];

13 int[] data_addr = new int[0];

15 int[] data_exp = new int[0];

17 int[] data_new = new int[0];

19 int main() {
20   havoc MAX_REGS;
21   havoc mem_data;
22   havoc mem_control;

```

```

23  havoc MAX_TID;
24  havoc control_addr;
25  havoc control_exp;
26  havoc data_addr;
27  havoc data_exp;
28  havoc data_new;
29  int r_size = 0;
30  havoc r_size;
31  int[] r = new int[r_size];
32  int ptr_size = 0;
33  havoc ptr_size;
34  int[] ptr = new int[ptr_size];
35  int t = 1;
36  havoc t;
37  int tid_size = 0;
38  havoc tid_size;
39  int[] tid = new int[tid_size];
40  assume(mem_data[ptr[t]] == data_exp[tid[t]]);
41  havoc r[t];
42  assume(mem_data[ptr[t]] == data_exp[tid[t]]);
43  assume(r[t] == mem_data[ptr[t]]);
44  mem_data[ptr[t]] = tid[t];
45  assert(exists old__mem_data_ptr_t__ : int . old__mem_data_ptr_t__ ==
46                                             data_exp[tid[t]] &&
47                                             mem_data[ptr[t]] == tid[t] &&
48                                             r[t] == old__mem_data_ptr_t__);
49  return 0;
50}

```

Subs [FAIL].

```

1 int MAX_REGS = 0;

3 int[] mem_data = new int[0];

5 int[] mem_control = new int[0];

7 int MAX_TID = 0;

9 int[] control_addr = new int[0];

11 int[] control_exp = new int[0];

13 int[] data_addr = new int[0];

15 int[] data_exp = new int[0];

17 int[] data_new = new int[0];

19 int main() {
20   havoc MAX_REGS;
21   havoc mem_data;
22   havoc mem_control;

```

```

23 havoc MAX_TID;
24 havoc control_addr;
25 havoc control_exp;
26 havoc data_addr;
27 havoc data_exp;
28 havoc data_new;
29 int ptr_size = 0;
30 havoc ptr_size;
31 int[] ptr = new int[ptr_size];
32 int r_size = 0;
33 havoc r_size;
34 int[] r = new int[r_size];
35 int ti = 1;
36 havoc ti;
37 int tid_size = 0;
38 havoc tid_size;
39 int[] tid = new int[tid_size];
40 int tj = 1;
41 havoc tj;
42 assume(exists old__mem_data_ptr_ti__ : int . ptr[tj] == data_addr[tid[tj]]
43                                     &&
44                                     ptr[ti] == data_addr[tid[ti]]
45                                     &&
46                                     old__mem_data_ptr_ti__ ==
47                                     data_exp[tid[ti]] &&
48                                     mem_data[ptr[ti]] == tid[ti]
49                                     &&
50                                     r[ti] ==
51                                     old__mem_data_ptr_ti__);
52 assert(mem_data[ptr[tj]] ≠ data_exp[tid[tj]] && r[tj] == mem_data[ptr[tj]]);
53 return 0;
54}

```

```

LMov [OK].
1int MAX_REGS = 0;

3int[] mem_data = new int[0];

5int[] mem_control = new int[0];

7int MAX_TID = 0;

9int[] control_addr = new int[0];

11int[] control_exp = new int[0];

13int[] data_addr = new int[0];

15int[] data_exp = new int[0];

17int[] data_new = new int[0];

```

```

19 int main() {
20   havoc MAX_REGS;
21   havoc mem_data;
22   havoc mem_control;
23   havoc MAX_TID;
24   havoc control_addr;
25   havoc control_exp;
26   havoc data_addr;
27   havoc data_exp;
28   havoc data_new;
29   int cptr3_size = 0;
30   havoc cptr3_size;
31   int[] cptr3 = new int[cptr3_size];
32   int t = 1;
33   havoc t;
34   int tid_size = 0;
35   havoc tid_size;
36   int[] tid = new int[tid_size];
37   int cptr1_size = 0;
38   havoc cptr1_size;
39   int[] cptr1 = new int[cptr1_size];
40   int cptr2_size = 0;
41   havoc cptr2_size;
42   int[] cptr2 = new int[cptr2_size];
43   int ptr_size = 0;
44   havoc ptr_size;
45   int[] ptr = new int[ptr_size];
46   int r_size = 0;
47   havoc r_size;
48   int[] r = new int[r_size];
49   commute_left (exists old__mem_data_cptr3_t__ : int . old__mem_data_cptr3_t__
50                                     == tid[t] &&
51                                     mem_data[cptr3[t]] ==
52                                     data_new[tid[t]]) {
53     {
54       int steps = 1;
55       havoc steps;
56       while (steps > 0) {
57         steps = steps - 1;
58         int tk = 1;
59         havoc tk;
60         assume(tk ≠ t);
61         int nd = 1;
62         havoc nd;
63         if (nd == 1) {
64           havoc mem_data[ptr[tk]];
65           assume(mem_data[ptr[tk]] == tid[tk]);
66           r[tk] = data_exp[tid[tk]];
67         } else {
68           if (nd == 2) {
69             havoc mem_data[cptr1[tk]];
70             assume(mem_data[cptr1[tk]] == data_new[r[tk]]);

```

```

71     } else {
72         if (nd == 3) {
73             havoc mem_data[cptr1[tk]];
74             assume(mem_data[cptr1[tk]] == data_exp[r[tk]]);
75         } else {
76             if (nd == 4) {
77                 havoc mem_data[ptr[tk]];
78                 assume(mem_data[ptr[tk]] == tid[tk]);
79                 r[tk] = data_exp[tid[tk]];
80             } else {
81                 if (nd == 5) {
82                     havoc mem_data[cptr2[tk]];
83                     assume(mem_data[cptr2[tk]] == data_new[r[tk]]);
84                 } else {
85                     if (nd == 6) {
86                         havoc mem_data[cptr2[tk]];
87                         assume(mem_data[cptr2[tk]] == data_exp[r[tk]]);
88                     } else {
89                         if (nd == 7) {
90                             havoc mem_data[cptr3[tk]];
91                             assume(mem_data[cptr3[tk]] == data_new[tid[tk]]);
92                         } else {
93                             if (true) {
94                                 havoc mem_data[cptr3[tk]];
95                                 assume(mem_data[cptr3[tk]] == data_exp[tid[tk]]);
96                             }
97                         }
98                     }
99                 }
100             }
101         }
102     }
103 }
104 }
105 }
106 {
107     return r[t];
108 }

110 }
111 return 0;
112}

```

## B.5 treiber

30 obligations across 9 specs: atom::aEmptyS, atom::aPopRd, atom::aPopRd.N, atom::aPopS, atom::aPushS, atom::aPushS.R, coop::cPopF, coop::cPushF, coop::cPushF.R.

### B.5.1 Sources.

#### Implementation (.cl)

```

1/* Treiber Stack */

3tloc Top = NULL;

5invariant Top ≠ NULL;

7int push(int val) {
8  while (true) {
9    tloc top = Top;
10   tloc ntop = new_cell();
11   ntop→val = val;
12   ntop→next = top;
13   if CAS(Top, top, ntop) {
14     return 1;
15   }
16 }
17}

19int pop() {
20  while (true) {
21    tloc top = Top;
22    if (top == NULL) {
23      return 0;
24    }
25    else {
26      tloc next = top→next;
27      int pval = top→val;
28      if CAS(Top, top, next) {
29        return pval;
30      }
31    }
32  }
33}

```

### Labeled program (.c1l)

```

1/*****
2/* [generated] KAT-friendly program with label annotations */
3/* -- */
4/* treiber.cl */
5/*****

7tloc Top = NULL;

9invariant Top ≠ NULL;

11int push(int val) {
12  while (true) {
13    [@pua1] tloc top = Top;
14    [@pua2] tloc ntop = new {0 {} NULL};
15    [@pua3:L] ntop→val = val;

```

```

16  [@pua4:L] ntop→next = top;
17  if * {
18      [@pucas5t] CAS(Top, top, ntop);
19      [@pua6] return 1;
20  }
21  else {
22      [@pucas5f] assume(Top ≠ top);
23  }
24  }
25}

27int pop() {
28  while (true) {
29      [@poa1] tloc top = Top;
30      if * {
31          [@pob2t] assume(top == NULL);
32          [@poa3] return 0;
33      }
34      else {
35          [@pob2f] assume(top ≠ NULL);
36          [@poa4] tloc next = top→next;
37          [@poa5] int pval = top→val;
38          if * {
39              [@pocas6t] CAS(Top, top, next);
40              [@poa7] return pval;
41          }
42          else {
43              [@pocas6f] assume(Top ≠ top);
44          }
45      }
46  }
47}

```

### B.5.2 *IRSpec.*

#### IR specifications (.irspecs)

```

1/* OBJECT: Treiber Stack */
2/* Push happy path */
3aPushS = atom{
4  ind = pucas5t,
5  red = [pua1; pua2; pua3; pua4] |> pucas5t <| pua6
6}

8/* Reading the top value */
9aPopRd = atom{
10 ind = poa4,
11 red = EPS |> [poa1; pob2f] <| poa4; poa5
12}

14/* Pop happy path */
15aPopS = atom{

```

```

16 ind = pocas6t,
17 red = [aPopRd] |> pocas6t <| poa7
18}

20/* Empty-pop path */
21aEmptyS = atom{
22 ind = poa3,
23 red = [poa1; pob2t; poa3]
24}

26/* Push observes a stale Top and fails */
27cPushF = coop{
28 ind = pucas5t + pocas6t,
29 red = [pua1; pua2; pua3; pua4] |> [aPushS] + [aPopS] <| pucas5f
30}

32/* Pop observes a stale Top and fails */
33cPopF = coop{
34 ind = pucas5t + pocas6t,
35 red = [aPopRd] |> [aPushS] + [aPopS] <| pocas6f
36}

```

### B.5.3 Contextual Commutativity (20).

| #  | Kind | Spec           | Obligation                                                                                                                                                                                                                                                                                            | Result |
|----|------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1  | Inv  |                | Top $\neq$ NULL                                                                                                                                                                                                                                                                                       | OK     |
| 2  | Atom | atom::aPushS.R | [tloc top = Top; tloc ntop = new {0 {} NULL}; ntop->val = val; ntop->next = top]                                                                                                                                                                                                                      | OK     |
| 3  | RMov | atom::aPushS   | $\forall t:\mathbb{N}. \text{top}[t] = \text{Top}; \text{ntop}[t] = \text{tloc\_pua2\_t}; \text{ntop}[t]\text{->val} = \text{val}[t]; \text{ntop}[t]\text{->next} = \text{top}[t] \triangleright \{(\text{Top} == \text{top}[t] \wedge \text{ntop}[t] \neq \text{NULL})\}$                            | OK     |
| 4  | HTrp | atom::aPushS   | $\forall t:\mathbb{N}. \{(\text{Top} == \text{top}[t] \wedge \text{ntop}[t] \neq \text{NULL})\} \text{CAS}(\text{Top}, \text{top}[t], \text{ntop}[t]) \{ \exists \text{old\_Top}. (\text{old\_Top} == \text{top}[t] \wedge \text{ntop}[t] \neq \text{NULL} \wedge \text{Top} == \text{ntop}[t]) \}$   | OK     |
| 5  | LMov | atom::aPushS   | $\forall t:\mathbb{N}. \{ \exists \text{old\_Top}. (\text{old\_Top} == \text{top}[t] \wedge \text{ntop}[t] \neq \text{NULL} \wedge \text{Top} == \text{ntop}[t]) \} \triangleleft \text{return } 1$                                                                                                   | OK     |
| 6  | Atom | atom::aPopRd.N | [tloc top = Top; assume(top $\neq$ NULL)]                                                                                                                                                                                                                                                             | OK     |
| 7  | HTrp | atom::aPopRd   | $\forall t:\mathbb{N}. \{ \text{Top} \neq \text{NULL} \} [\text{top}[t] = \text{Top}; \text{assume}(\text{top}[t] \neq \text{NULL})] \{ (\text{Top} \neq \text{NULL} \wedge \text{top}[t] == \text{Top} \wedge \text{top}[t] \neq \text{NULL}) \}$                                                    | OK     |
| 8  | LMov | atom::aPopRd   | $\forall t:\mathbb{N}. \{ (\text{Top} \neq \text{NULL} \wedge \text{top}[t] == \text{Top} \wedge \text{top}[t] \neq \text{NULL}) \} \triangleleft \text{next}[t] = \text{top}[t]\text{->next}$                                                                                                        | OK     |
| 9  | LMov | atom::aPopRd   | $\forall t:\mathbb{N}. \{ (\text{Top} \neq \text{NULL} \wedge \text{top}[t] == \text{Top} \wedge \text{top}[t] \neq \text{NULL} \wedge \text{next}[t] == \text{top}[t]\text{->next}) \} \triangleleft \text{pval}[t] = \text{top}[t]\text{->val}$                                                     | OK     |
| 10 | RMov | atom::aPopS    | $\forall t:\mathbb{N}. \text{top}[t] = \text{Top}; \text{assume}(\text{top}[t] \neq \text{NULL}); \text{next}[t] = \text{top}[t]\text{->next}; \text{pval}[t] = \text{top}[t]\text{->val} \triangleright \{ (\text{Top} == \text{top}[t] \wedge \text{next}[t] \neq \text{NULL}) \}$                  | OK     |
| 11 | HTrp | atom::aPopS    | $\forall t:\mathbb{N}. \{ (\text{Top} == \text{top}[t] \wedge \text{next}[t] \neq \text{NULL}) \} \text{CAS}(\text{Top}, \text{top}[t], \text{next}[t]) \{ \exists \text{old\_Top}. (\text{old\_Top} == \text{top}[t] \wedge \text{next}[t] \neq \text{NULL} \wedge \text{Top} == \text{next}[t]) \}$ | OK     |



| #  | Kind | Spec           | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Result |
|----|------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 12 | LMov | atom::aPopS    | $\forall t:\mathbb{N}. \{\exists \text{old\_Top}. (\text{old\_Top} == \text{top}[t] \wedge \text{next}[t] \neq \text{NULL} \wedge \text{Top} == \text{next}[t])\} \triangleleft \text{return pval}[t]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | OK     |
| 13 | Atom | atom::aEmptyS  | $[\text{tloc top} = \text{Top}; \text{assume}(\text{top} == \text{NULL}); \text{return } 0]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | OK     |
| 14 | Atom | coop::cPushF.R | $[\text{tloc top} = \text{Top}; \text{tloc ntop} = \text{new } \{0\} \text{ NULL}; \text{ntop} \rightarrow \text{val} = \text{val}; \text{ntop} \rightarrow \text{next} = \text{top}]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | OK     |
| 15 | RMov | coop::cPushF   | $\forall tj:\mathbb{N}. \text{top}[tj] = \text{Top}; \text{ntop}[tj] = \text{tloc\_pua2\_tj}; \text{ntop}[tj] \rightarrow \text{val} = \text{val}[tj]; \text{ntop}[tj] \rightarrow \text{next} = \text{top}[tj] \triangleright \{(\text{Top} \neq \text{NULL} \wedge \text{top}[tj] == \text{Top} \wedge \text{ntop}[tj] == \text{tloc\_pua2\_tj} \wedge \text{ntop}[tj] \rightarrow \text{val} == \text{val}[tj] \wedge \text{ntop}[tj] \rightarrow \text{next} == \text{top}[tj])\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | OK     |
| 16 | HTrp | coop::cPushF   | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{(\text{Top} \neq \text{NULL} \wedge \text{top}[tj] == \text{Top} \wedge \text{ntop}[tj] == \text{tloc\_pua2\_tj} \wedge \text{ntop}[tj] \rightarrow \text{val} == \text{val}[tj] \wedge \text{ntop}[tj] \rightarrow \text{next} == \text{top}[tj])\} ([[\text{top}[ti] = \text{Top}; \text{ntop}[ti] = \text{tloc\_pua2\_ti}; \text{ntop}[ti] \rightarrow \text{val} = \text{val}[ti]; \text{ntop}[ti] \rightarrow \text{next} = \text{top}[ti]]; \text{CAS}(\text{Top}, \text{top}[ti], \text{ntop}[ti]); \text{return } 1] + [[\text{skip}; [\text{top}[ti] = \text{Top}; \text{assume}(\text{top}[ti] \neq \text{NULL})]; \text{next}[ti] = \text{top}[ti] \rightarrow \text{next}; \text{pval}[ti] = \text{top}[ti] \rightarrow \text{val}; \text{CAS}(\text{Top}, \text{top}[ti], \text{next}[ti]); \text{return pval}[ti]]) \{(\exists \text{old\_Top}. (\text{old\_Top} \neq \text{NULL} \wedge \text{top}[tj] == \text{old\_Top} \wedge \text{ntop}[tj] == \text{tloc\_pua2\_tj} \wedge \text{ntop}[tj] \rightarrow \text{val} == \text{val}[tj] \wedge \text{ntop}[tj] \rightarrow \text{next} == \text{top}[tj] \wedge \text{top}[ti] == \text{old\_Top} \wedge \text{ntop}[ti] == \text{tloc\_pua2\_ti} \wedge \text{ntop}[ti] \rightarrow \text{val} == \text{val}[ti] \wedge \text{ntop}[ti] \rightarrow \text{next} == \text{top}[ti] \wedge \text{Top} == \text{ntop}[ti]) \vee \exists \text{old\_Top}. (\text{old\_Top} \neq \text{NULL} \wedge \text{top}[tj] == \text{old\_Top} \wedge \text{ntop}[tj] == \text{tloc\_pua2\_tj} \wedge \text{ntop}[tj] \rightarrow \text{val} == \text{val}[tj] \wedge \text{ntop}[tj] \rightarrow \text{next} == \text{top}[tj] \wedge \text{top}[ti] == \text{old\_Top} \wedge \text{top}[ti] \neq \text{NULL} \wedge \text{next}[ti] == \text{top}[ti] \rightarrow \text{next} \wedge \text{pval}[ti] == \text{top}[ti] \rightarrow \text{val} \wedge \text{Top} == \text{next}[ti]))\}$ | OK     |
| 17 | Subs | coop::cPushF   | $\forall ti, tj:\mathbb{N}, ti \neq tj. \{(\exists \text{old\_Top}. (\text{old\_Top} \neq \text{NULL} \wedge \text{top}[tj] == \text{old\_Top} \wedge \text{ntop}[tj] == \text{tloc\_pua2\_tj} \wedge \text{ntop}[tj] \rightarrow \text{val} == \text{val}[tj] \wedge \text{ntop}[tj] \rightarrow \text{next} == \text{top}[tj] \wedge \text{top}[ti] == \text{old\_Top} \wedge \text{ntop}[ti] == \text{tloc\_pua2\_ti} \wedge \text{ntop}[ti] \rightarrow \text{val} == \text{val}[ti] \wedge \text{ntop}[ti] \rightarrow \text{next} == \text{top}[ti] \wedge \text{Top} == \text{ntop}[ti]) \vee \exists \text{old\_Top}. (\text{old\_Top} \neq \text{NULL} \wedge \text{top}[tj] == \text{old\_Top} \wedge \text{ntop}[tj] == \text{tloc\_pua2\_tj} \wedge \text{ntop}[tj] \rightarrow \text{val} == \text{val}[tj] \wedge \text{ntop}[tj] \rightarrow \text{next} == \text{top}[tj] \wedge \text{top}[ti] == \text{old\_Top} \wedge \text{top}[ti] \neq \text{NULL} \wedge \text{next}[ti] == \text{top}[ti] \rightarrow \text{next} \wedge \text{pval}[ti] == \text{top}[ti] \rightarrow \text{val} \wedge \text{Top} == \text{next}[ti]))\} \Rightarrow \{\text{Top} \neq \text{top}[tj]\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | OK     |
| 18 | RMov | coop::cPopF    | $\forall tj:\mathbb{N}. \text{top}[tj] = \text{Top}; \text{assume}(\text{top}[tj] \neq \text{NULL}); \text{next}[tj] = \text{top}[tj] \rightarrow \text{next}; \text{pval}[tj] = \text{top}[tj] \rightarrow \text{val} \triangleright \{(\text{Top} \neq \text{NULL} \wedge \text{top}[tj] == \text{Top} \wedge \text{top}[tj] \neq \text{NULL} \wedge \text{next}[tj] == \text{top}[tj] \rightarrow \text{next} \wedge \text{pval}[tj] == \text{top}[tj] \rightarrow \text{val})\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | OK     |

| #  | Kind | Spec        | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Result |
|----|------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 19 | HTrp | coop::cPopF | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ (Top \neq \text{NULL} \wedge top[tj] == Top \wedge top[tj] \neq \text{NULL} \wedge next[tj] == top[tj] \rightarrow next \wedge pval[tj] == top[tj] \rightarrow val) \} ( [[top[ti] = Top; ntop[ti] = tloc\_pua2\_ti; ntop[ti] \rightarrow val = val[ti]; ntop[ti] \rightarrow next = top[ti]]; CAS(Top, top[ti], ntop[ti]); return 1] + [[skip; [top[ti] = Top; assume(top[ti] \neq \text{NULL})]; next[ti] = top[ti] \rightarrow next; pval[ti] = top[ti] \rightarrow val; CAS(Top, top[ti], next[ti]); return pval[ti]]] ) \{ (\exists old\_Top. (old\_Top \neq \text{NULL} \wedge top[tj] == old\_Top \wedge top[tj] \neq \text{NULL} \wedge next[tj] == top[tj] \rightarrow next \wedge pval[tj] == top[tj] \rightarrow val \wedge top[ti] == old\_Top \wedge ntop[ti] == tloc\_pua2\_ti \wedge ntop[ti] \rightarrow val == val[ti] \wedge ntop[ti] \rightarrow next == top[ti] \wedge Top == ntop[ti]) \vee \exists old\_Top. (old\_Top \neq \text{NULL} \wedge top[tj] == old\_Top \wedge top[tj] \neq \text{NULL} \wedge next[tj] == top[tj] \rightarrow next \wedge pval[tj] == top[tj] \rightarrow val \wedge top[ti] == old\_Top \wedge top[ti] \neq \text{NULL} \wedge next[ti] == top[ti] \rightarrow next \wedge pval[ti] == top[ti] \rightarrow val \wedge Top == next[ti])) \} \Rightarrow \{ Top \neq top[tj] \}$ | OK     |
| 20 | Subs | coop::cPopF | $\forall ti, tj: \mathbb{N}, ti \neq tj. \{ (\exists old\_Top. (old\_Top \neq \text{NULL} \wedge top[tj] == old\_Top \wedge top[tj] \neq \text{NULL} \wedge next[tj] == top[tj] \rightarrow next \wedge pval[tj] == top[tj] \rightarrow val \wedge top[ti] == old\_Top \wedge ntop[ti] == tloc\_pua2\_ti \wedge ntop[ti] \rightarrow val == val[ti] \wedge ntop[ti] \rightarrow next == top[ti] \wedge Top == ntop[ti]) \vee \exists old\_Top. (old\_Top \neq \text{NULL} \wedge top[tj] == old\_Top \wedge top[tj] \neq \text{NULL} \wedge next[tj] == top[tj] \rightarrow next \wedge pval[tj] == top[tj] \rightarrow val \wedge top[ti] == old\_Top \wedge top[ti] \neq \text{NULL} \wedge next[ti] == top[ti] \rightarrow next \wedge pval[ti] == top[ti] \rightarrow val \wedge Top == next[ti])) \} \Rightarrow \{ Top \neq top[tj] \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | OK     |

#### B.5.4 Automata (53).

coverage.

| Check         | Result |
|---------------|--------|
| push: COVERED | OK     |
| pop: COVERED  | OK     |

non-overlapping.

1 all distinct pairs are non-overlapping

non-self-overlapping.

1 no spec self-overlaps

bijection part I: well-definedness.

| Check                                                     | Result |
|-----------------------------------------------------------|--------|
| cPushF: induction_pattern $\subseteq$ exactly_one_ais: OK | OK     |
| aPopS: induction_pattern $\subseteq$ exactly_one_ais: OK  | OK     |
| aPopRd: induction_pattern $\subseteq$ exactly_one_ais: OK | OK     |

| Check                                                      | Result |
|------------------------------------------------------------|--------|
| aEmptyS: induction_pattern $\subseteq$ exactly_one_ais: OK | OK     |
| aPushS: induction_pattern $\subseteq$ exactly_one_ais: OK  | OK     |
| cPopF: induction_pattern $\subseteq$ exactly_one_ais: OK   | OK     |

*bijection part II: surjectivity.*

| Check                                                      | Result |
|------------------------------------------------------------|--------|
| aPopRd $\cap$ at_least_one_ais(aEmptyS) = $\emptyset$ : OK | OK     |
| aPopS $\cap$ at_least_one_ais(aEmptyS) = $\emptyset$ : OK  | OK     |
| aPushS $\cap$ at_least_one_ais(aEmptyS) = $\emptyset$ : OK | OK     |
| cPopF $\cap$ at_least_one_ais(aEmptyS) = $\emptyset$ : OK  | OK     |
| cPushF $\cap$ at_least_one_ais(aEmptyS) = $\emptyset$ : OK | OK     |
| aEmptyS $\cap$ at_least_one_ais(aPopRd) = $\emptyset$ : OK | OK     |
| aPopS $\subseteq$ at_most_one_ais(aPopRd): OK              | OK     |
| aPushS $\cap$ at_least_one_ais(aPopRd) = $\emptyset$ : OK  | OK     |
| cPopF $\subseteq$ at_most_one_ais(aPopRd): OK              | OK     |
| cPushF $\subseteq$ at_most_one_ais(aPopRd): OK             | OK     |
| aEmptyS $\cap$ at_least_one_ais(aPopS) = $\emptyset$ : OK  | OK     |
| aPushS $\cap$ at_least_one_ais(aPopS) = $\emptyset$ : OK   | OK     |
| cPopF $\subseteq$ at_most_one_ais(aPopS): OK               | OK     |
| cPushF $\subseteq$ at_most_one_ais(aPopS): OK              | OK     |
| aEmptyS $\cap$ at_least_one_ais(aPushS) = $\emptyset$ : OK | OK     |
| aPopRd $\cap$ at_least_one_ais(aPushS) = $\emptyset$ : OK  | OK     |
| aPopS $\cap$ at_least_one_ais(aPushS) = $\emptyset$ : OK   | OK     |
| cPopF $\subseteq$ at_most_one_ais(aPushS): OK              | OK     |
| cPushF $\subseteq$ at_most_one_ais(aPushS): OK             | OK     |
| aEmptyS $\cap$ at_least_one_ais(cPopF) = $\emptyset$ : OK  | OK     |
| cPushF $\cap$ at_least_one_ais(cPopF) = $\emptyset$ : OK   | OK     |
| aEmptyS $\cap$ at_least_one_ais(cPushF) = $\emptyset$ : OK | OK     |
| cPopF $\cap$ at_least_one_ais(cPushF) = $\emptyset$ : OK   | OK     |

*bijection support (IRSspecs).*

```

1 aPopS: {poa4, poa5, poa7, pob2f, pocas6t}
2 aPopRd: {poa4, poa5, pob2f}
3 aEmptyS: {poa3, pob2t}
4 aPushS: {pua1, pua2, pua3, pua4, pua6, pucas5t}
5 cPushF: {poa4, poa5, poa7, pob2f, pocas6t, pua1, pua2, pua3, pua4, pua6, pucas5t}
6 cPopF: {poa4, poa5, poa7, pob2f, pocas6t, pua1, pua2, pua3, pua4, pua6, pucas5t}

```

### B.5.5 Interference Stability (10).

| # | Kind  | Spec         | Obligation                                                                                                                                                                                                                                                                                                                                                                                               | Result |
|---|-------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1 | IStbl | coop::cPushF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Top \neq NULL \wedge top[tj] == Top \wedge ntop[tj] == tloc\_pua2\_tj \wedge ntop[tj] \rightarrow val == val[tj] \wedge ntop[tj] \rightarrow next == top[tj])\} top[tk] = Top \{(Top \neq NULL \wedge top[tj] == Top \wedge ntop[tj] == tloc\_pua2\_tj \wedge ntop[tj] \rightarrow val == val[tj] \wedge ntop[tj] \rightarrow next == top[tj])\}$            | OK     |
| 2 | IStbl | coop::cPushF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Top \neq NULL \wedge top[tj] == Top \wedge ntop[tj] == tloc\_pua2\_tj \wedge ntop[tj] \rightarrow val == val[tj] \wedge ntop[tj] \rightarrow next == top[tj])\} return \emptyset \{(Top \neq NULL \wedge top[tj] == Top \wedge ntop[tj] == tloc\_pua2\_tj \wedge ntop[tj] \rightarrow val == val[tj] \wedge ntop[tj] \rightarrow next == top[tj])\}$         | OK     |
| 3 | IStbl | coop::cPushF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Top \neq NULL \wedge top[tj] == Top \wedge ntop[tj] == tloc\_pua2\_tj \wedge ntop[tj] \rightarrow val == val[tj] \wedge ntop[tj] \rightarrow next == top[tj])\} assume(top[tk] == NULL) \{(Top \neq NULL \wedge top[tj] == Top \wedge ntop[tj] == tloc\_pua2\_tj \wedge ntop[tj] \rightarrow val == val[tj] \wedge ntop[tj] \rightarrow next == top[tj])\}$  | OK     |
| 4 | IStbl | coop::cPushF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Top \neq NULL \wedge top[tj] == Top \wedge ntop[tj] == tloc\_pua2\_tj \wedge ntop[tj] \rightarrow val == val[tj] \wedge ntop[tj] \rightarrow next == top[tj])\} assume(Top \neq top[tk]) \{(Top \neq NULL \wedge top[tj] == Top \wedge ntop[tj] == tloc\_pua2\_tj \wedge ntop[tj] \rightarrow val == val[tj] \wedge ntop[tj] \rightarrow next == top[tj])\}$ | OK     |
| 5 | IStbl | coop::cPushF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Top \neq NULL \wedge top[tj] == Top \wedge ntop[tj] == tloc\_pua2\_tj \wedge ntop[tj] \rightarrow val == val[tj] \wedge ntop[tj] \rightarrow next == top[tj])\} assume(Top \neq top[tk]) \{(Top \neq NULL \wedge top[tj] == Top \wedge ntop[tj] == tloc\_pua2\_tj \wedge ntop[tj] \rightarrow val == val[tj] \wedge ntop[tj] \rightarrow next == top[tj])\}$ | OK     |
| 6 | IStbl | coop::cPopF  | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Top \neq NULL \wedge top[tj] == Top \wedge top[tj] \neq NULL \wedge next[tj] == top[tj] \rightarrow next \wedge pval[tj] == top[tj] \rightarrow val)\} top[tk] = Top \{(Top \neq NULL \wedge top[tj] == Top \wedge top[tj] \neq NULL \wedge next[tj] == top[tj] \rightarrow next \wedge pval[tj] == top[tj] \rightarrow val)\}$                              | OK     |
| 7 | IStbl | coop::cPopF  | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Top \neq NULL \wedge top[tj] == Top \wedge top[tj] \neq NULL \wedge next[tj] == top[tj] \rightarrow next \wedge pval[tj] == top[tj] \rightarrow val)\} return \emptyset \{(Top \neq NULL \wedge top[tj] == Top \wedge top[tj] \neq NULL \wedge next[tj] == top[tj] \rightarrow next \wedge pval[tj] == top[tj] \rightarrow val)\}$                           | OK     |
| 8 | IStbl | coop::cPopF  | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Top \neq NULL \wedge top[tj] == Top \wedge top[tj] \neq NULL \wedge next[tj] == top[tj] \rightarrow next \wedge pval[tj] == top[tj] \rightarrow val)\} assume(top[tk] == NULL) \{(Top \neq NULL \wedge top[tj] == Top \wedge top[tj] \neq NULL \wedge next[tj] == top[tj] \rightarrow next \wedge pval[tj] == top[tj] \rightarrow val)\}$                    | OK     |
| 9 | IStbl | coop::cPopF  | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{(Top \neq NULL \wedge top[tj] == Top \wedge top[tj] \neq NULL \wedge next[tj] == top[tj] \rightarrow next \wedge pval[tj] == top[tj] \rightarrow val)\} assume(Top \neq top[tk]) \{(Top \neq NULL \wedge top[tj] == Top \wedge top[tj] \neq NULL \wedge next[tj] == top[tj] \rightarrow next \wedge pval[tj] == top[tj] \rightarrow val)\}$                   | OK     |

| #  | Kind  | Spec        | Obligation                                                                                                                                                                                                                                                                                                                                                                                                                      | Result |
|----|-------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 10 | ISTbl | coop::cPopF | $\forall tj, tk: \mathbb{N}, tj \neq tk. \{ (Top \neq \text{NULL} \wedge top[tj] == Top \wedge top[tj] \neq \text{NULL} \wedge next[tj] == top[tj] \rightarrow next \wedge pval[tj] == top[tj] \rightarrow val) \}$ $\text{assume}(Top \neq top[tk]) \{ (Top \neq \text{NULL} \wedge top[tj] == Top \wedge top[tj] \neq \text{NULL} \wedge next[tj] == top[tj] \rightarrow next \wedge pval[tj] == top[tj] \rightarrow val) \}$ | OK     |

### Veracity Program Examples (one per obligation kind)

```

RMov [OK].
1 tloc Top = null;

3 int main() {
4   havoc Top;
5   assume(Top ≠ null);
6   int top_size = 0;
7   havoc top_size;
8   tloc[] top = new tloc[top_size];
9   int ntop_size = 0;
10  havoc ntop_size;
11  tloc[] ntop = new tloc[ntop_size];
12  int t = 1;
13  havoc t;
14  int next_size = 0;
15  havoc next_size;
16  tloc[] next = new tloc[next_size];
17  int val_size = 0;
18  havoc val_size;
19  int[] val = new int[val_size];
20  tloc tloc_pua2_t = null;
21  havoc tloc_pua2_t;
22  tloc_pua2_t = new {0{}null};
23  commute_right_ctx (Top ≠ null && ntop[t] ≠ null) {
24    {
25      havoc top[t];
26      assume(top[t] ≠ null);
27      ntop[t] = tloc_pua2_t;
28      ntop[t]→value = val[t];
29      ntop[t]→next = top[t];
30    }
31    {
32      havoc Top;
33    }

35    context: Top ≠ null && Top == top[t] && ntop[t] ≠ null
36  }
37  return 0;
38}

```

```

HTrp [OK].
1 tloc Top = null;

```

```

3int main() {
4  havoc Top;
5  int ntop_size = 0;
6  havoc ntop_size;
7  tloc[] ntop = new tloc[ntop_size];
8  int t = 1;
9  havoc t;
10 int top_size = 0;
11 havoc top_size;
12 tloc[] top = new tloc[top_size];
13 assume(Top == top[t] && ntop[t] ≠ null);
14 assume(Top == top[t]);
15 Top = ntop[t];
16 assert(exists old__Top : tloc . old__Top == top[t] && ntop[t] ≠ null &&
17        Top == ntop[t]);
18 return 0;
19}

```

```

LMov [OK].
1tloc Top = null;

3int main() {
4  havoc Top;
5  assume(Top ≠ null);
6  int ntop_size = 0;
7  havoc ntop_size;
8  tloc[] ntop = new tloc[ntop_size];
9  int t = 1;
10 havoc t;
11 int top_size = 0;
12 havoc top_size;
13 tloc[] top = new tloc[top_size];
14 int next_size = 0;
15 havoc next_size;
16 tloc[] next = new tloc[next_size];
17 commute_left (exists old__Top : tloc . old__Top == top[t] &&
18               ntop[t] ≠ null && Top == ntop[t]) {
19   {
20     int steps = 1;
21     havoc steps;
22     while (steps > 0) invariant Top ≠ null {
23       steps = steps - 1;
24       int tk = 1;
25       havoc tk;
26       assume(tk ≠ t);
27       int nd = 1;
28       havoc nd;
29       if (nd == 1) {
30         havoc Top;
31         assume(Top == ntop[tk]);
32       } else {
33         if (true) {

```

```

34         havoc Top;
35         assume(Top == next[tk]);
36     }
37 }
38 }
39 }
40 {
41     return 1;
42 }

44 }
45 return 0;
46 }

```

Subs [OK].

```

1 tloc Top = null;

3 int main() {
4     havoc Top;
5     int next_size = 0;
6     havoc next_size;
7     tloc[] next = new tloc[next_size];
8     int ntop_size = 0;
9     havoc ntop_size;
10    tloc[] ntop = new tloc[ntop_size];
11    int pval_size = 0;
12    havoc pval_size;
13    int[] pval = new int[pval_size];
14    int ti = 1;
15    havoc ti;
16    int tj = 1;
17    havoc tj;
18    int top_size = 0;
19    havoc top_size;
20    tloc[] top = new tloc[top_size];
21    int val_size = 0;
22    havoc val_size;
23    int[] val = new int[val_size];
24    tloc tloc_pua2_ti = null;
25    havoc tloc_pua2_ti;
26    tloc tloc_pua2_tj = null;
27    havoc tloc_pua2_tj;
28    tloc_pua2_ti = new {0}{null};
29    tloc_pua2_tj = new {0}{null};
30    assume(exists old__Top : tloc . old__Top ≠ null && top[tj] == old__Top &&
31          ntop[tj] == tloc_pua2_tj &&
32          ntop[tj]→value == val[tj] &&
33          ntop[tj]→next == top[tj] &&
34          top[ti] == old__Top &&
35          ntop[ti] == tloc_pua2_ti &&
36          ntop[ti]→value == val[ti] &&
37          ntop[ti]→next == top[ti] &&

```

```

38                                     Top == ntop[ti]
39     ||
40     exists old__Top : tloc . old__Top ≠ null && top[tj] == old__Top &&
41                             ntop[tj] == tloc_pua2_tj &&
42                             ntop[tj]→value == val[tj] &&
43                             ntop[tj]→next == top[tj] &&
44                             top[ti] == old__Top && top[ti] ≠ null &&
45                             next[ti] == top[ti]→next &&
46                             pval[ti] == top[ti]→value &&
47                             Top == next[ti]);
48 assert(Top ≠ top[tj]);
49 return 0;
50 }

```